The data file AMMONIAEX (based on AMUJEL format): Additional Atomic and Molecular Data for EIRENE for AMMONIA

S. Touchard
LSPM – 99 avenue J.B. Clement
93430 Villetaneuse
FR

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Available via e-mail from sylvain.touchard@lspm.cnrs.fr

Contents

Introduction

Record:

- April 17: First version

to be done:

- To be done

Additional atomic data fits, read by EIRENE Specific reaction for AMMONIAC
Format as HYDHEL DATEN [2] or METHANE DATEN [1] or AMJUEL DATEN
SPECIFIC BIBLIOGAPHY IS AVIABLE

End of preface:

This next string is searched by EIRENE in subroutine SLREAC to initialize search for a particular set of fit coefficients. From here on, a character string **'H.n'** , n an integer, must only appear in the section title, but not in the text. Likewise: identifiers p0, a0, b0, ...,h0, k0 are used in SLREAC and must not appear in the text elsewhere, from here on.

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.                                      .
.       ##BEGIN DATA HERE##            .
.                                      .
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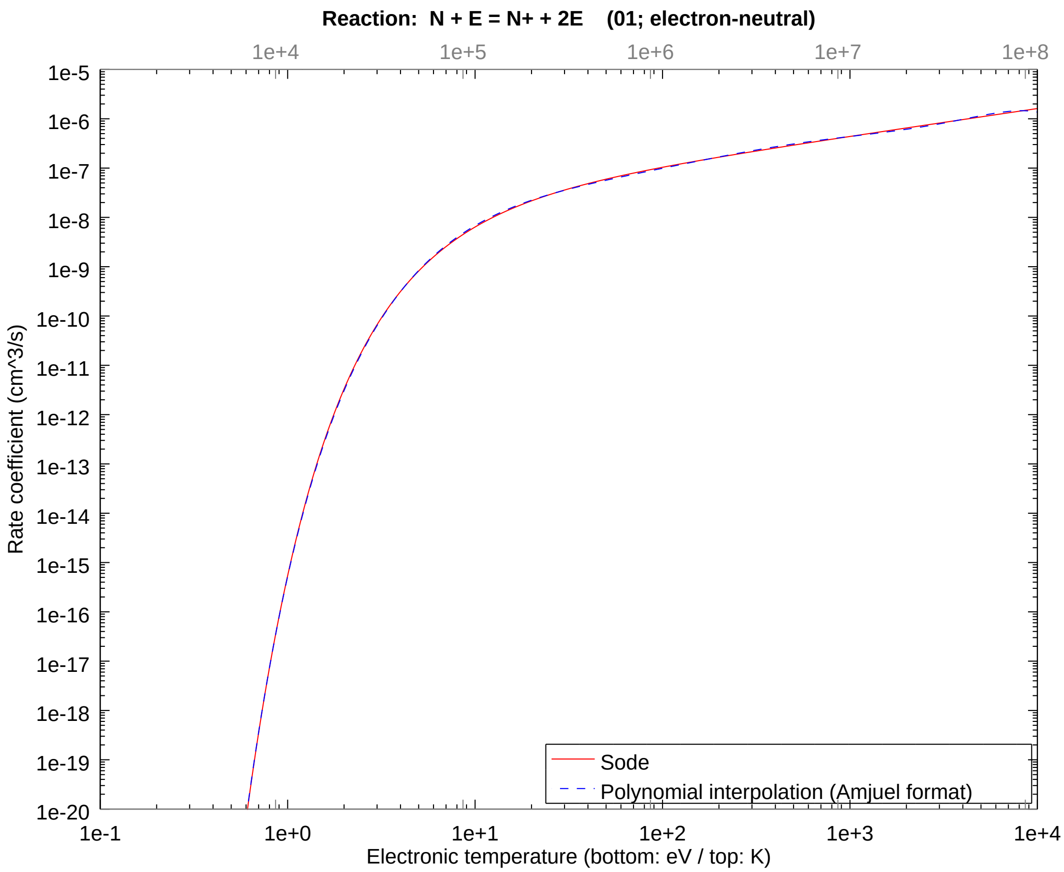
2 H.2 : Fits for rate coefficient

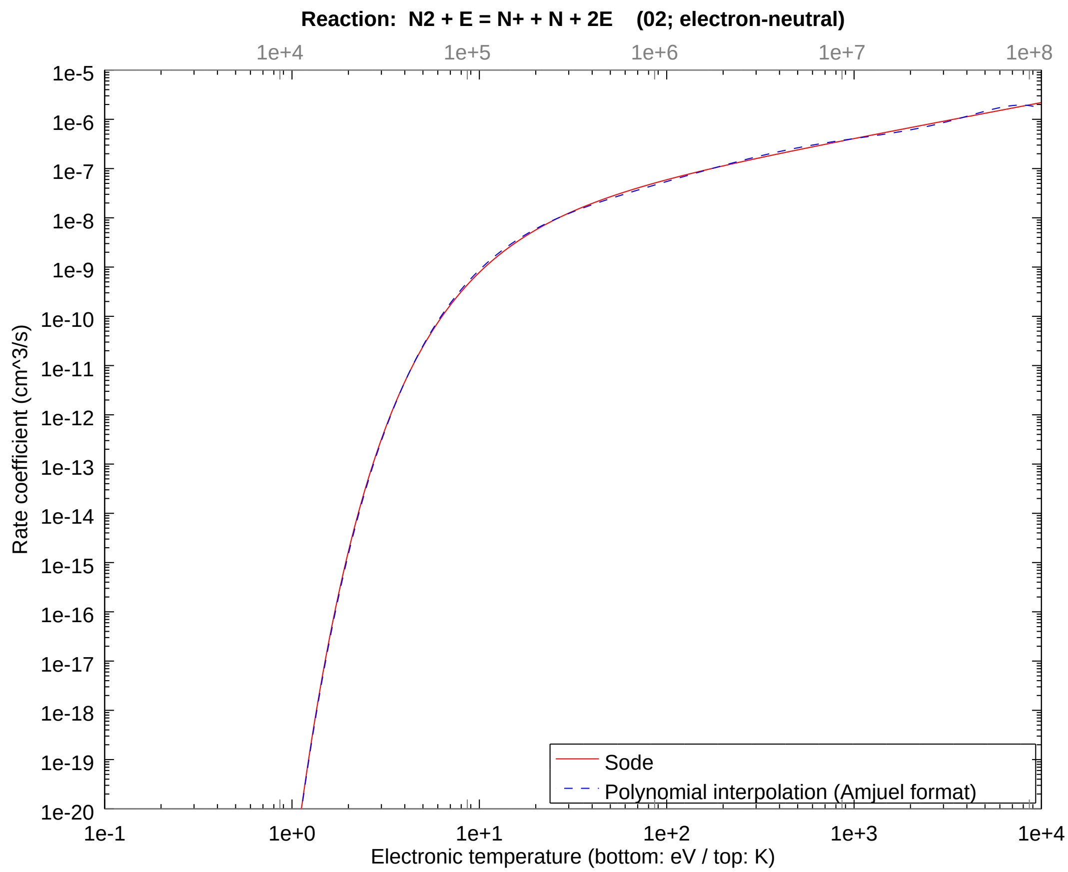
2.1 Reaction 01 N + E = N+ + 2E

b0	-3.518518351008E+01	b1	1.707576198208E+01	b2	-8.227861906601E+00
b3	2.853001756746E+00	b4	-7.513126739699E-01	b5	1.408230464943E-01
b6	-1.675815431763E-02	b7	1.103977300446E-03	b8	-3.033294428047E-05



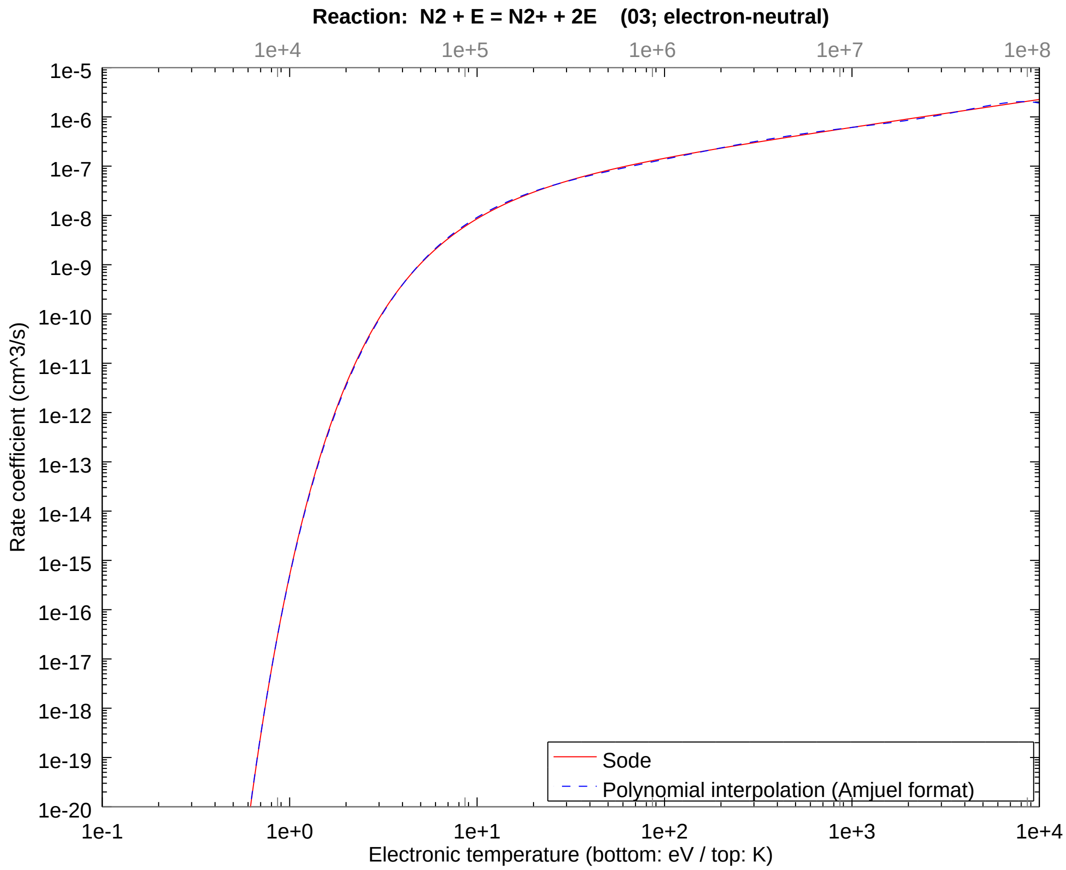
2.2 Reaction 02 N2 + E = N+ + N + 2E

b0 -4.942549999176E+01	b1 3.017277842063E+01	b2 -1.467285577702E+01
b3 5.087796049996E+00	b4 -1.339825902980E+00	b5 2.511316153268E-01
b6 -2.988503990258E-02	b7 1.968737430747E-03	b8 -5.409314373186E-05



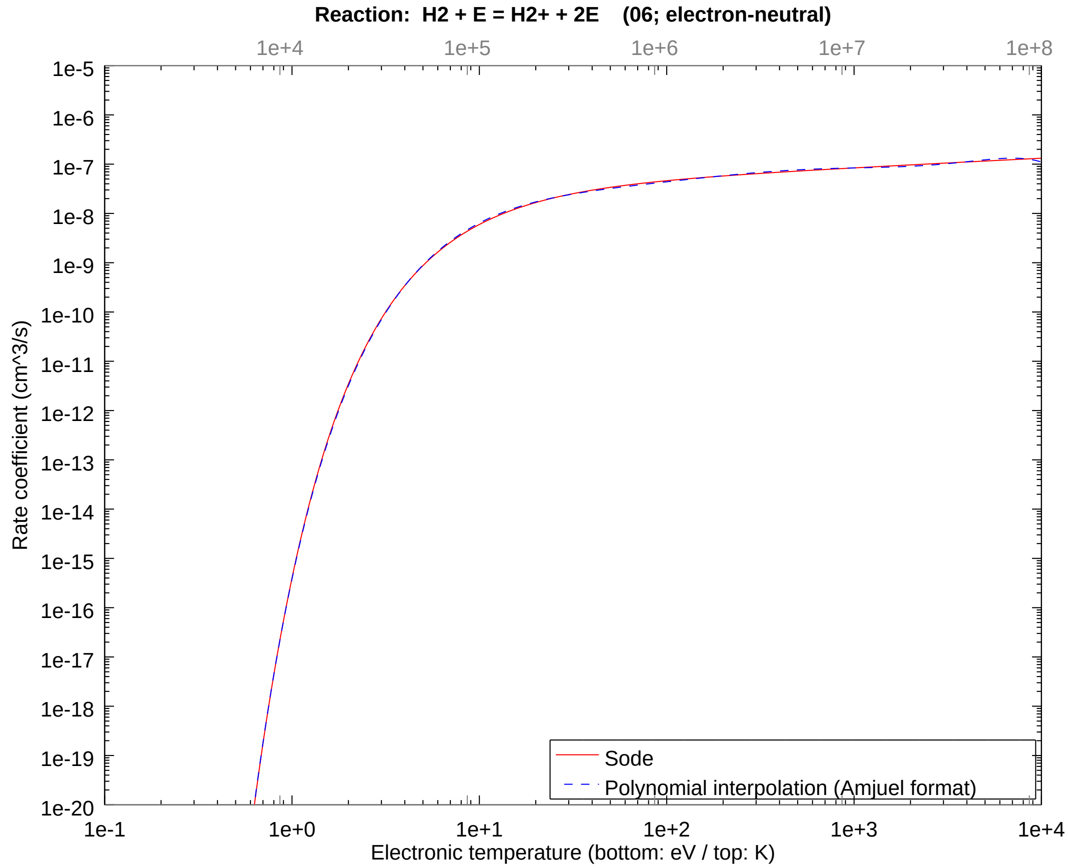
2.3 Reaction 03 N2 + E = N2+ + 2E

b0	-3.526103439612E+01	b1	1.748221230697E+01	b2	-8.430348304062E+00
b3	2.923213684733E+00	b4	-7.698023616246E-01	b5	1.442886796913E-01
b6	-1.717056988007E-02	b7	1.131146009520E-03	b8	-3.107943330538E-05



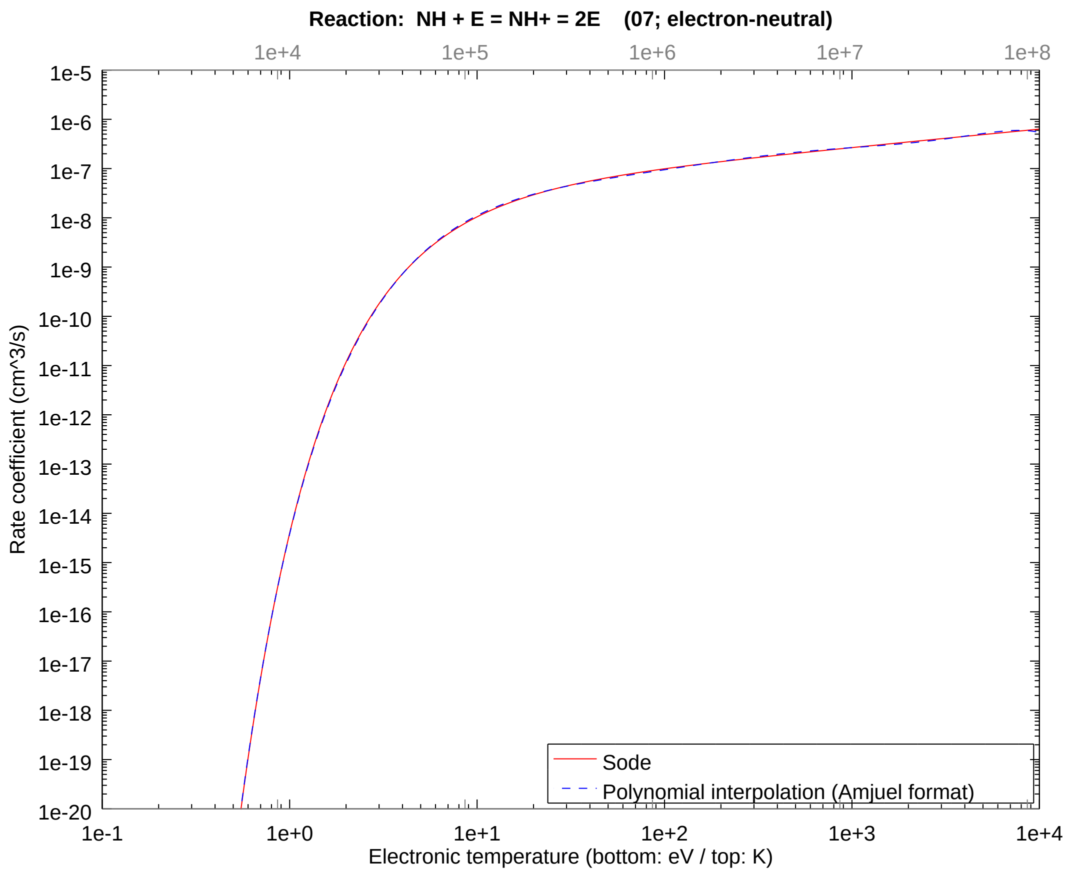
2.4 Reaction 06 H2 + E = H2+ + 2E

b0 -3.549202289051E+01	b1 1.790528611163E+01	b2 -8.825443713743E+00
b3 3.060212568611E+00	b4 -8.058798009509E-01	b5 1.510508908075E-01
b6 -1.797528317264E-02	b7 1.184158124788E-03	b8 -3.253599725642E-05



2.5 Reaction 07 NH + E = NH+ = 2E

b0	-3.319843294396E+01	b1	1.572589154276E+01	b2	-7.650034869943E+00
b3	2.652640889075E+00	b4	-6.985494189552E-01	b5	1.309333127369E-01
b6	-1.558126112726E-02	b7	1.026447081867E-03	b8	-2.820271950207E-05



2.6 Reaction 08 NH + E = N+ + H +2E

```
b0 -3.554735633352E+01    b1  1.696437674301E+01    b2 -8.306880988537E+00
b3  2.880401533522E+00    b4 -7.585281618352E-01    b5  1.421754887175E-01
b6 -1.691909697615E-02    b7  1.114579723499E-03    b8 -3.062425707068E-05
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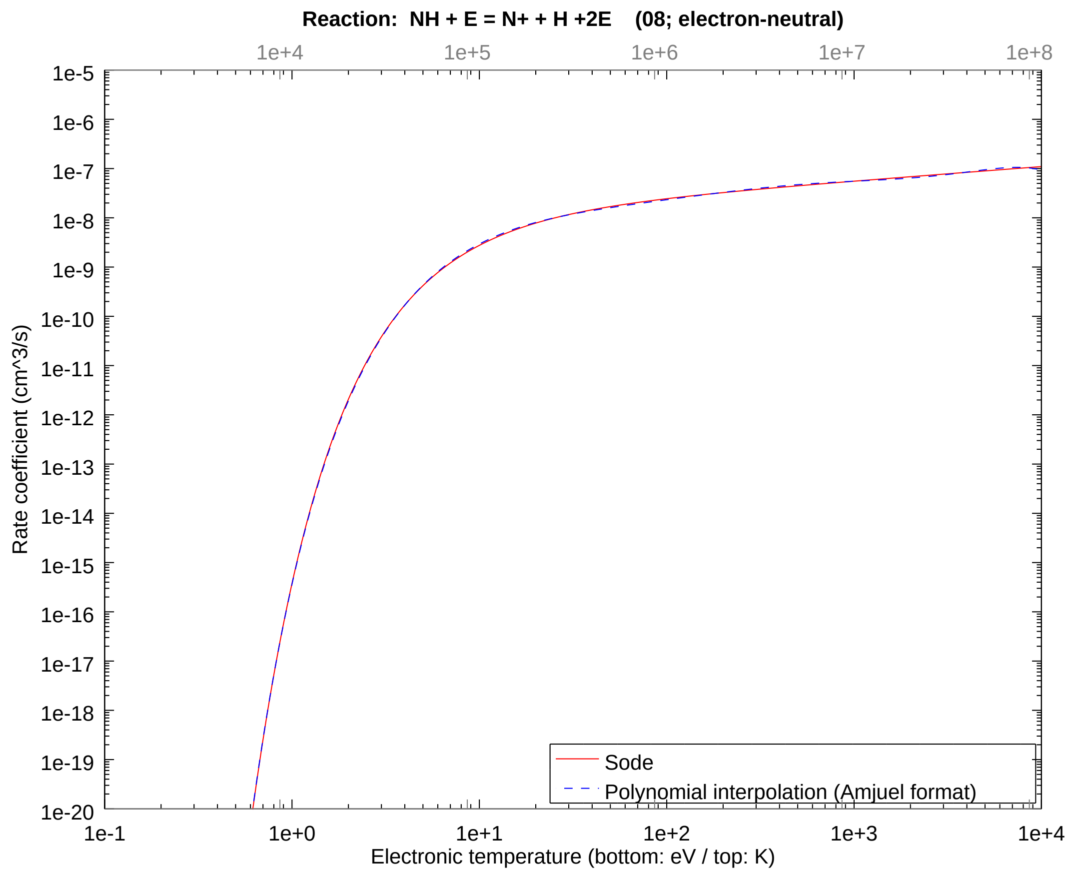
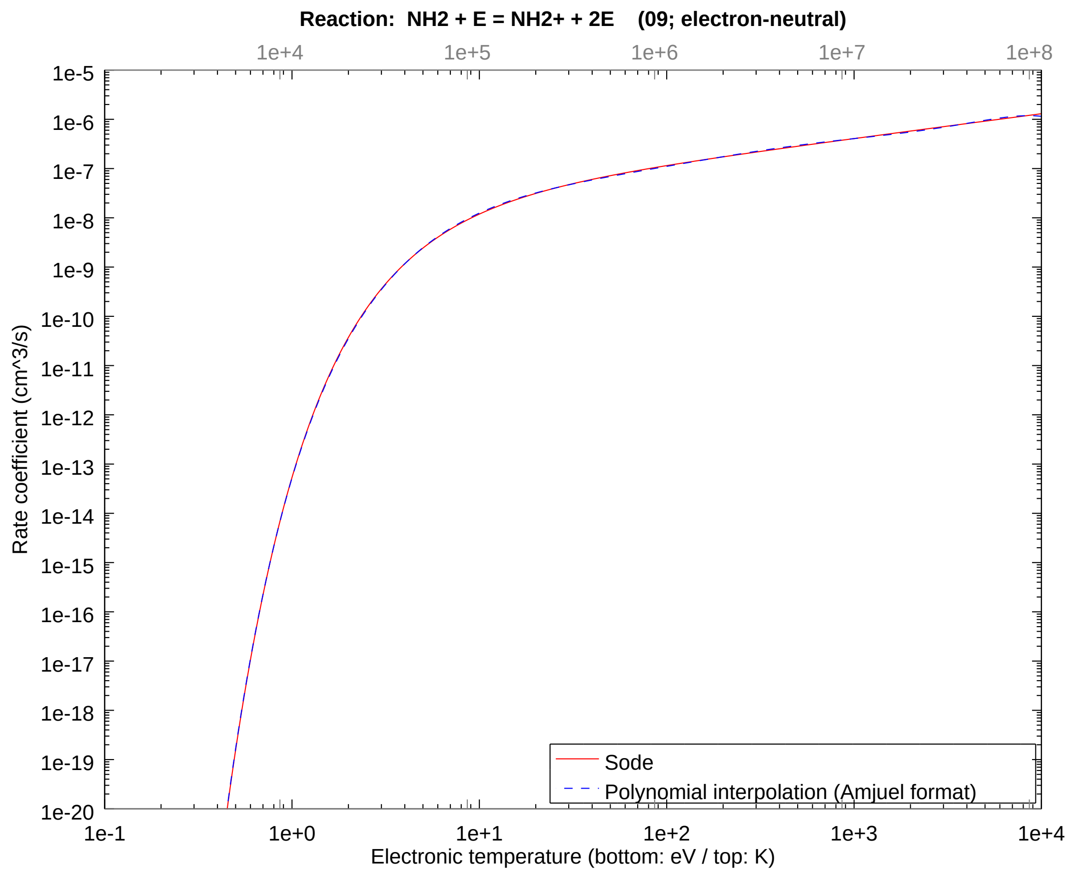


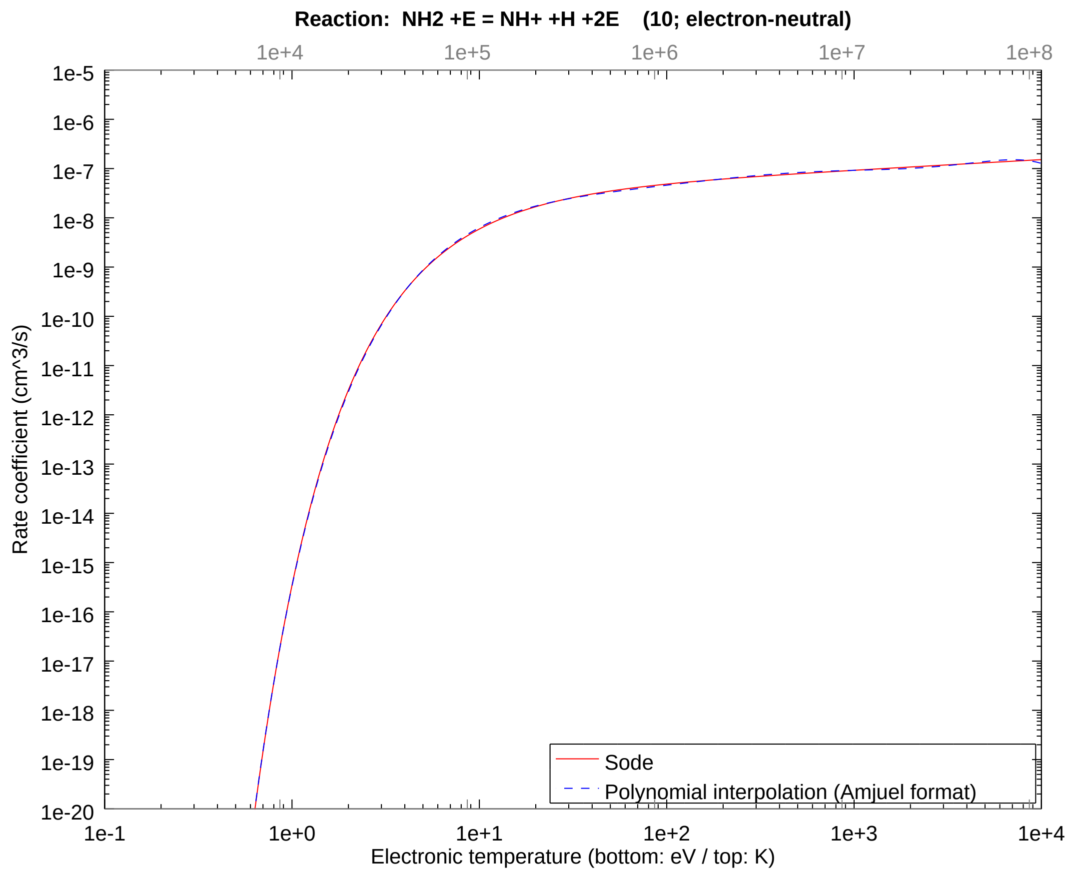
2.7 Reaction 09 NH2 + E = NH2+ + 2E

b0	-3.058208344880E+01	b1	1.279264397226E+01	b2	-6.123978850051E+00
b3	2.123482700099E+00	b4	-5.592003095574E-01	b5	1.048142723007E-01
b6	-1.247305603473E-02	b7	8.216877866463E-04	b8	-2.257674124117E-05



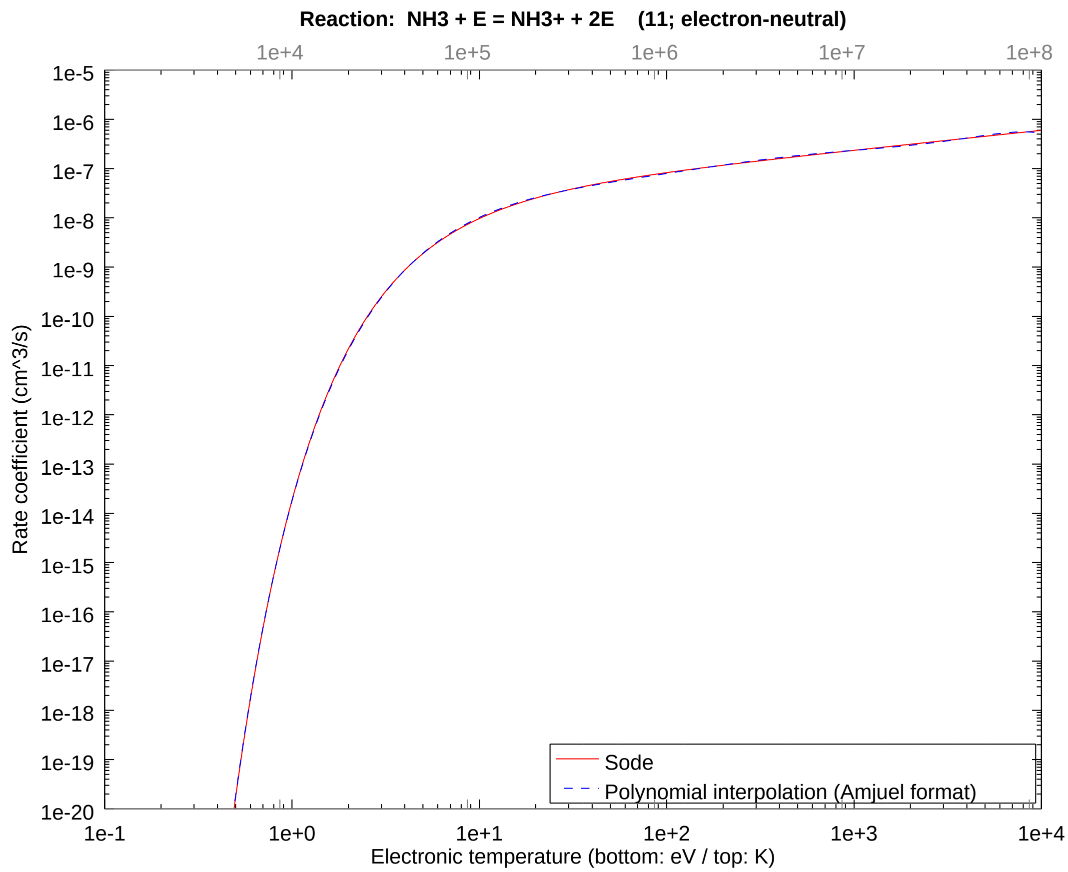
2.8 Reaction 10 NH2 +E = NH+ +H +2E

b0	-3.563666632219E+01	b1	1.802442033722E+01	b2	-8.874830639953E+00
b3	3.077337429095E+00	b4	-8.103894808667E-01	b5	1.518961671970E-01
b6	-1.807587233421E-02	b7	1.190784639196E-03	b8	-3.271806775030E-05



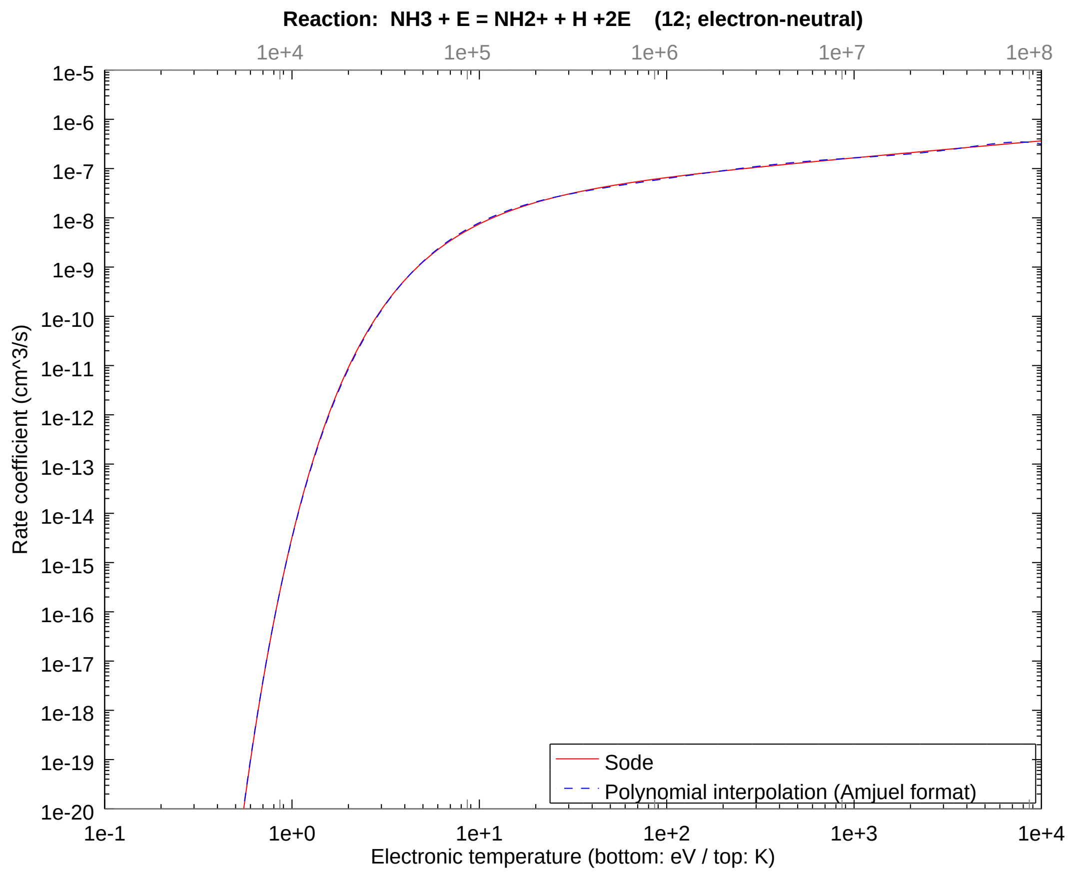
2.9 Reaction 11 NH3 + E = NH3+ + 2E

b0	-3.165130180136E+01	b1	1.389216810181E+01	b2	-6.721560657193E+00
b3	2.330693511964E+00	b4	-6.137674365384E-01	b5	1.150421166139E-01
b6	-1.369018488974E-02	b7	9.018686109884E-04	b8	-2.477979421712E-05



2.10 Reaction 12 NH3 + E = NH2+ + H +2E

b0	-3.339021332416E+01	b1	1.561658416230E+01	b2	-7.610525328975E+00
b3	2.638941000688E+00	b4	-6.949416750226E-01	b5	1.302570916252E-01
b6	-1.550078979800E-02	b7	1.021145870340E-03	b8	-2.805706310696E-05



2.11 Reaction ZA NH3 + E = NH+ + 2H + 2E

```
b0 -4.744940094576E+01    b1  2.620437918686E+01    b2 -1.287023297035E+01
b3  4.462738642305E+00    b4 -1.175222586054E+00    b5  2.202790271093E-01
b6 -2.621353550525E-02    b7  1.726869654839E-03    b8 -4.744757070522E-05
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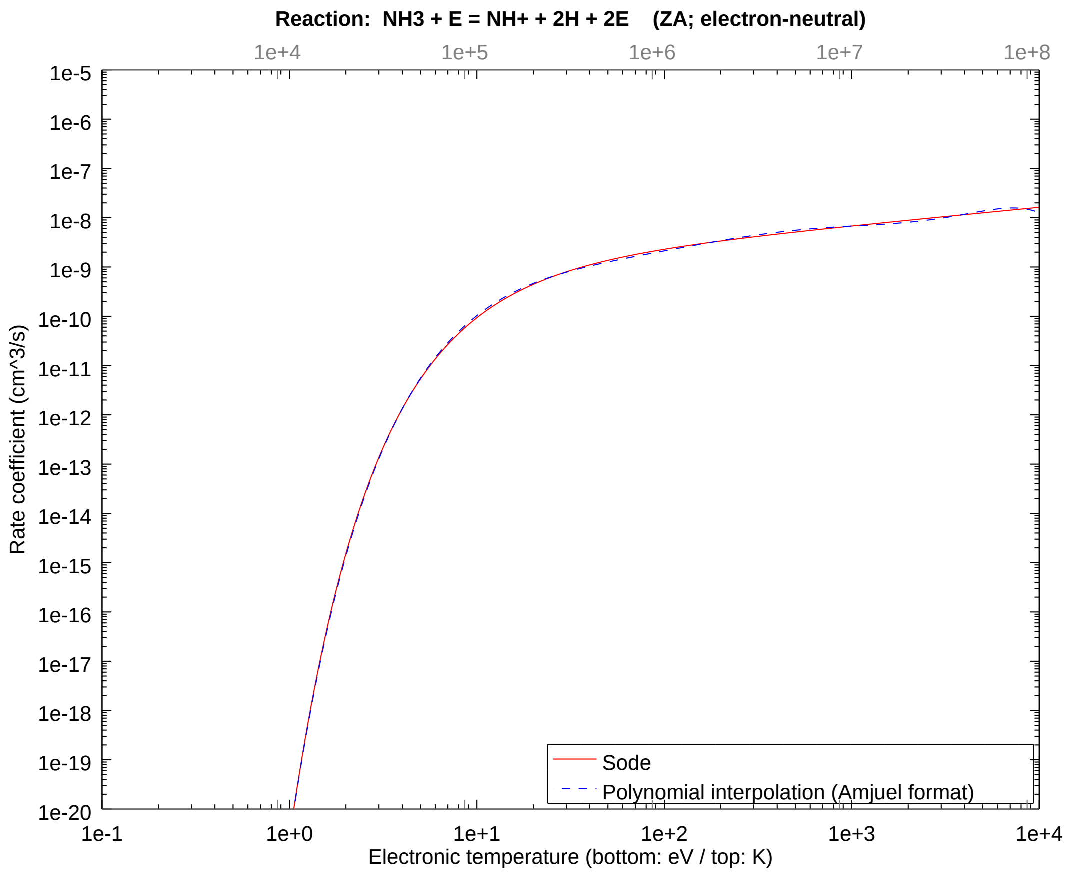
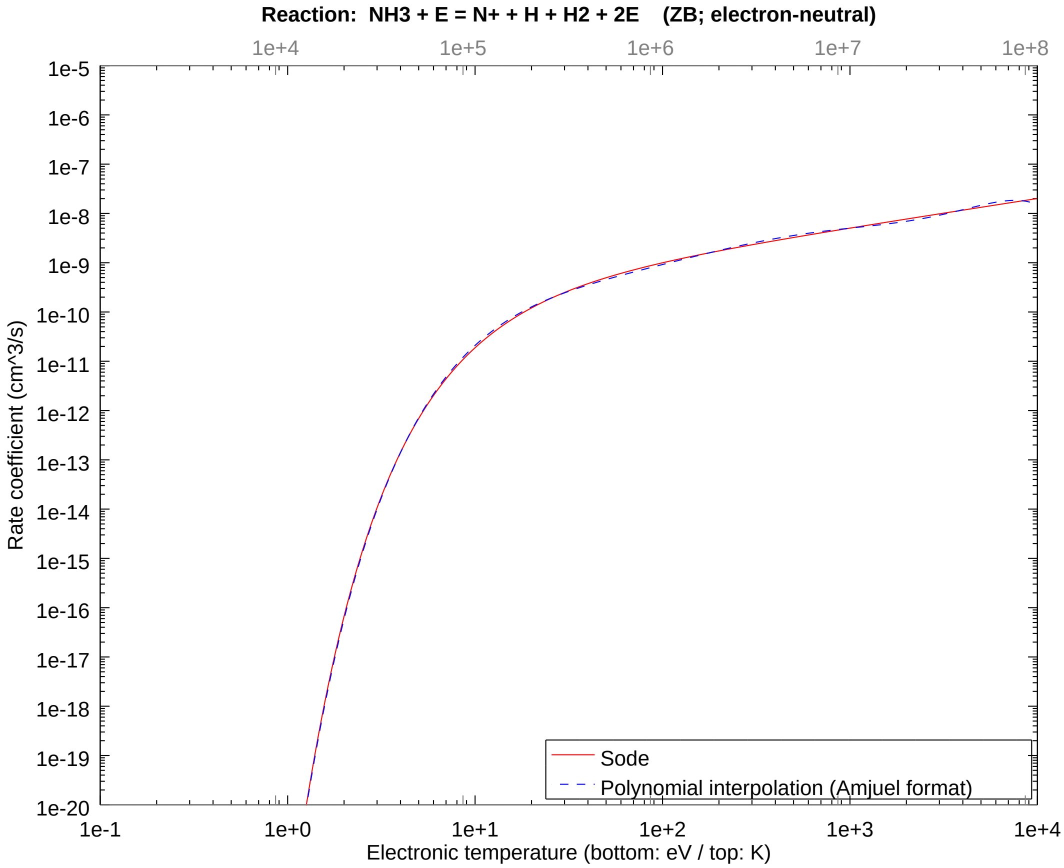


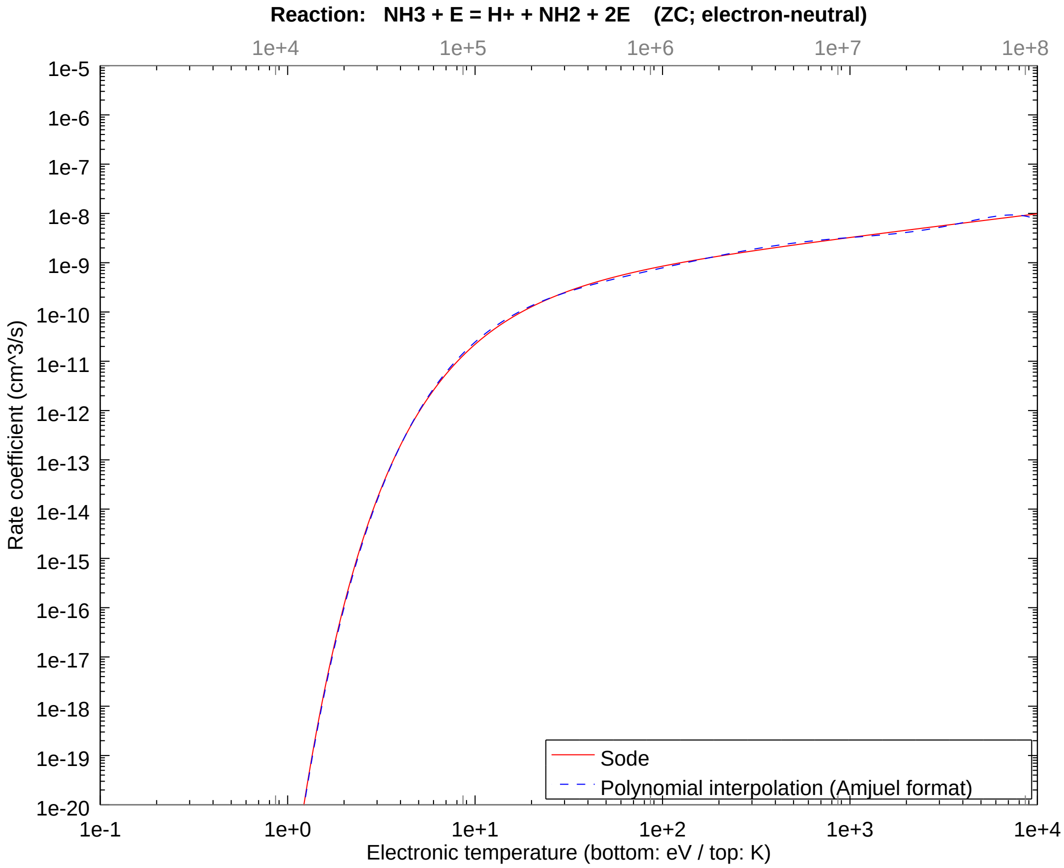
2.12 Reaction ZB NH3 + E = N+ + H + H2 + 2E

b0	-5.220926834260E+01	b1	2.933892541899E+01	b2	-1.432220860093E+01
b3	4.966209540554E+00	b4	-1.307807175578E+00	b5	2.451301529612E-01
b6	-2.917085685543E-02	b7	1.921689178447E-03	b8	-5.280044322530E-05



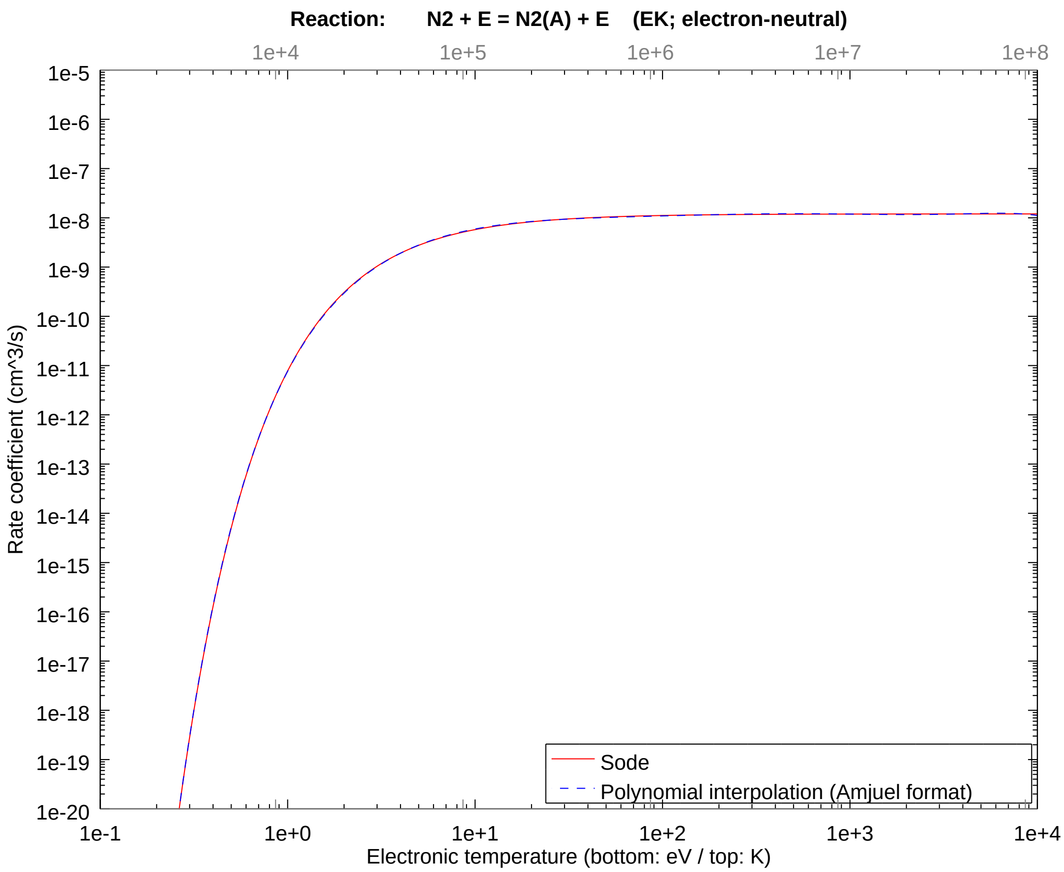
2.13 Reaction ZC NH3 + E = H+ + NH2 + 2E

b0	-5.136820819564E+01	b1	2.877282140387E+01	b2	-1.409996743298E+01
b3	4.889147668373E+00	b4	-1.287513615957E+00	b5	2.413264092084E-01
b6	-2.871820562836E-02	b7	1.891869863609E-03	b8	-5.198112600285E-05



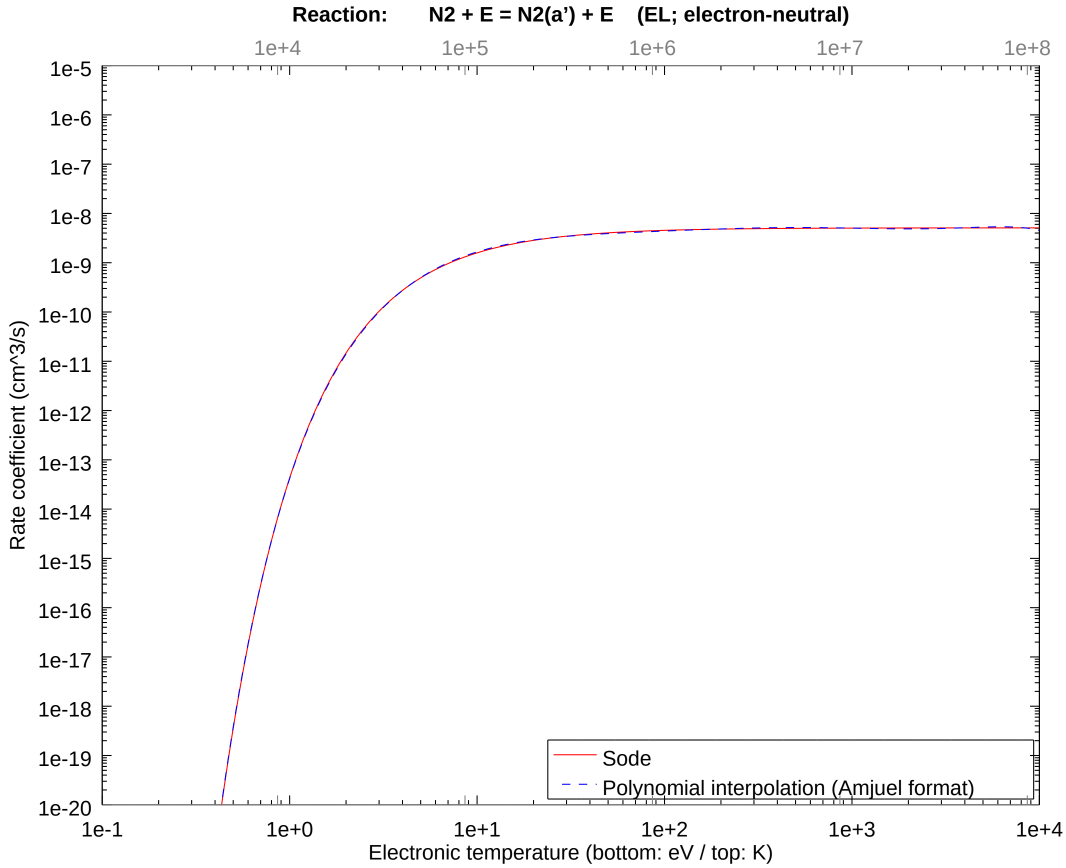
2.14 Reaction EK N2 + E = N2(A) + E

b0 -2.559242769964E+01	b1 7.276452157772E+00	b2 -3.625000383821E+00
b3 1.256964759575E+00	b4 -3.310105058187E-01	b5 6.204328699087E-02
b6 -7.383244459270E-03	b7 4.863861575793E-04	b8 -1.336397425082E-05



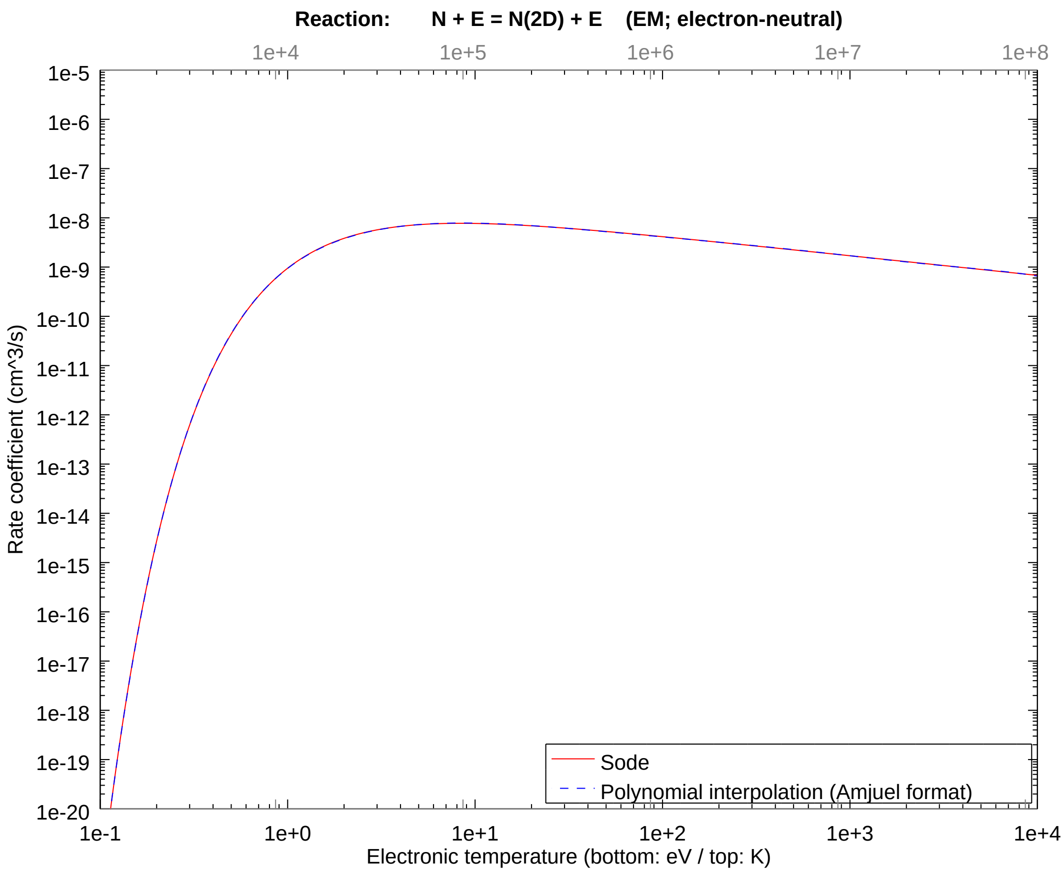
2.15 Reaction EL N2 + E = N2(a') + E

b0	-3.080643141587E+01	b1	1.158879097062E+01	b2	-5.773331673960E+00
b3	2.001896190658E+00	b4	-5.271815821553E-01	b5	9.881280993505E-02
b6	-1.175887298758E-02	b7	7.746395343463E-04	b8	-2.128404073461E-05



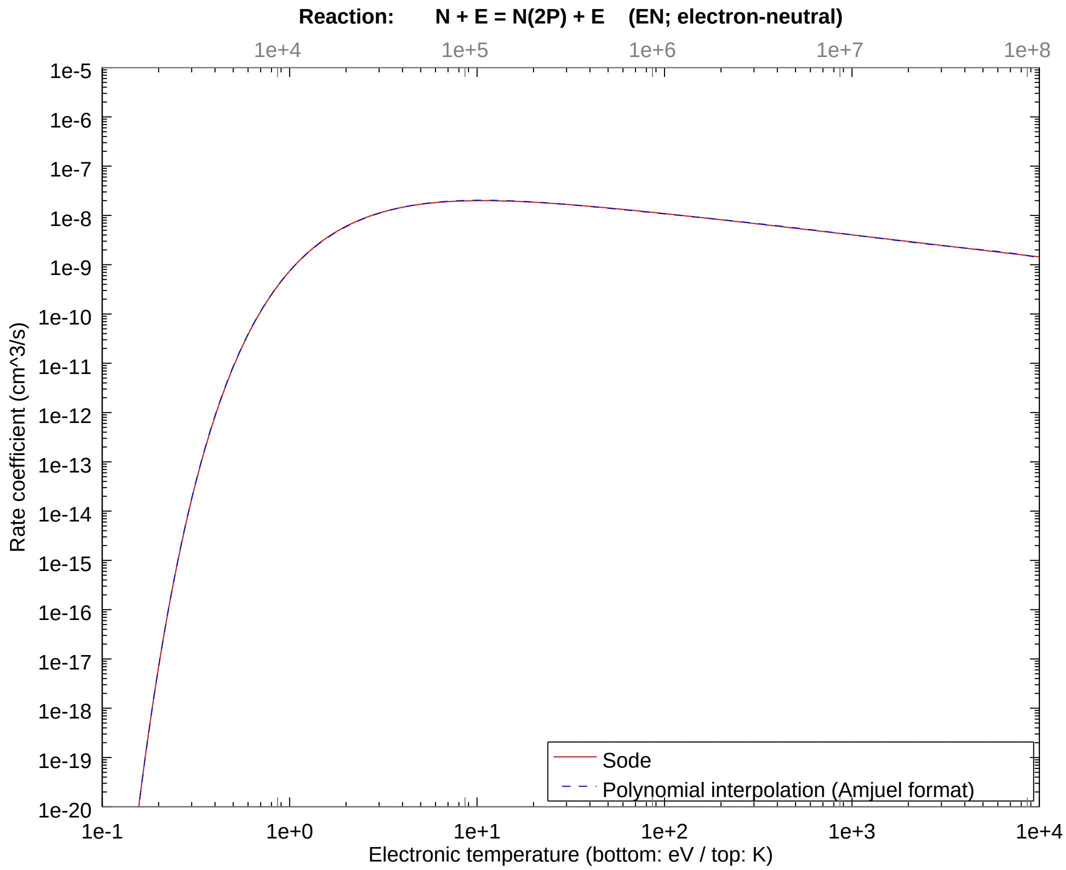
2.16 Reaction EM N + E = N(2D) + E

b0	-2.078384988604E+01	b1	2.920996557021E+00	b2	-1.654462028038E+00
b3	5.736828262365E-01	b4	-1.510742771788E-01	b5	2.831675904897E-02
b6	-3.369736912610E-03	b7	2.219882326827E-04	b8	-6.099361544993E-06



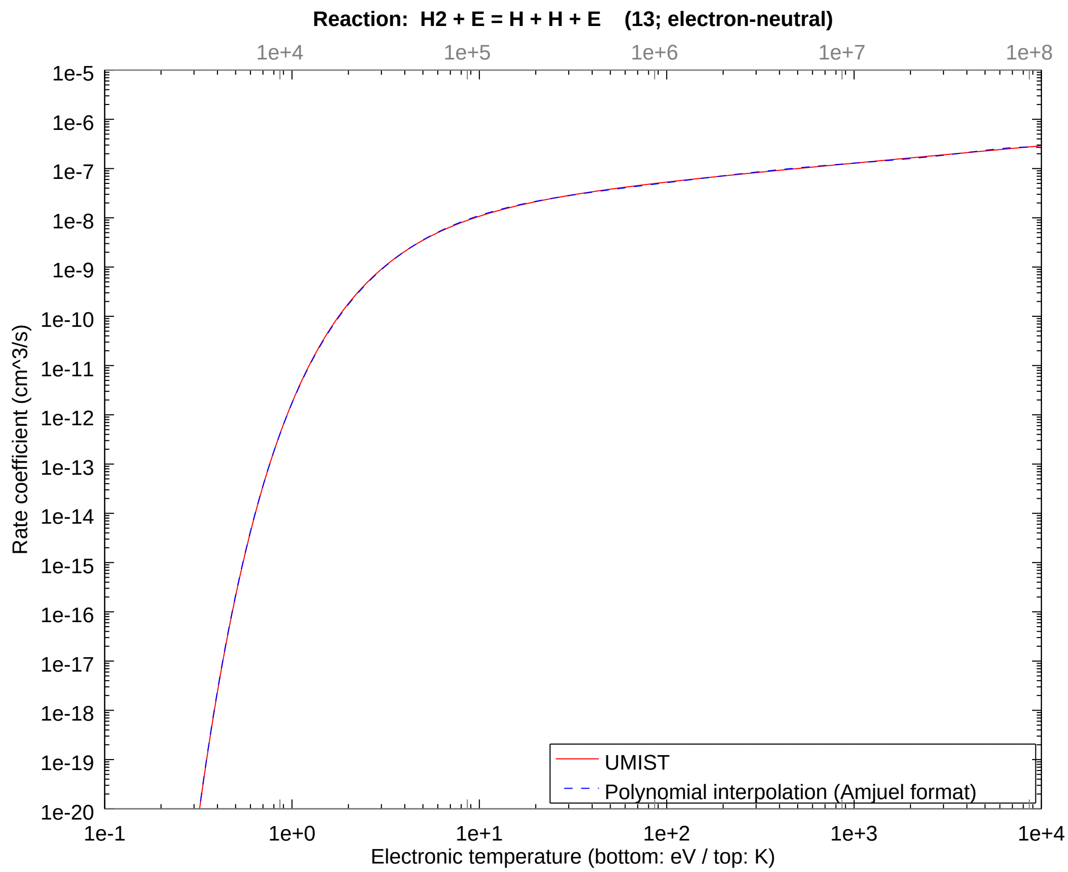
2.17 Reaction EN N + E = N(2P) + E

b0	-2.102160644758E+01	b1	4.308442827971E+00	b2	-2.370572458084E+00
b3	8.219933032642E-01	b4	-2.164646359577E-01	b5	4.057326669703E-02
b6	-4.828279755381E-03	b7	3.180726916050E-04	b8	-8.739383706258E-06



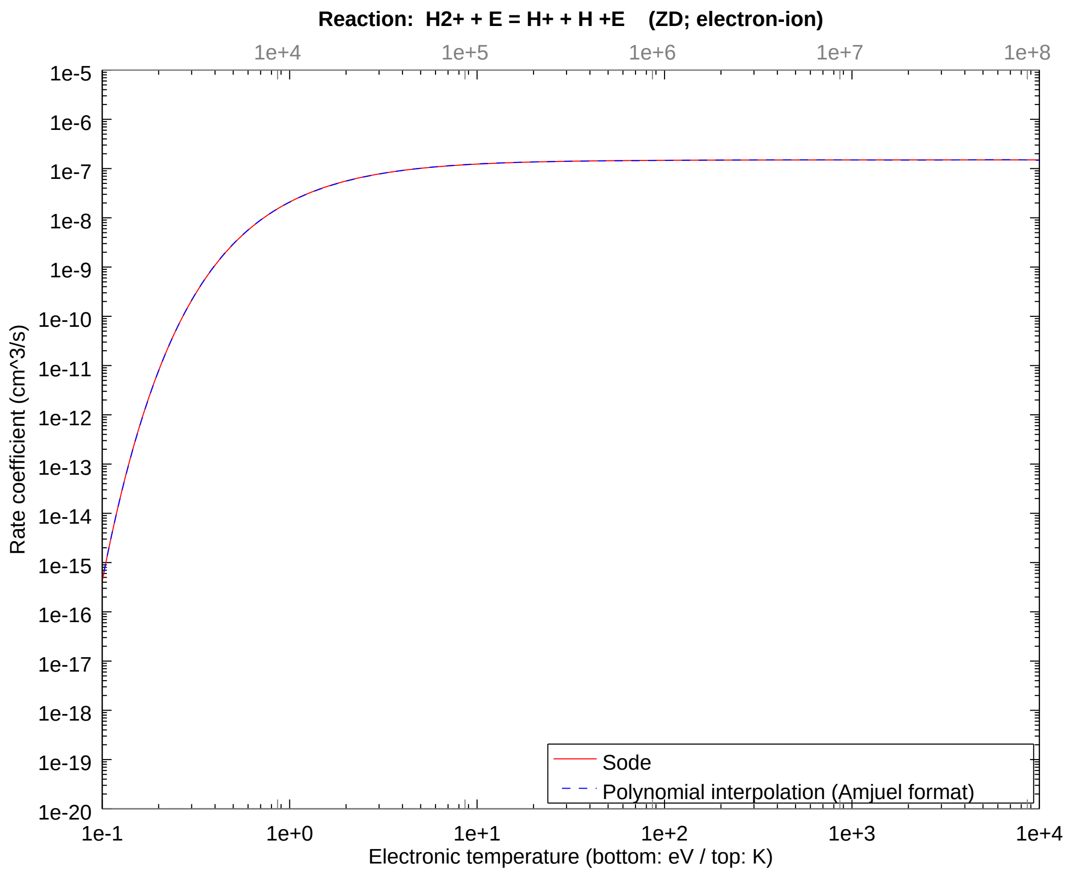
2.18 Reaction 13 H2 + E = H + H + E

b0	-2.708571324418E+01	b1	9.068035062033E+00	b2	-4.343171611769E+00
b3	1.505989815931E+00	b4	-3.965890427172E-01	b5	7.433506599401E-02
b6	-8.845984646340E-03	b7	5.827471250443E-04	b8	-1.601159377683E-05



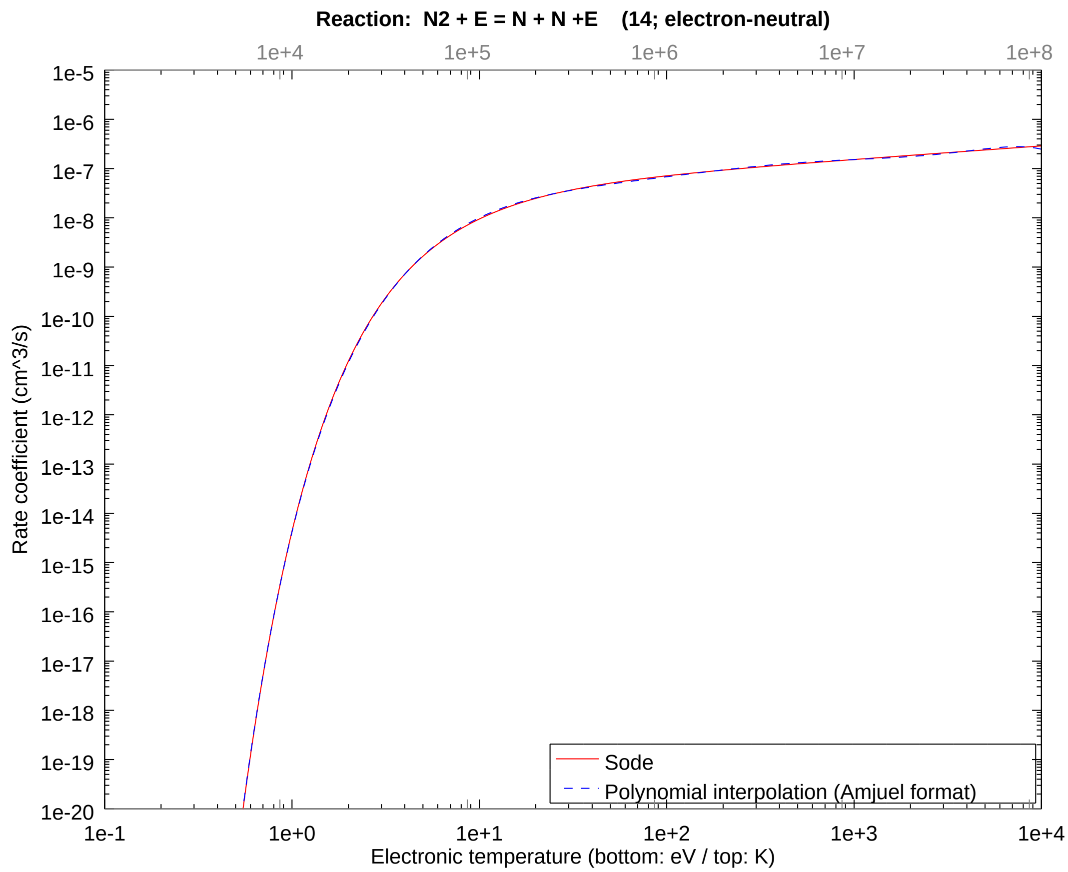
2.19 Reaction ZD H2+ + E = H+ + H +E

b0	-1.768640642427E+01	b1	1.952944243980E+00	b2	-9.729224463388E-01
b3	3.373597515480E-01	b4	-8.884069434098E-02	b5	1.665194487357E-02
b6	-1.981606482938E-03	b7	1.305423338462E-04	b8	-3.586788729444E-06



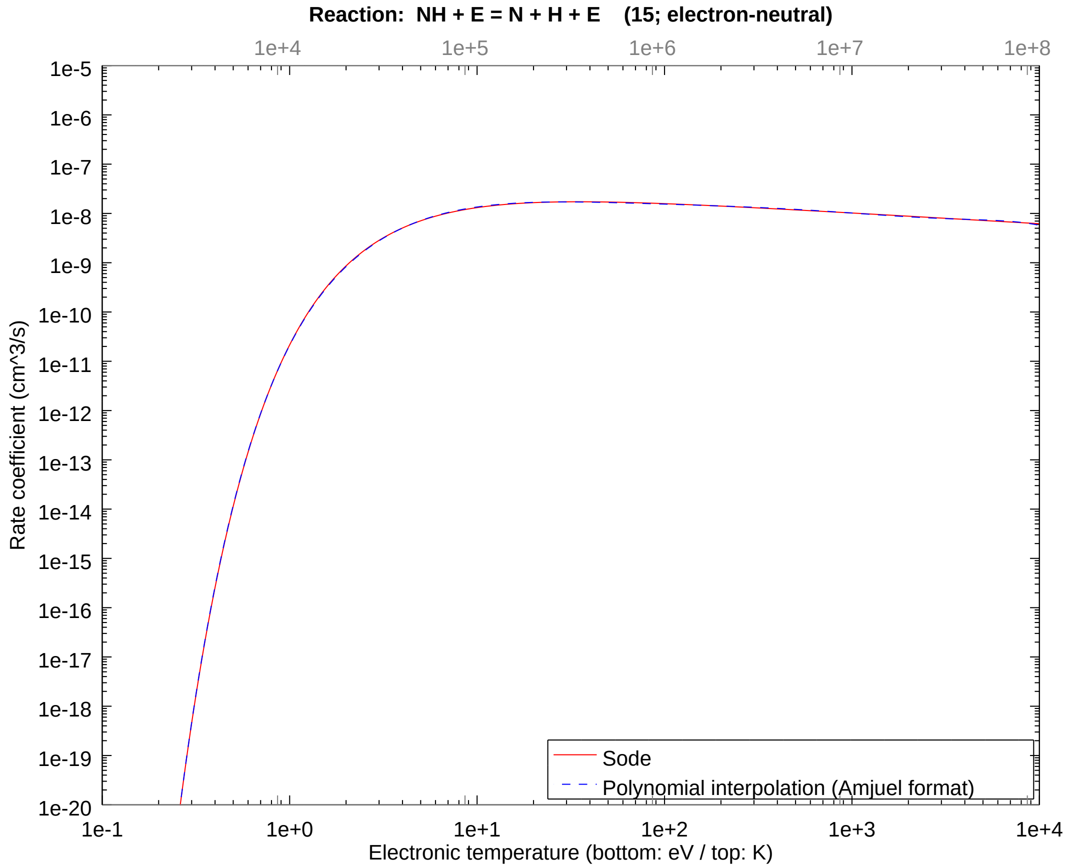
2.20 Reaction 14 N2 + E = N + N +E

b0 -3.310497821898E+01	b1 1.566554523300E+01	b2 -7.669789640427E+00
b3 2.659490833269E+00	b4 -7.003532909215E-01	b5 1.312714232927E-01
b6 -1.562149679189E-02	b7 1.029097687630E-03	b8 -2.827554769962E-05



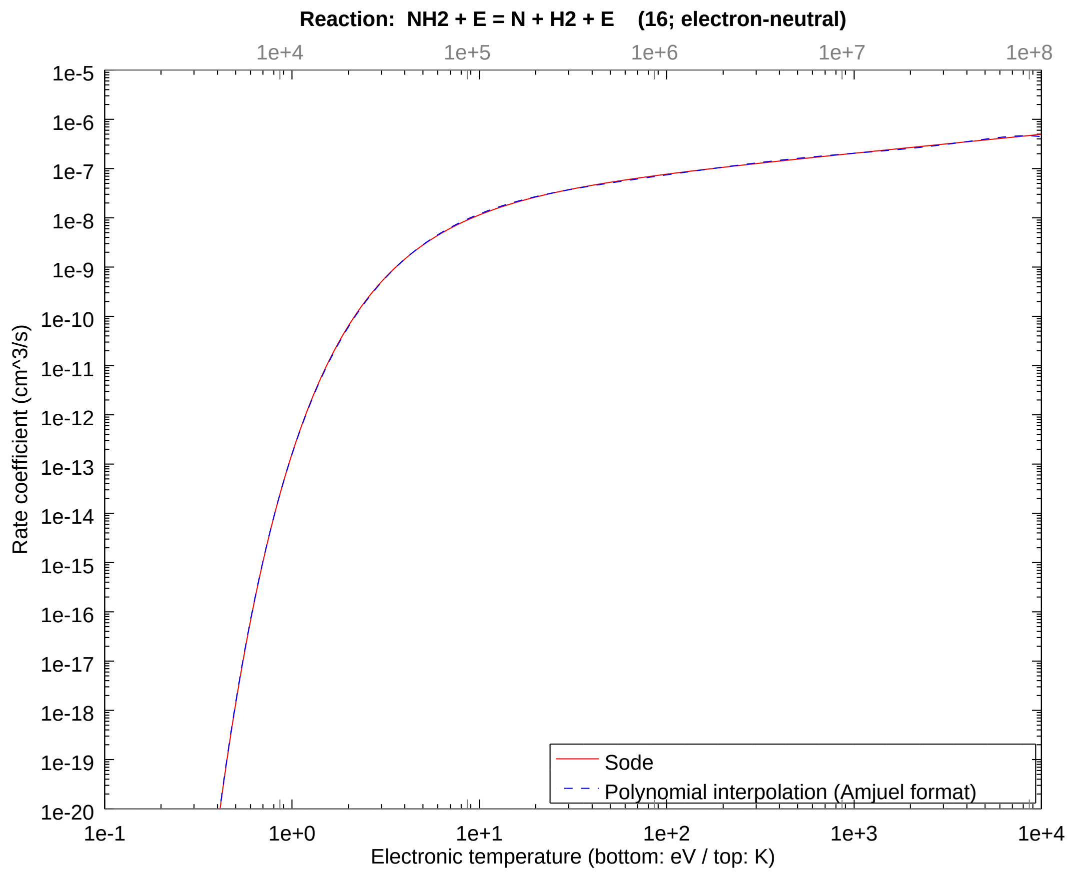
2.21 Reaction 15 NH + E = N + H + E

b0 -2.457785758960E+01	b1 7.403421947312E+00	b2 -3.797854625556E+00
b3 1.316901771271E+00	b4 -3.467943855239E-01	b5 6.500175435420E-02
b6 -7.735306524768E-03	b7 5.095789580089E-04	b8 -1.400122097940E-05



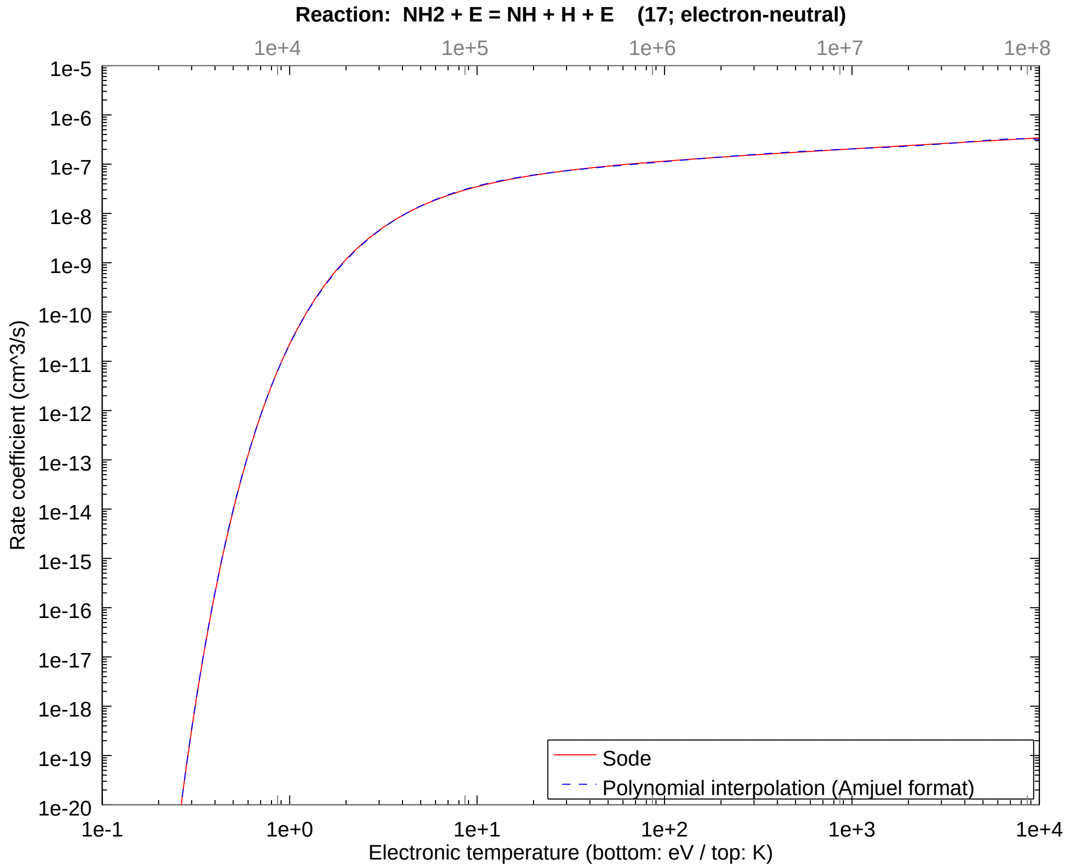
2.22 Reaction 16 NH2 + E = N + H2 + E

b0	-2.947714258173E+01	b1	1.172095540666E+01	b2	-5.649864358434E+00
b3	1.959084039446E+00	b4	-5.159073823659E-01	b5	9.669961896125E-02
b6	-1.150740008366E-02	b7	7.580732483252E-04	b8	-2.082886449991E-05



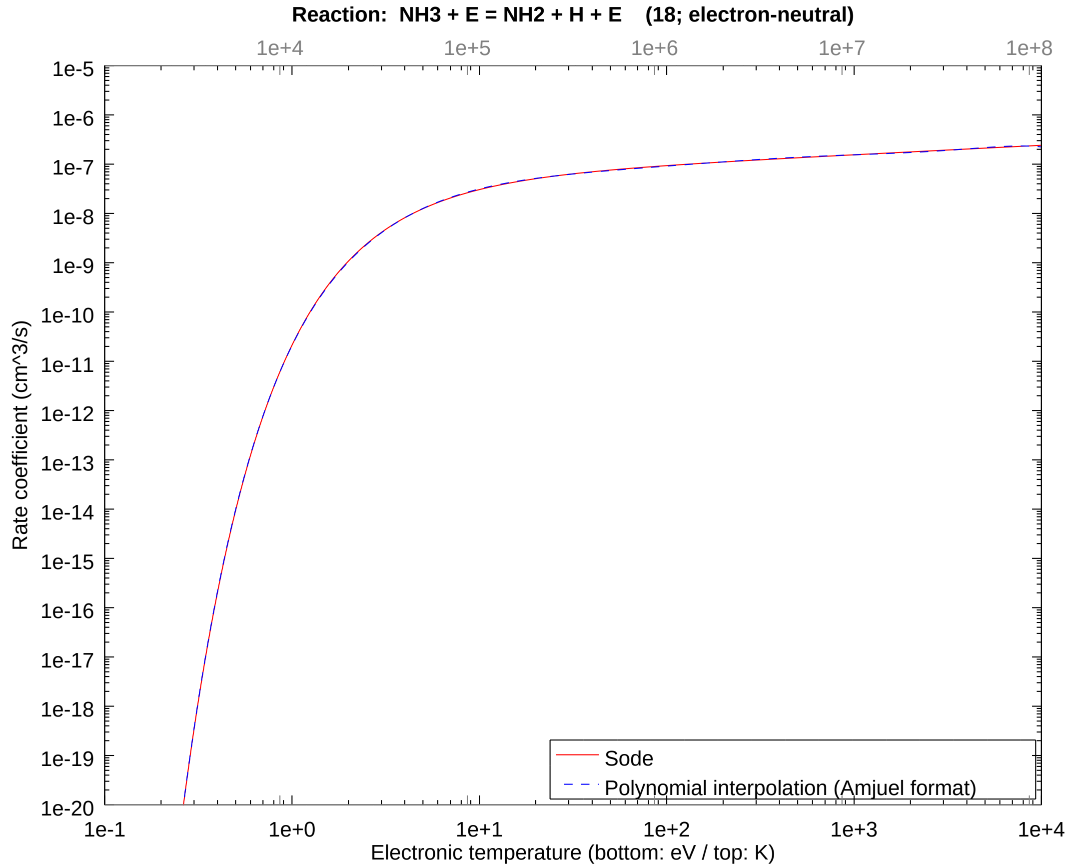
2.23 Reaction 17 NH2 + E = NH + H + E

b0	-2.454118936625E+01	b1	7.764114566845E+00	b2	-3.758345084588E+00
b3	1.303201882883E+00	b4	-3.431866415913E-01	b5	6.432553324258E-02
b6	-7.654835195510E-03	b7	5.042777464821E-04	b8	-1.385556458430E-05



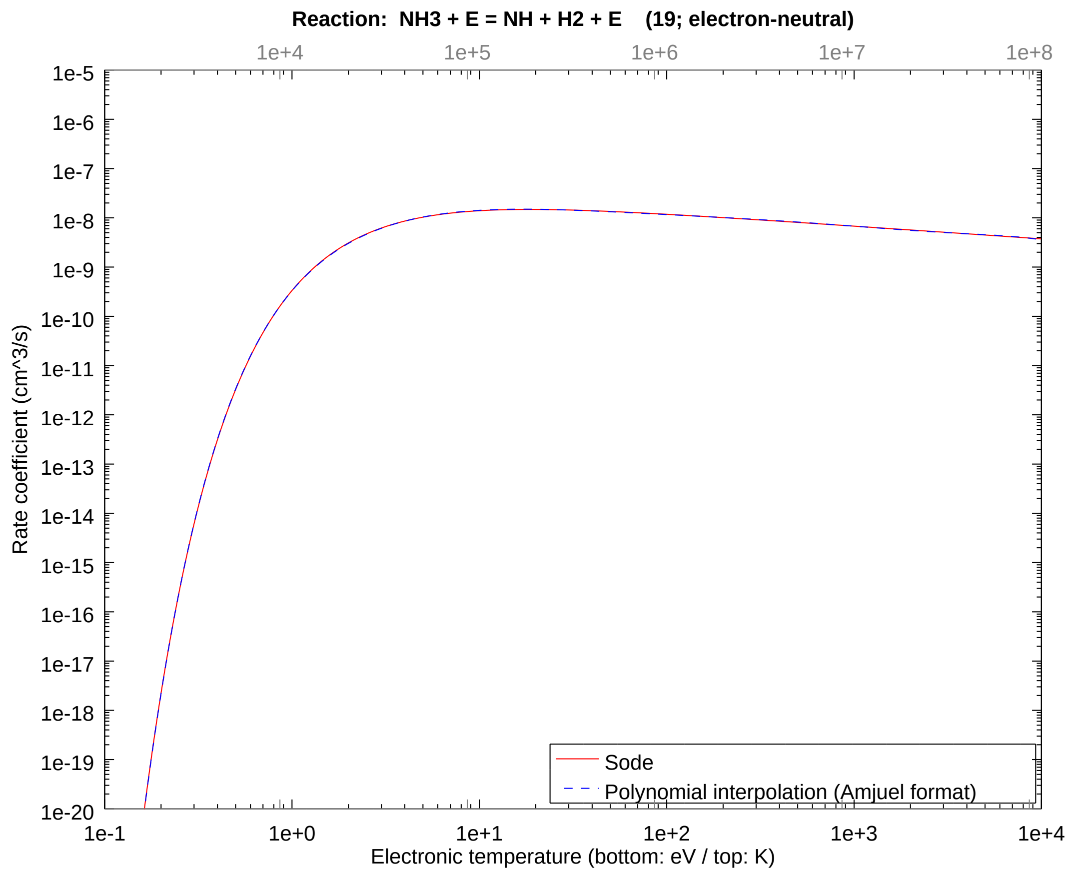
2.24 Reaction 18 NH3 + E = NH2 + H + E

b0 -2.459014390392E+01	b1 7.714287721729E+00	b2 -3.748467699346E+00
b3 1.299776910786E+00	b4 -3.422847056081E-01	b5 6.415647796467E-02
b6 -7.634717363196E-03	b7 5.029524436004E-04	b8 -1.381915048552E-05



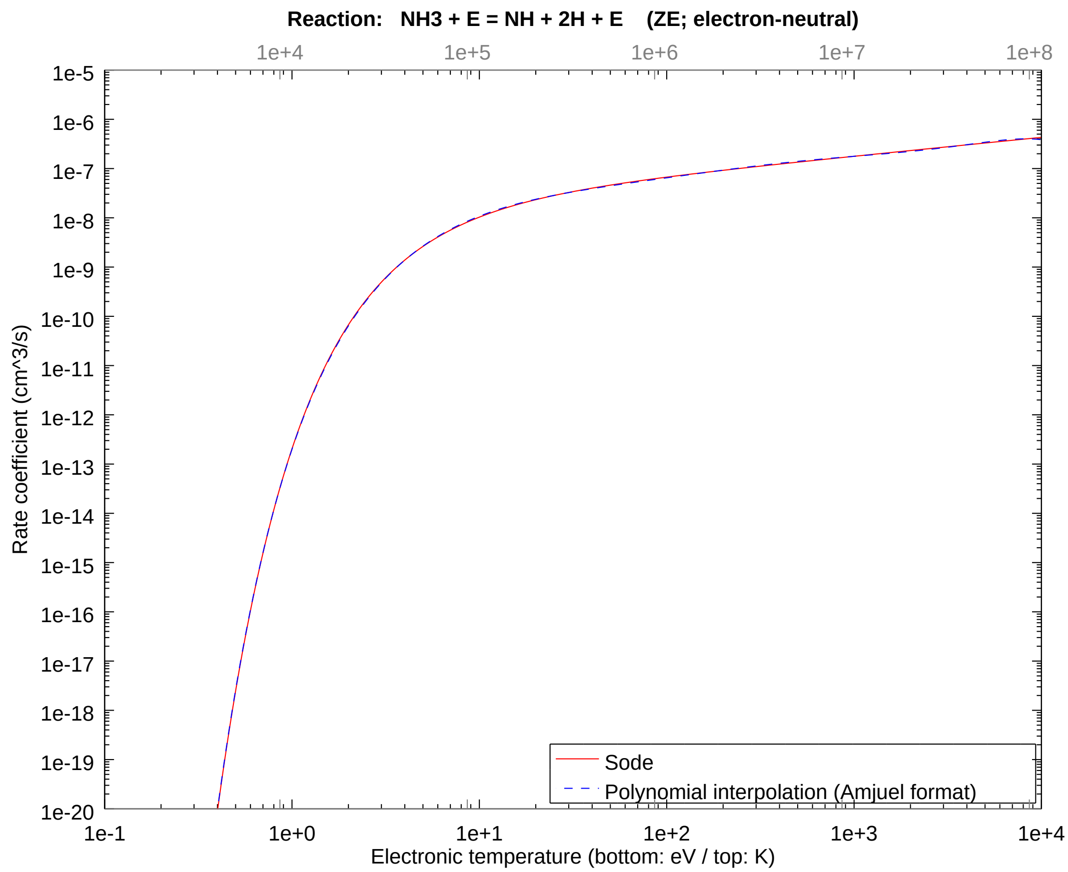
2.25 Reaction 19 NH3 + E = NH + H2 + E

b0	-2.182891305431E+01	b1	4.508356250529E+00	b2	-2.375511150705E+00
b3	8.237057893126E-01	b4	-2.169156039493E-01	b5	4.065779433598E-02
b6	-4.838338671538E-03	b7	3.187353430458E-04	b8	-8.757590755646E-06



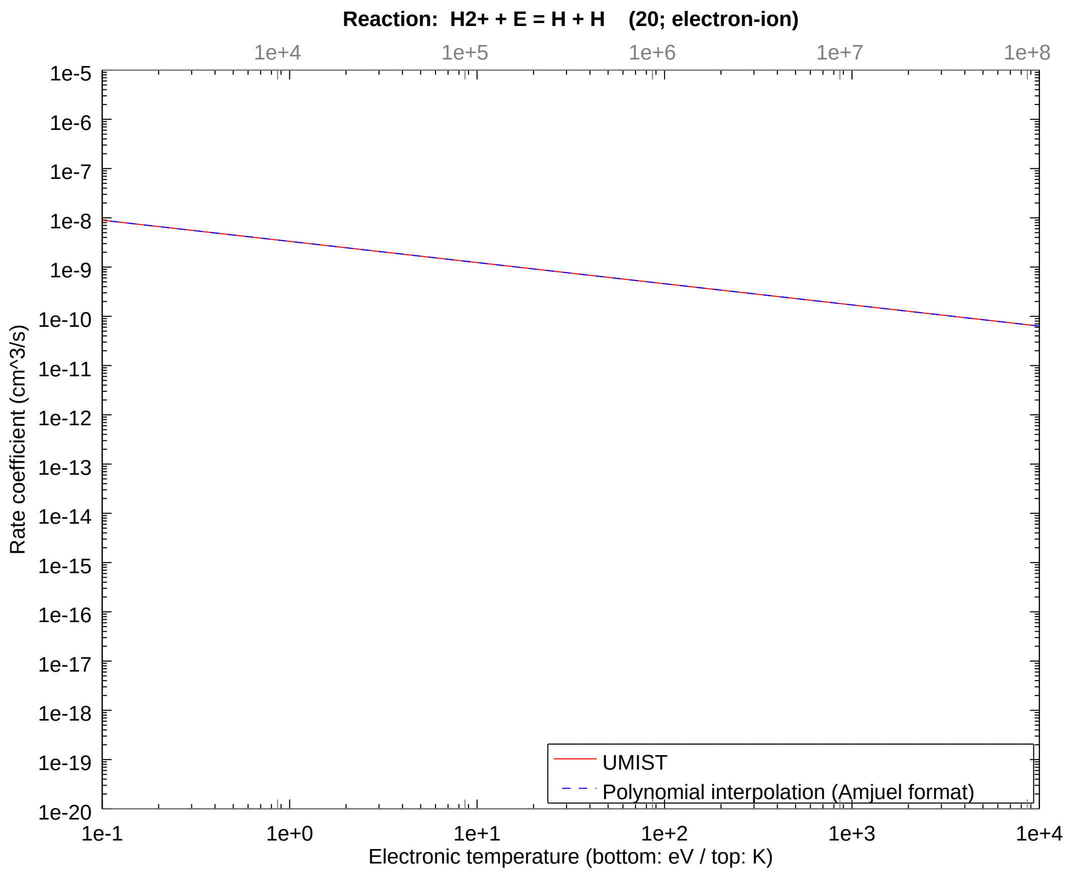
2.26 Reaction ZE NH3 + E = NH + 2H + E

b0	-2.923951508276E+01	b1	1.134424534945E+01	b2	-5.462194038836E+00
b3	1.894009569605E+00	b4	-4.987705986859E-01	b5	9.348756868107E-02
b6	-1.112516126969E-02	b7	7.328924935732E-04	b8	-2.013699662317E-05



2.27 Reaction 20 H2+ + E = H + H

b0	-1.952226532449E+01	b1	-4.300000000000E-01	b2	1.176699630083E-14
b3	-4.306129821113E-15	b4	-1.976279697331E-15	b5	1.188464274130E-15
b6	-2.107638125050E-16	b7	1.540936682618E-17	b8	-3.878633051697E-19



2.28 Reaction 21 H3+ + E = 3H

```
b0 -1.886152596060E+01    b1 -5.235000000000E-01    b2  1.372816235097E-14
b3  2.371151876342E-16    b4 -2.671124322317E-15    b5  6.436036735214E-16
b6 -3.609478803178E-17    b7 -2.932772922030E-18    b8  2.671970254938E-19
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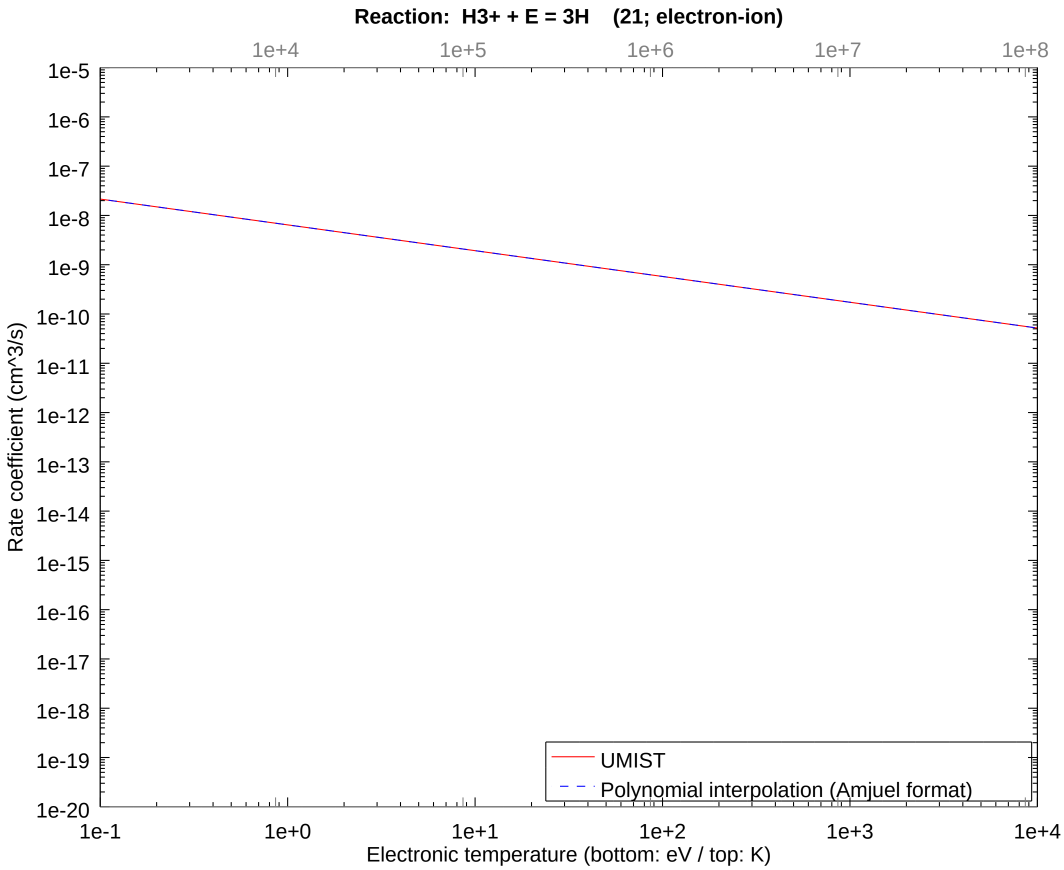
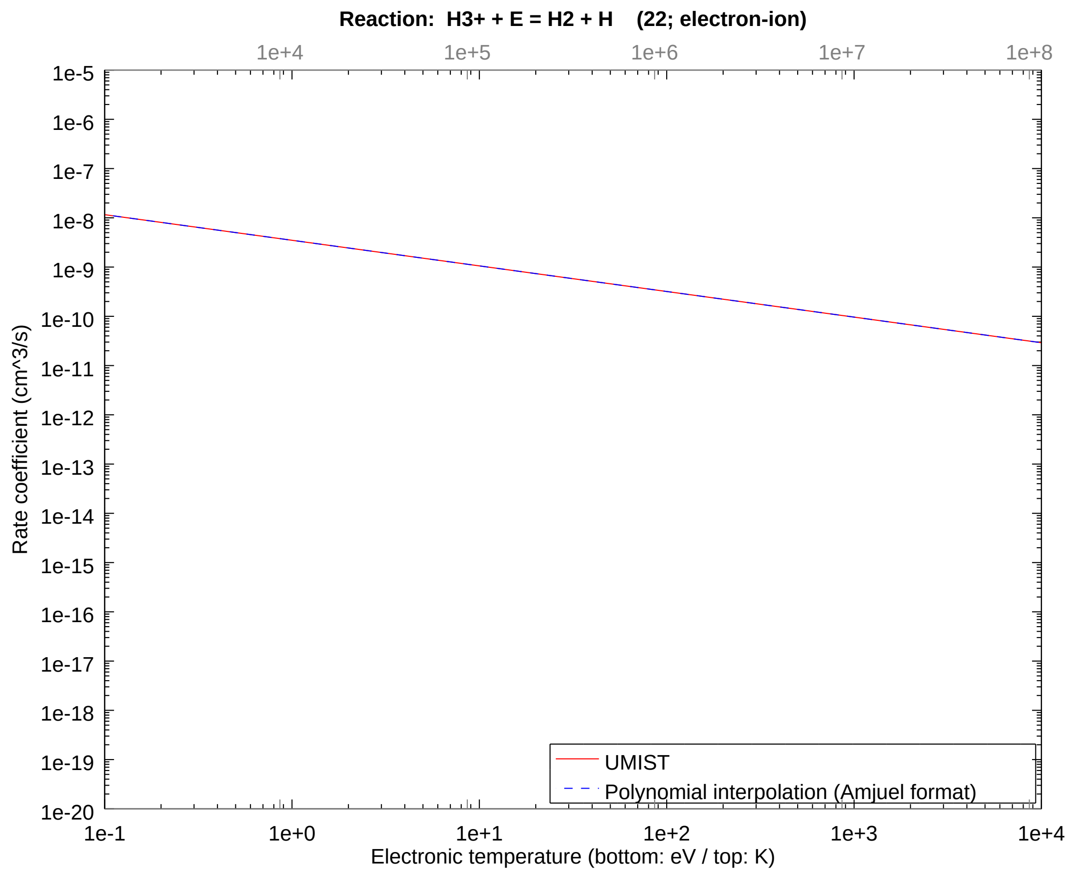


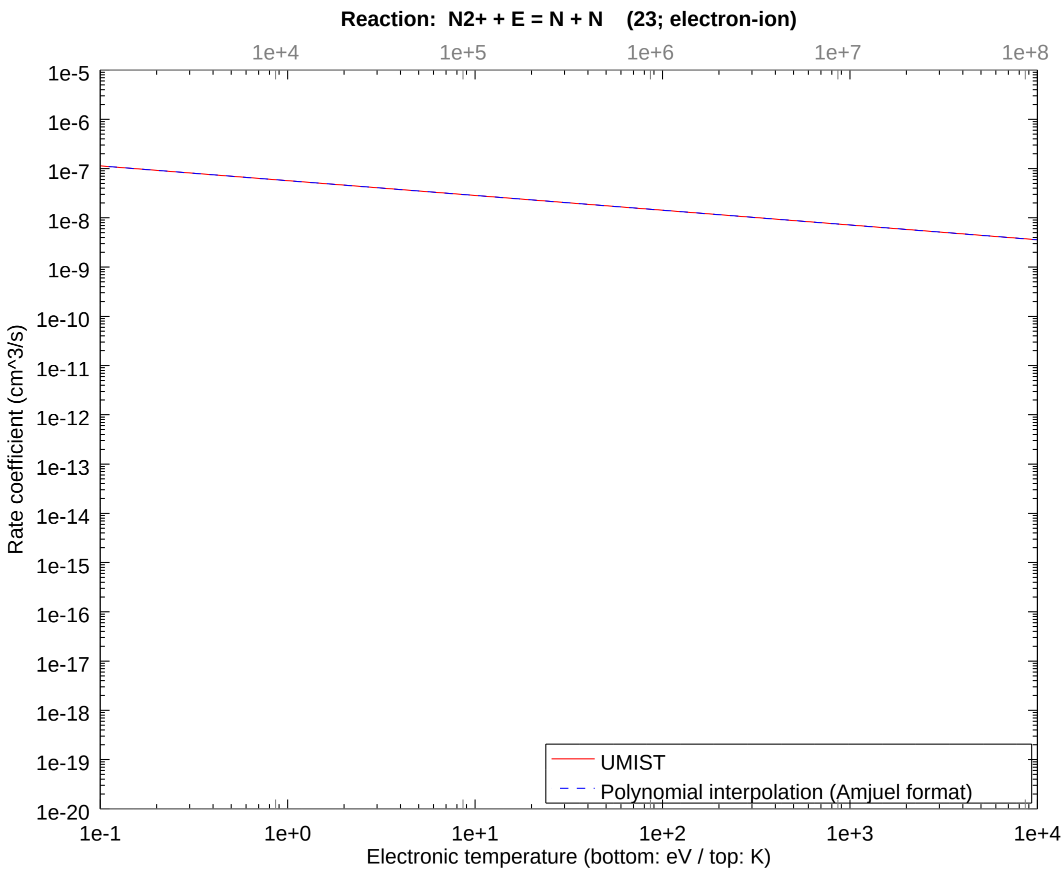
2.29 Reaction 22 H3+ + E = H2 + H

b0	-1.947105509154E+01	b1	-5.200000000000E-01	b2	1.176699630083E-14
b3	3.446824533805E-15	b4	-2.138917803282E-15	b5	-2.721988437046E-16
b6	2.082826560054E-16	b7	-2.858807325302E-17	b8	1.225739194722E-18



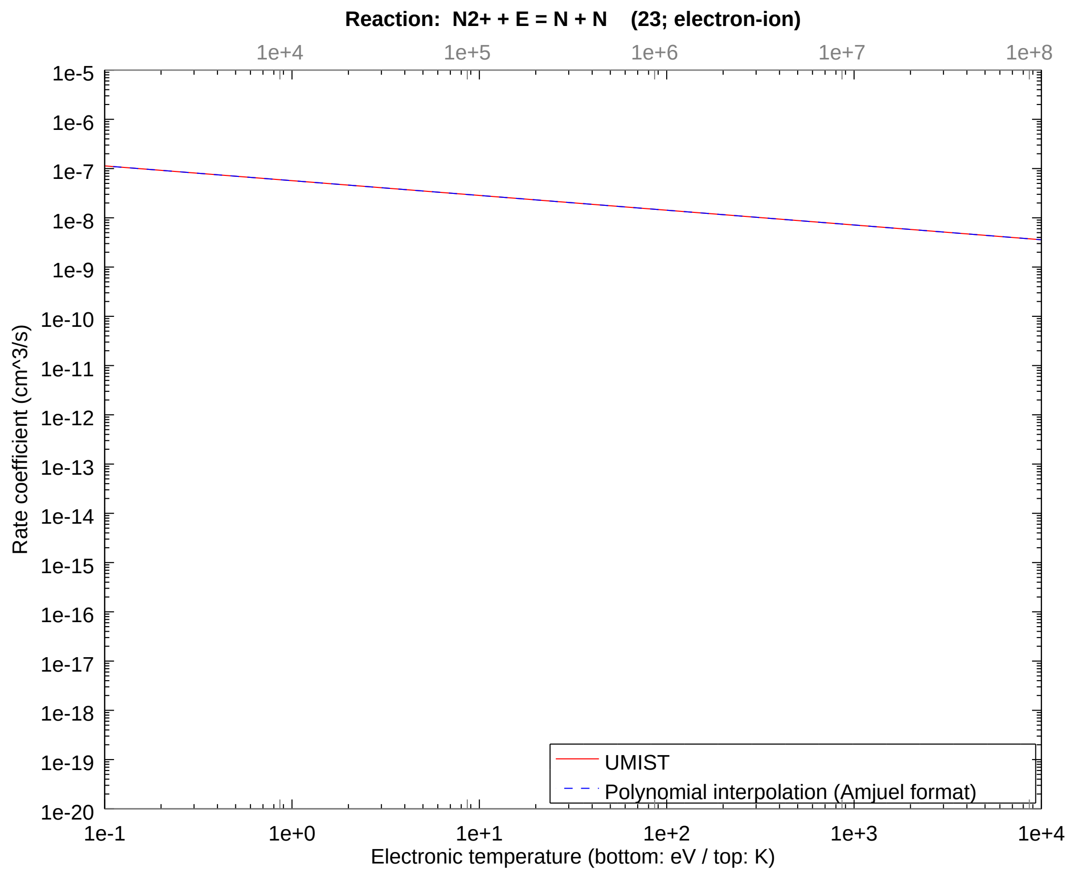
2.30 Reaction 23 N2+ + E = N(2D) + N

b0	-1.668392429045E+01	b1	-3.000000000000E-01	b2	1.961166050138E-14
b3	-5.930872513862E-15	b4	-2.419718156833E-15	b5	1.337121931740E-15
b6	-2.261907631853E-16	b7	1.639312063691E-17	b8	-4.303619611140E-19



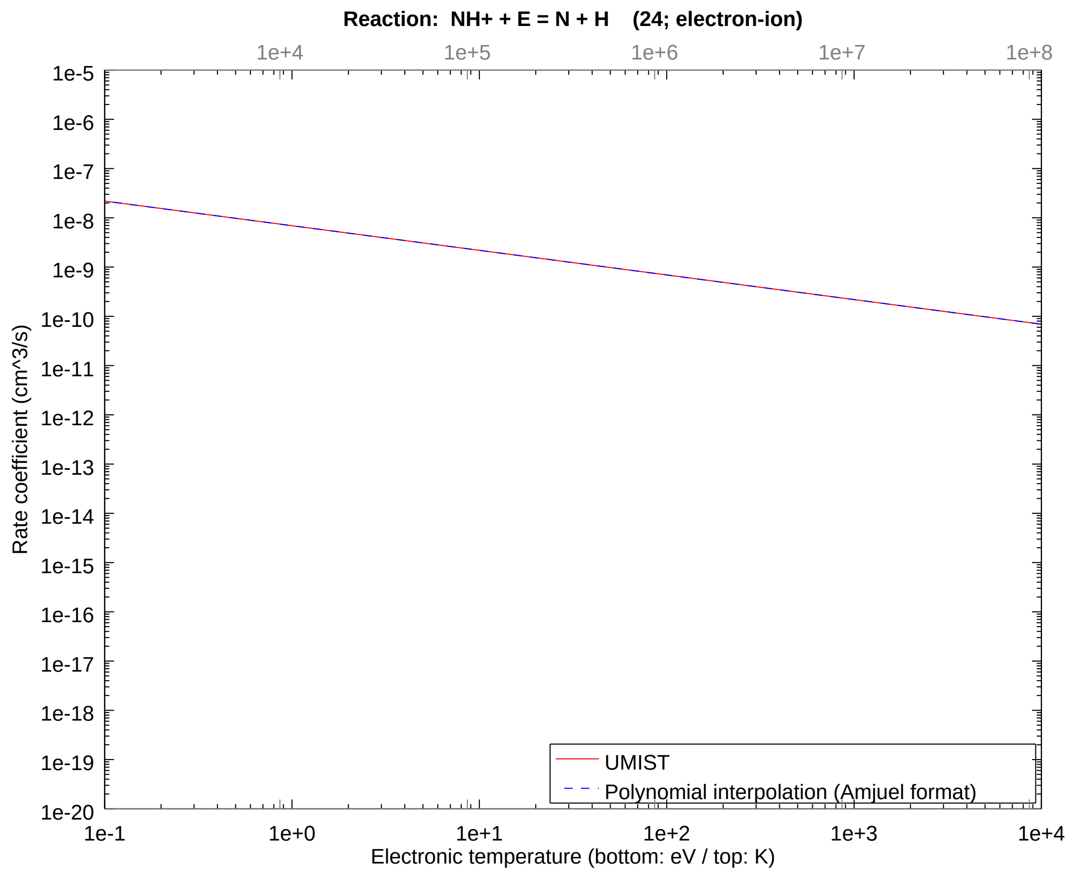
2.31 Reaction 23s N2+ + E = N + N

b0 -1.668392429045E+01	b1 -3.000000000000E-01	b2 1.961166050138E-14
b3 -5.930872513862E-15	b4 -2.419718156833E-15	b5 1.337121931740E-15
b6 -2.261907631853E-16	b7 1.639312063691E-17	b8 -4.303619611140E-19



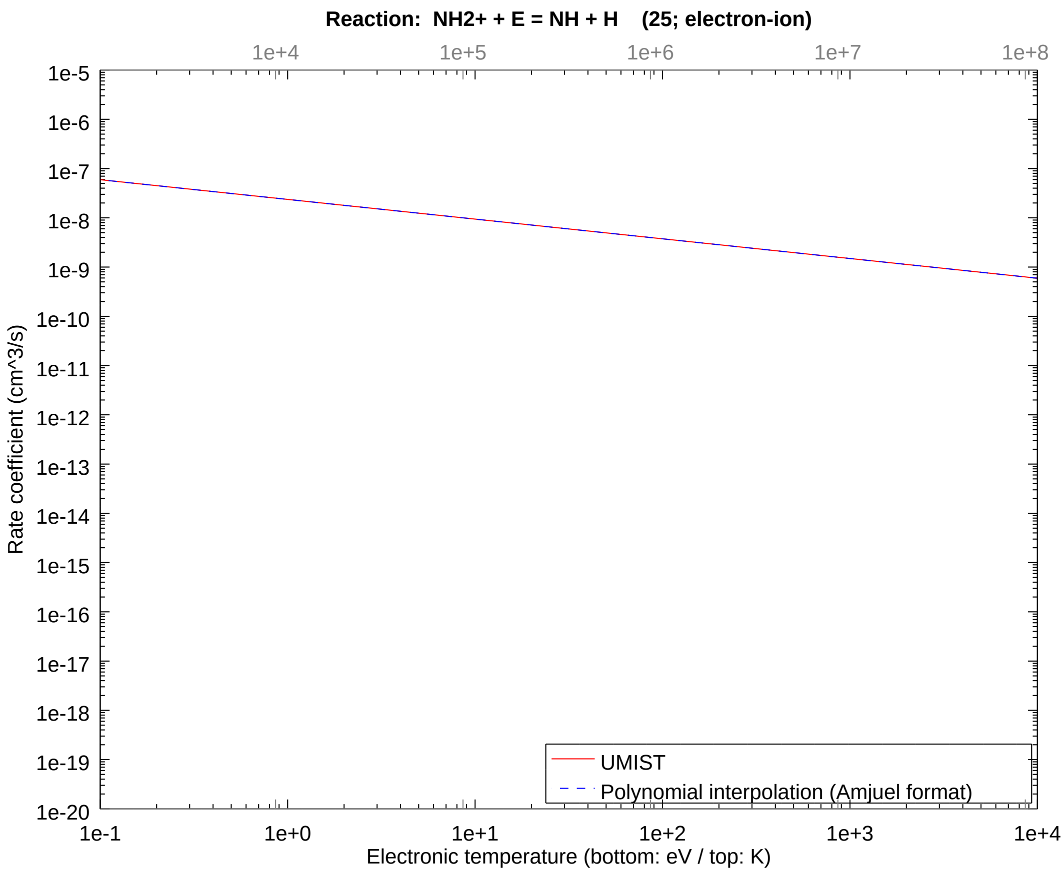
2.32 Reaction 24 NH+ + E = N + H

b0	-1.878949387217E+01	b1	-5.000000000000E-01	b2	3.726215495262E-14
b3	-8.098253227630E-15	b4	-4.407151722654E-15	b5	2.057457077627E-15
b6	-3.302568924012E-16	b7	2.375034314407E-17	b8	-6.443341406356E-19



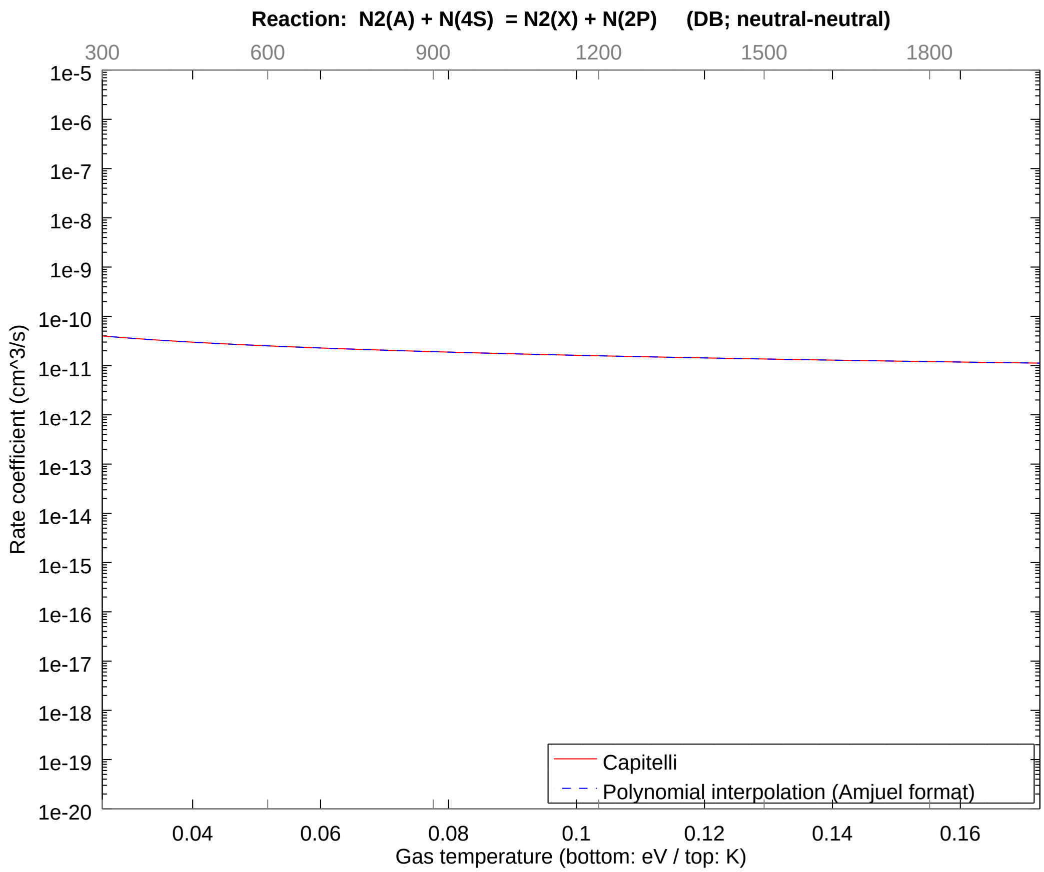
2.33 Reaction 25 NH2+ + E = NH + H

b0	-1.756023554439E+01	b1	-4.000000000000E-01	b2	3.137865680221E-14
b3	-4.837623409228E-15	b4	-5.267214245188E-15	b5	1.760403301084E-15
b6	-1.625073412217E-16	b7	-2.977237729308E-19	b8	4.512501894627E-19



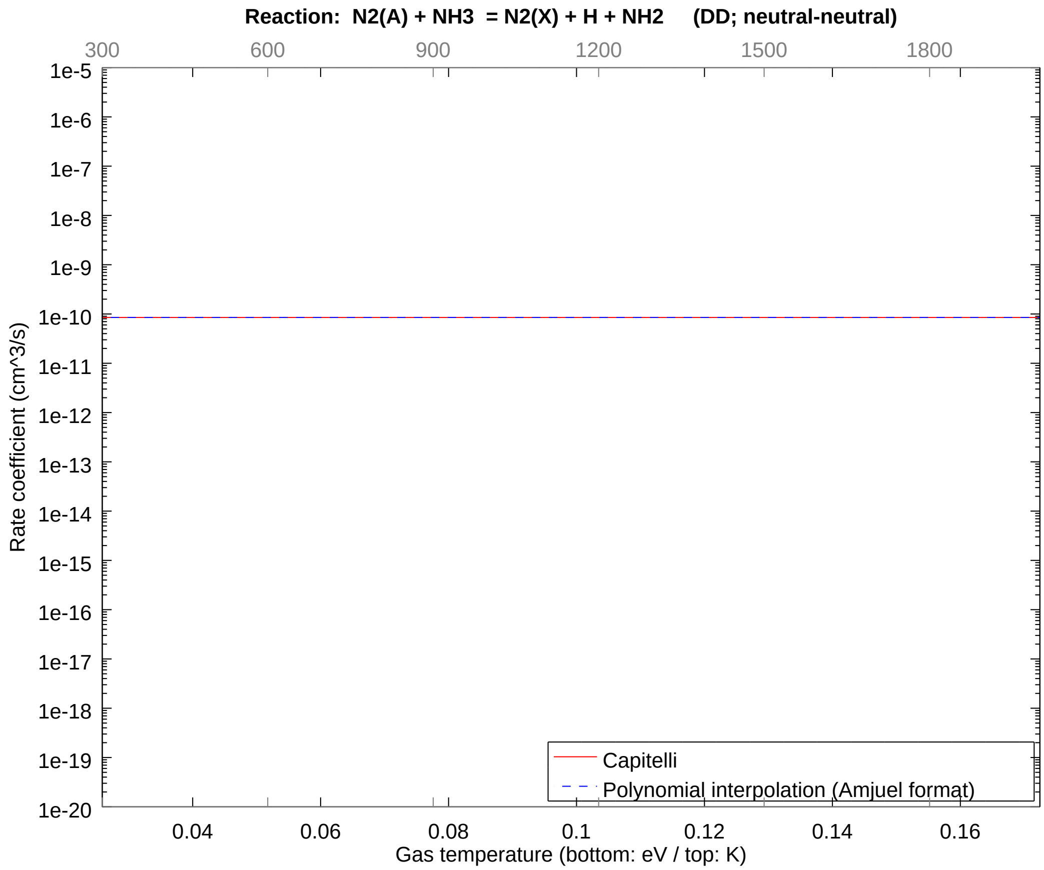
2.34 Reaction DB N2(A) + N(4S) = N2(X) + N(2P)

b0	-2.639089538605E+01	b1	-6.700000063335E-01	b2	-8.624699809858E-09
b3	-6.633570679721E-09	b4	-3.151551870313E-09	b5	-9.470599538949E-10
b6	-1.758156215834E-10	b7	-1.843916759489E-11	b8	-8.367108693434E-13



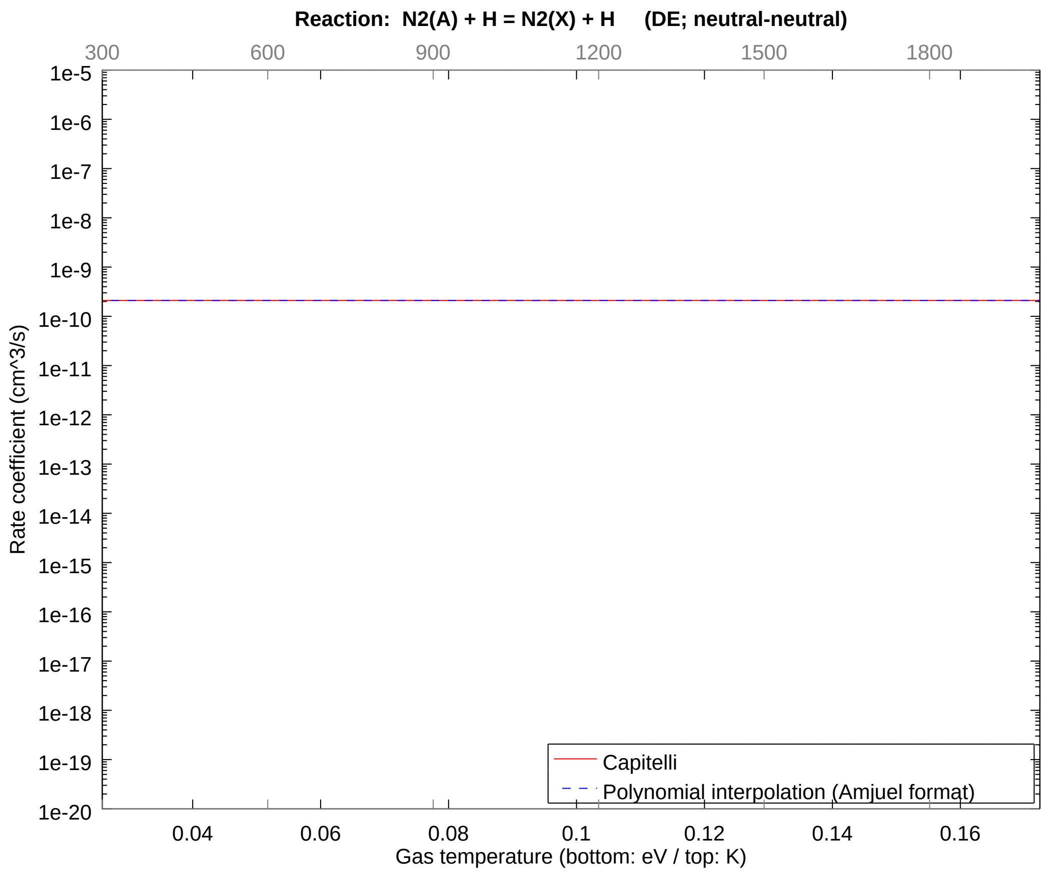
2.35 Reaction DD N2(A) + NH3 = N2(X) + H + NH2

b0	-2.318836985944E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



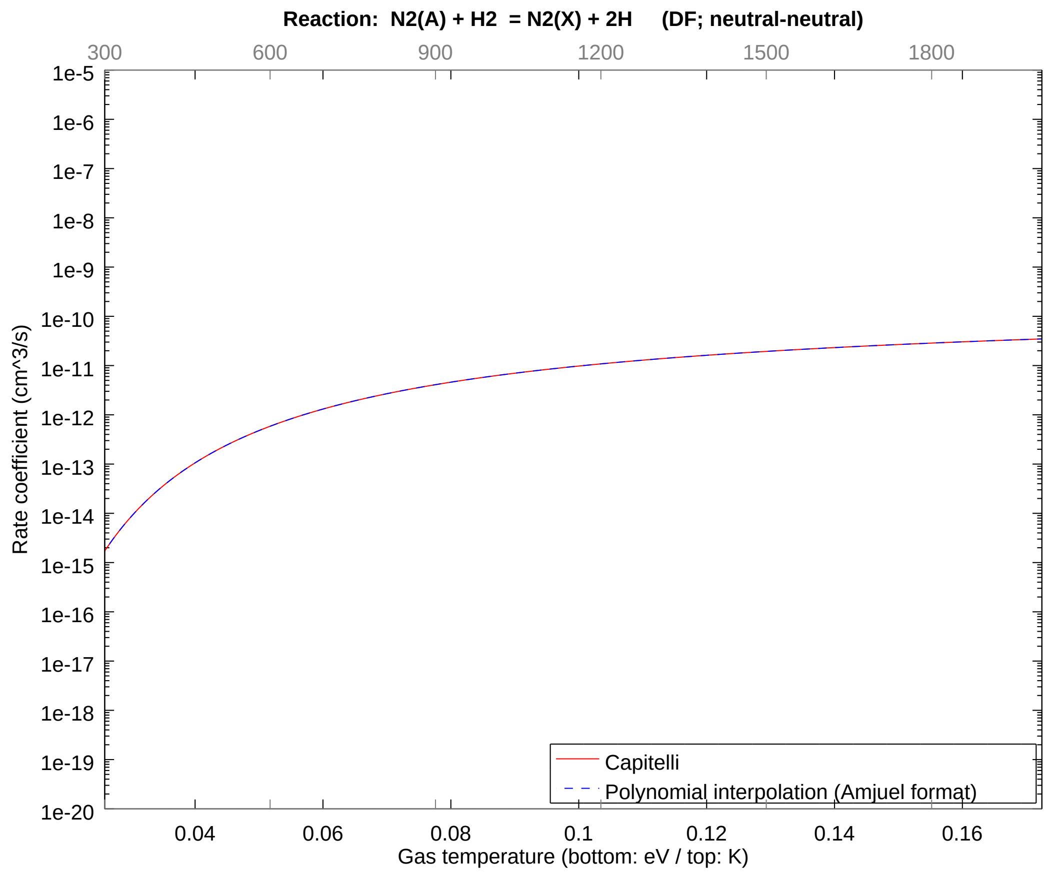
2.36 Reaction DE N2(A) + H = N2(X) + H

b0	-2.228391358521E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



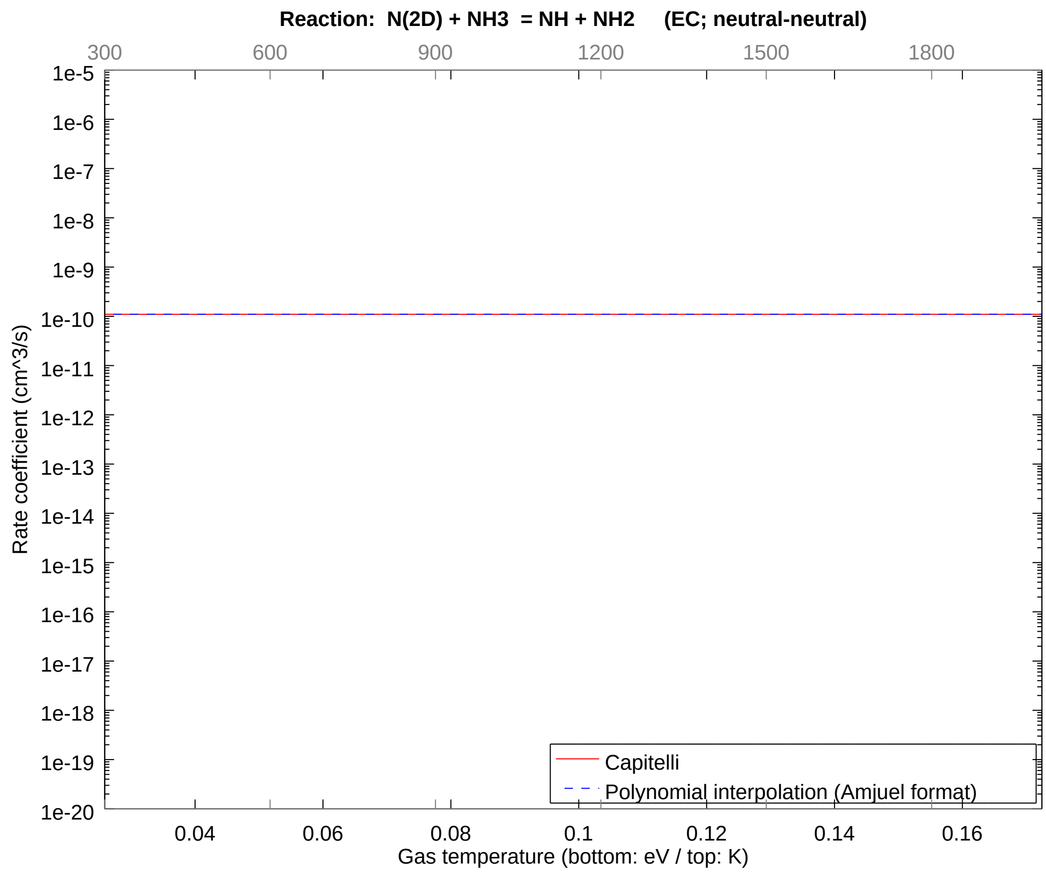
2.37 Reaction DF N2(A) + H2 = N2(X) + 2H

b0	-2.268826321125E+01	b1	1.143045937131E-01	b2	-4.372098380053E-01
b3	-2.007521826120E-01	b4	-1.512801491418E-01	b5	-4.722928220100E-02
b6	-1.184186593315E-02	b7	-1.507809868440E-03	b8	-1.121565090195E-04



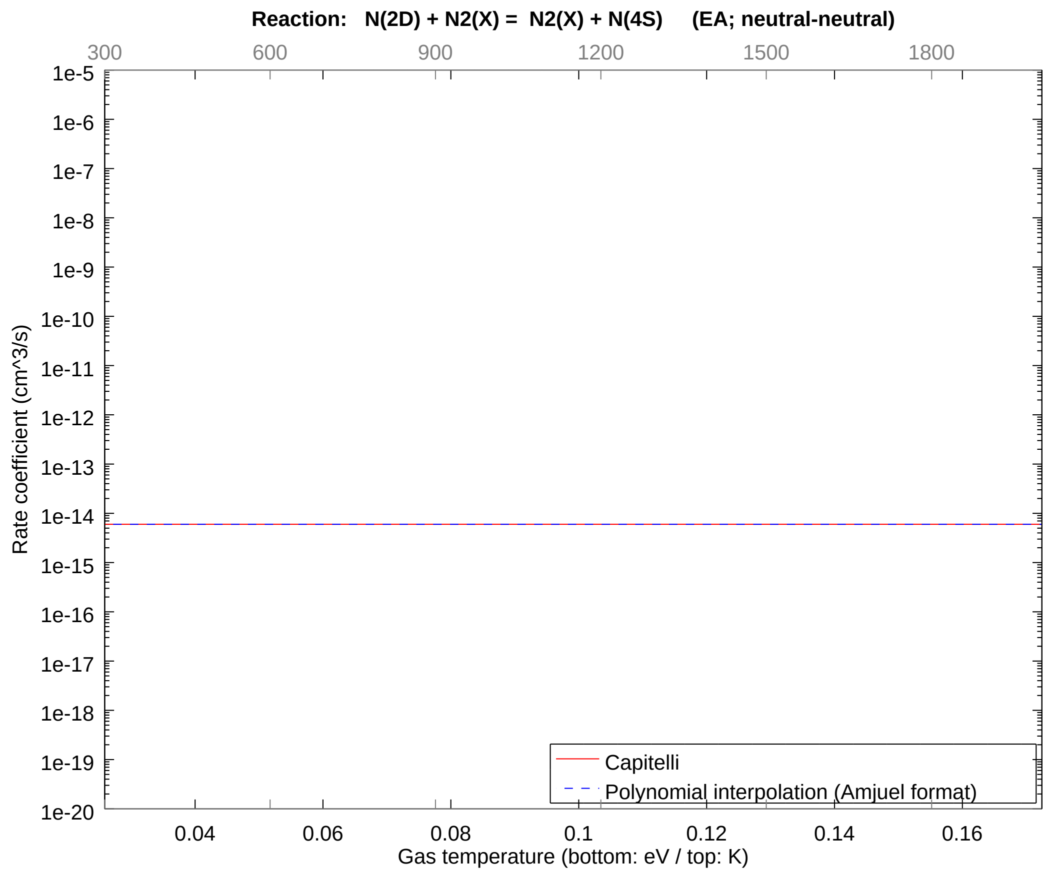
2.38 Reaction EC N(2D) + NH3 = NH + NH2

b0	-2.293054075014E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



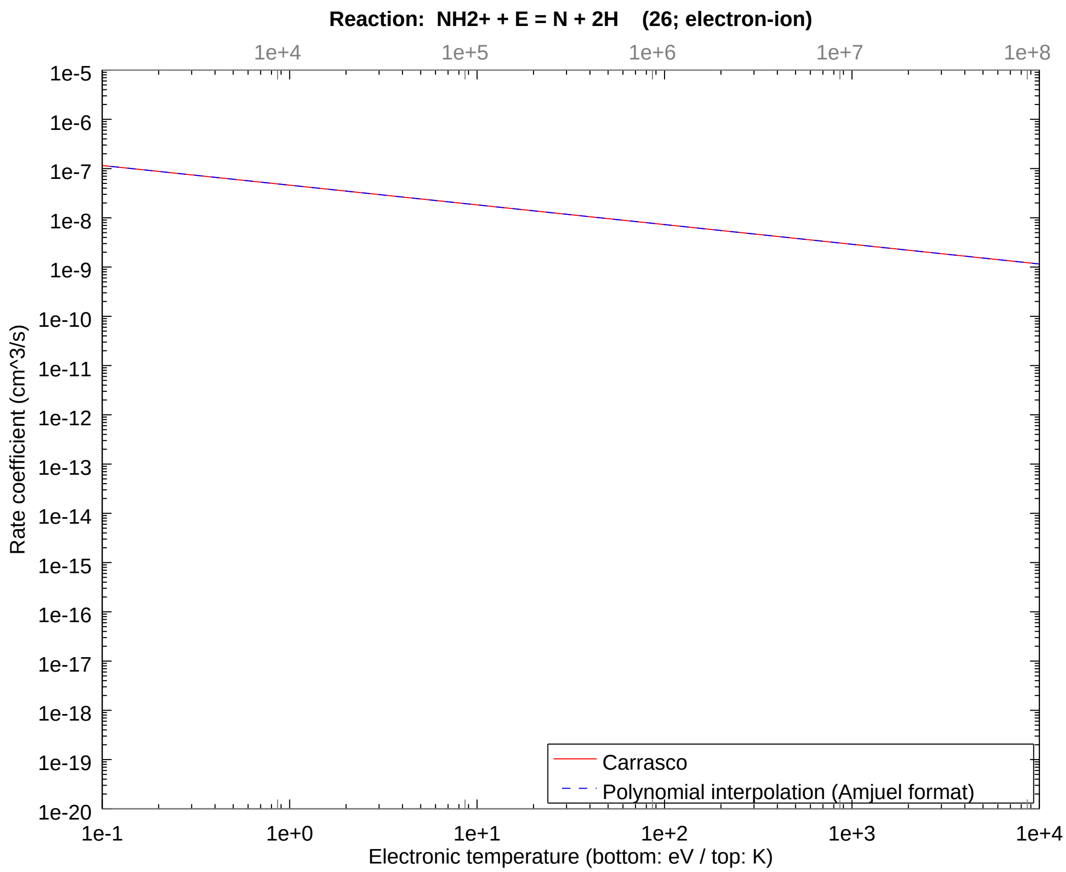
2.39 Reaction EA N(2D) + N2(X) = N2(X) + N(4S)

b0	-3.274701692568E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



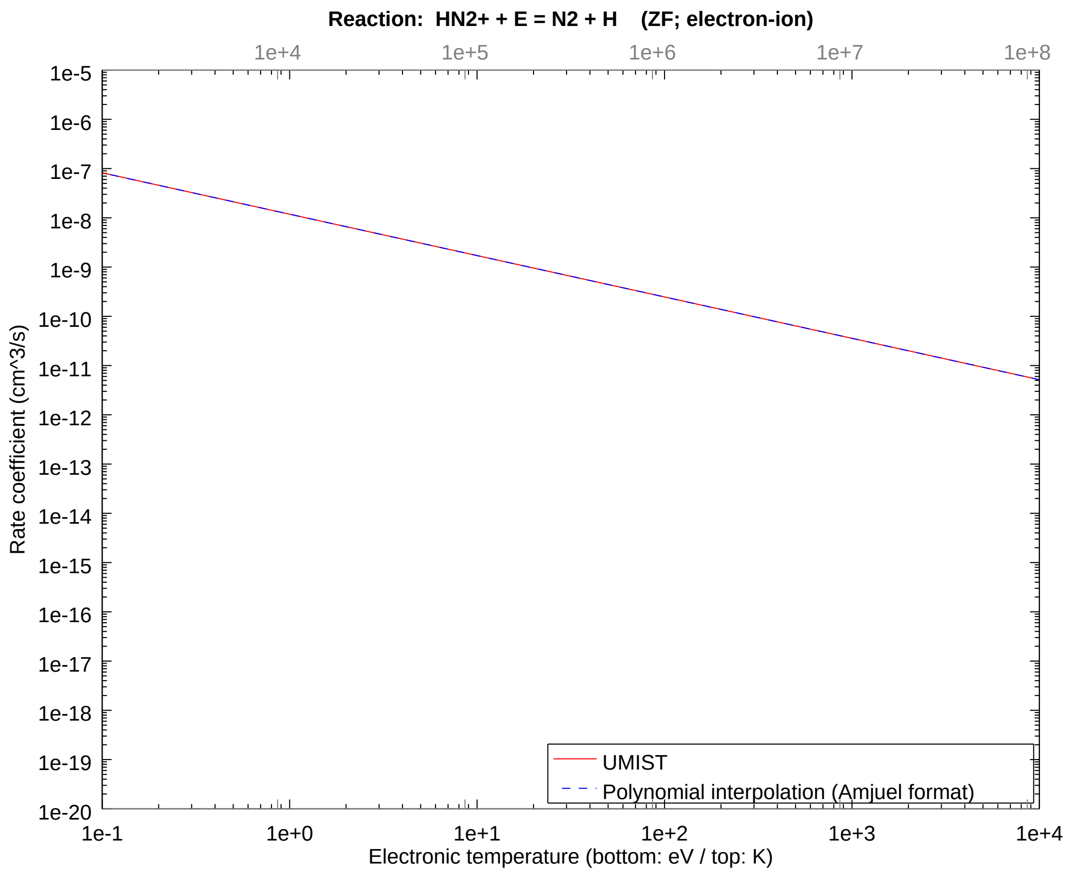
2.40 Reaction 26 NH2+ + E = N + 2H

b0	-1.689486230264E+01	b1	-4.000000000000E-01	b2	3.922332100276E-15
b3	4.933121613073E-15	b4	-2.597700526548E-15	b5	1.035377292669E-16
b6	1.120423733007E-16	b7	-1.903463952049E-17	b8	8.964767471941E-19



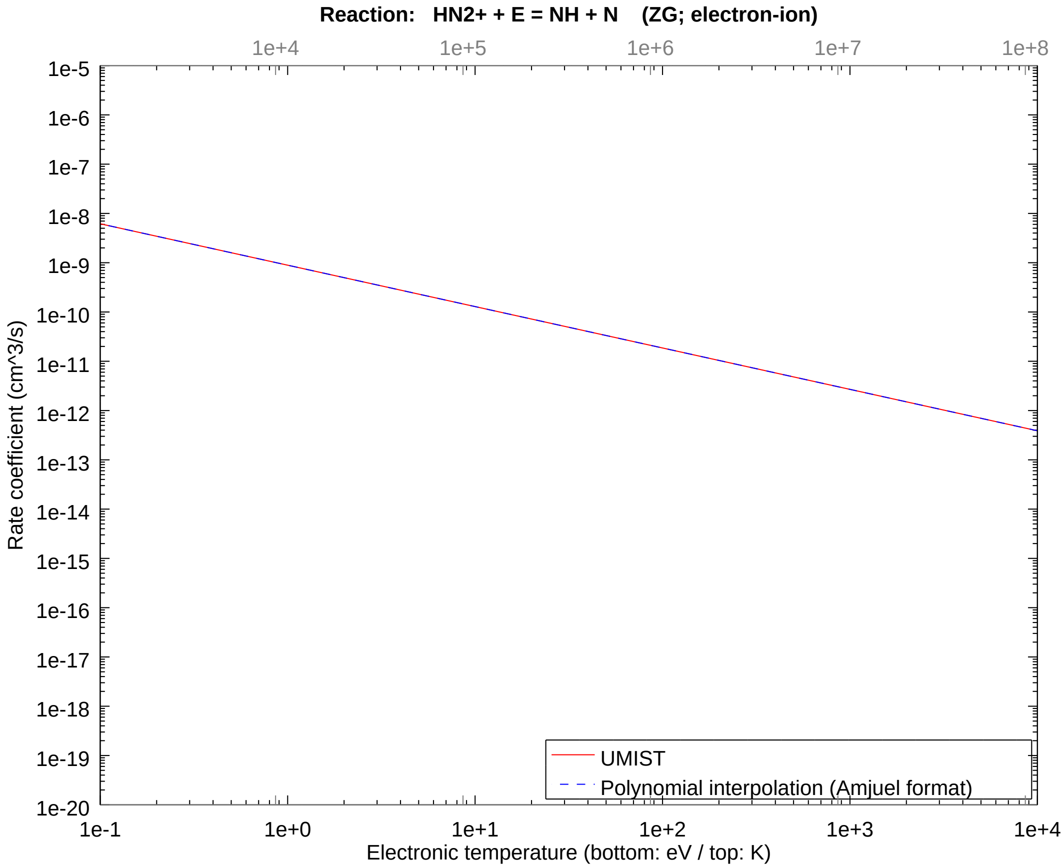
2.41 Reaction ZF HN2+ + E = N2 + H

b0	-1.825208158533E+01	b1	-8.400000000000E-01	b2	1.961166050138E-15
b3	6.667678312880E-16	b4	4.608871172109E-18	b5	-8.196062952573E-17
b6	1.886554408333E-17	b7	-1.936160340987E-18	b8	8.166550190105E-20



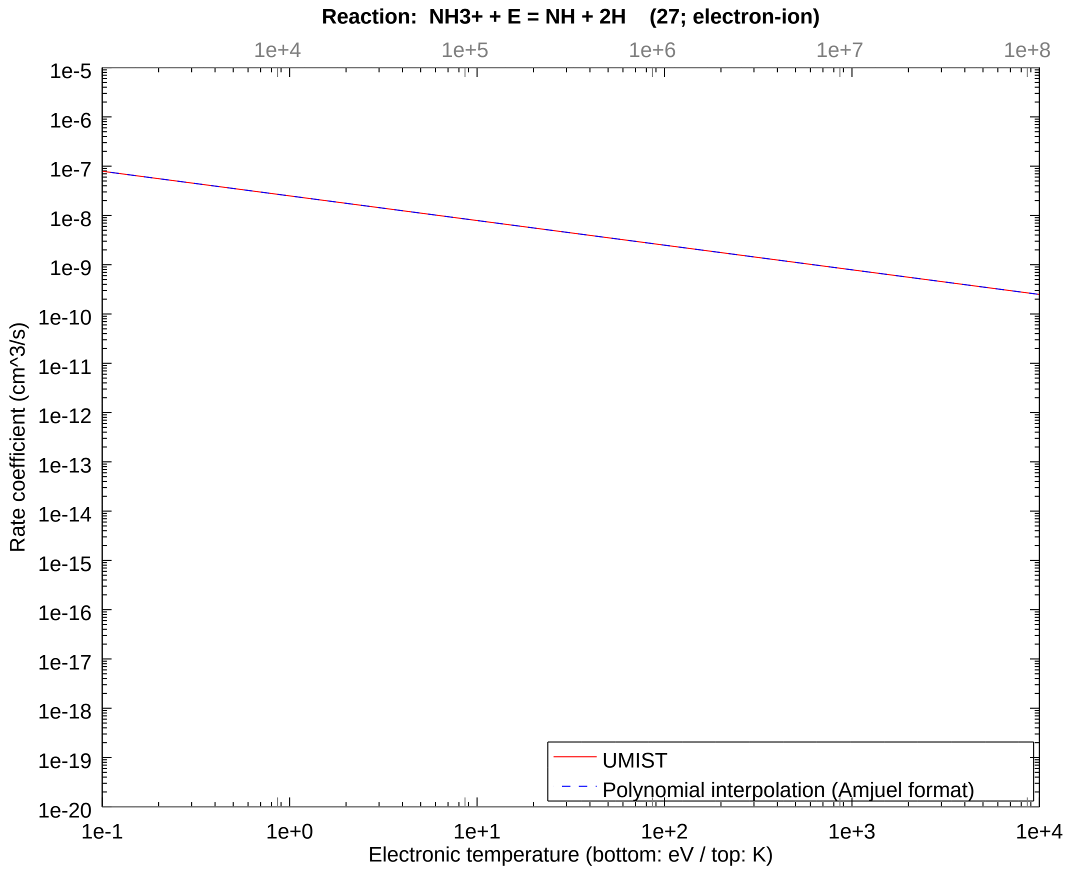
2.42 Reaction ZG HN2+ + E = NH + N

b0	-2.083843485145E+01	b1	-8.400000000000E-01	b2	3.530098890249E-14
b3	-8.211238604995E-15	b4	-4.923772150637E-15	b5	2.244719695150E-15
b6	-3.385173756315E-16	b7	2.160520453501E-17	b8	-4.775349462097E-19



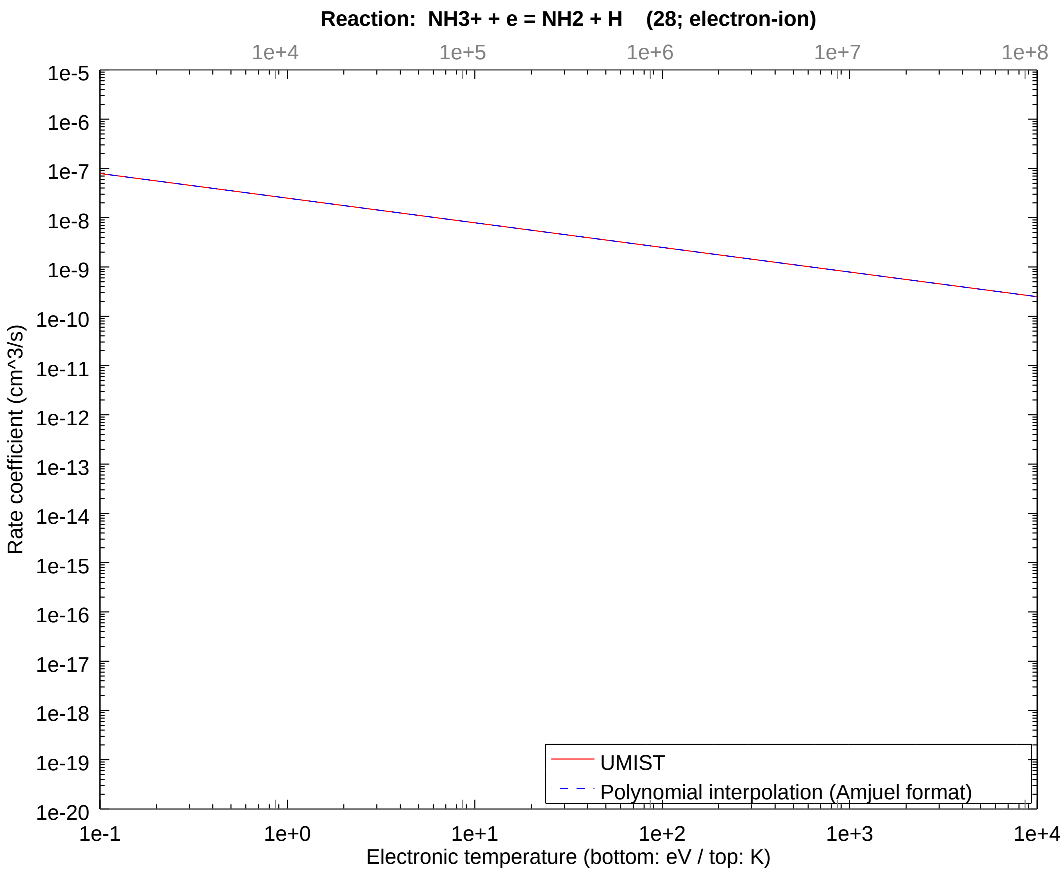
2.43 Reaction 27 NH3+ + E = NH + 2H

b0	-1.750726887094E+01	b1	-5.000000000000E-01	b2	2.745632470193E-14
b3	-1.004763624926E-14	b4	-2.722575161826E-15	b5	1.938931165252E-15
b6	-3.680508755642E-16	b7	2.957064295090E-17	b8	-8.674063015304E-19



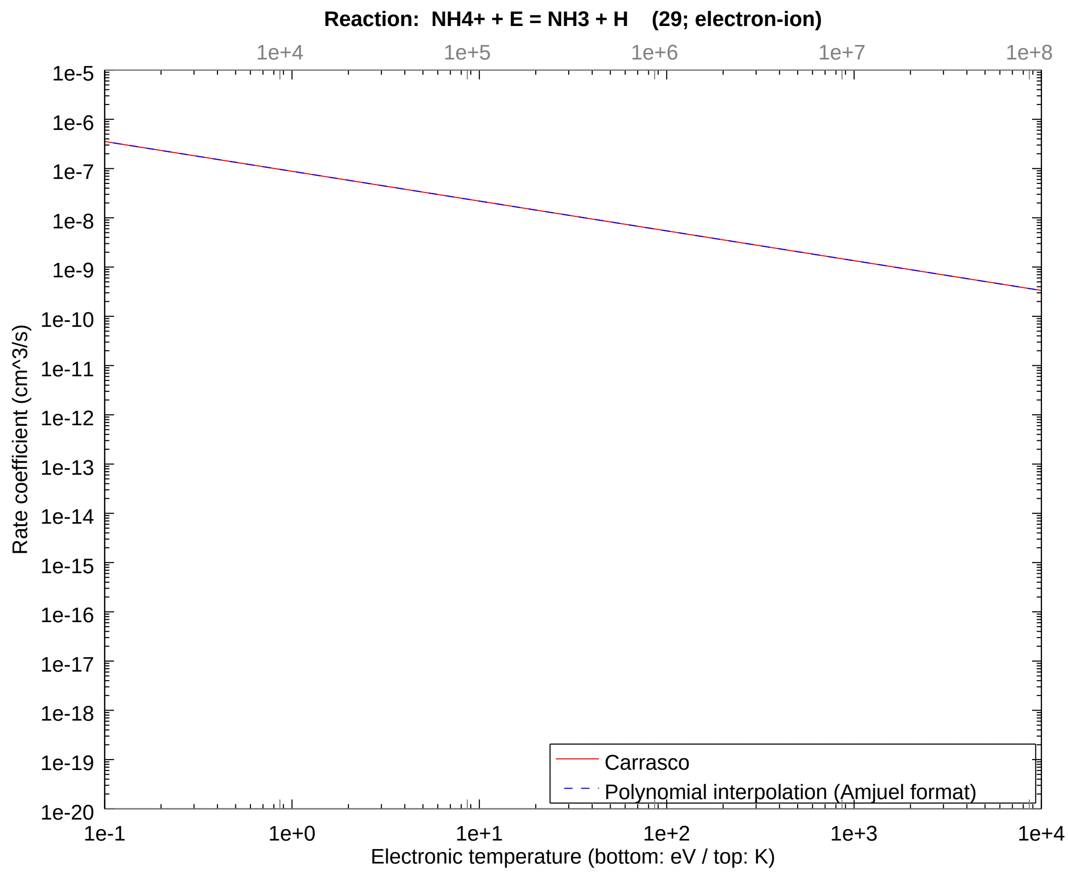
2.44 Reaction 28 NH3+ + e = NH2 + H

b0	-1.750726887094E+01	b1	-5.000000000000E-01	b2	2.745632470193E-14
b3	-1.004763624926E-14	b4	-2.722575161826E-15	b5	1.938931165252E-15
b6	-3.680508755642E-16	b7	2.957064295090E-17	b8	-8.674063015304E-19



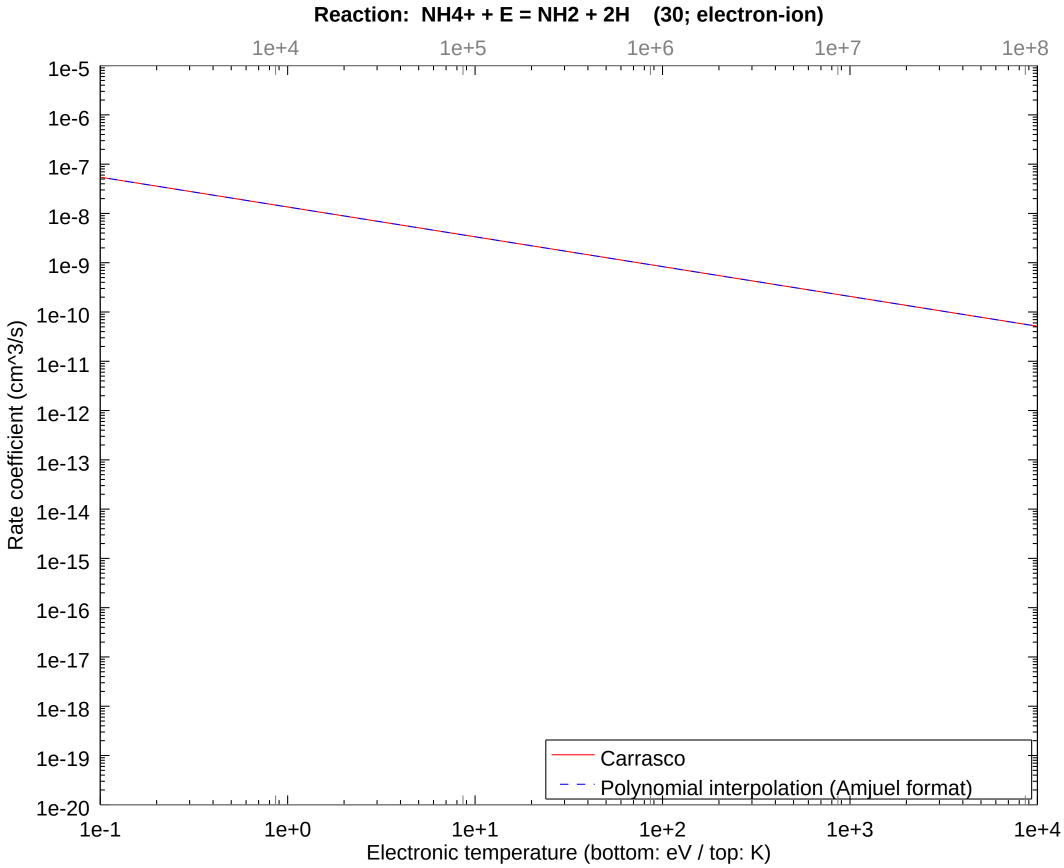
2.45 Reaction 29 NH4+ + E = NH3 + H

b0	-1.624544842816E+01	b1	-6.050000000000E-01	b2	0.000000000000E+00
b3	8.306736808840E-16	b4	-4.686048483672E-17	b5	-2.950510937902E-16
b6	1.050373201078E-16	b7	-1.304766889864E-17	b8	5.502683070338E-19



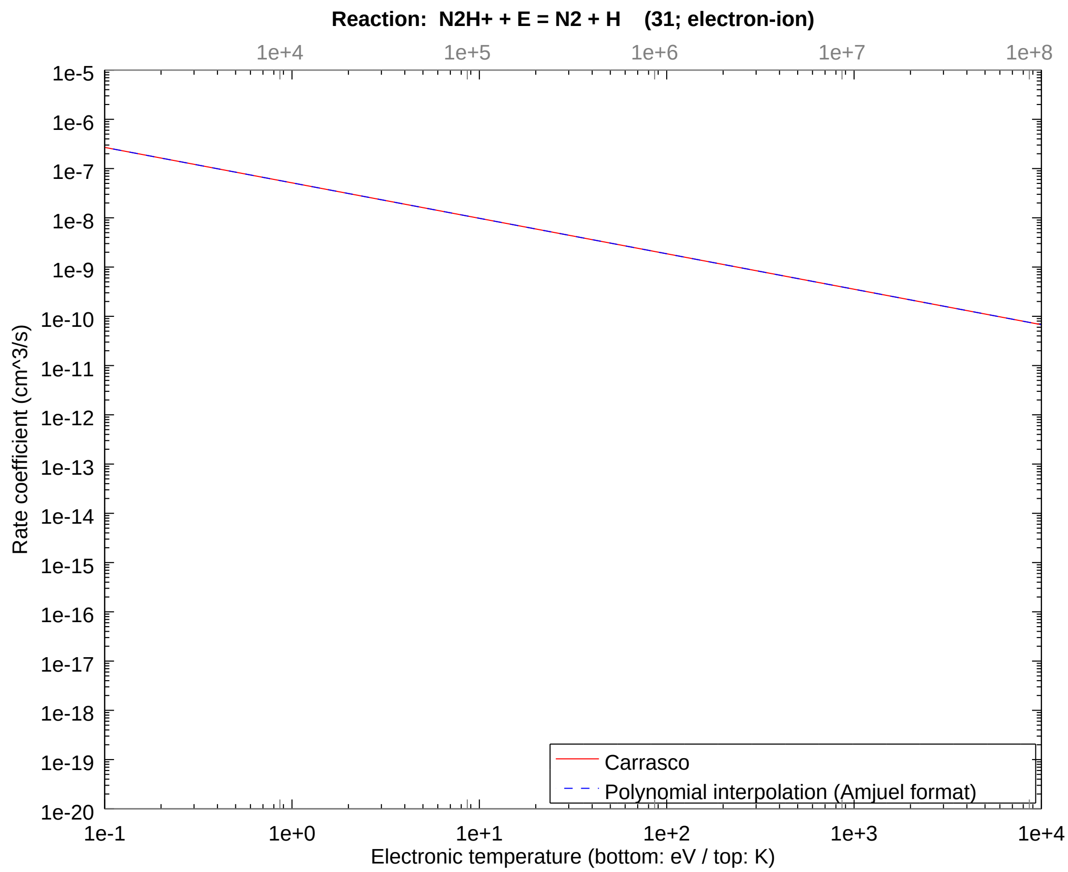
2.46 Reaction 30 NH4+ + E = NH2 + 2H

b0	-1.811912501986E+01	b1	-6.050000000000E-01	b2	2.941749075207E-14
b3	-9.380868417975E-15	b4	-3.748190362625E-15	b5	2.200689446104E-15
b6	-3.784562783185E-16	b7	2.695309824768E-17	b8	-6.649703260841E-19



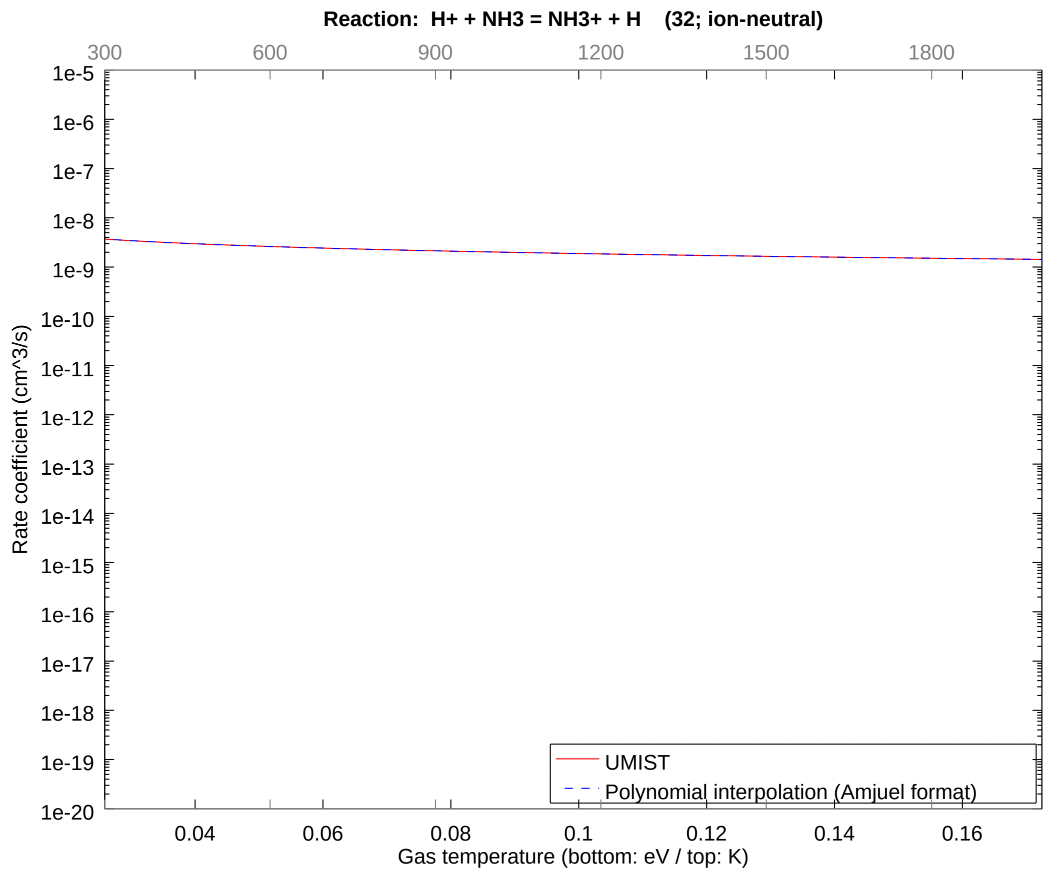
2.47 Reaction 31 N2H+ + E = N2 + H

b0	-1.678575516040E+01	b1	-7.200000000000E-01	b2	2.941749075207E-14
b3	-1.766359676538E-15	b4	-4.297937615358E-15	b5	1.044560914165E-15
b6	-3.956040344312E-17	b7	-8.226431945192E-18	b8	6.016644534124E-19



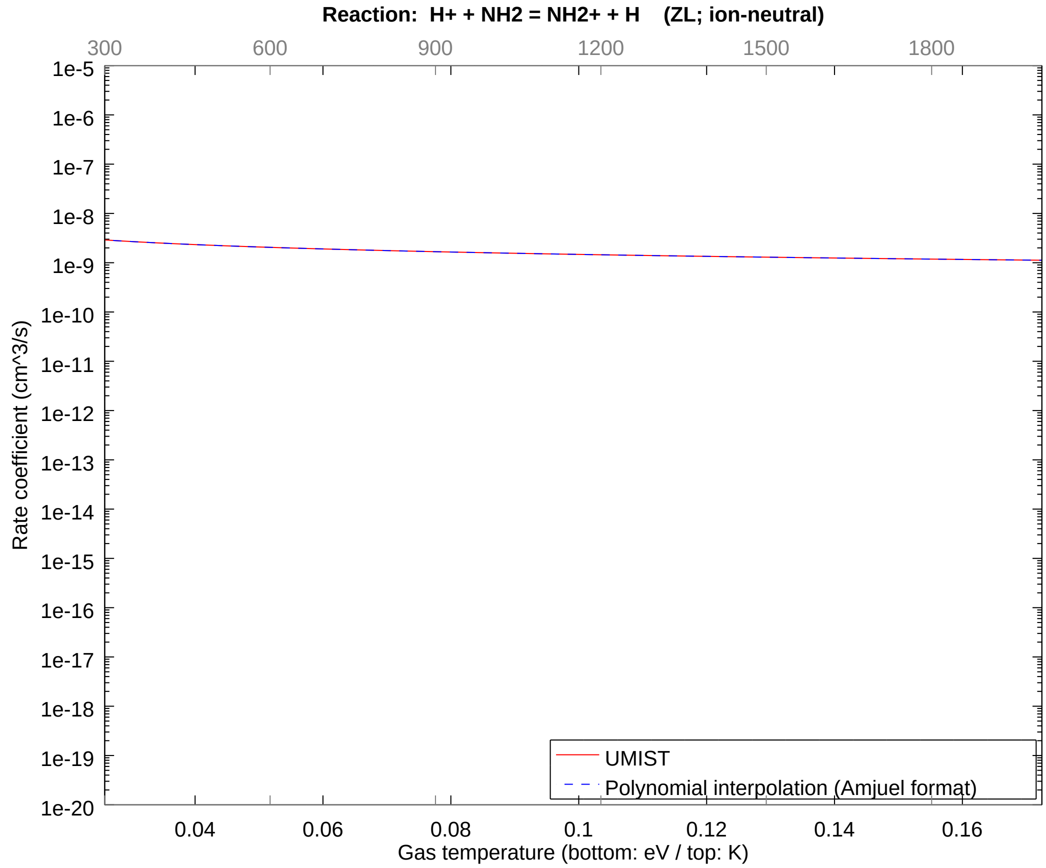
2.48 Reaction 32 H+ + NH3 = NH3+ + H

b0 -2.124236116983E+01	b1 -5.000000049540E-01	b2 -6.581921337736E-09
b3 -4.941457323182E-09	b4 -2.292305524667E-09	b5 -6.727088691196E-10
b6 -1.219481205389E-10	b7 -1.248479911283E-11	b8 -5.526669688591E-13



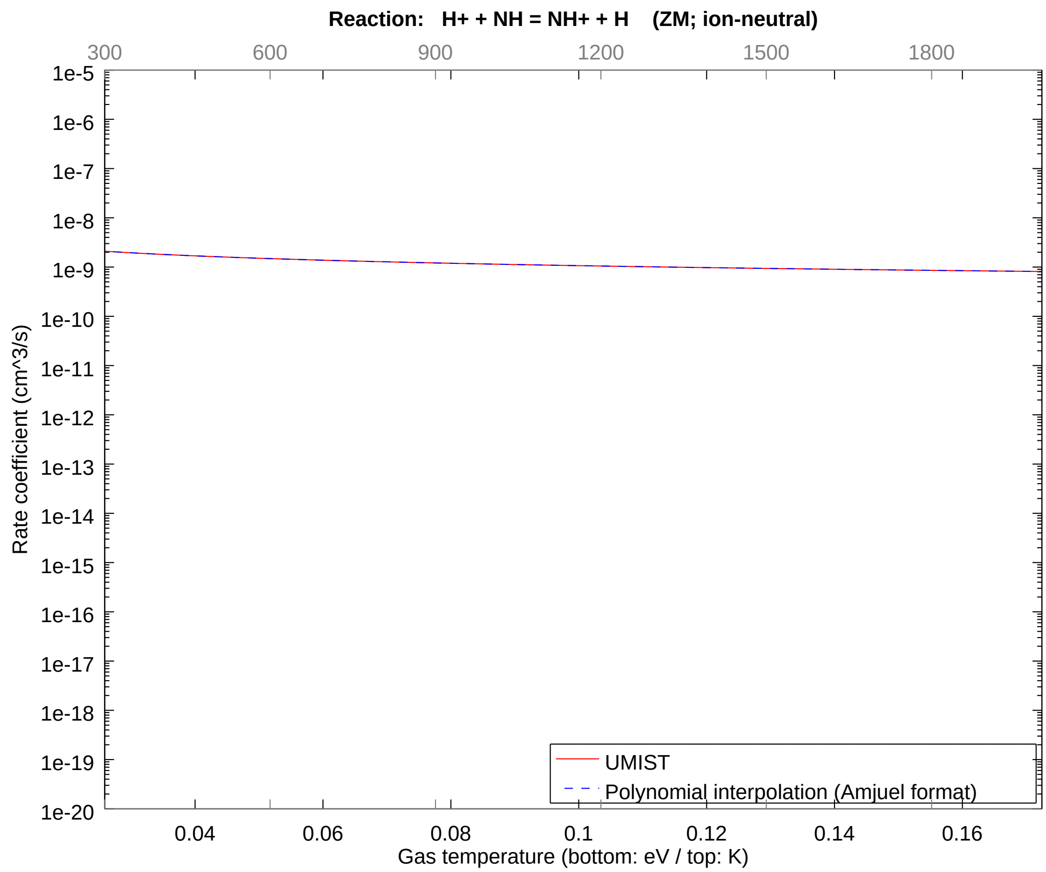
2.49 Reaction ZL H+ + NH2 = NH2+ + H

b0 -2.148598325252E+01	b1 -5.000000052634E-01	b2 -7.274686331173E-09
b3 -5.686468493740E-09	b4 -2.749065636477E-09	b5 -8.416173449789E-10
b6 -1.593528795054E-10	b7 -1.706404308800E-11	b8 -7.914454258785E-13



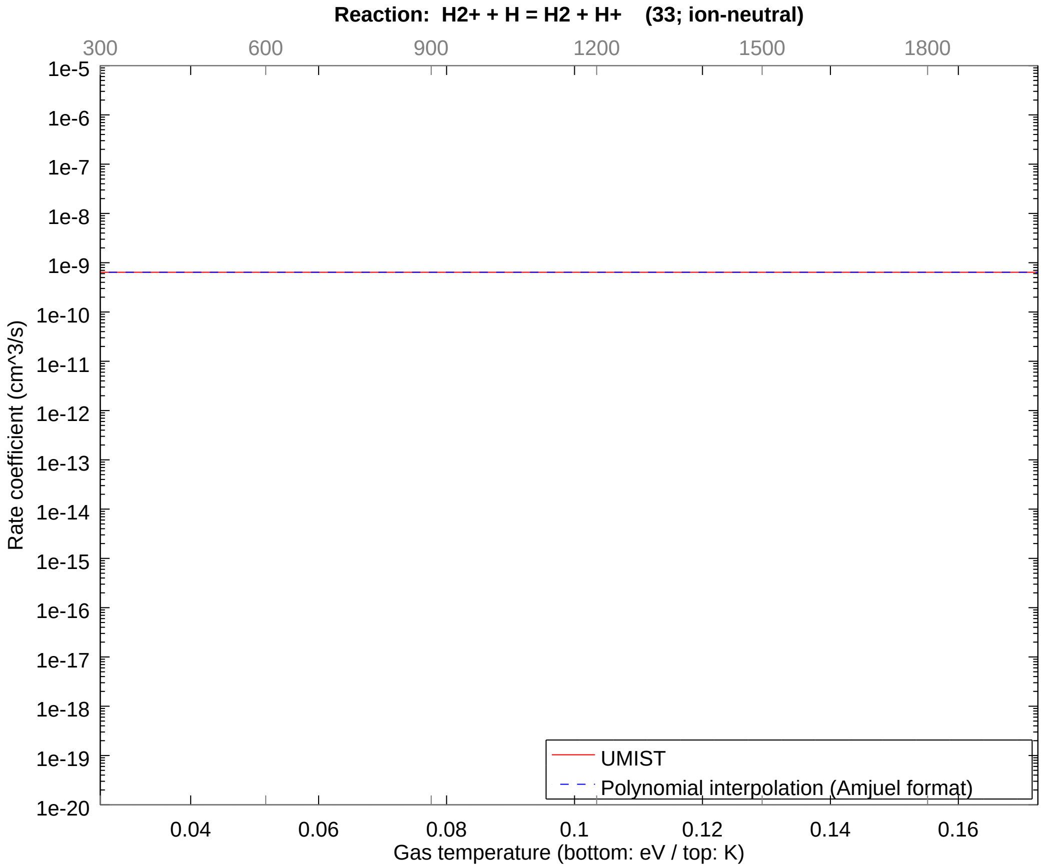
2.50 Reaction ZM H+ + NH = NH+ + H

b0 -2.180875664609E+01	b1 -5.000000092849E-01	b2 -1.261517857558E-08
b3 -9.699489564652E-09	b4 -4.615551314815E-09	b5 -1.391879349947E-09
b6 -2.597688881768E-10	b7 -2.743376818213E-11	b8 -1.255322335567E-12



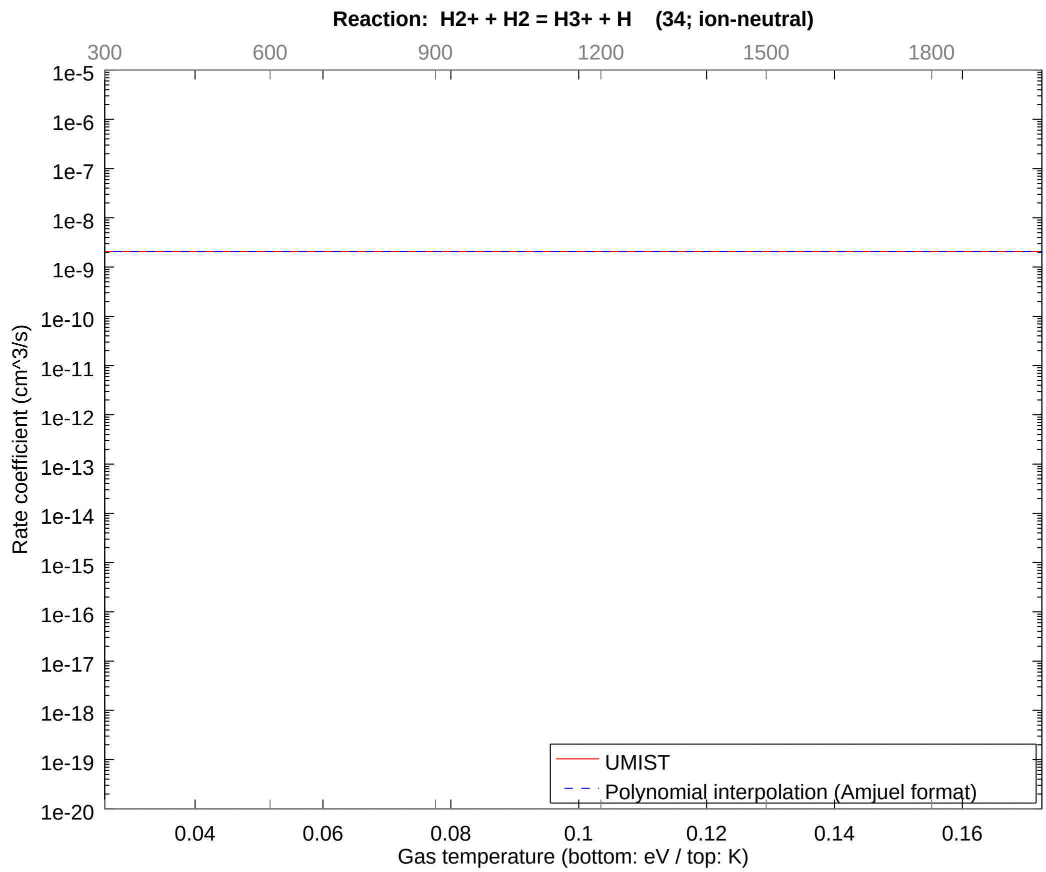
2.51 Reaction 33 H2+ + H = H2 + H+

b0	-2.116955293957E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



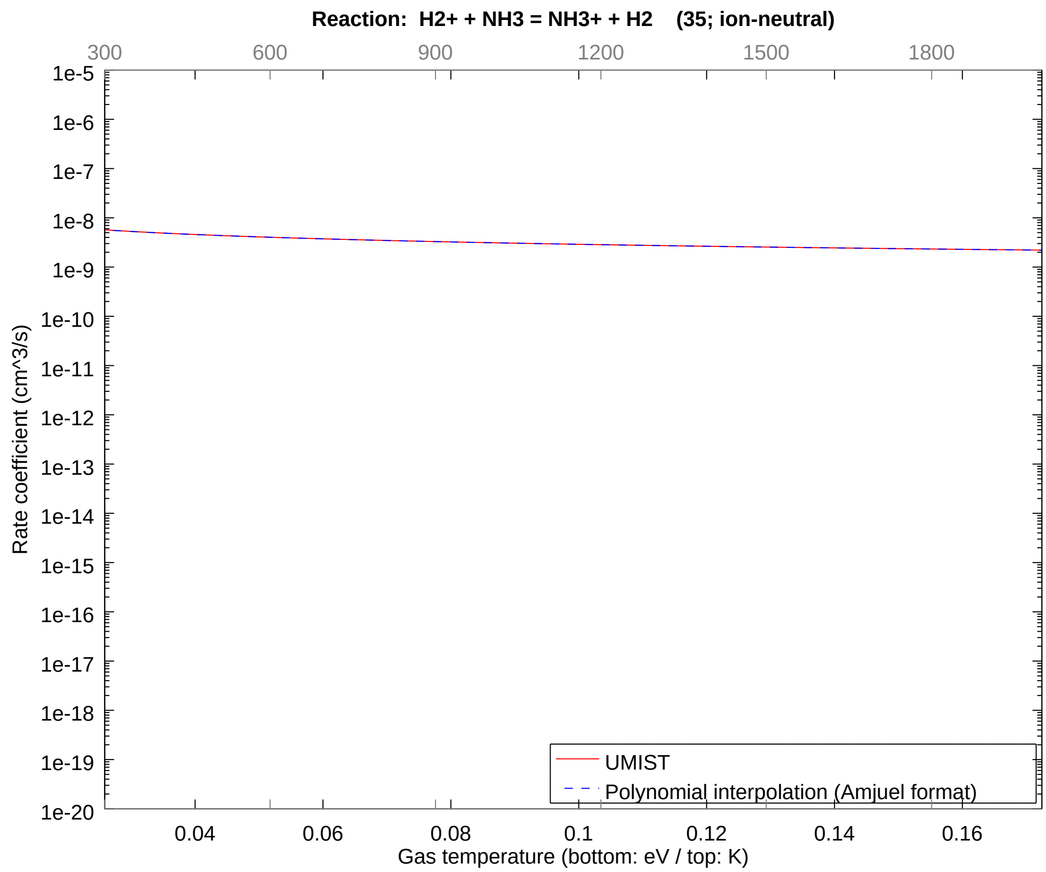
2.52 Reaction 34 H2+ + H2 = H3+ + H

b0	-1.999089794323E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



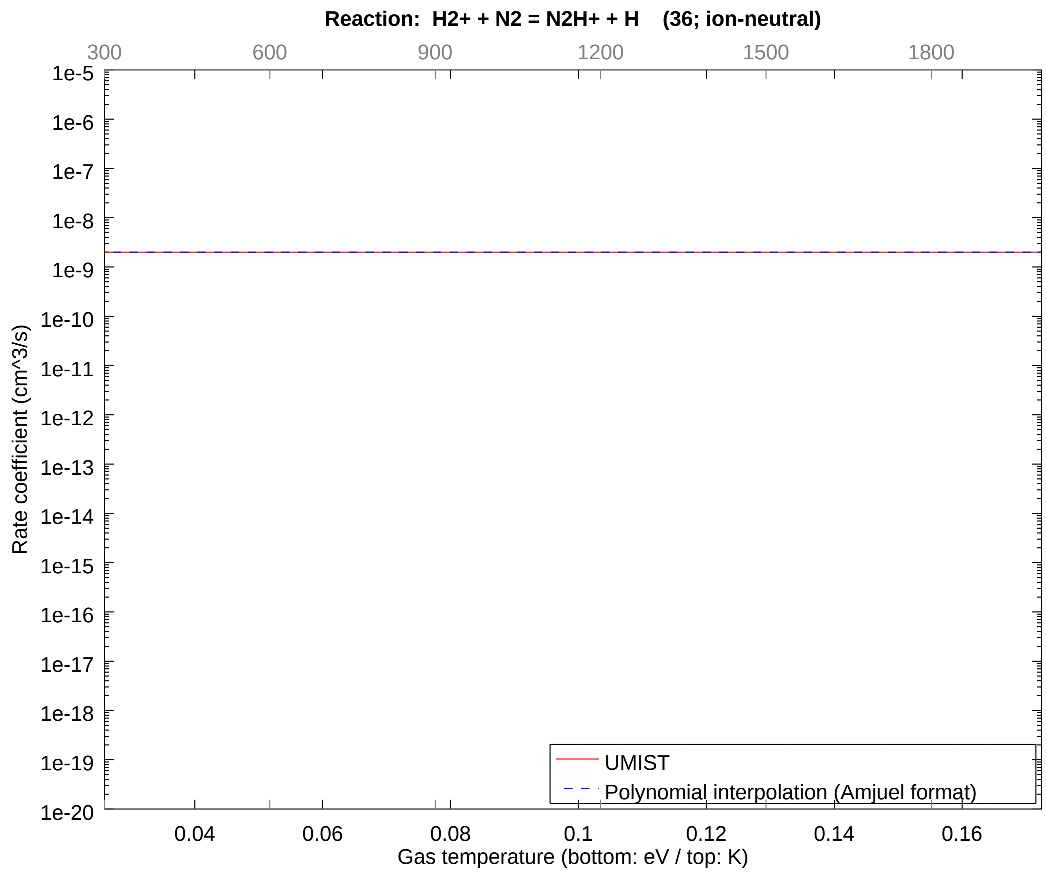
2.53 Reaction 35 H2+ + NH3 = NH3+ + H2

b0 -2.081022781504E+01	b1 -5.000000063682E-01	b2 -8.716351898605E-09
b3 -6.744375376163E-09	b4 -3.226162467434E-09	b5 -9.769045577405E-10
b6 -1.828822359175E-10	b7 -1.935531934856E-11	b8 -8.868789870933E-13



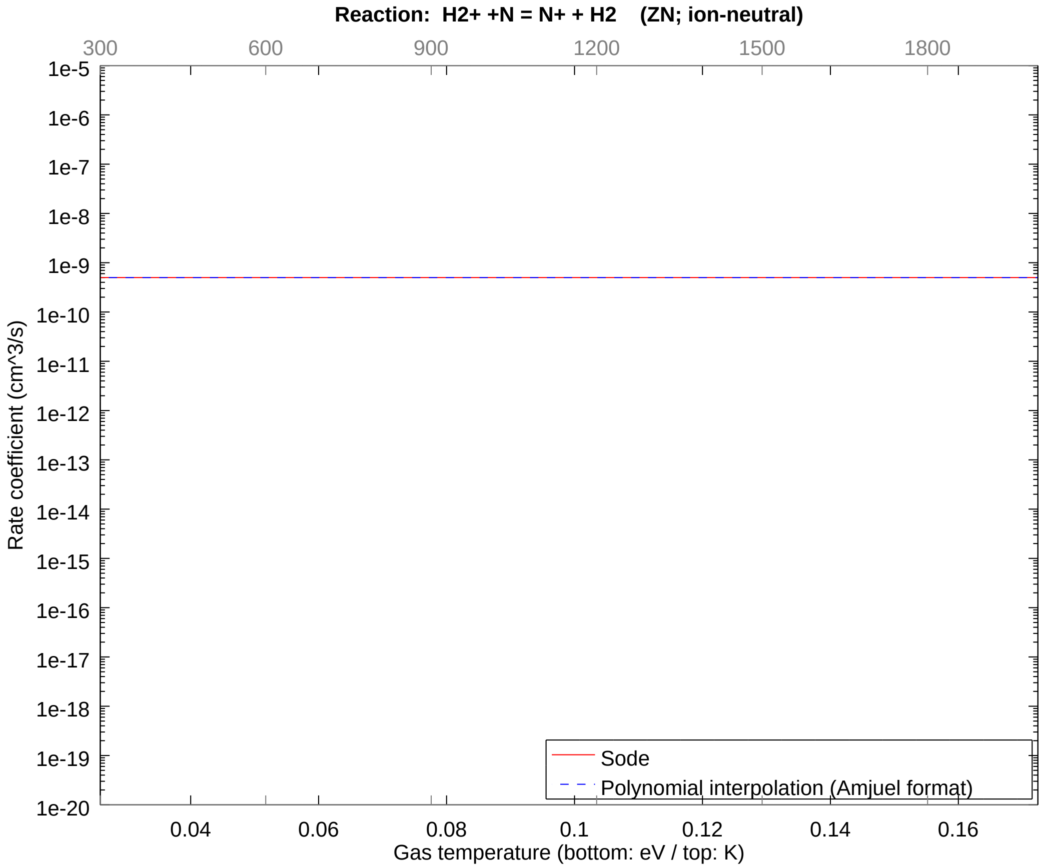
2.54 Reaction 36 H2+ + N2 = N2H+ + H

b0	-2.003011865639E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



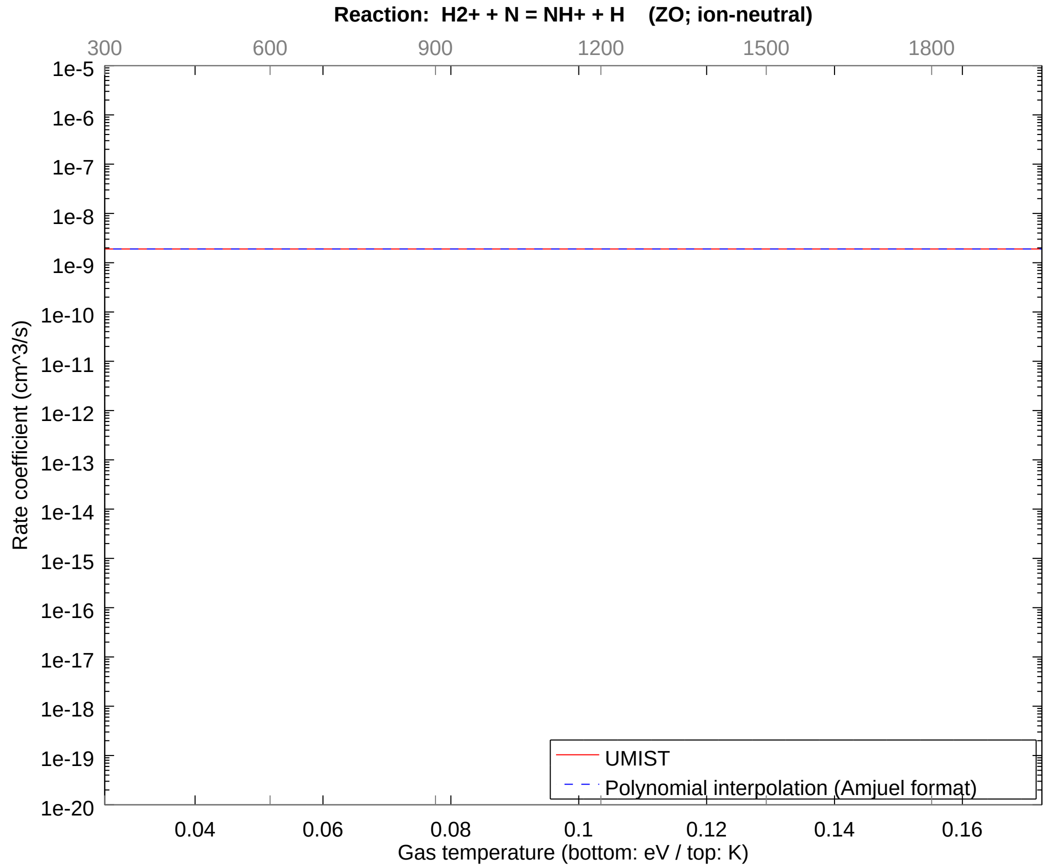
2.55 Reaction ZN H2+ +N = N+ + H2

b0	-2.141641301751E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



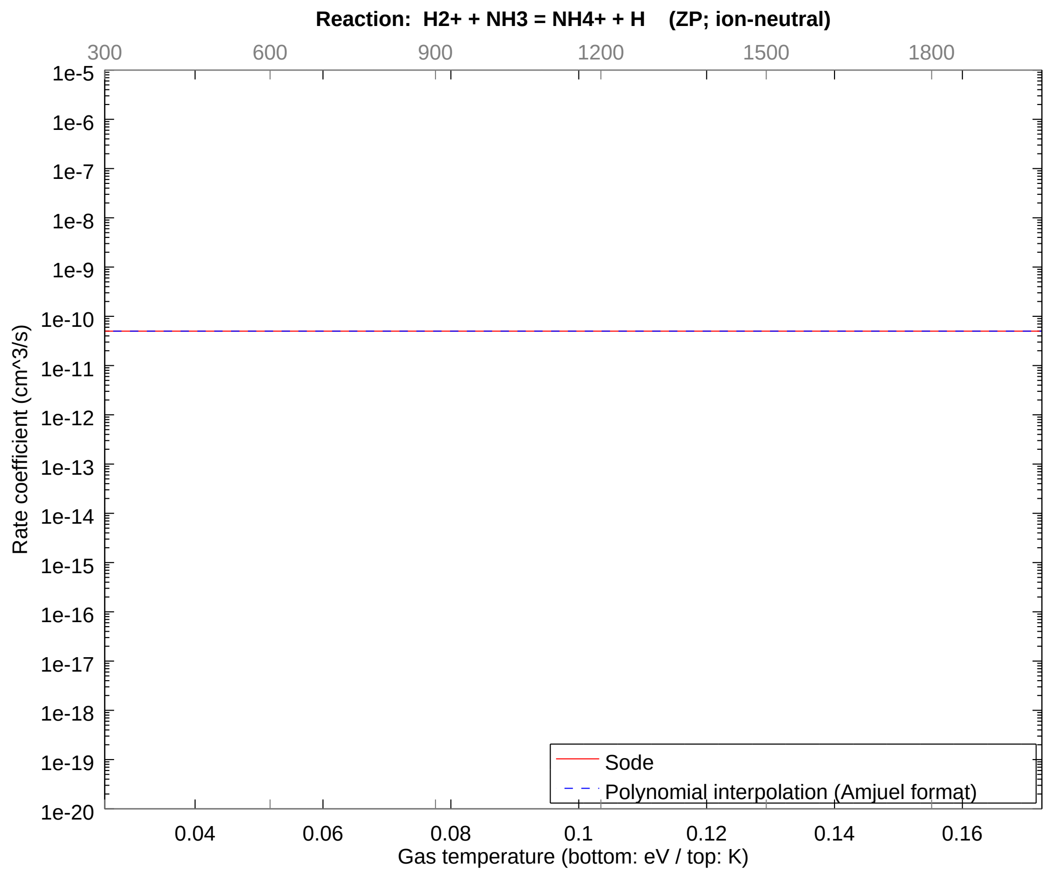
2.56 Reaction ZO H2+ + N = NH+ + H

b0	-2.008141195077E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



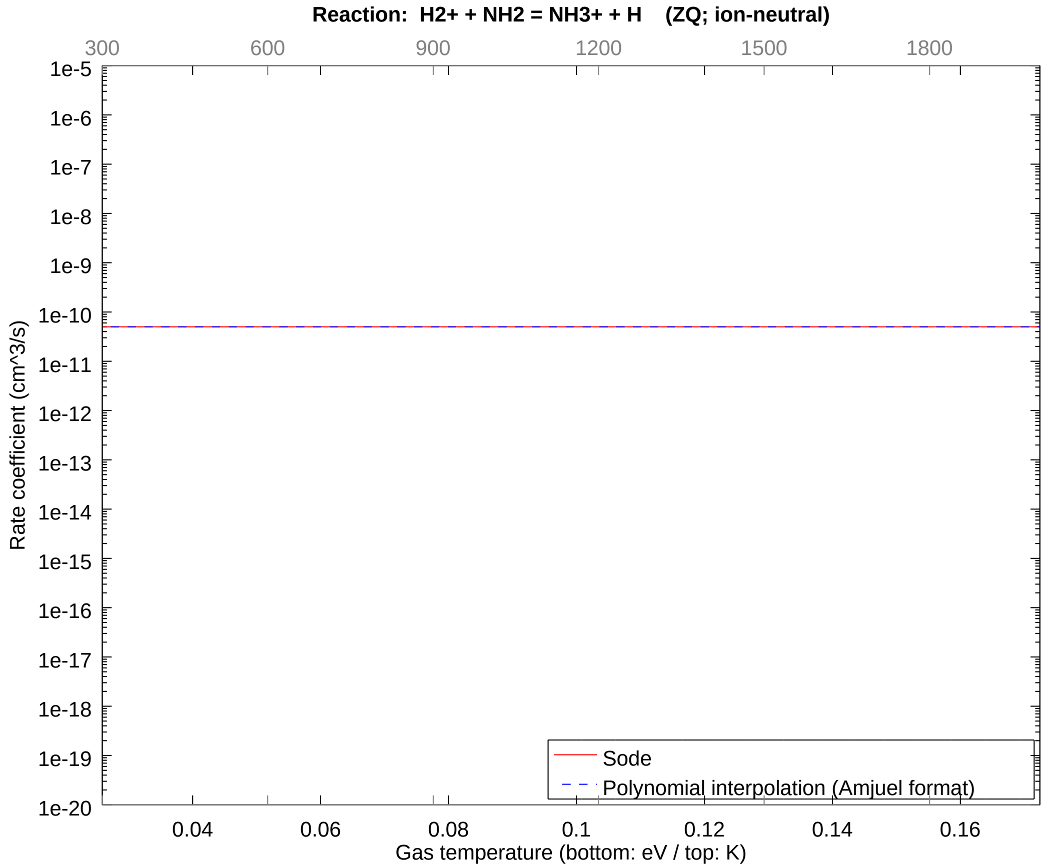
2.57 Reaction ZP H2+ + NH3 = NH4+ + H

b0	-2.371899811050E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



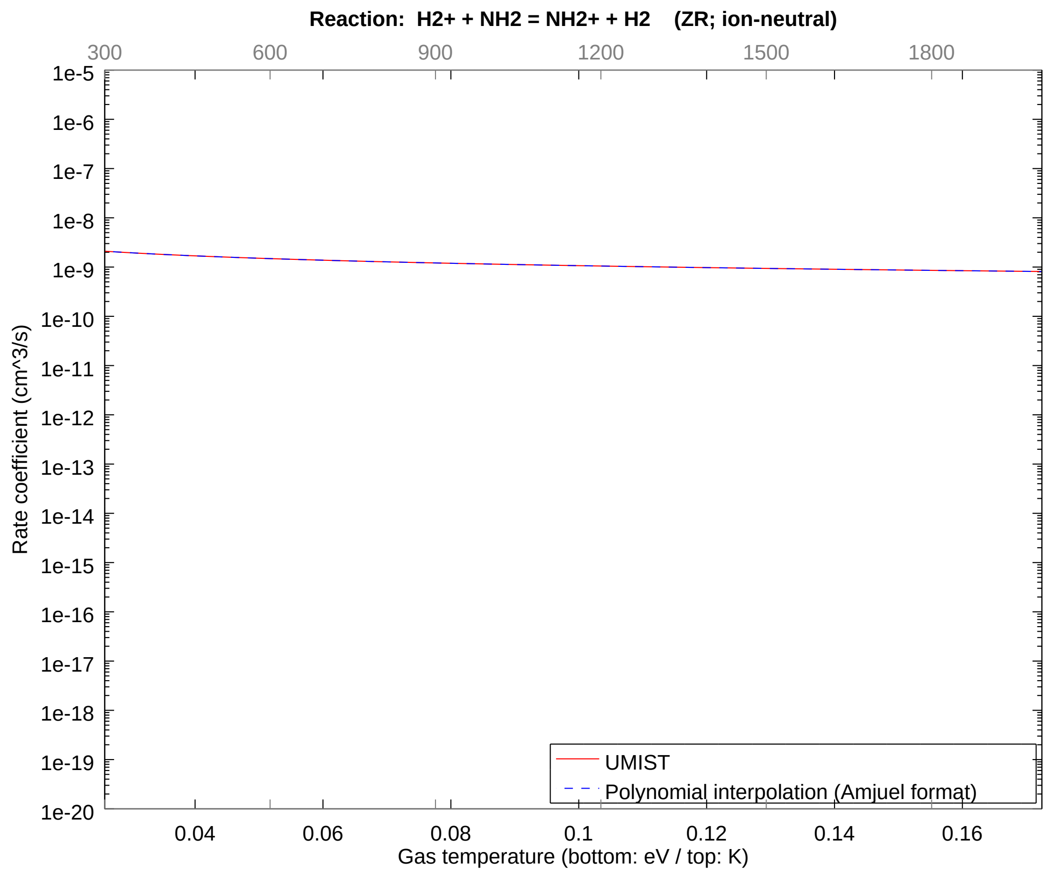
2.58 Reaction ZQ H2+ + NH2 = NH3+ + H

b0	-2.371899811050E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



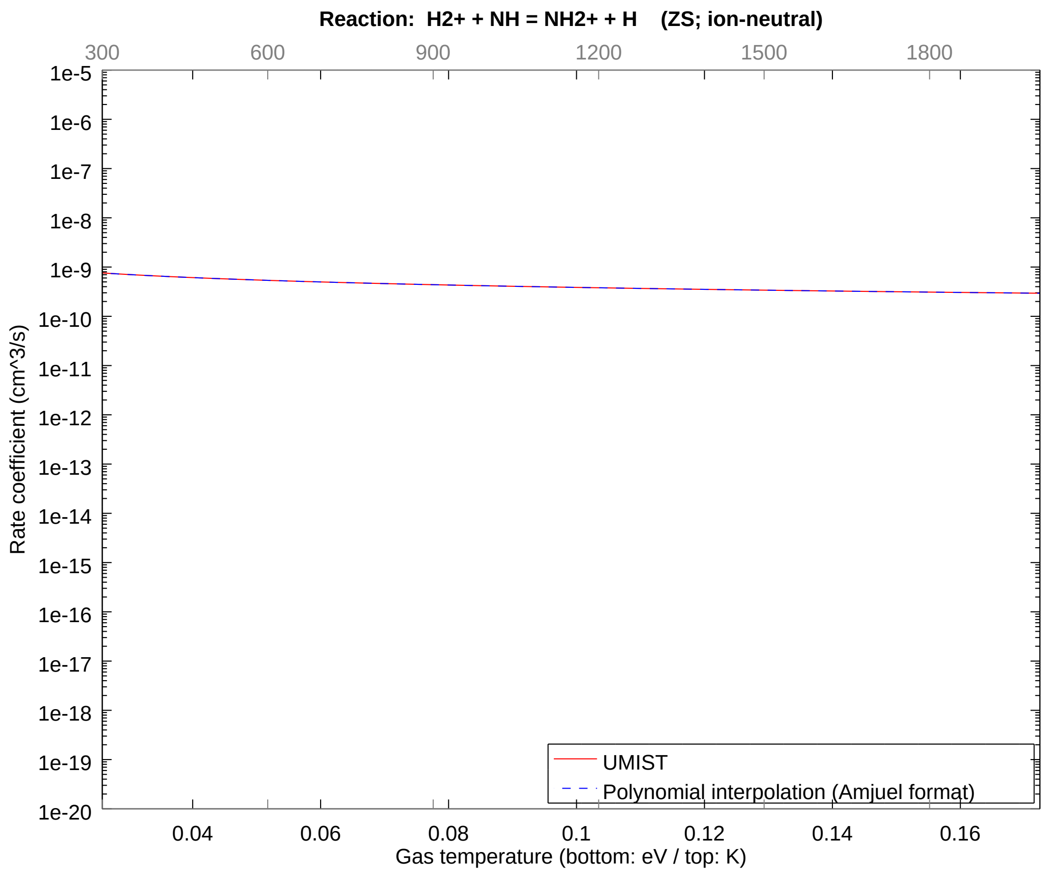
2.59 Reaction ZR H2+ + NH2 = NH2+ + H2

```
b0 -2.180875664609E+01    b1 -5.000000092849E-01    b2 -1.261517857558E-08
b3 -9.699489564652E-09    b4 -4.615551314815E-09    b5 -1.391879349947E-09
b6 -2.597688881768E-10    b7 -2.743376818213E-11    b8 -1.255322335567E-12
```



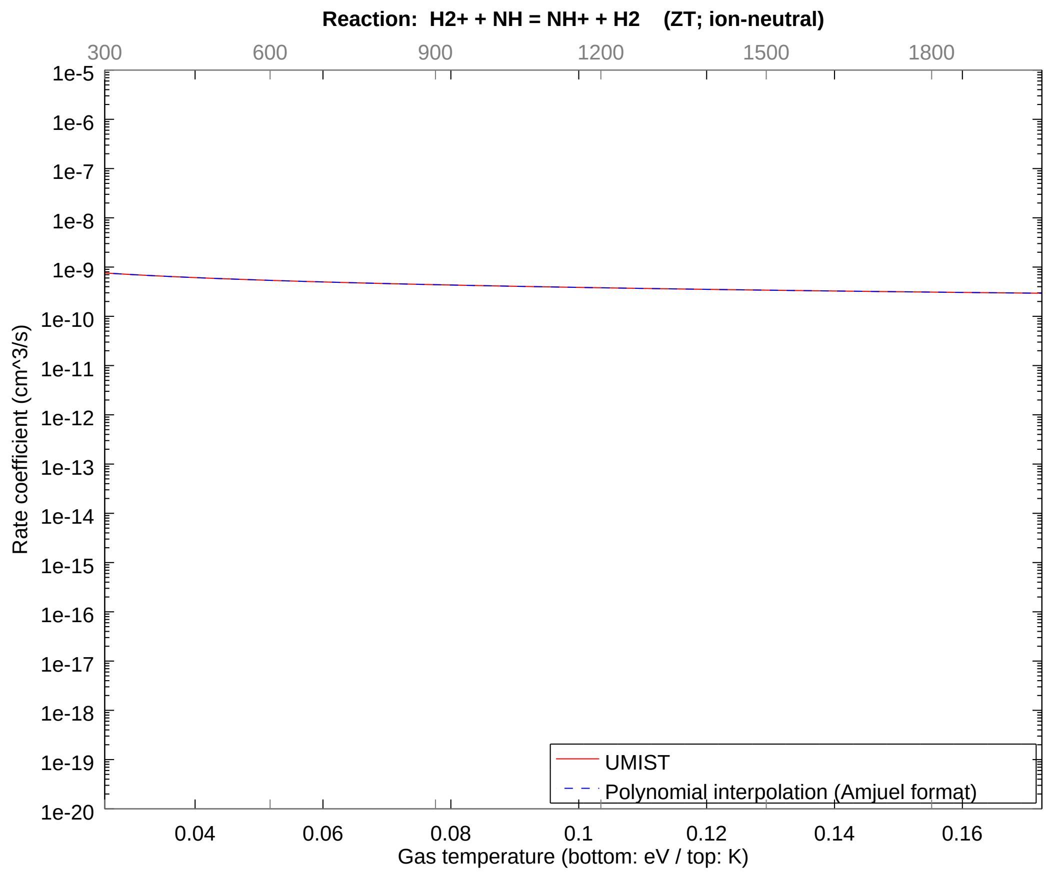
2.60 Reaction ZS H2+ + NH = NH2+ + H

b0 -2.282513083456E+01	b1 -5.000000030980E-01	b2 -4.172533795604E-09
b3 -3.178279948201E-09	b4 -1.497002555752E-09	b5 -4.463695735110E-10
b6 -8.227558117594E-11	b7 -8.571561358925E-12	b8 -3.865197085583E-13



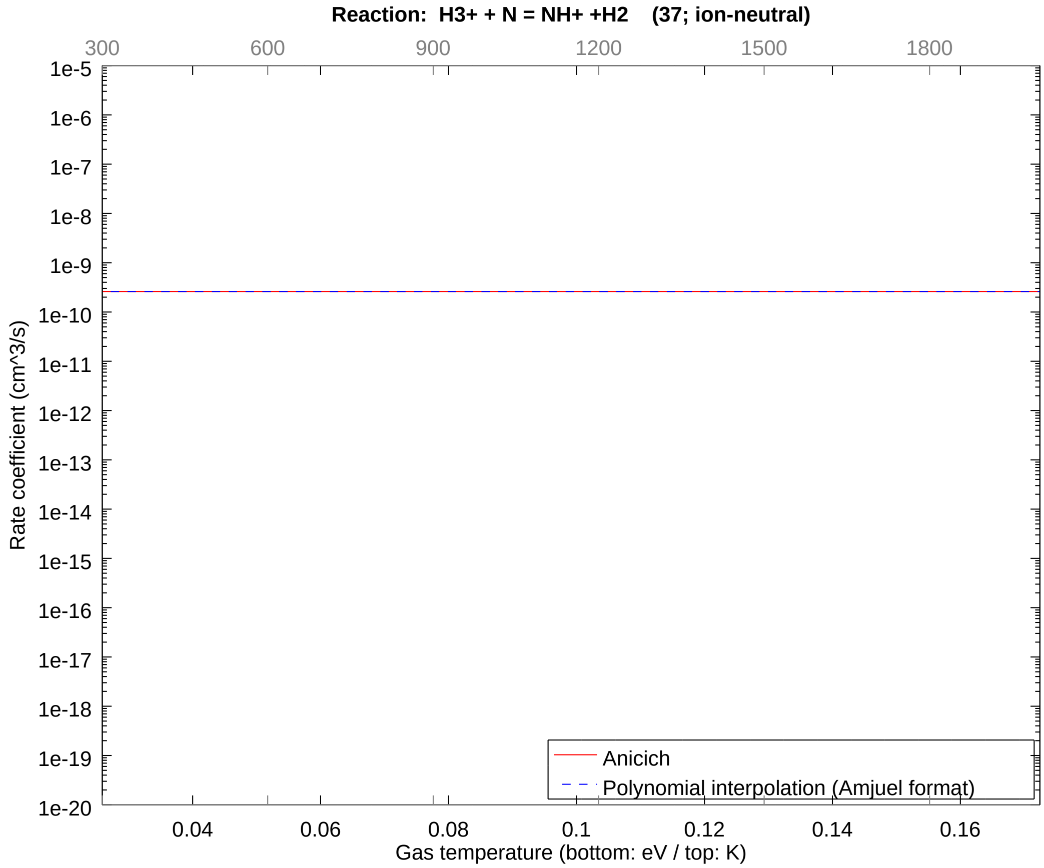
2.61 Reaction ZT H2+ + NH = NH+ + H2

b0 -2.282513083456E+01	b1 -5.000000030980E-01	b2 -4.172533795604E-09
b3 -3.178279948201E-09	b4 -1.497002555752E-09	b5 -4.463695735110E-10
b6 -8.227558117594E-11	b7 -8.571561358925E-12	b8 -3.865197085583E-13



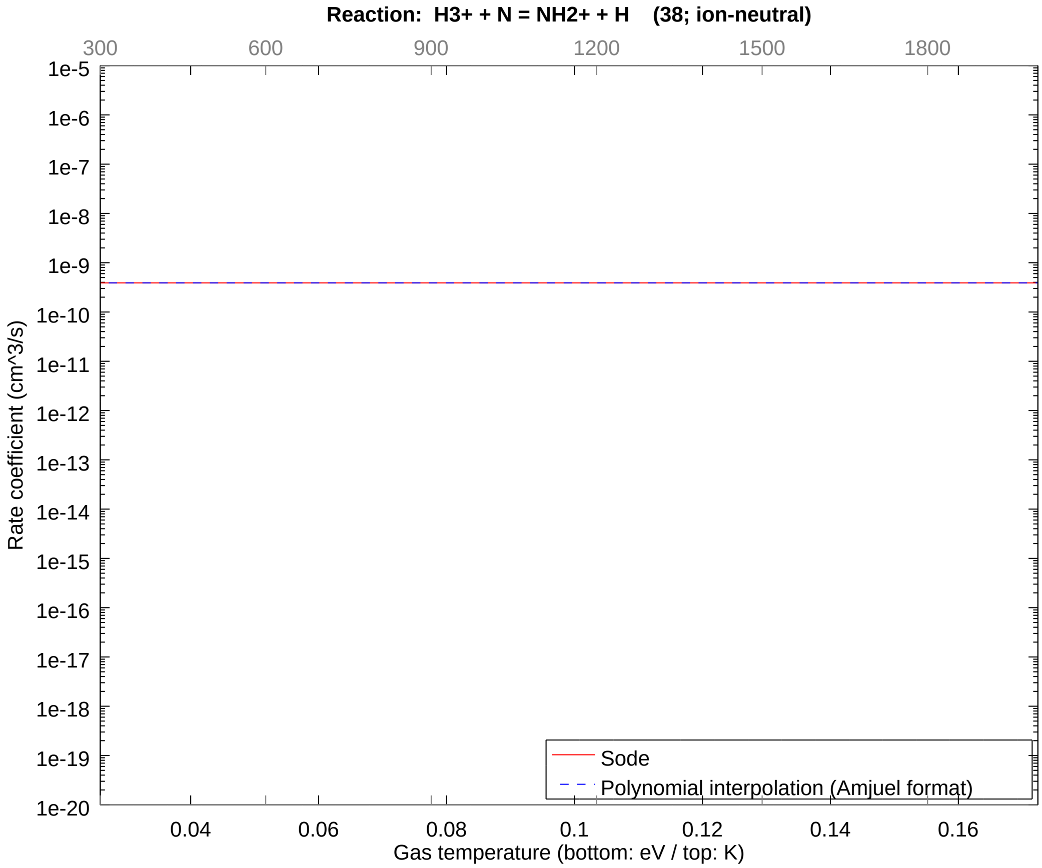
2.62 Reaction 37 H3+ + N = NH+ +H2

b0	-2.207033948491E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



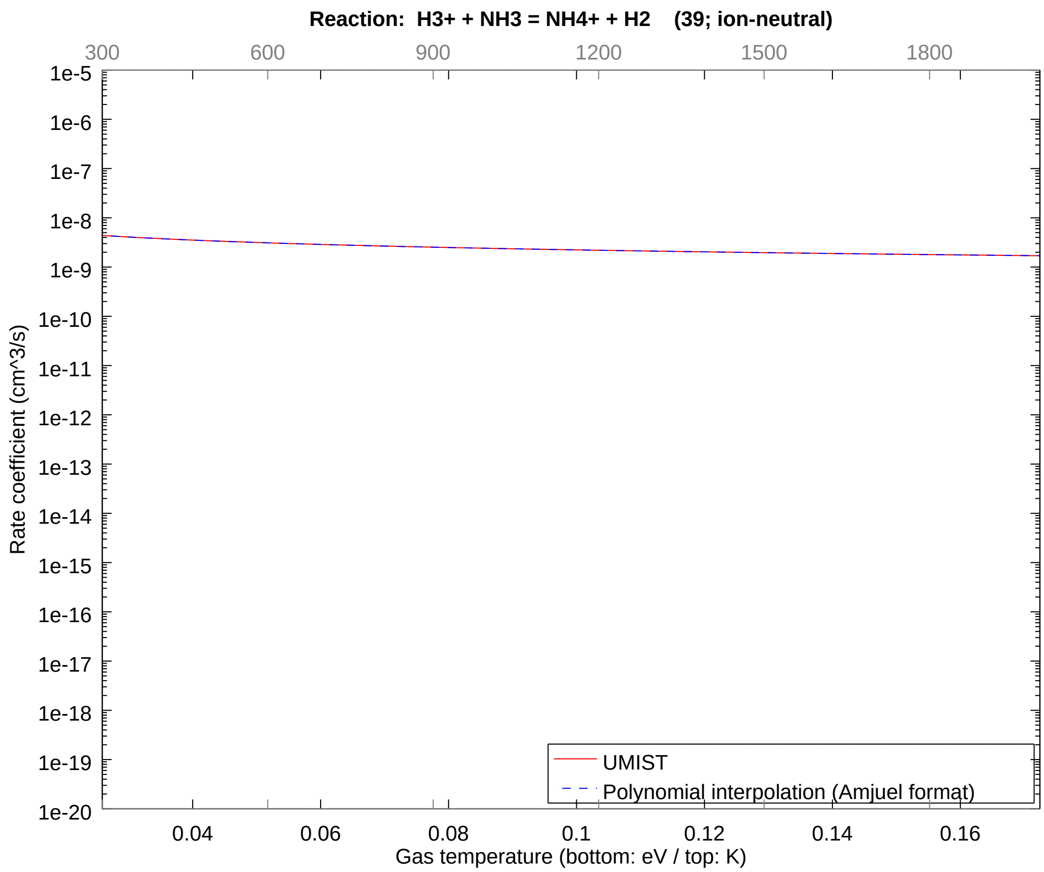
2.63 Reaction 38 H3+ + N = NH2+ + H

b0	-2.166487437680E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



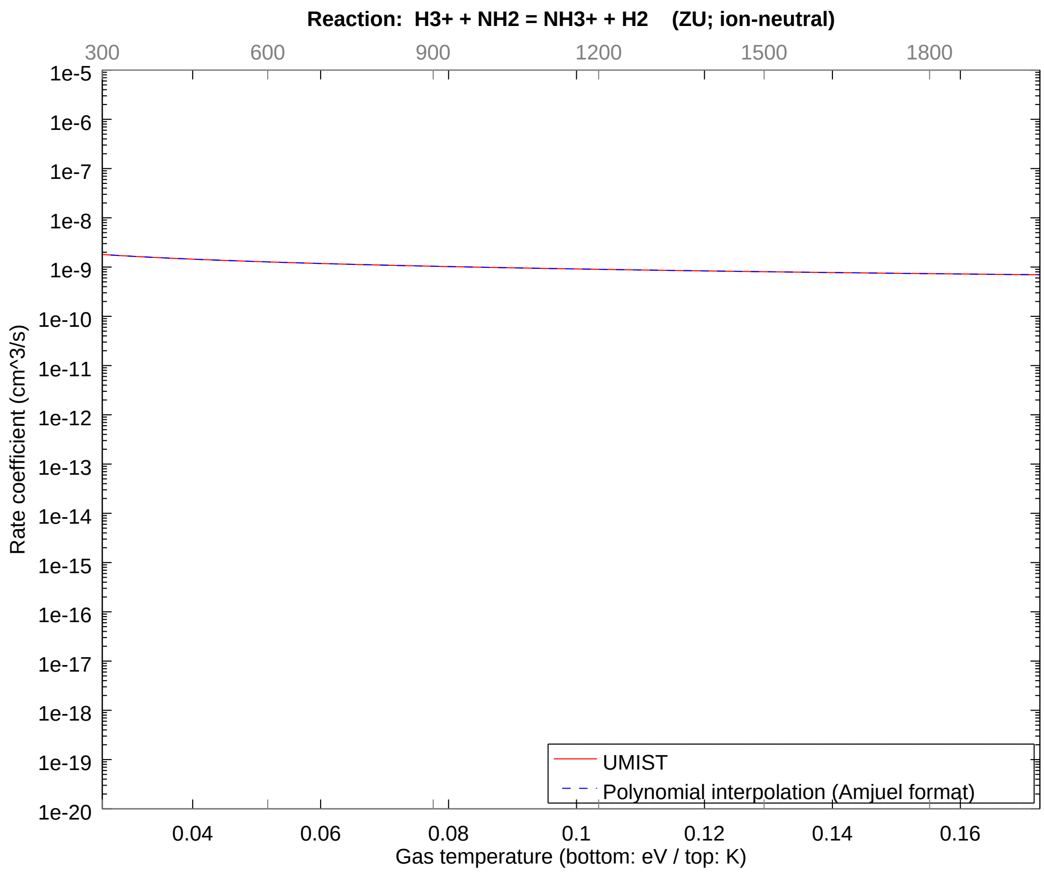
2.64 Reaction 39 H3+ + NH3 = NH4+ + H2

b0 -2.107136476258E+01	b1 -5.000000055727E-01	b2 -7.437148651905E-09
b3 -5.609819632118E-09	b4 -2.615584427083E-09	b5 -7.718869172379E-10
b6 -1.408075560704E-10	b7 -1.451862542892E-11	b8 -6.479800800281E-13



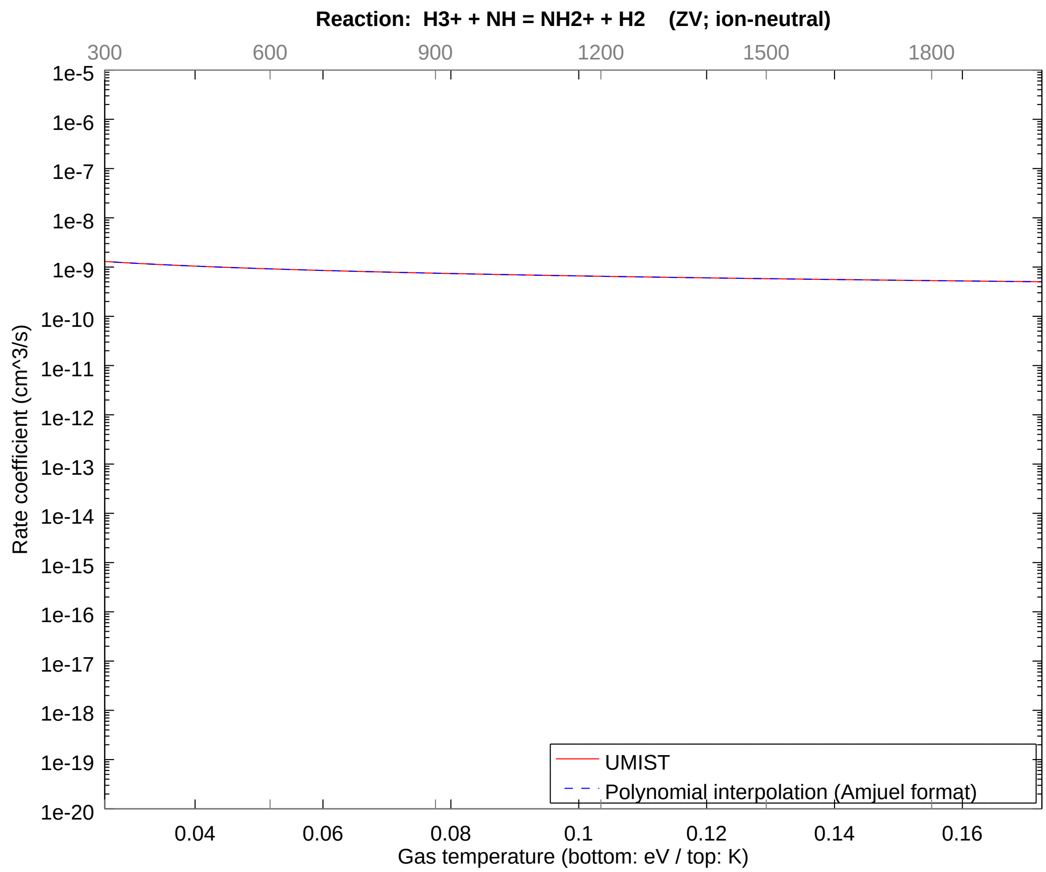
2.65 Reaction ZU H3+ + NH2 = NH3+ + H2

b0 -2.196290732349E+01	b1 -5.000000017280E-01	b2 -2.428579288053E-09
b3 -1.921777919375E-09	b4 -9.364377031497E-10	b5 -2.877434523747E-10
b6 -5.445759602353E-11	b7 -5.805294394799E-12	b8 -2.669522116361E-13



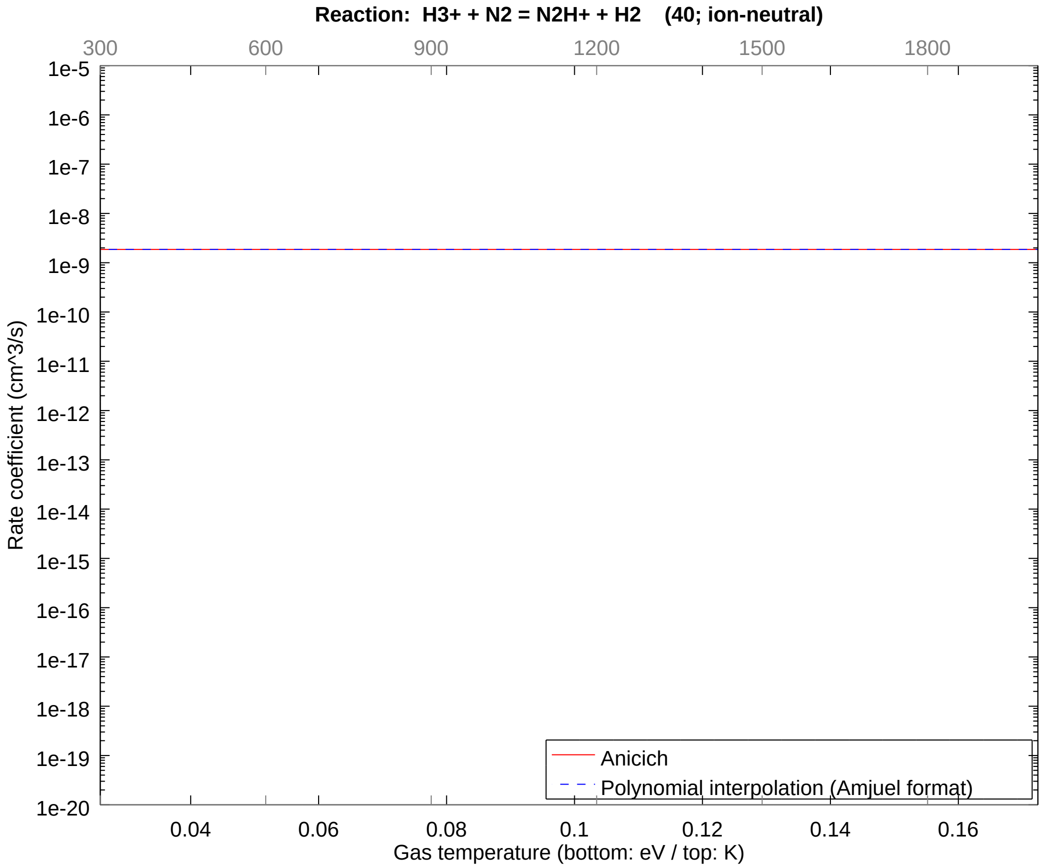
2.66 Reaction ZV H3+ + NH = NH2+ + H2

b0 -2.228832972476E+01	b1 -5.000000042028E-01	b2 -5.569379050783E-09
b3 -4.161520963593E-09	b4 -1.916979254164E-09	b5 -5.572768434265E-10
b6 -9.981865098488E-11	b7 -1.007015872766E-11	b8 -4.380051097582E-13



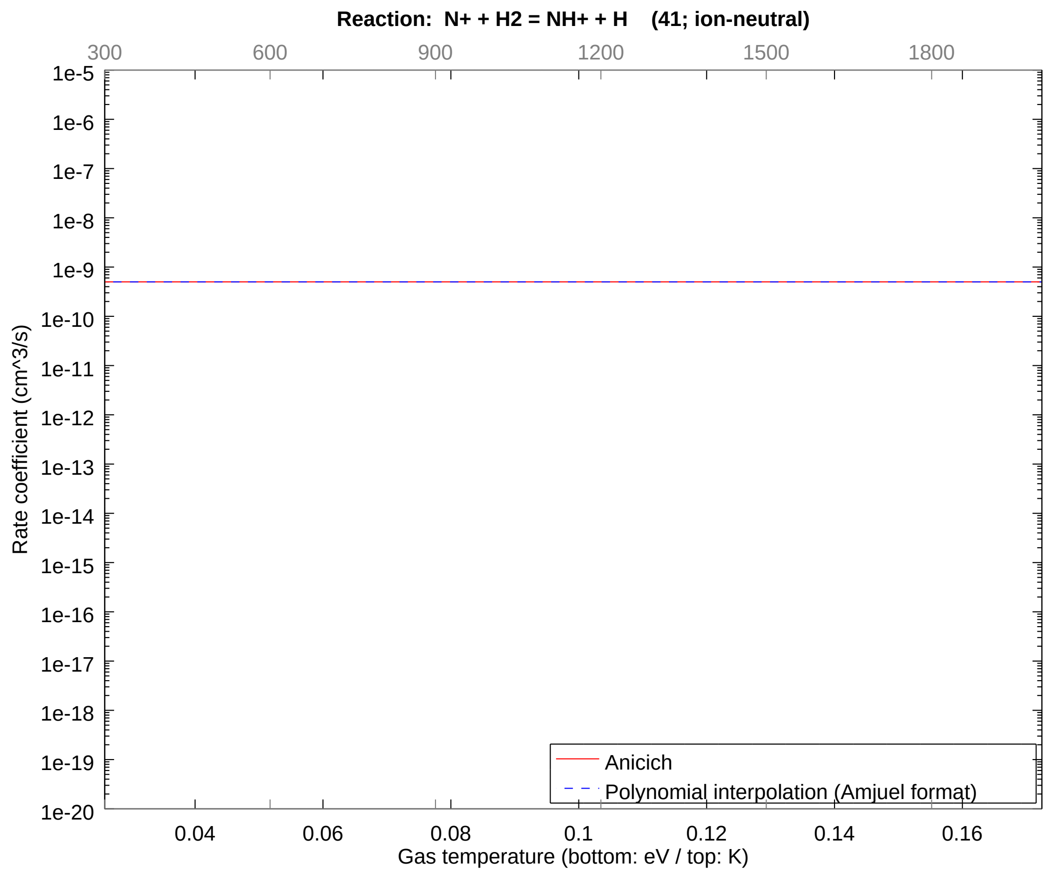
2.67 Reaction 40 H3+ + N2 = N2H+ + H2

b0	-2.010268934922E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



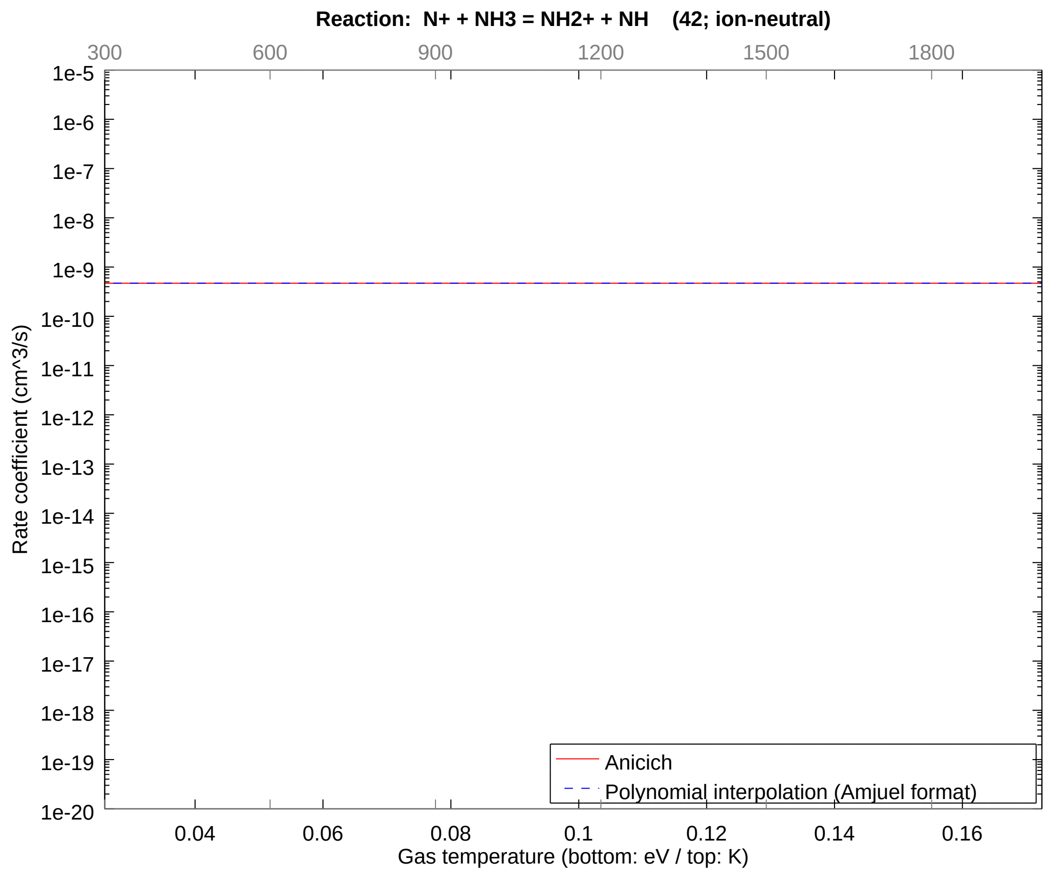
2.68 Reaction 41 N+ + H2 = NH+ + H

b0	-2.141641301751E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



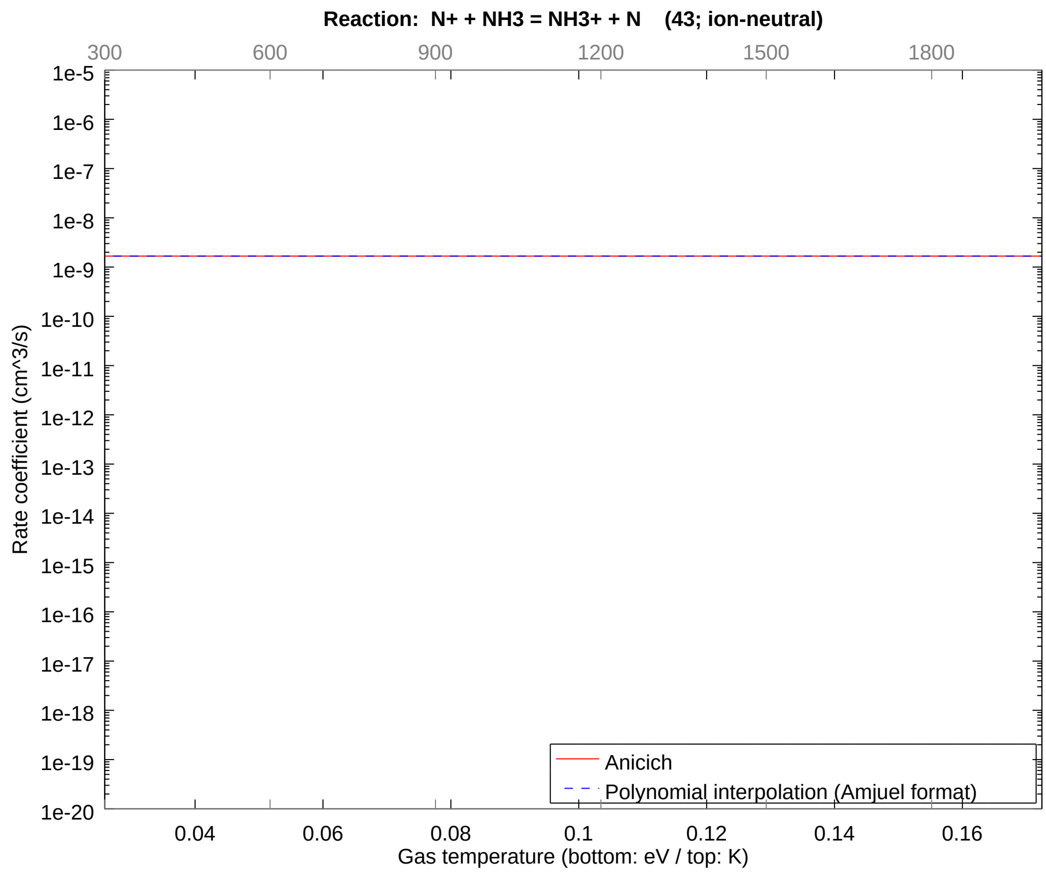
2.69 Reaction 42 N+ + NH3 = NH2+ + NH

b0	-2.147828842122E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



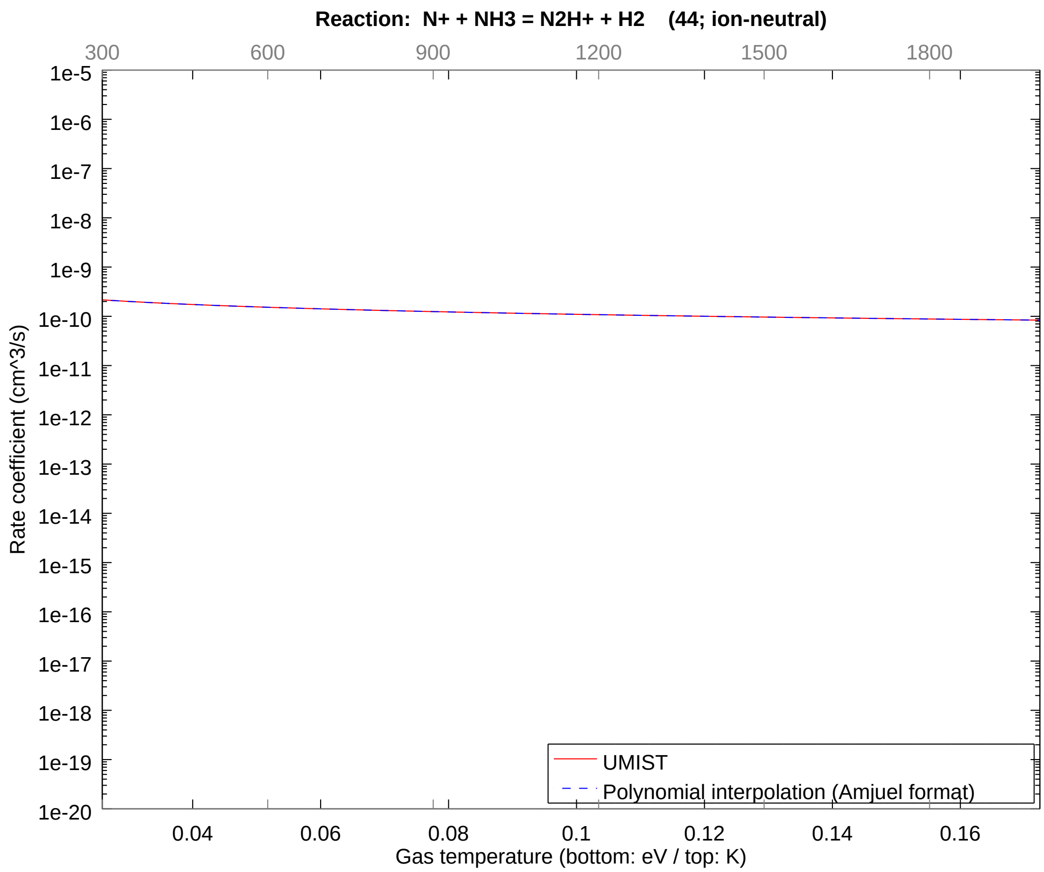
2.70 Reaction 43 N+ + NH3 = NH3+ + N

b0	-2.021044221052E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



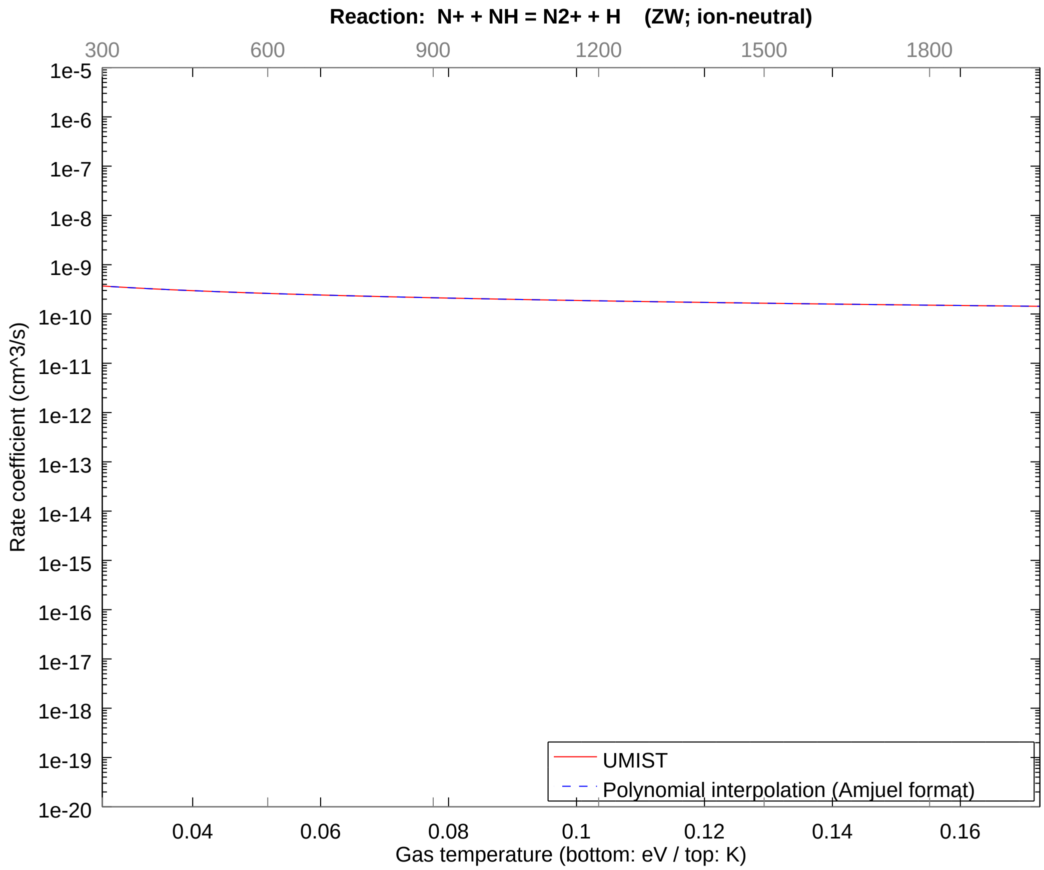
2.71 Reaction 44 N+ + NH3 = N2H+ + H2

b0 -2.408317086045E+01	b1 -5.000000039818E-01	b2 -5.322487393845E-09
b3 -4.009817581528E-09	b4 -1.860909331983E-09	b5 -5.443952121280E-10
b6 -9.796847626584E-11	b7 -9.908533882743E-12	b8 -4.308943182685E-13



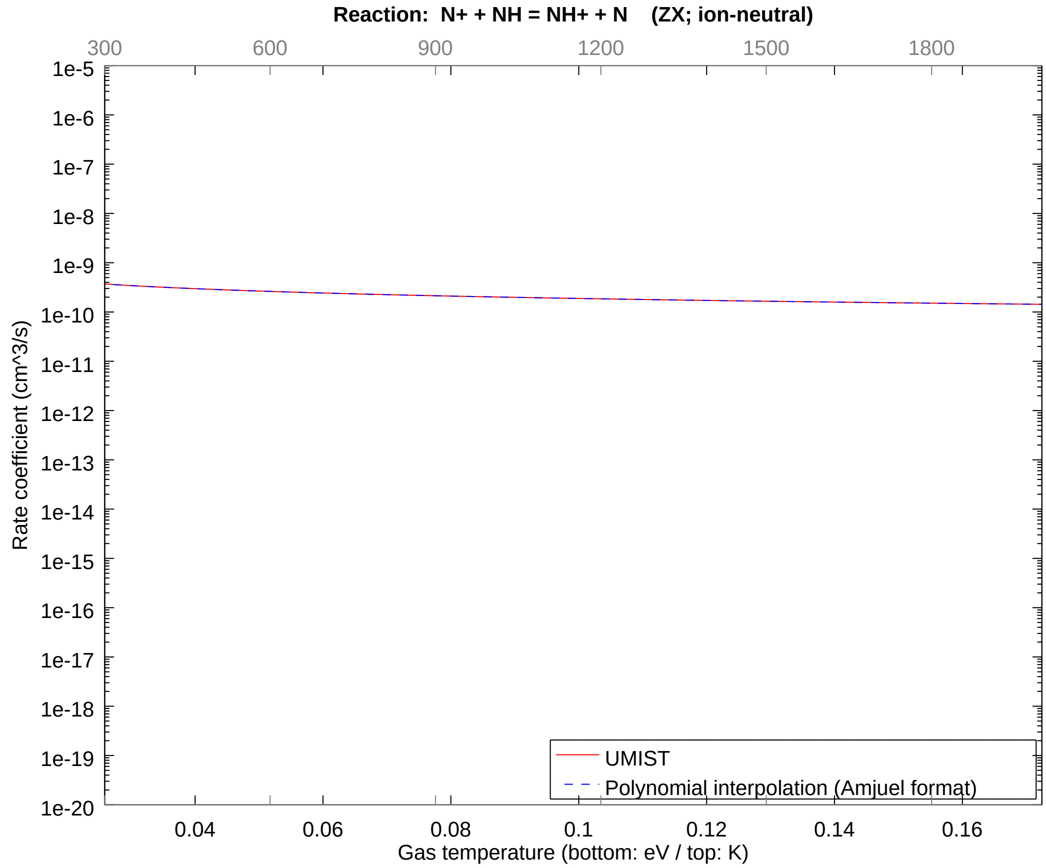
2.72 Reaction ZW N+ + NH = N2+ + H

b0 -2.354494626224E+01	b1 -5.000000032306E-01	b2 -4.347944933015E-09
b3 -3.299966277679E-09	b4 -1.544289864373E-09	b5 -4.561831738007E-10
b6 -8.305450654833E-11	b7 -8.519342746733E-12	b8 -3.768628601479E-13



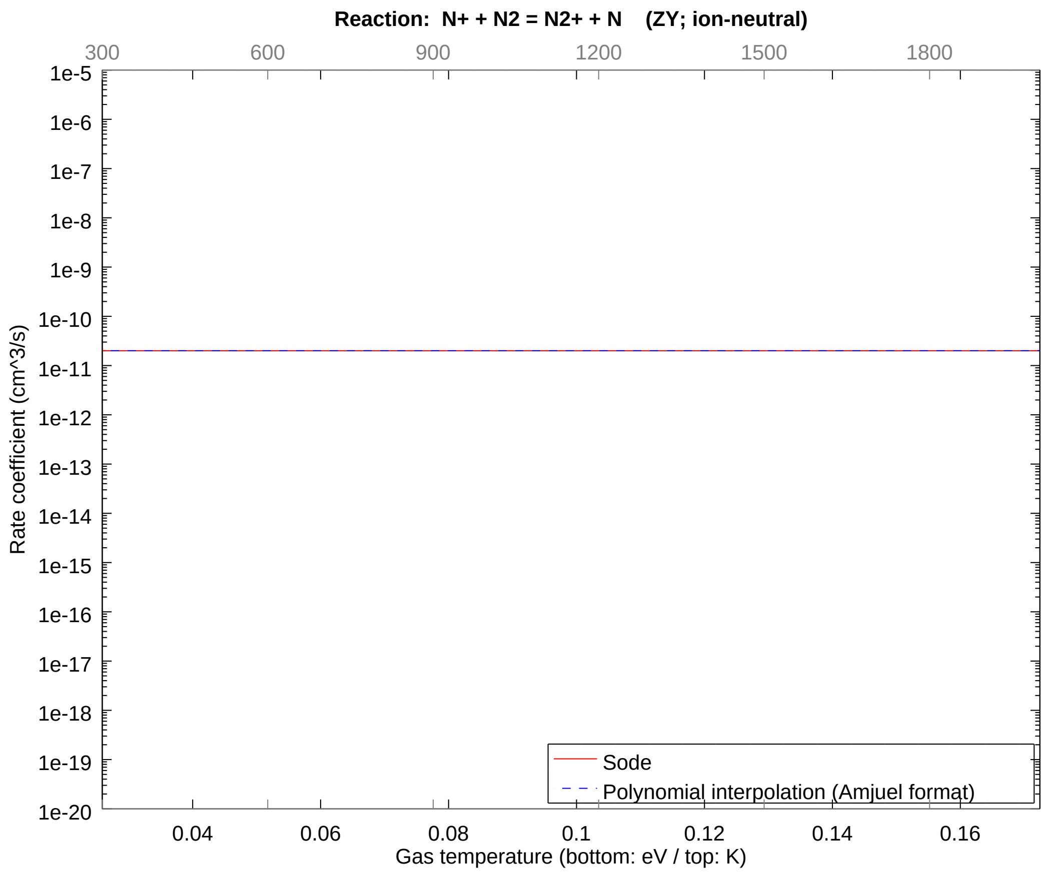
2.73 Reaction ZX N+ + NH = NH+ + N

b0	-2.354494626224E+01	b1	-5.000000032306E-01	b2	-4.347944933015E-09
b3	-3.299966277679E-09	b4	-1.544289864373E-09	b5	-4.561831738007E-10
b6	-8.305450654833E-11	b7	-8.519342746733E-12	b8	-3.768628601479E-13



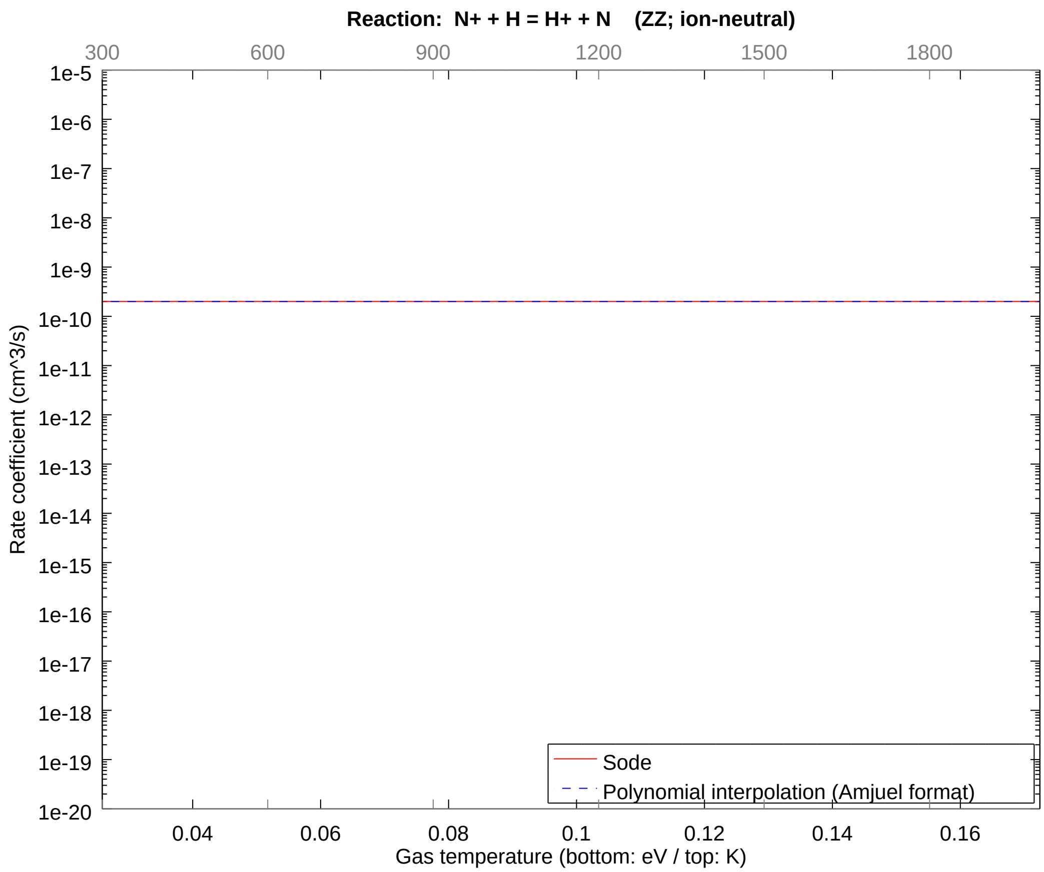
2.74 Reaction ZY N+ + N2 = N2+ + N

b0	-2.463528884237E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



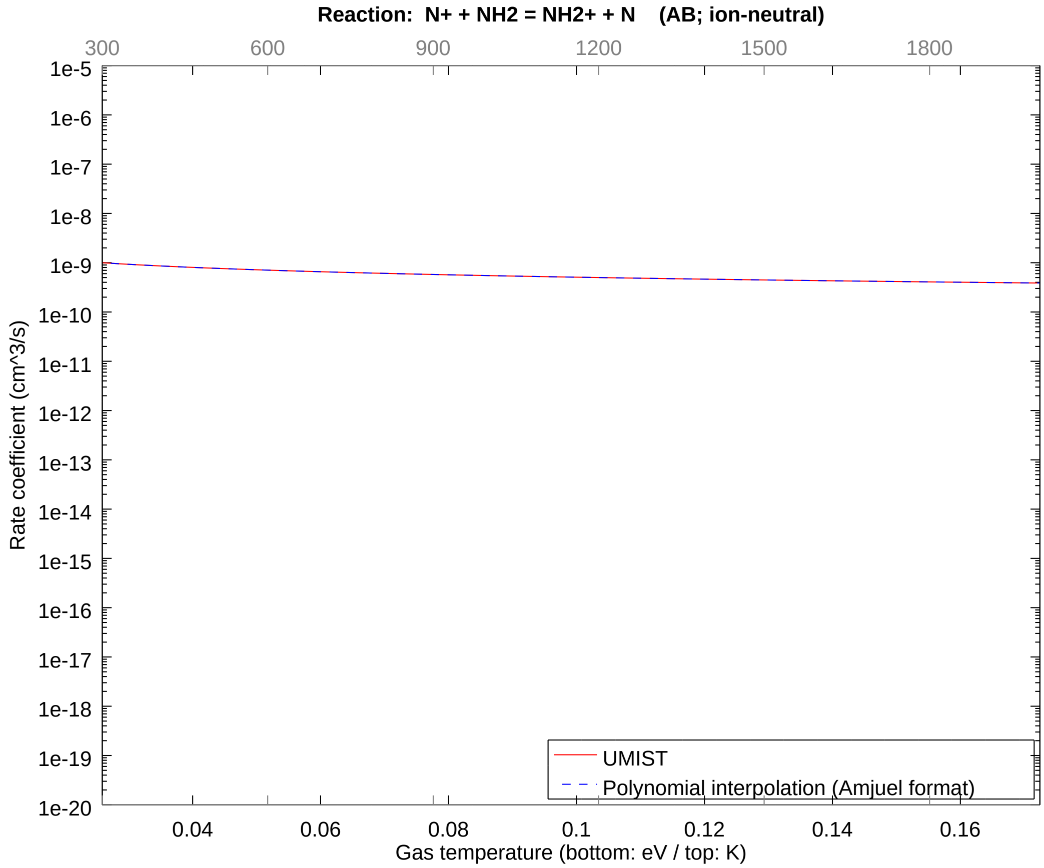
2.75 Reaction ZZ N+ + H = H+ + N

b0	-2.233270374938E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



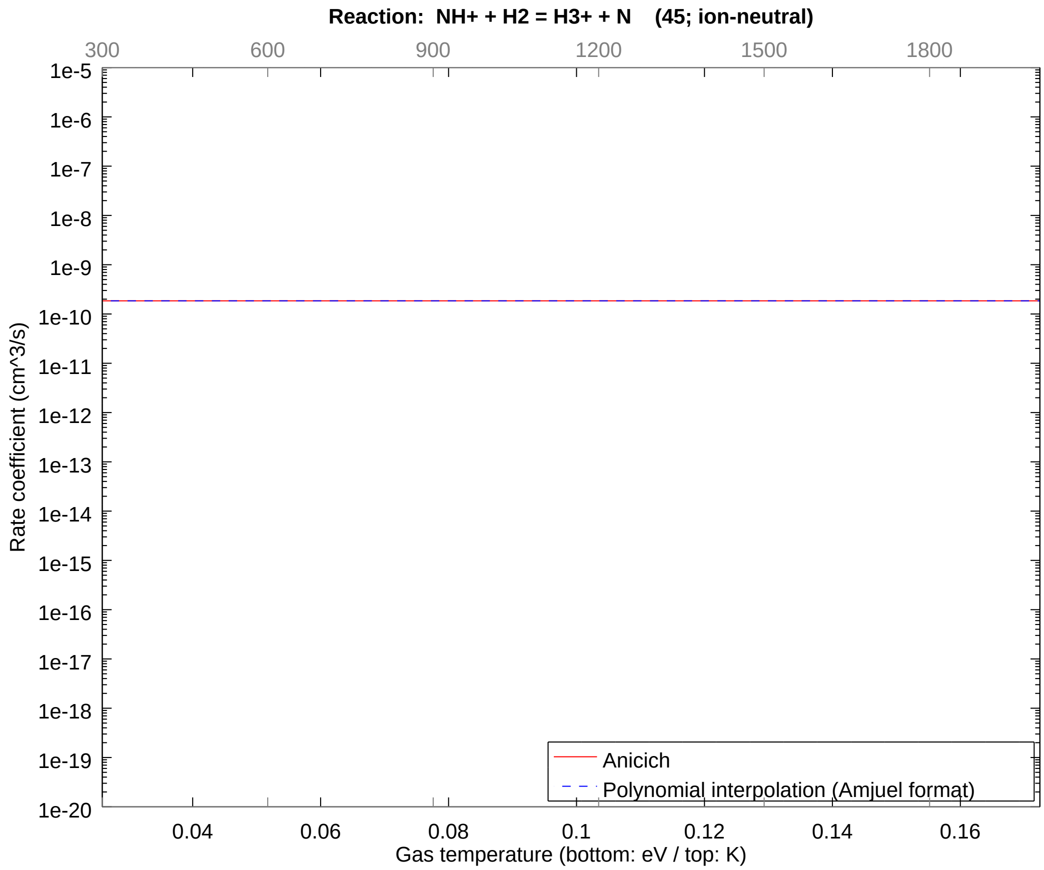
2.76 Reaction AB N+ + NH2 = NH2+ + N

b0 -2.255069399034E+01	b1 -5.000000077382E-01	b2 -1.048331289759E-08
b3 -8.030169368750E-09	b4 -3.804180693869E-09	b5 -1.141479368811E-09
b6 -2.119003834735E-10	b7 -2.225565586486E-11	b8 -1.012819638805E-12



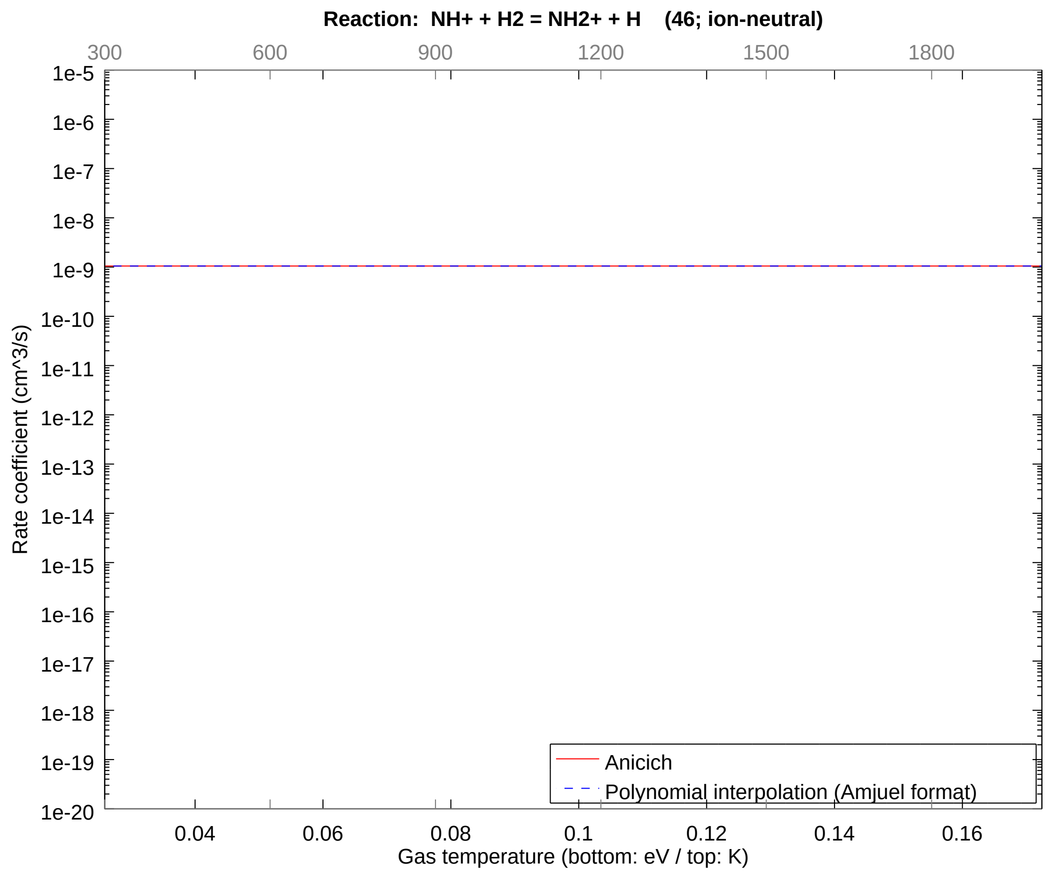
2.77 Reaction 45 NH+ + H2 = H3+ + N

b0	-2.241066529085E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



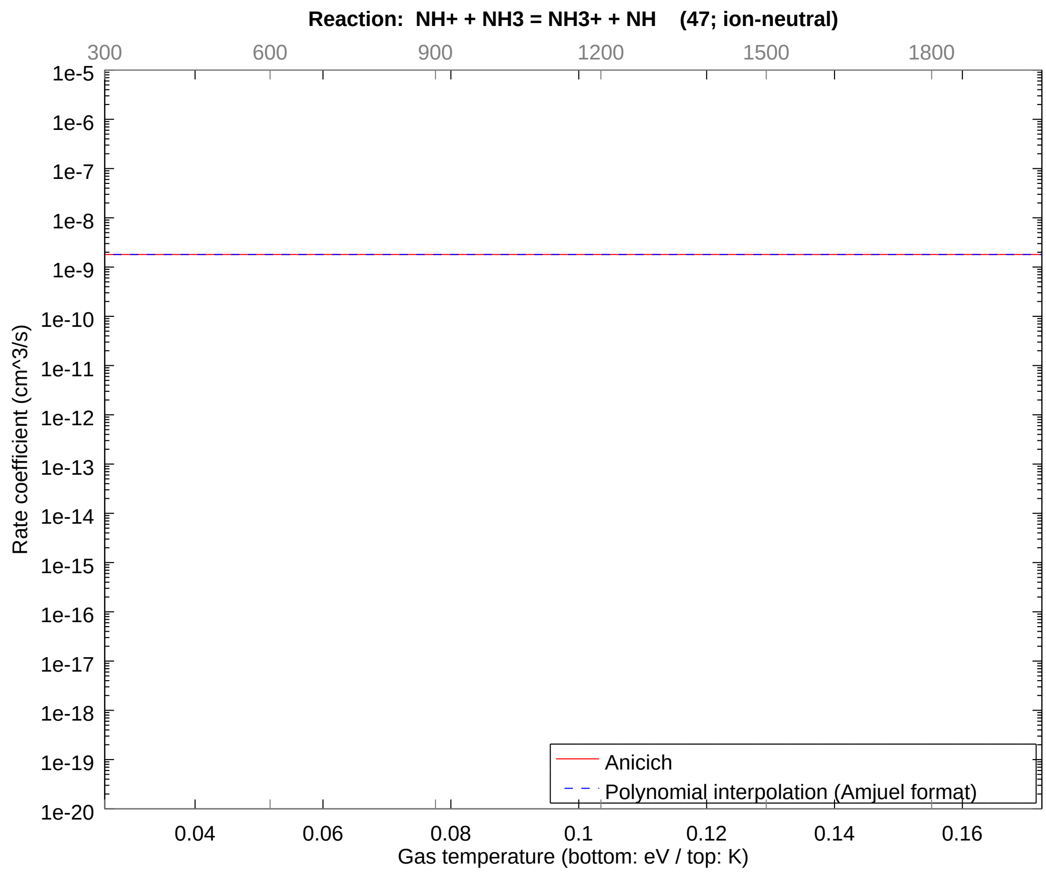
2.78 Reaction 46 NH+ + H2 = NH2+ + H

b0	-2.067447567278E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



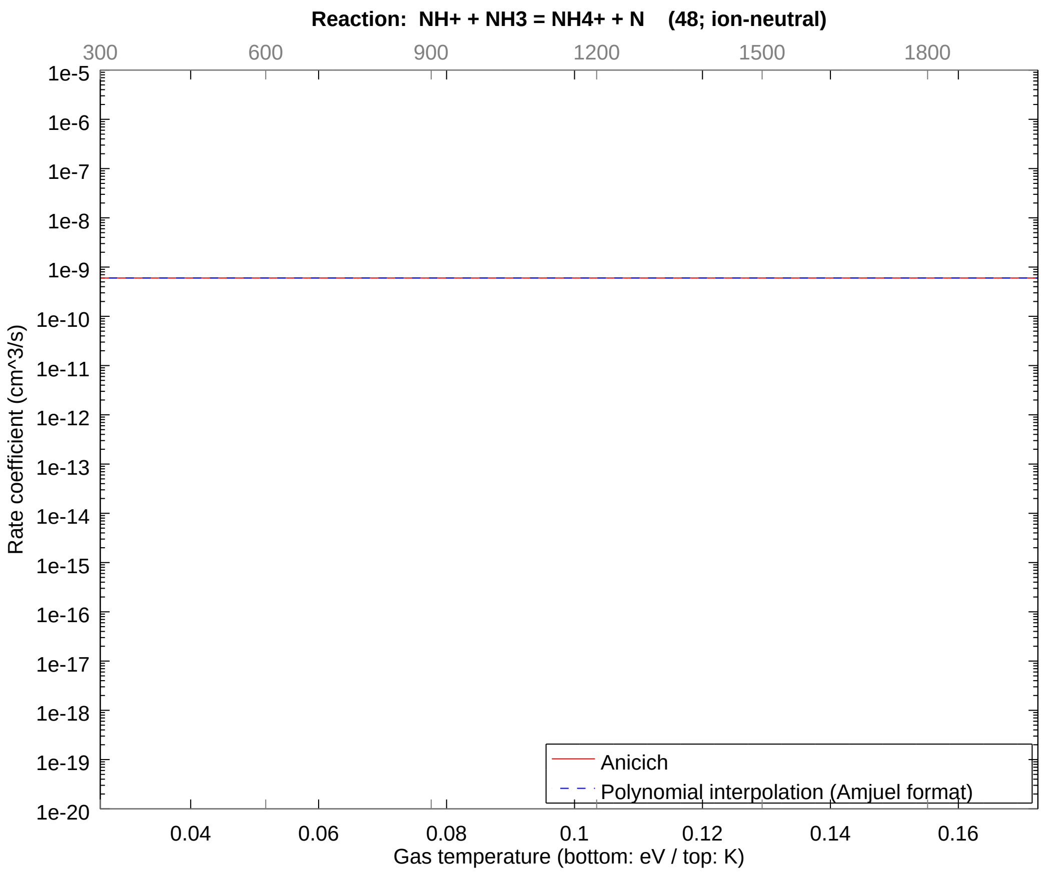
2.79 Reaction 47 NH+ + NH3 = NH3+ + NH

b0	-2.013547917204E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



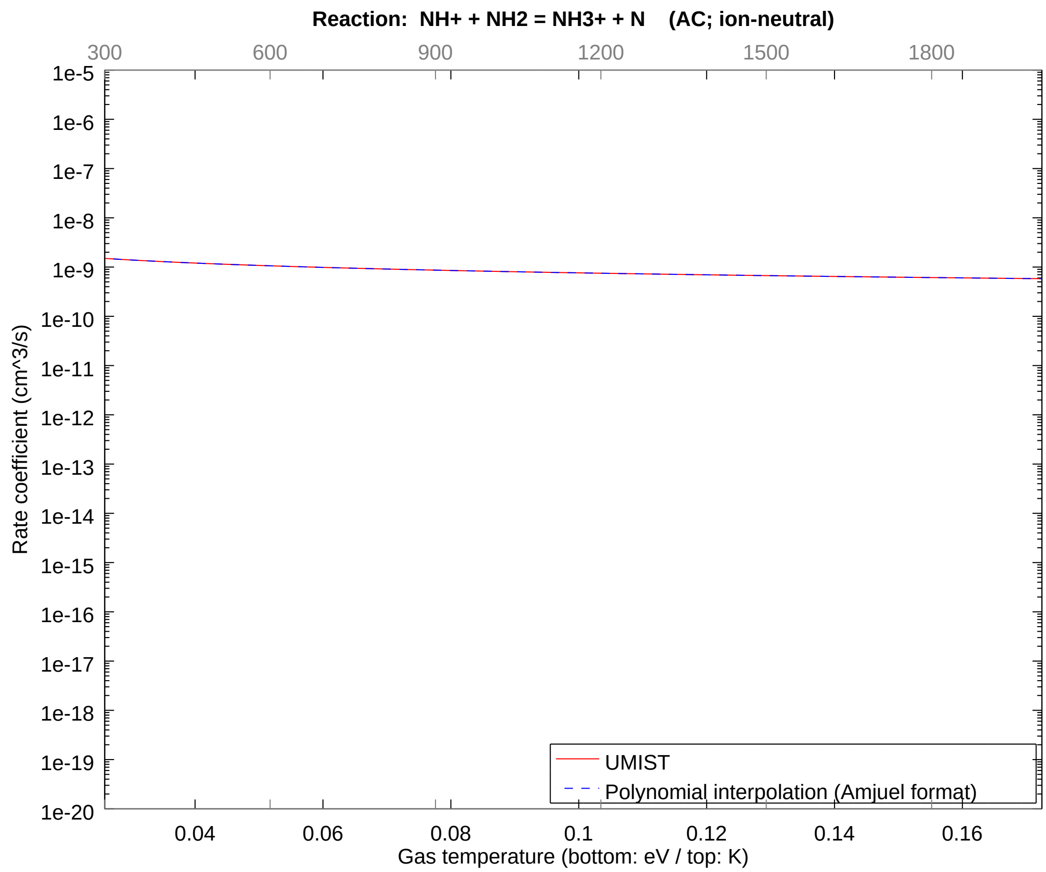
2.80 Reaction 48 NH+ + NH3 = NH4+ + N

b0	-2.123409146071E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



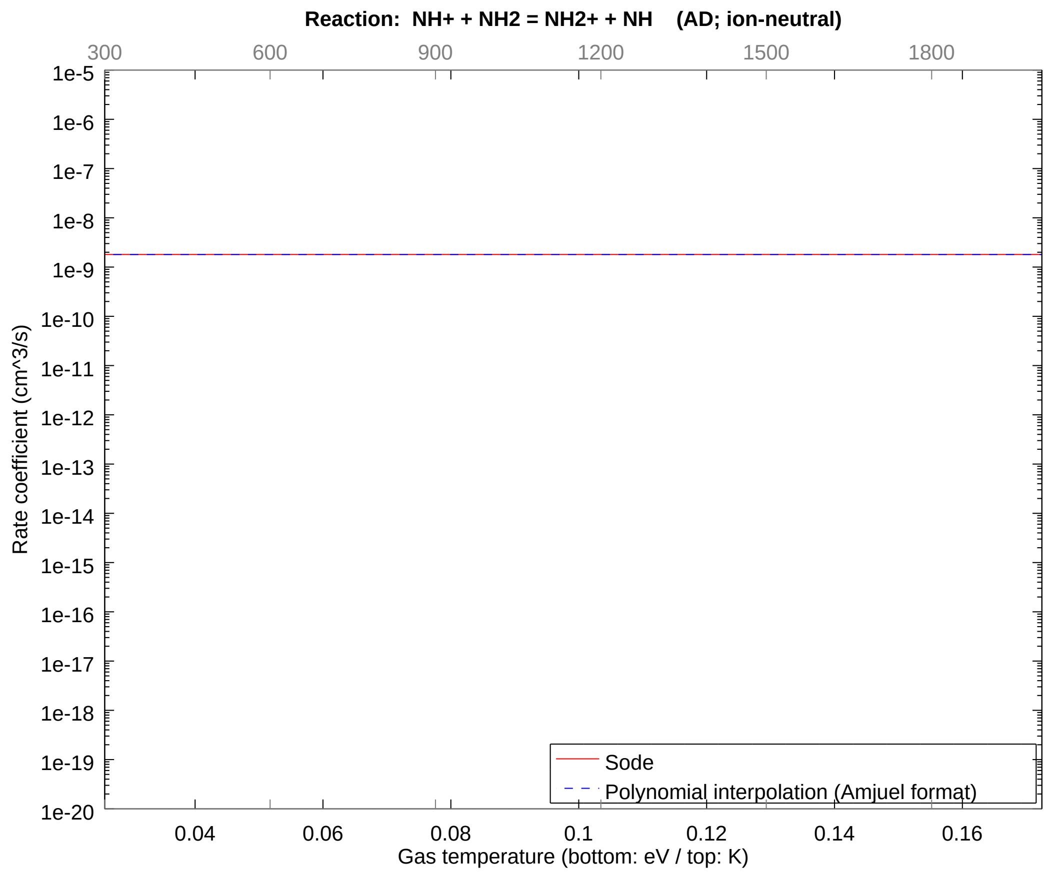
2.81 Reaction AC NH+ + NH2 = NH3+ + N

b0 -2.214522888071E+01	b1 -5.000000029212E-01	b2 -3.860018050693E-09
b3 -2.872424232405E-09	b4 -1.316010331162E-09	b5 -3.799779029463E-10
b6 -6.750141857656E-11	b7 -6.743736474333E-12	b8 -2.900439964921E-13



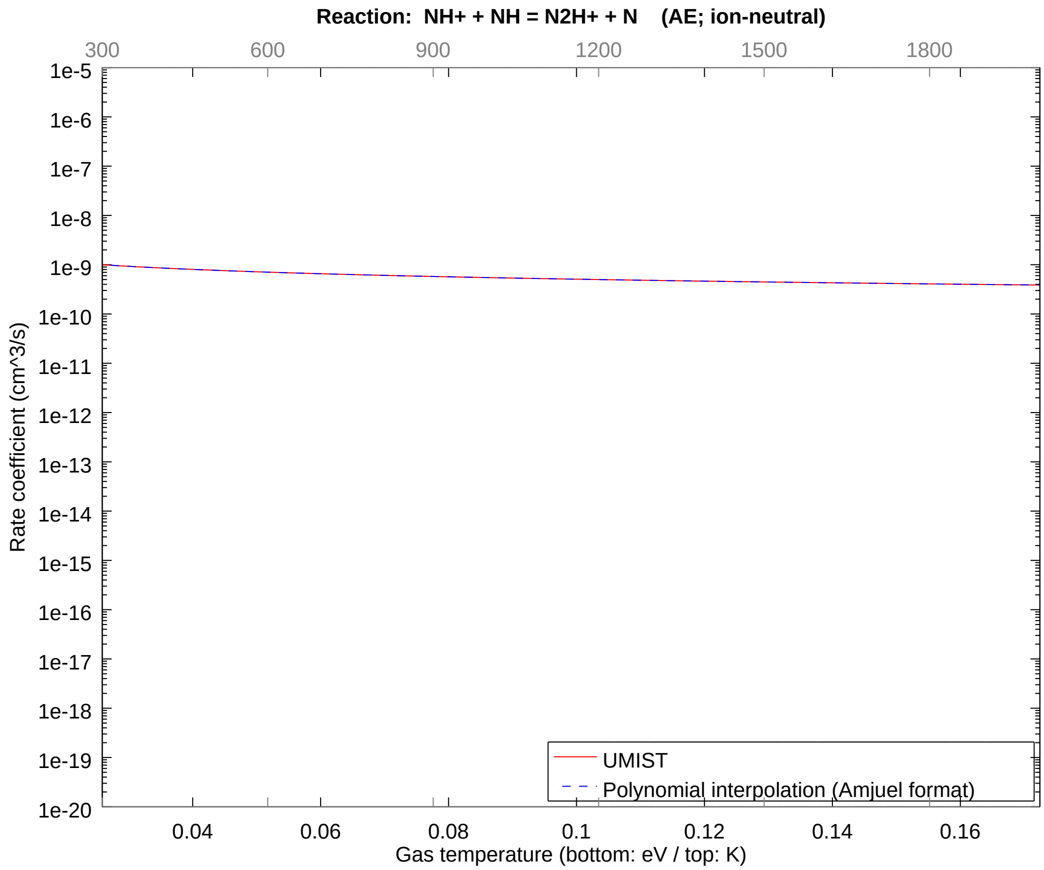
2.82 Reaction AD NH+ + NH2 = NH2+ + NH

b0	-2.013547917204E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



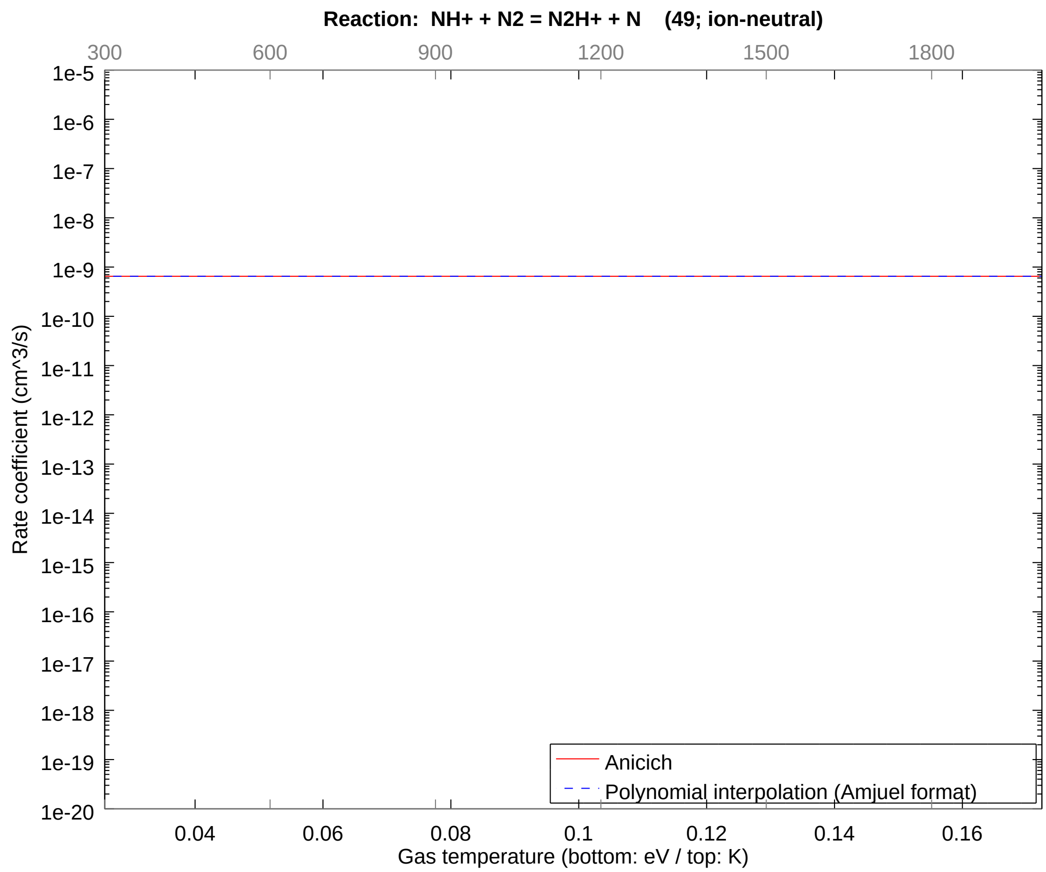
2.83 Reaction AE NH+ + NH = N2H+ + N

b0 -2.255069399034E+01	b1 -5.000000077382E-01	b2 -1.048331289759E-08
b3 -8.030169368750E-09	b4 -3.804180693869E-09	b5 -1.141479368811E-09
b6 -2.119003834735E-10	b7 -2.225565586486E-11	b8 -1.012819638805E-12



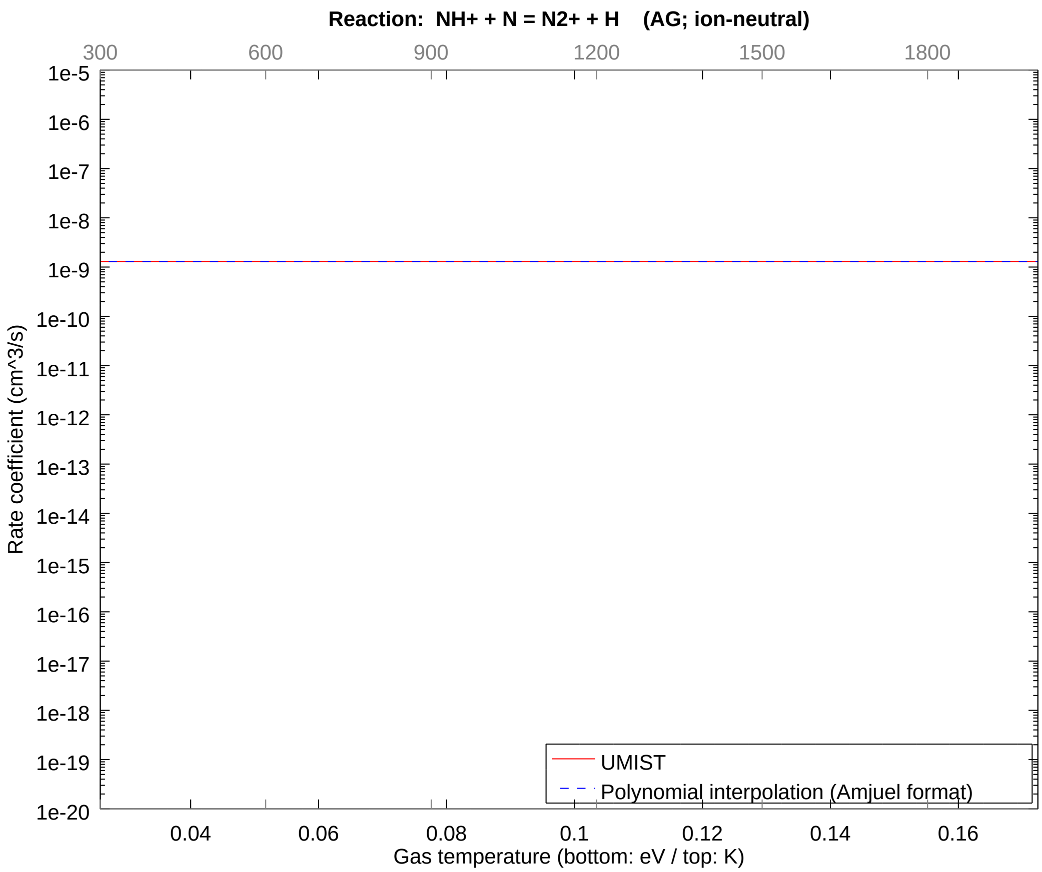
2.84 Reaction 49 NH+ + N2 = N2H+ + N

b0	-2.115404875304E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



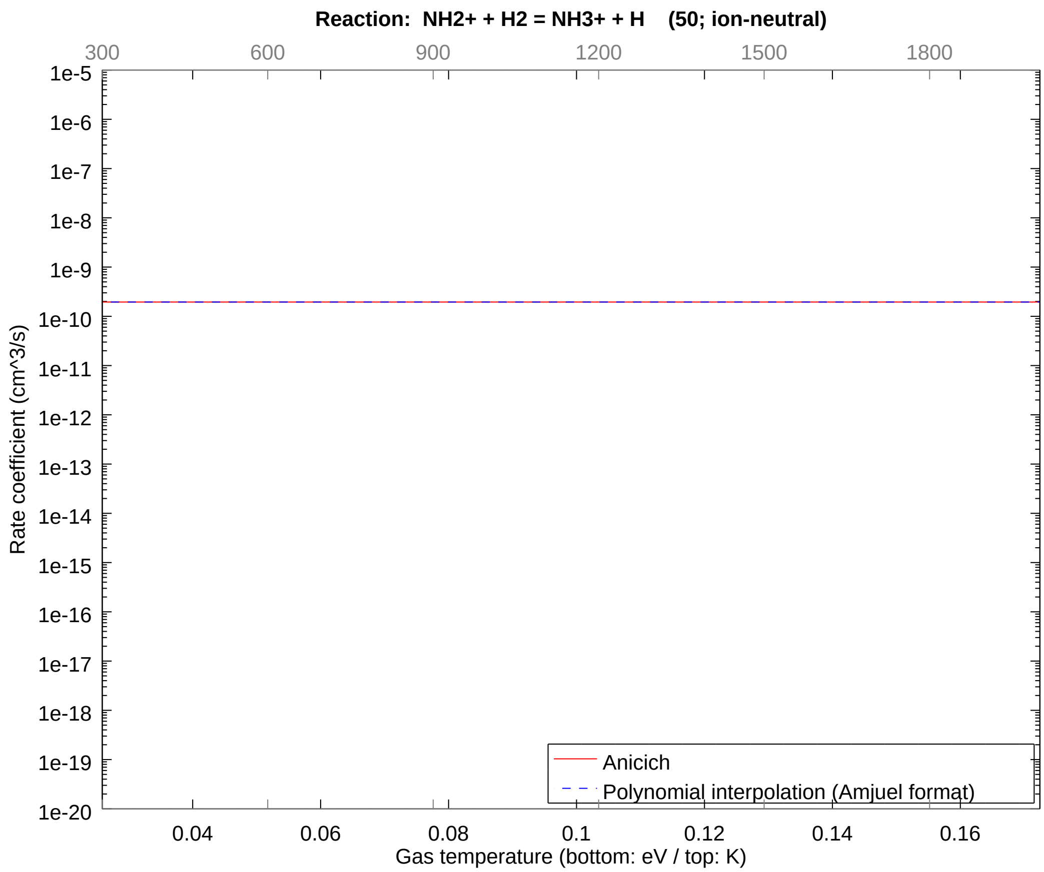
2.85 Reaction AG NH+ + N = N2+ + H

b0	-2.046090157248E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



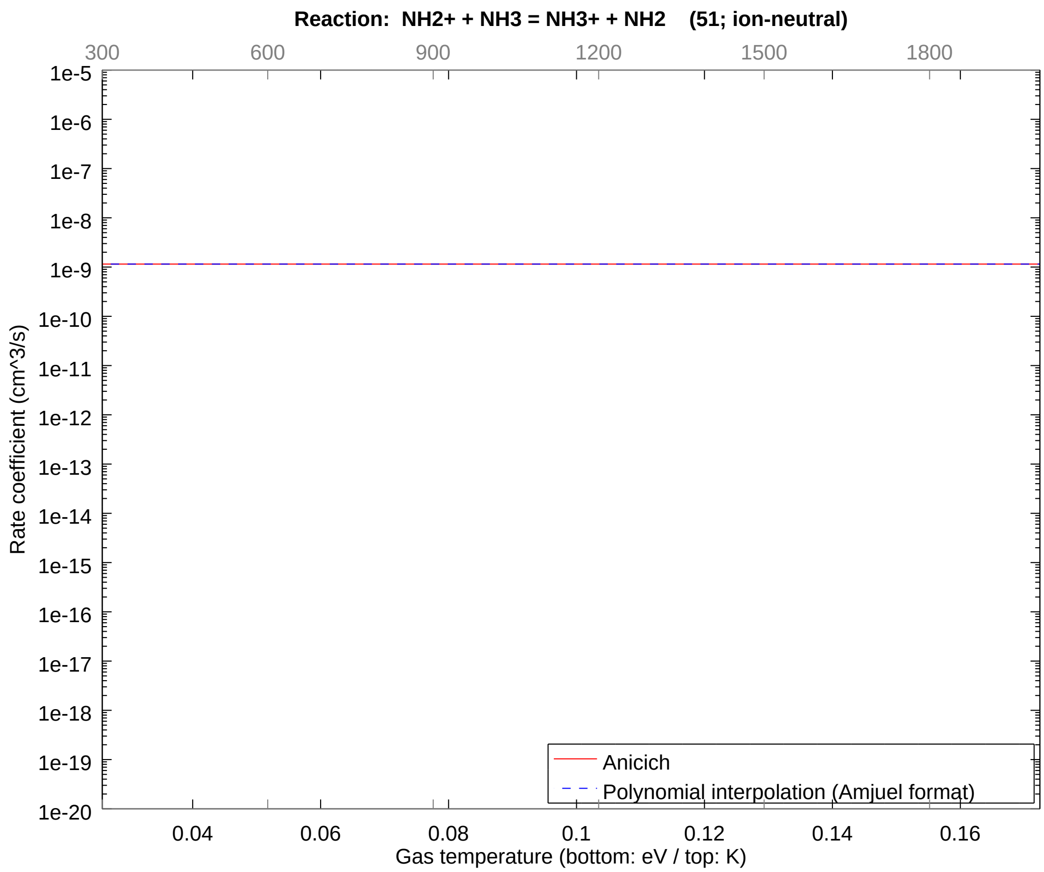
2.86 Reaction 50 NH2+ + H2 = NH3+ + H

b0	-2.235802155736E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



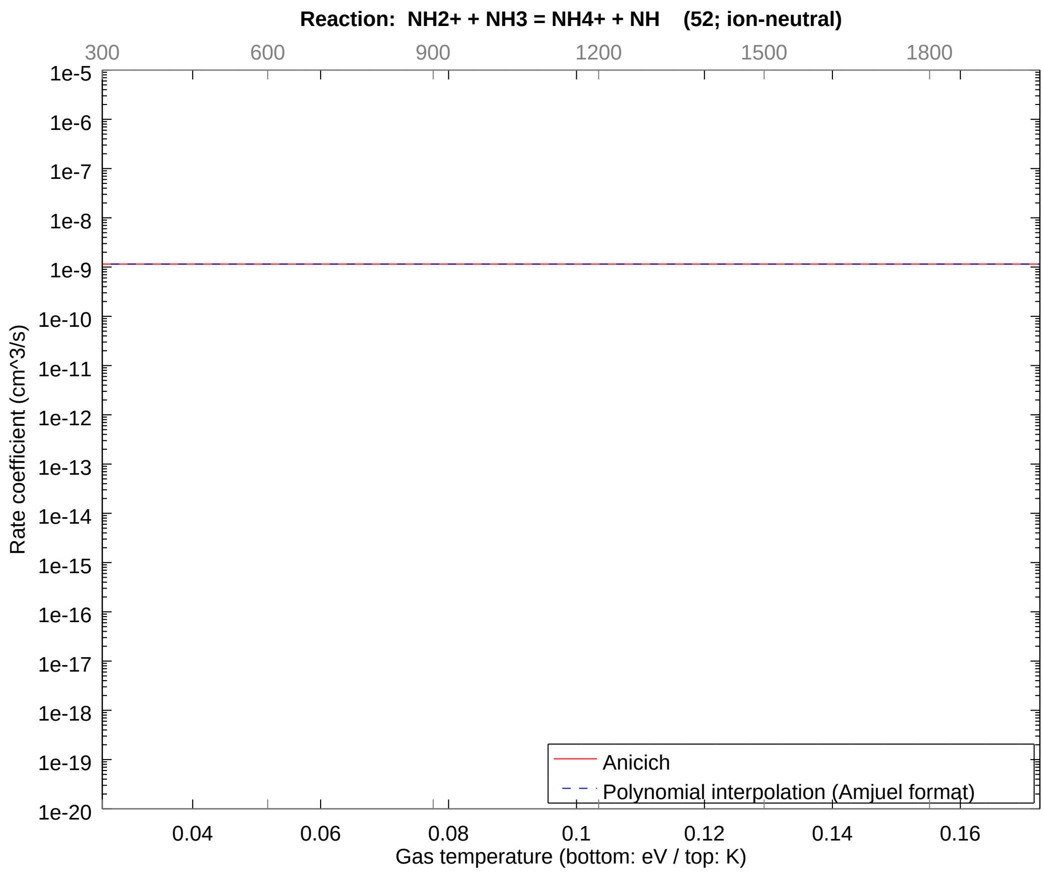
2.87 Reaction 51 NH2+ + NH3 = NH3+ + NH2

b0	-2.058350389457E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



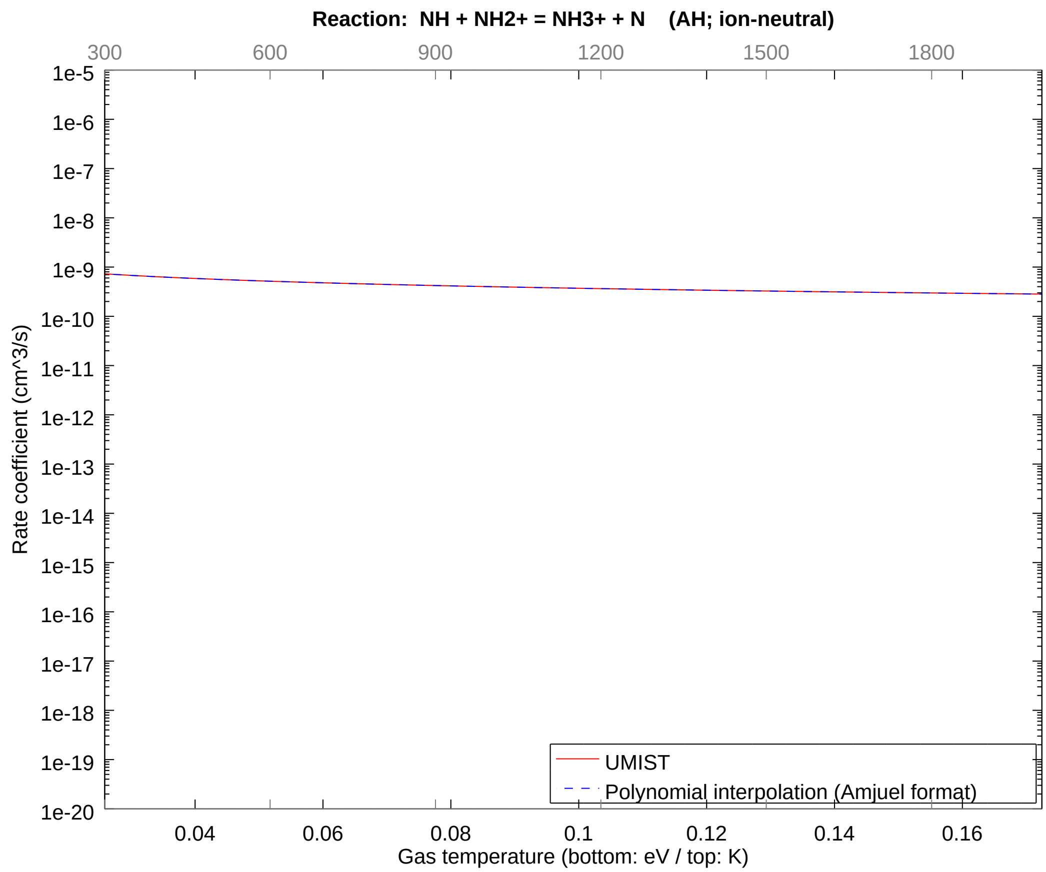
2.88 Reaction 52 NH2+ + NH3 = NH4+ + NH

b0	-2.058350389457E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



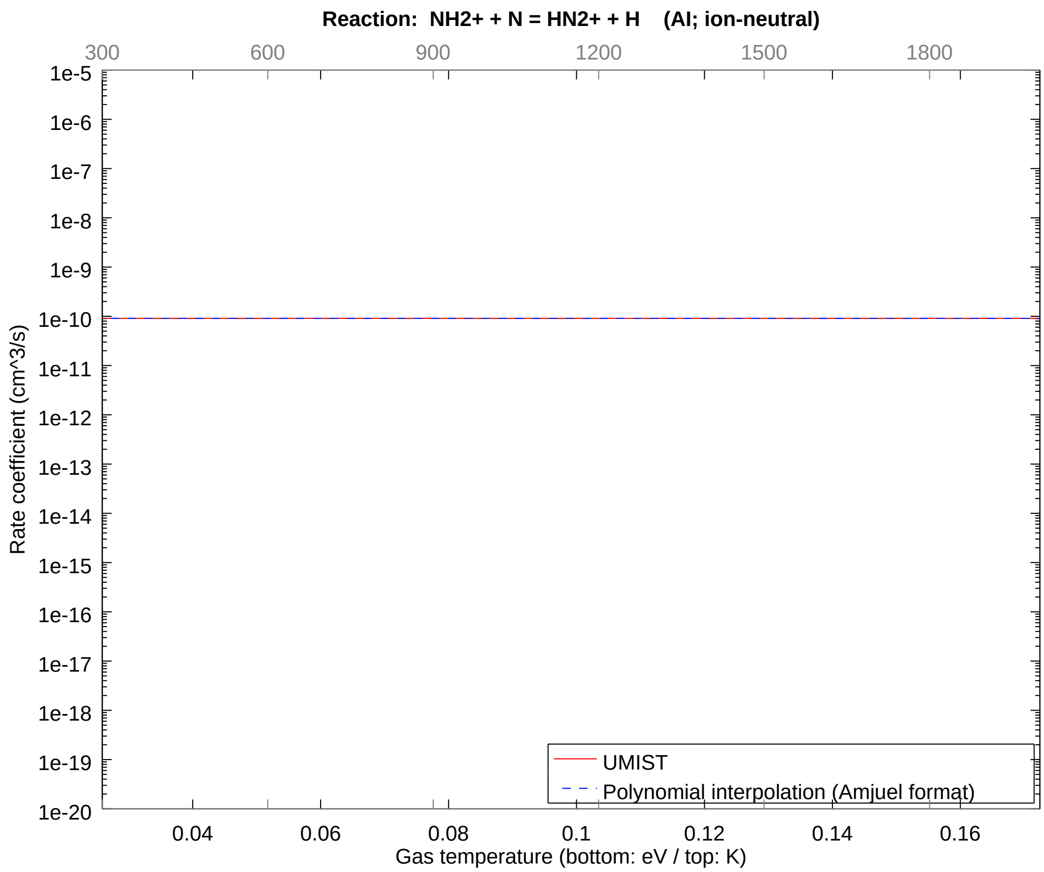
2.89 Reaction AH NH + NH2+ = NH3+ + N

b0 -2.286540473346E+01	b1 -5.000000022583E-01	b2 -2.903991793554E-09
b3 -2.092602780058E-09	b4 -9.224932782956E-10	b5 -2.541818400592E-10
b6 -4.262482993526E-11	b7 -3.961221014971E-12	b8 -1.552398934101E-13



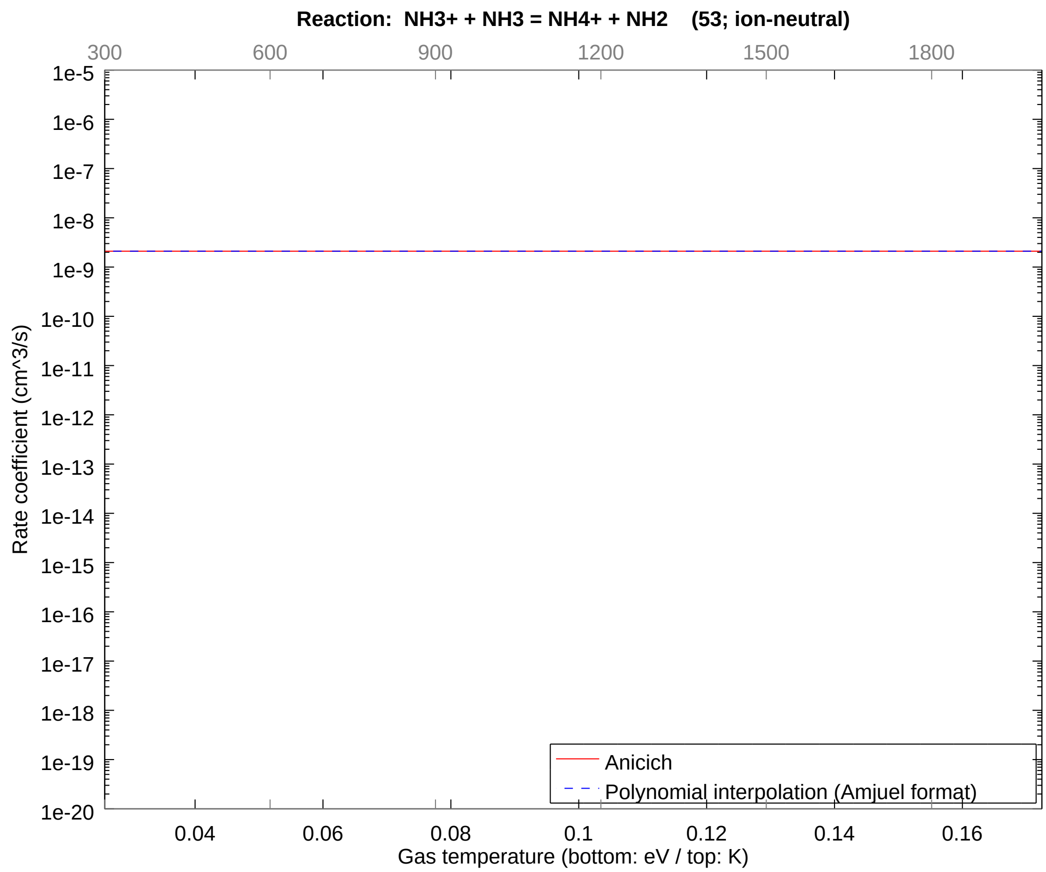
2.90 Reaction AI NH2+ + N = HN2+ + H

b0	-2.312016160941E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



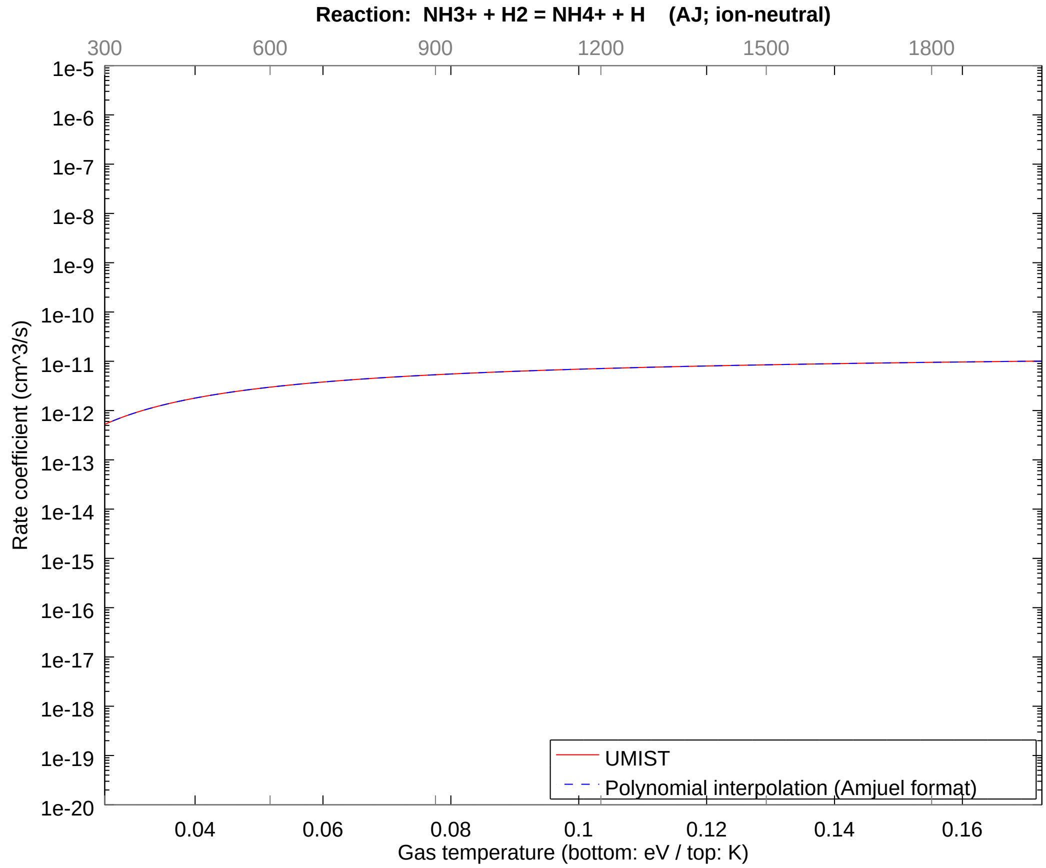
2.91 Reaction 53 NH3+ + NH3 = NH4+ + NH2

b0	-1.998132849222E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



2.92 Reaction AJ NH3+ + H2 = NH4+ + H

b0	-2.490386608232E+01	b1	3.409541987533E-02	b2	-1.304134590088E-01
b3	-5.988151600689E-02	b4	-4.512471106138E-02	b5	-1.408782129630E-02
b6	-3.532259647172E-03	b7	-4.497581656640E-04	b8	-3.345468541418E-05



2.93 Reaction AK NH3+ + NH = NH4+ + N

```
b0 -2.289318429743E+01    b1 -5.000000018164E-01    b2 -2.303568193971E-09
b3 -1.639801354397E-09    b4 -7.151531646047E-10    b5 -1.951520773247E-10
b6 -3.242435734298E-11    b7 -2.983863276617E-12    b8 -1.155432594465E-13
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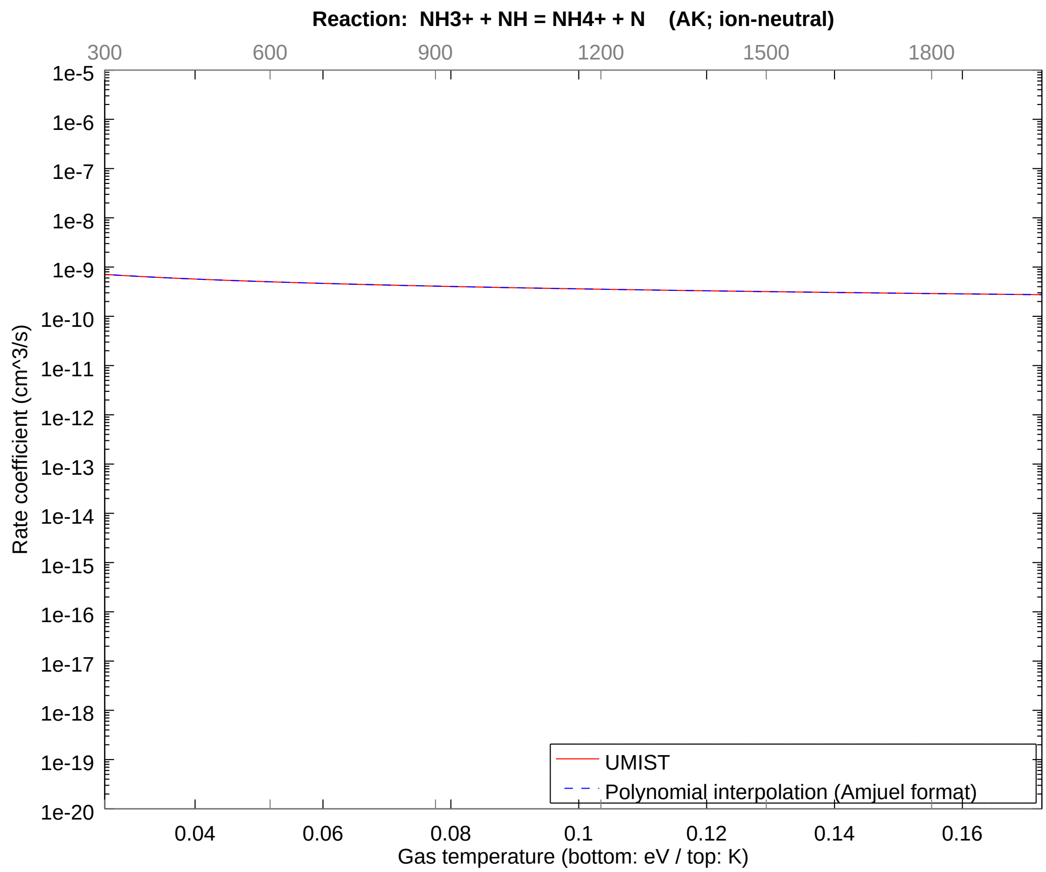
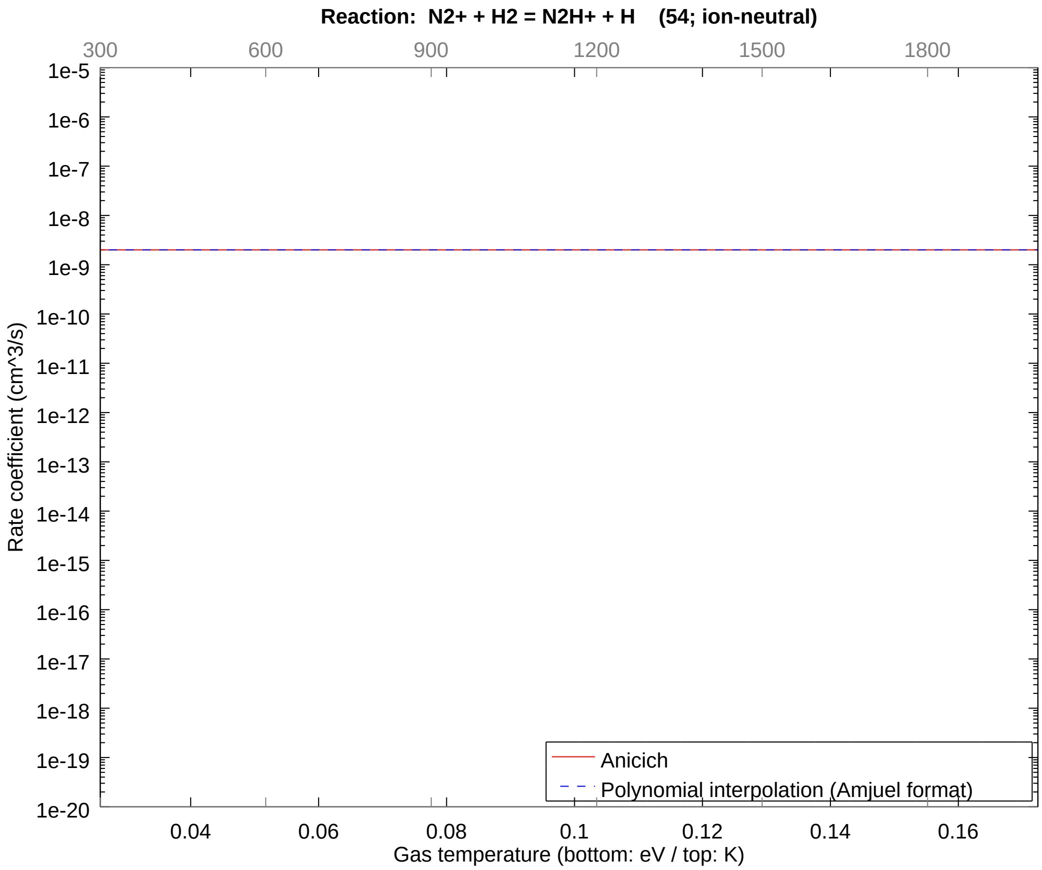


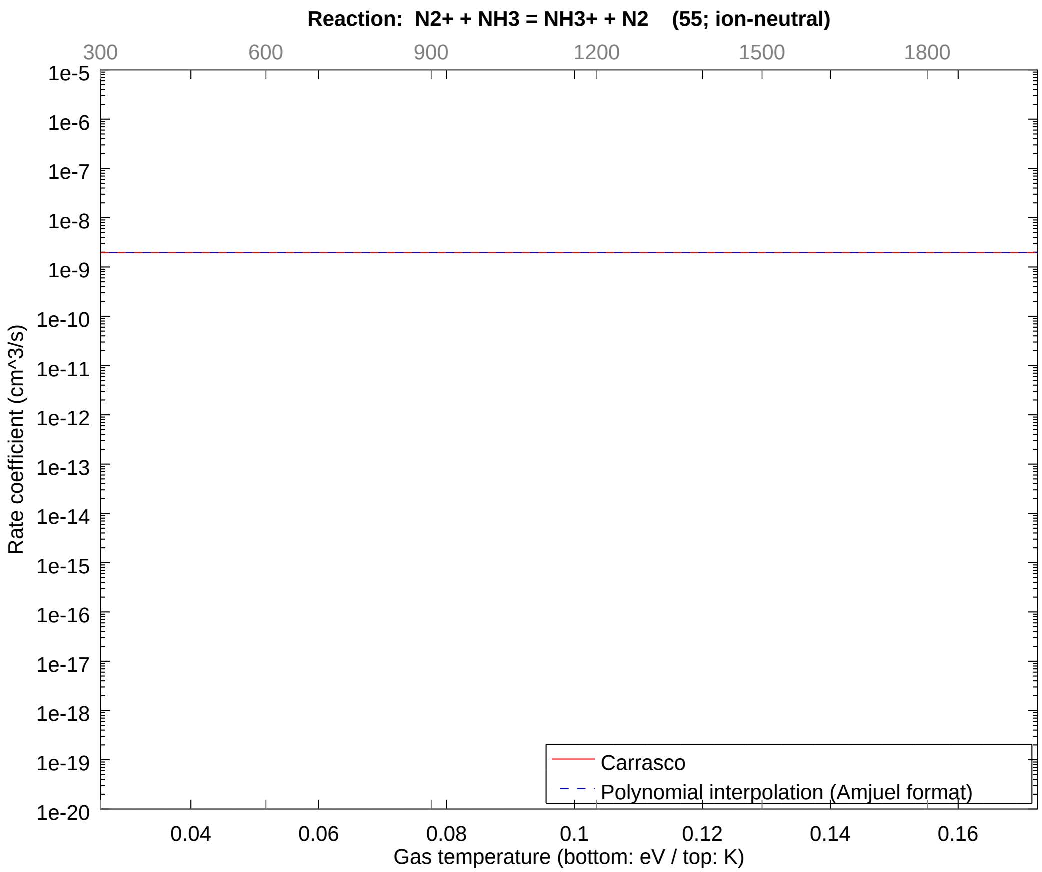
2.94 Reaction 54 N2+ + H2 = N2H+ + H

b0	-2.003011865639E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



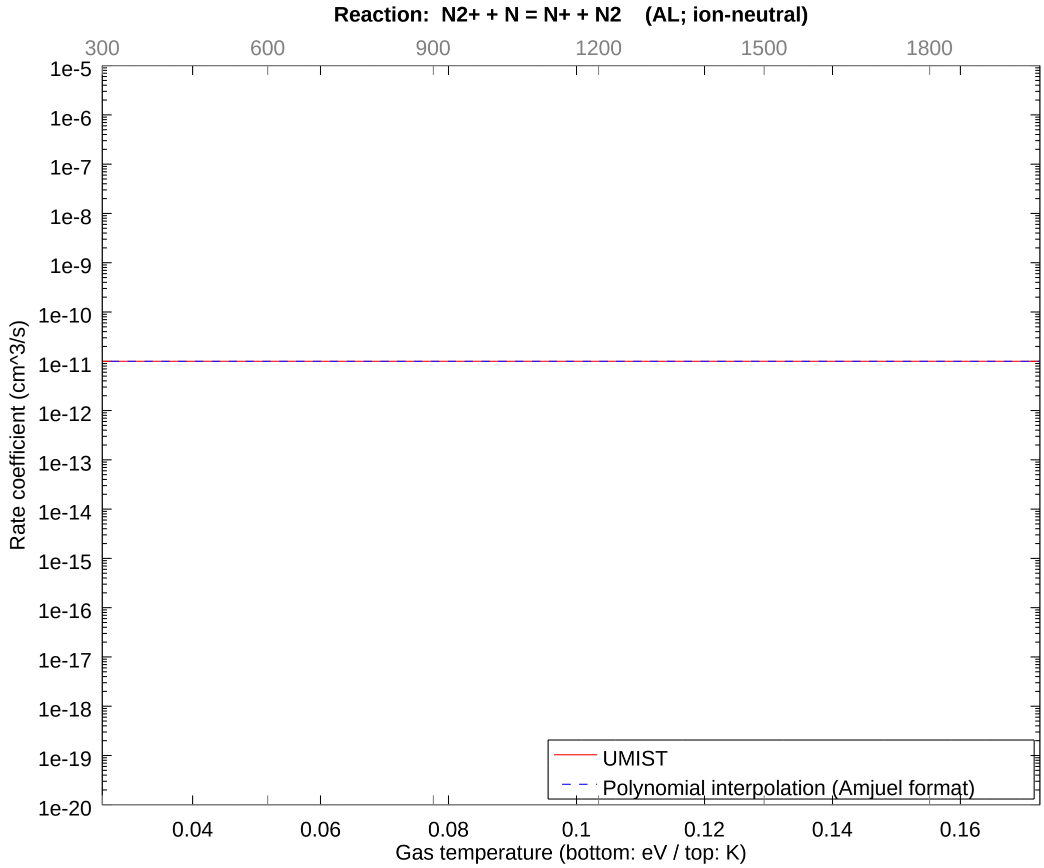
2.95 Reaction 55 N2+ + NH3 = NH3+ + N2

b0	-2.005543646437E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



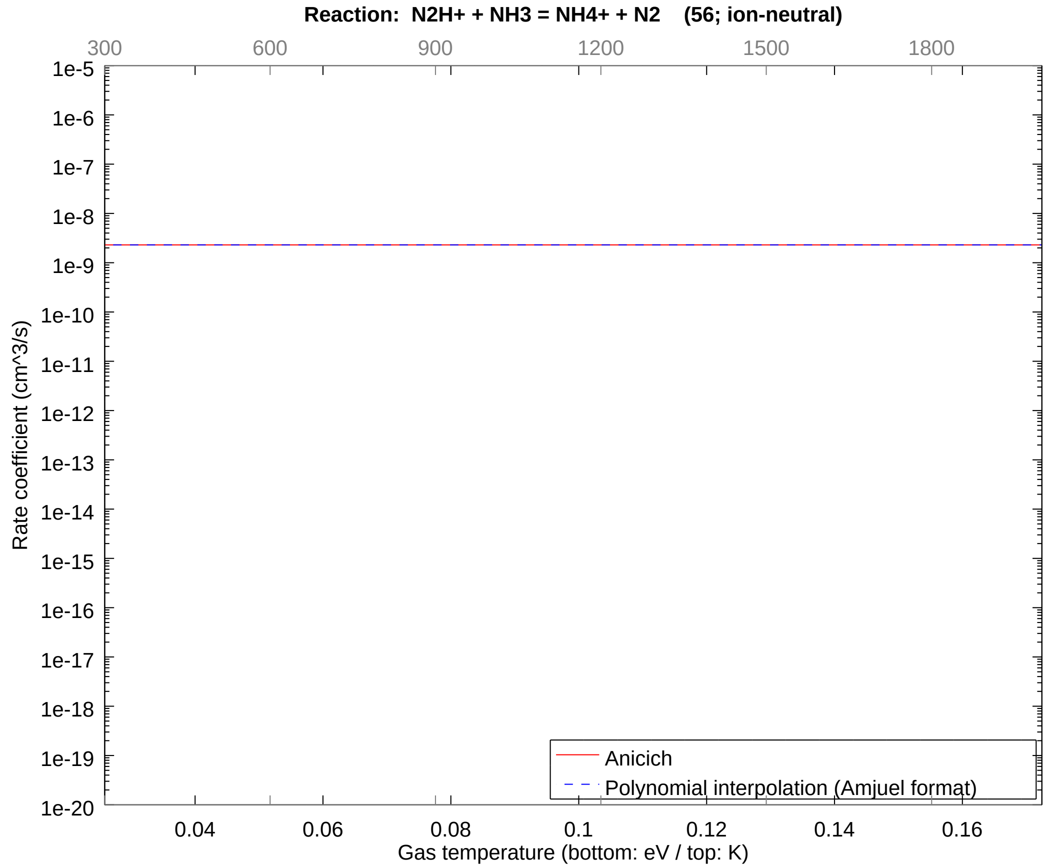
2.96 Reaction AL N2+ + N = N+ + N2

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



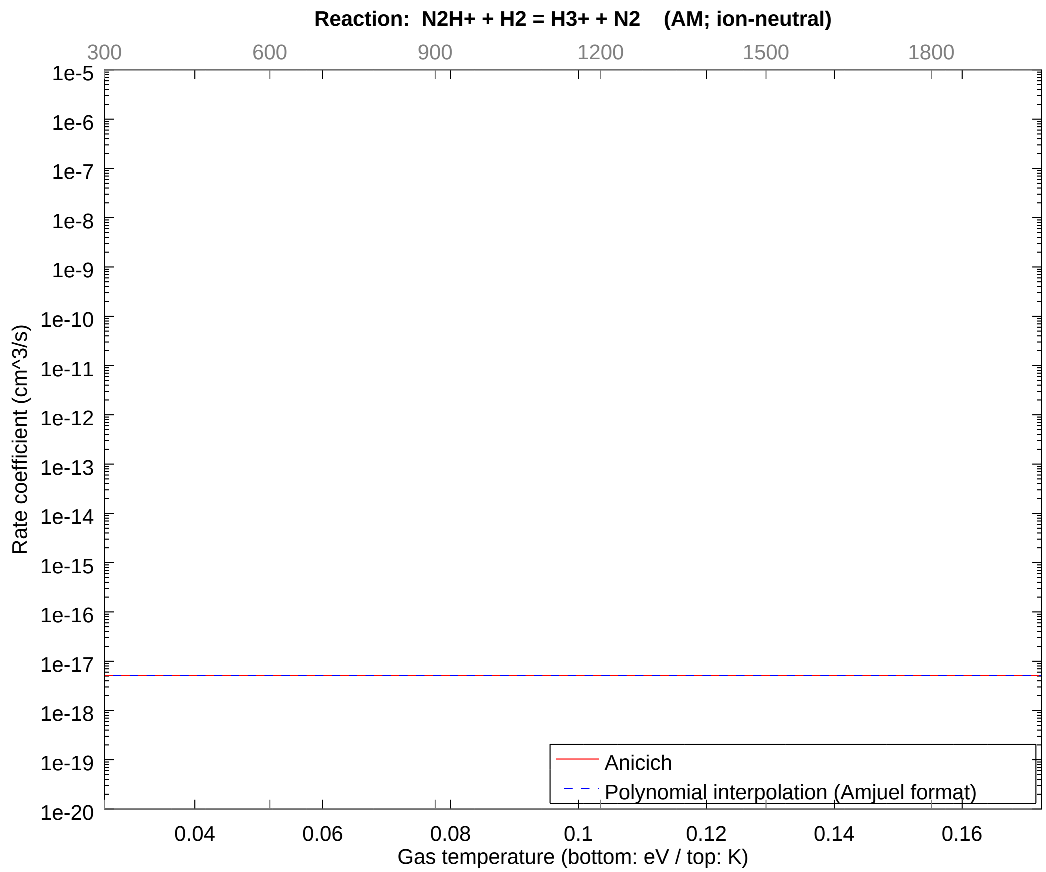
2.97 Reaction 56 N2H+ + NH3 = NH4+ + N2

b0	-1.989035671401E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



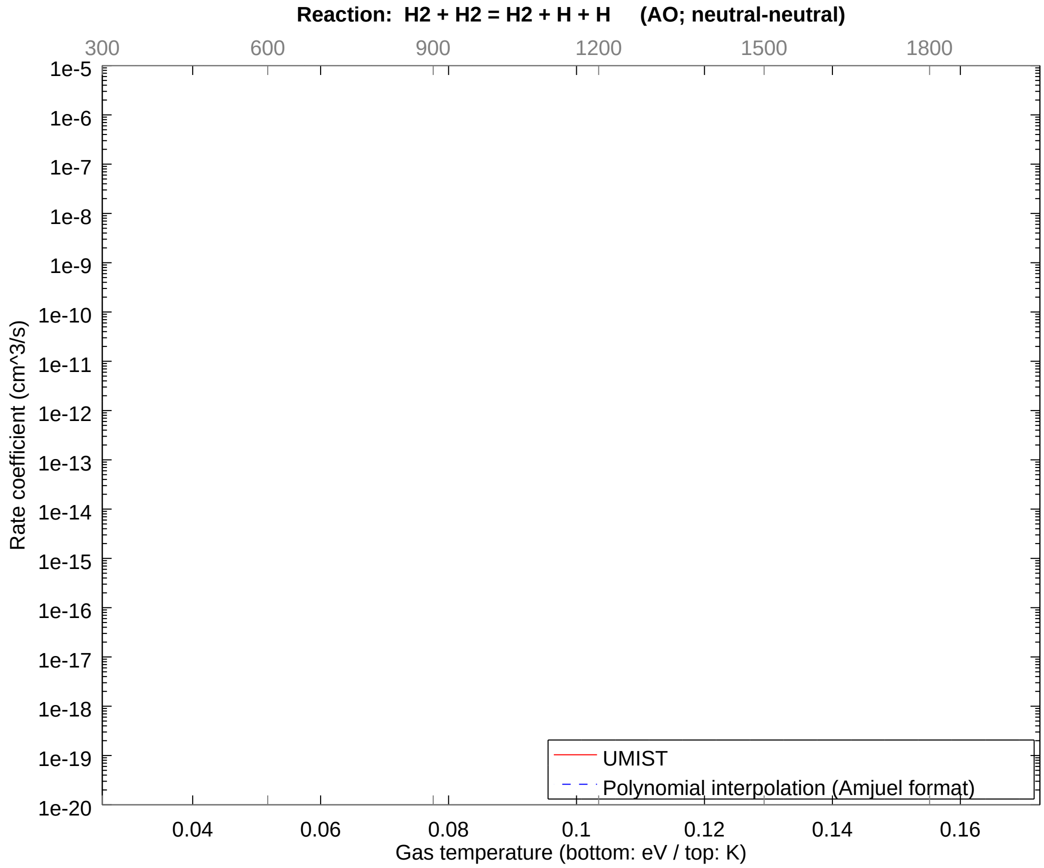
2.98 Reaction AM N2H+ + H2 = H3+ + N2

b0	-3.981729113416E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



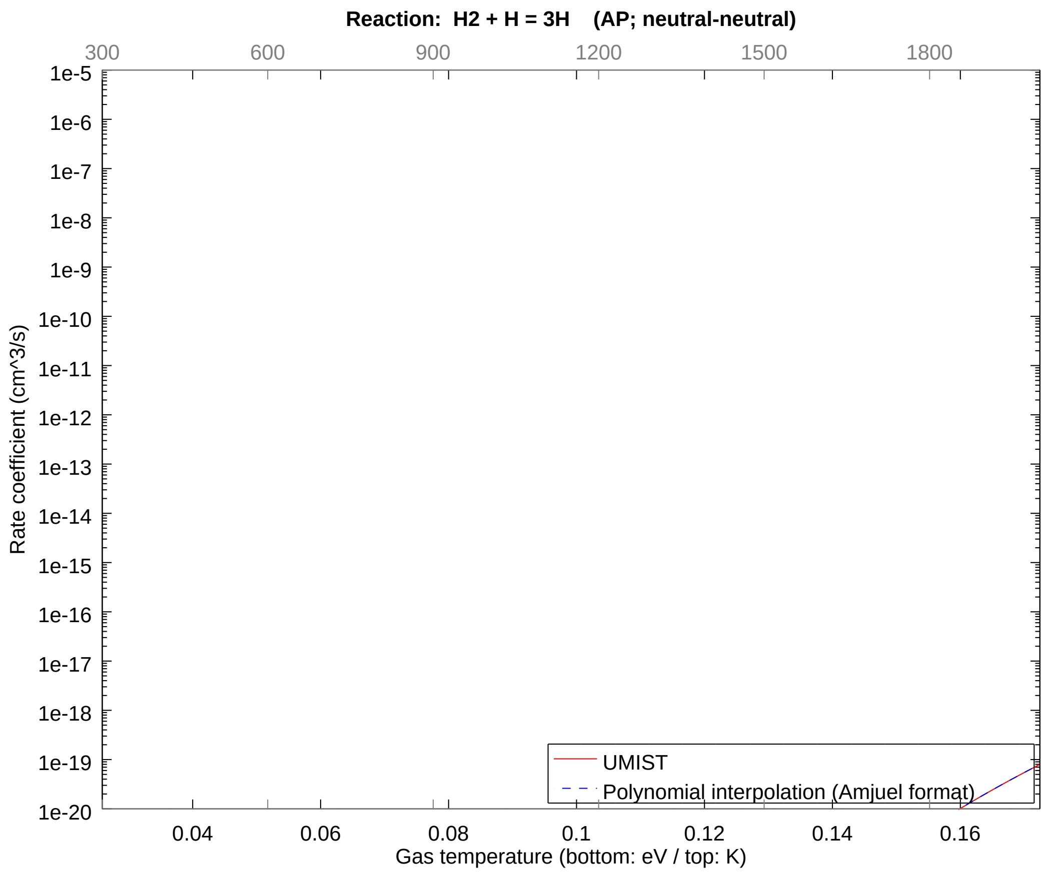
2.99 Reaction AO H2 + H2 = H2 + H + H

b0 -2.696426662024E+01	b1 2.746576251943E+00	b2 -1.050552760935E+01
b3 -4.823787997126E+00	b4 -3.635045792807E+00	b5 -1.134852158008E+00
b6 -2.845431171969E-01	b7 -3.623051668352E-02	b8 -2.694960668238E-03



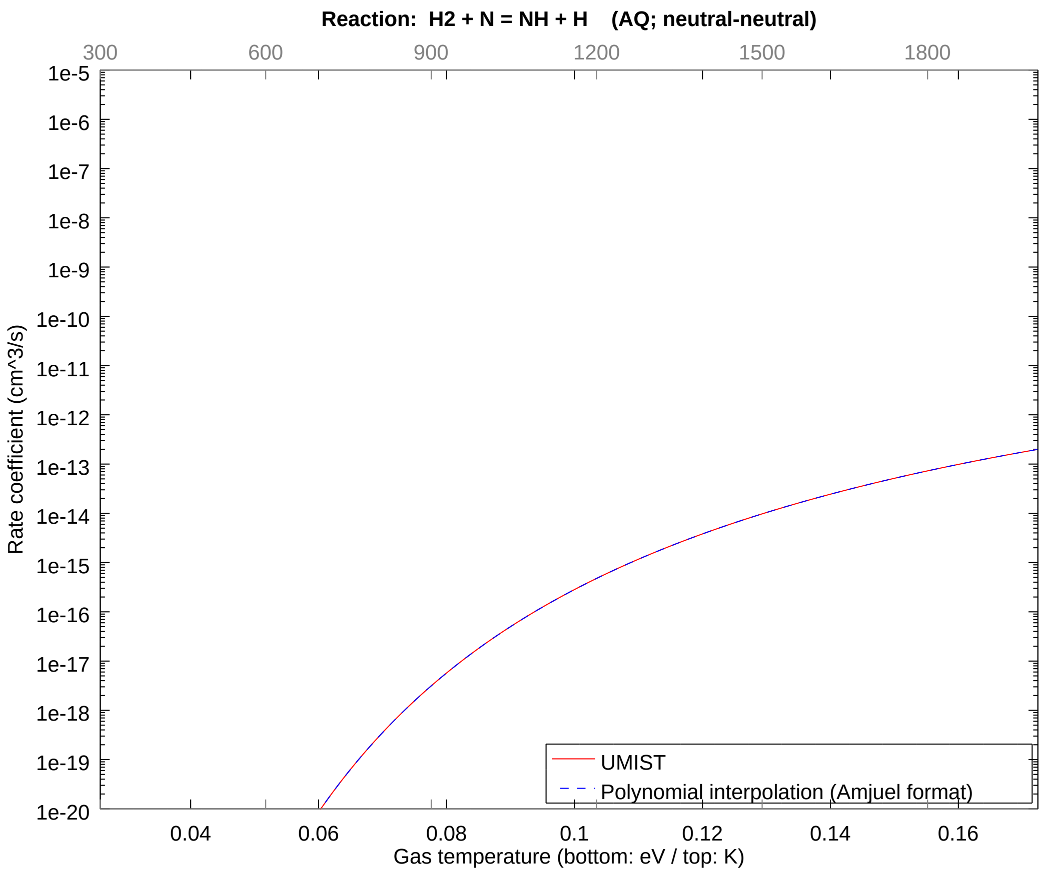
2.100 Reaction AP H2 + H = 3H

b0 -2.381915581932E+01	b1 7.962151505816E-01	b2 -6.870440167331E+00
b3 -3.154677044982E+00	b4 -2.377259434294E+00	b5 -7.421744189427E-01
b6 -1.860864617664E-01	b7 -2.369415477215E-02	b8 -1.762459413704E-03



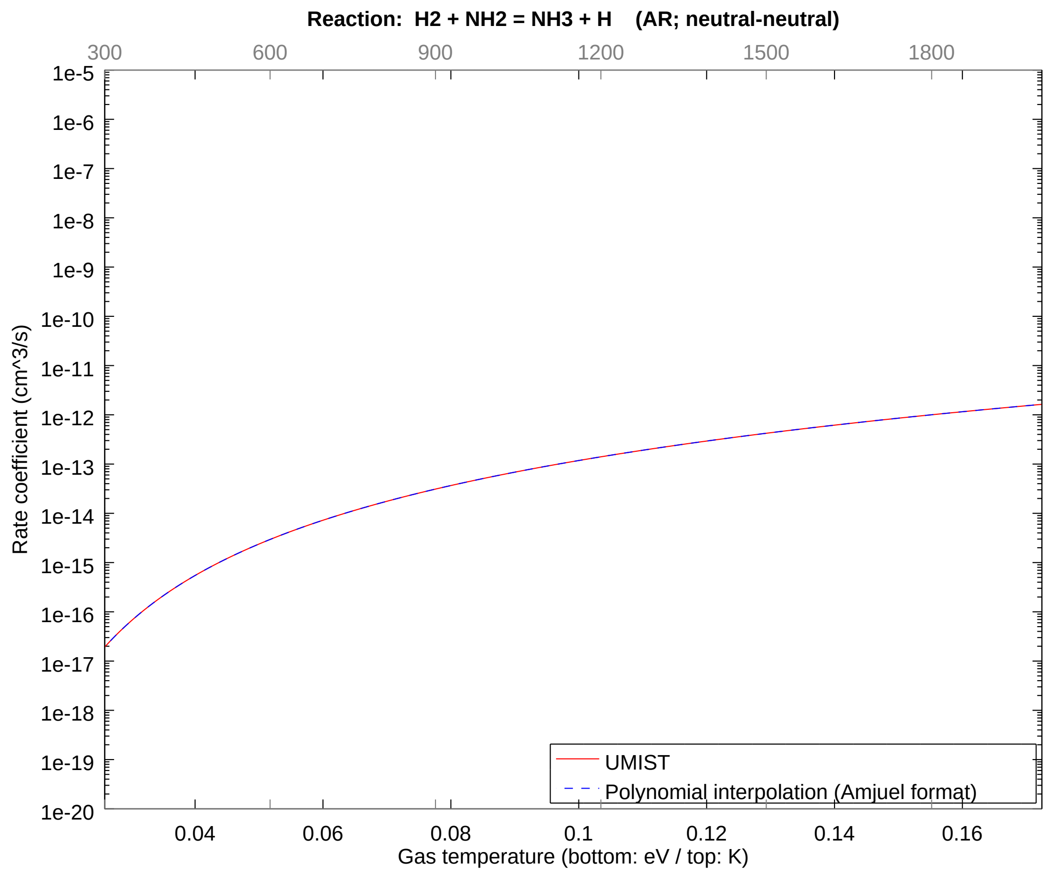
2.101 Reaction AQ H2 + N = NH + H

b0 -2.203677971440E+01	b1 5.909547855171E-01	b2 -2.260374813605E+00
b3 -1.037888746589E+00	b4 -7.821183532573E-01	b5 -2.441753836278E-01
b6 -6.122244587988E-02	b7 -7.795376915334E-03	b8 -5.798491468765E-04



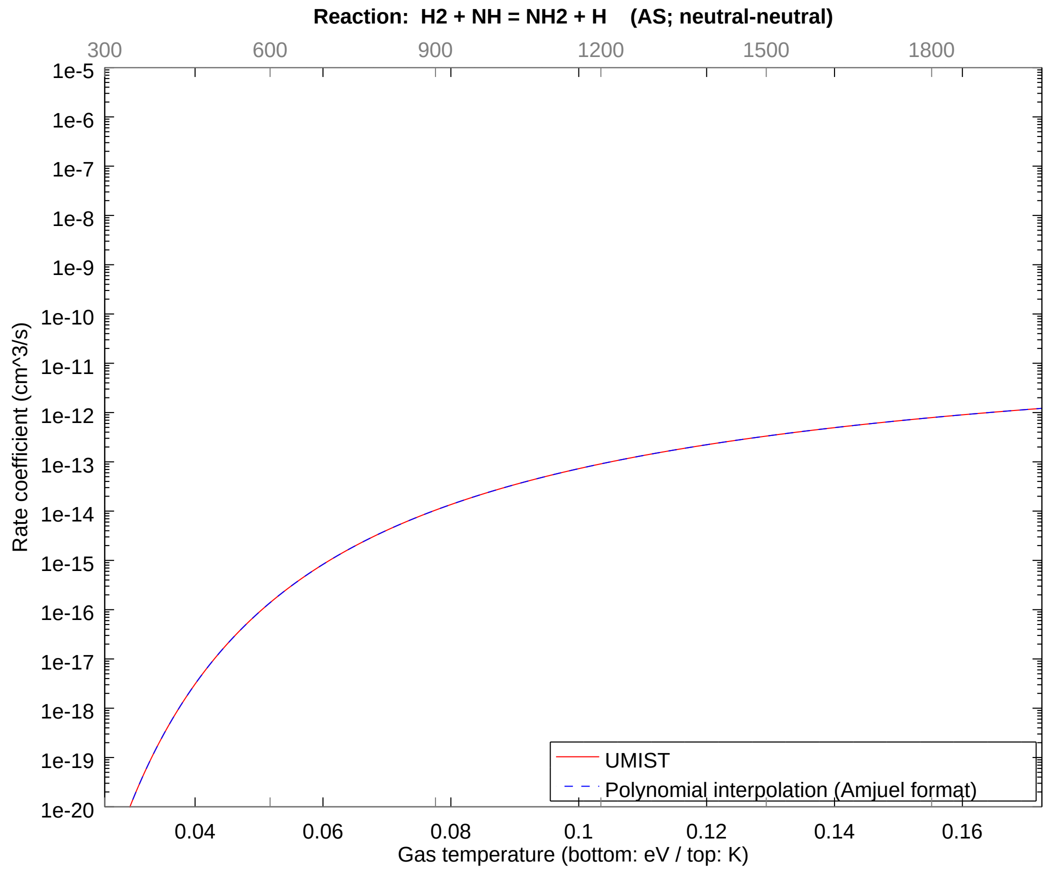
2.102 Reaction AR H2 + NH2 = NH3 + H

```
b0 -1.974576937221E+01    b1  3.935721838265E+00    b2 -1.748839337655E-01
b3 -8.030087161977E-02    b4 -6.051205881110E-02    b5 -1.889171257029E-02
b6 -4.736746304237E-03    b7 -6.031239388174E-04    b8 -4.486260315385E-05
```



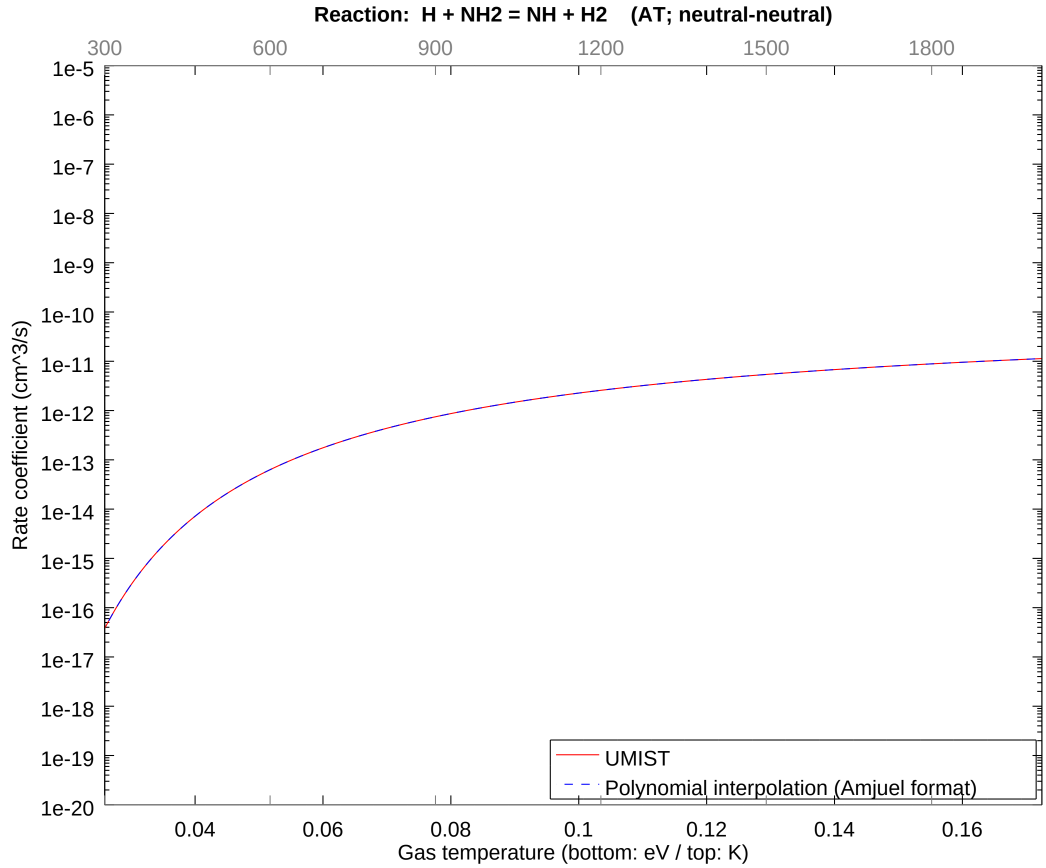
2.103 Reaction AS H2 + NH = NH2 + H

b0	-2.433392660574E+01	b1	2.541481071904E-01	b2	-9.721048352481E-01
b3	-4.463581308227E-01	b4	-3.363606022624E-01	b5	-1.050109343825E-01
b6	-2.632954285966E-02	b7	-3.352507521719E-03	b8	-2.493719857357E-04



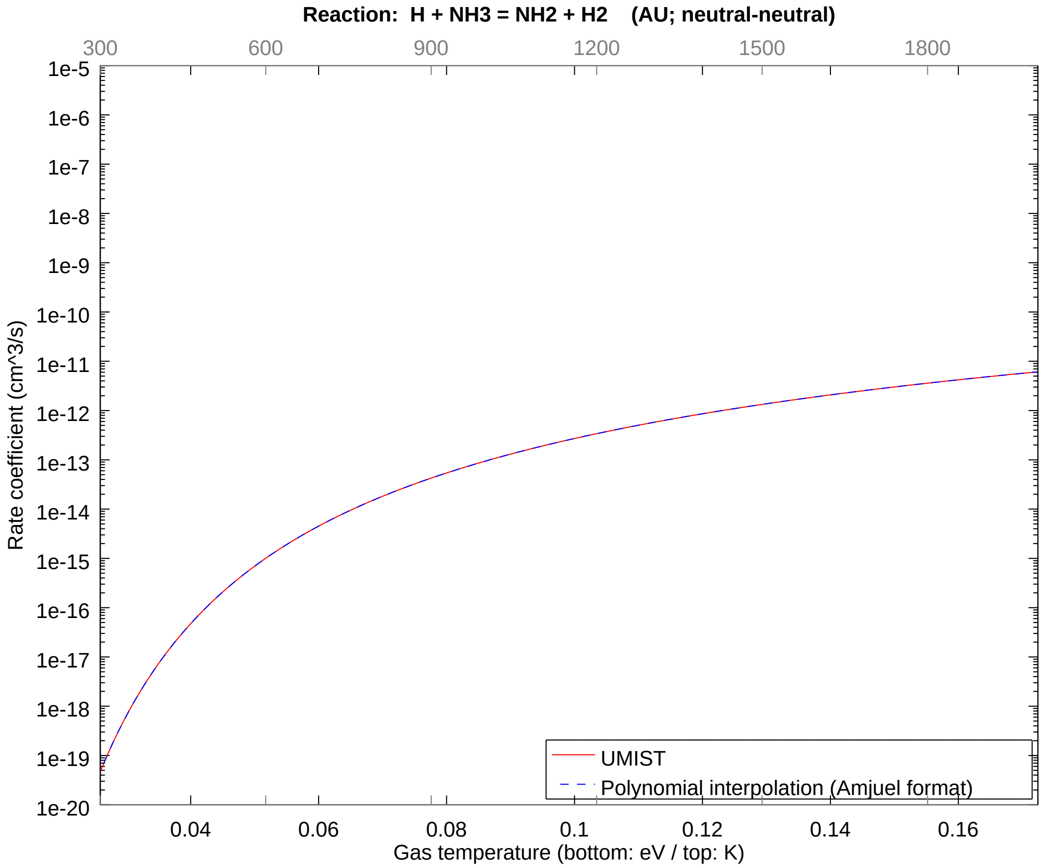
2.104 Reaction AT H + NH2 = NH + H2

b0	-2.342912922359E+01	b1	1.453301289331E-01	b2	-5.558810761840E-01
b3	-2.552420580269E-01	b4	-1.923419026233E-01	b5	-6.004865841419E-02
b6	-1.505608661547E-02	b7	-1.917072539614E-03	b8	-1.425989897052E-04



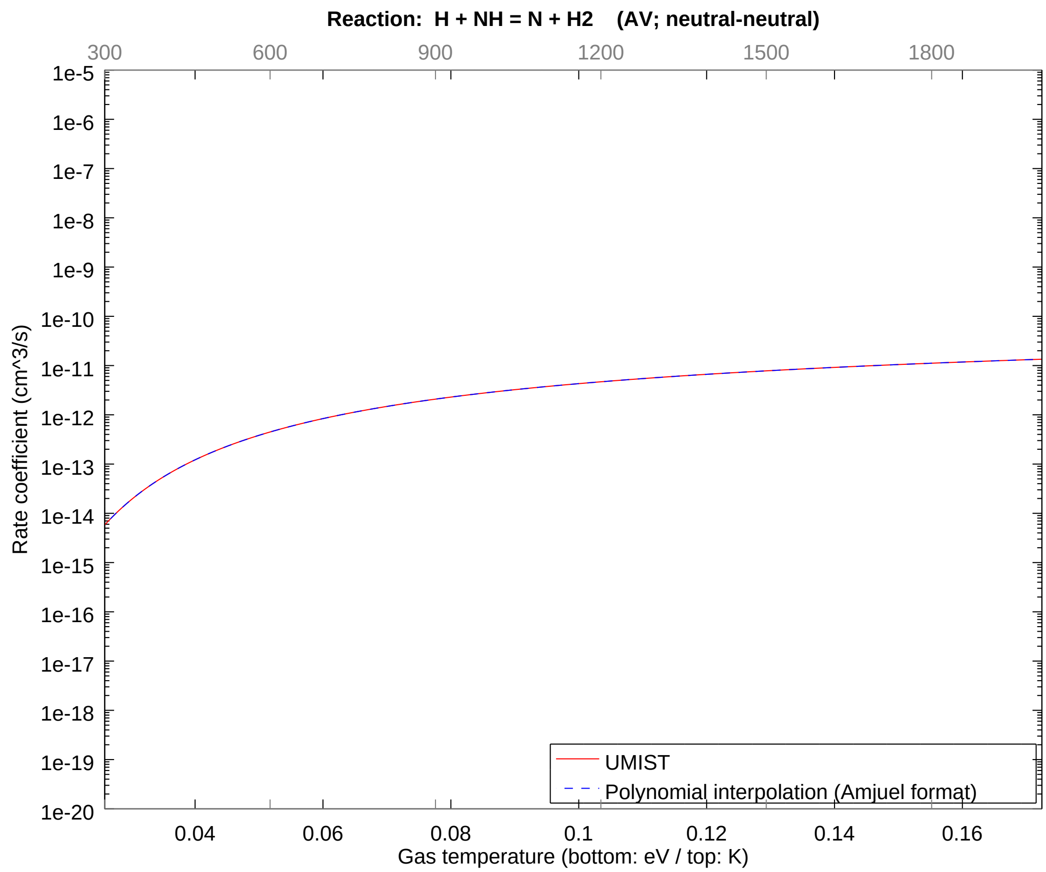
2.105 Reaction AU H + NH3 = NH2 + H2

b0 -1.961475355238E+01	b1 2.562965700633E+00	b2 -6.233363002700E-01
b3 -2.862152455775E-01	b4 -2.156822654030E-01	b5 -6.733546099889E-02
b6 -1.688311717638E-02	b7 -2.149706042274E-03	b8 -1.599031358719E-04



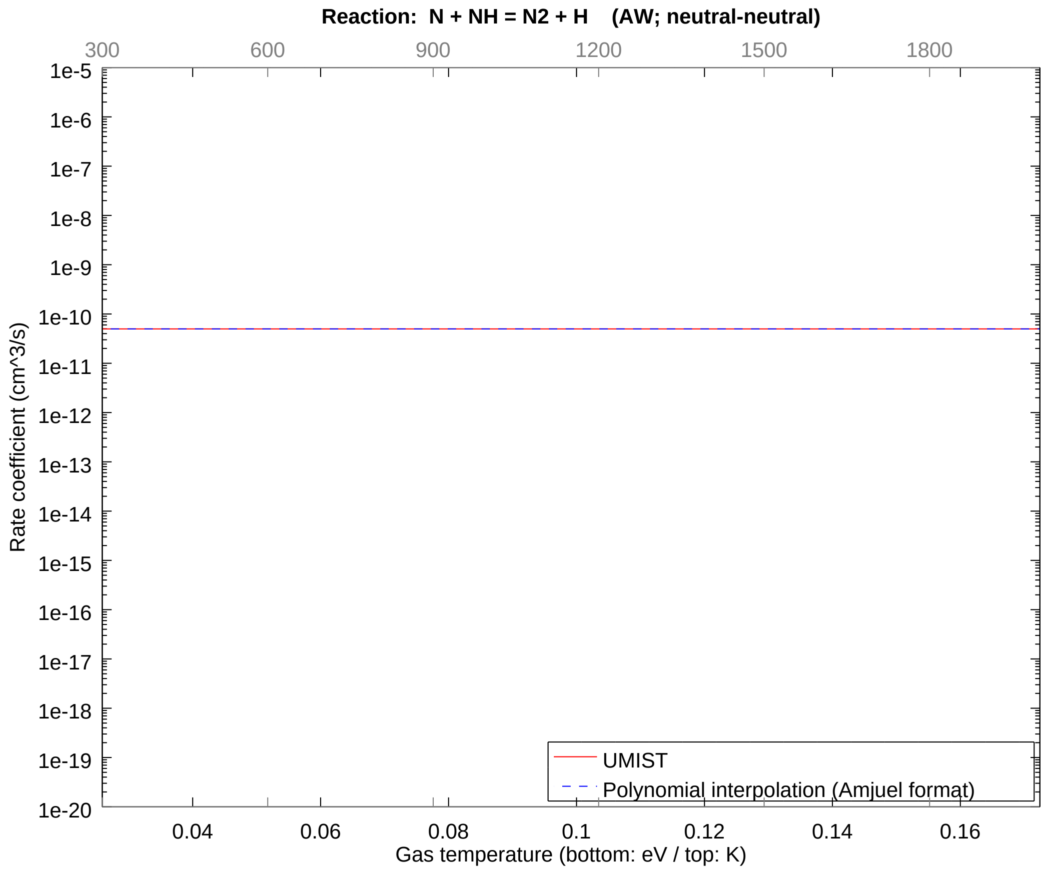
2.106 Reaction AV H + NH = N + H2

b0	-2.319669866637E+01	b1	5.783802913881E-01	b2	-2.998010338650E-01
b3	-1.376586412086E-01	b4	-1.037349602663E-01	b5	-3.238579377985E-02
b6	-8.120136692671E-03	b7	-1.033926772738E-03	b8	-7.690732074648E-05



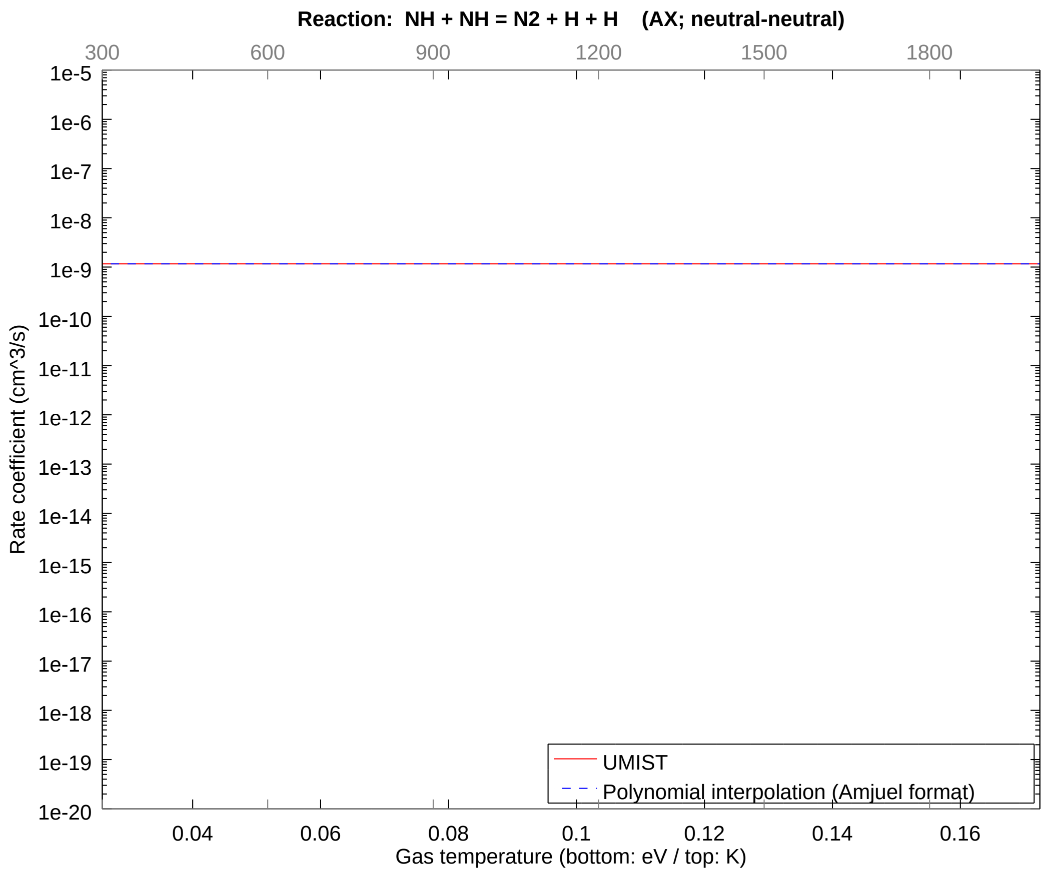
2.107 Reaction AW N + NH = N2 + H

b0	-2.372300613190E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



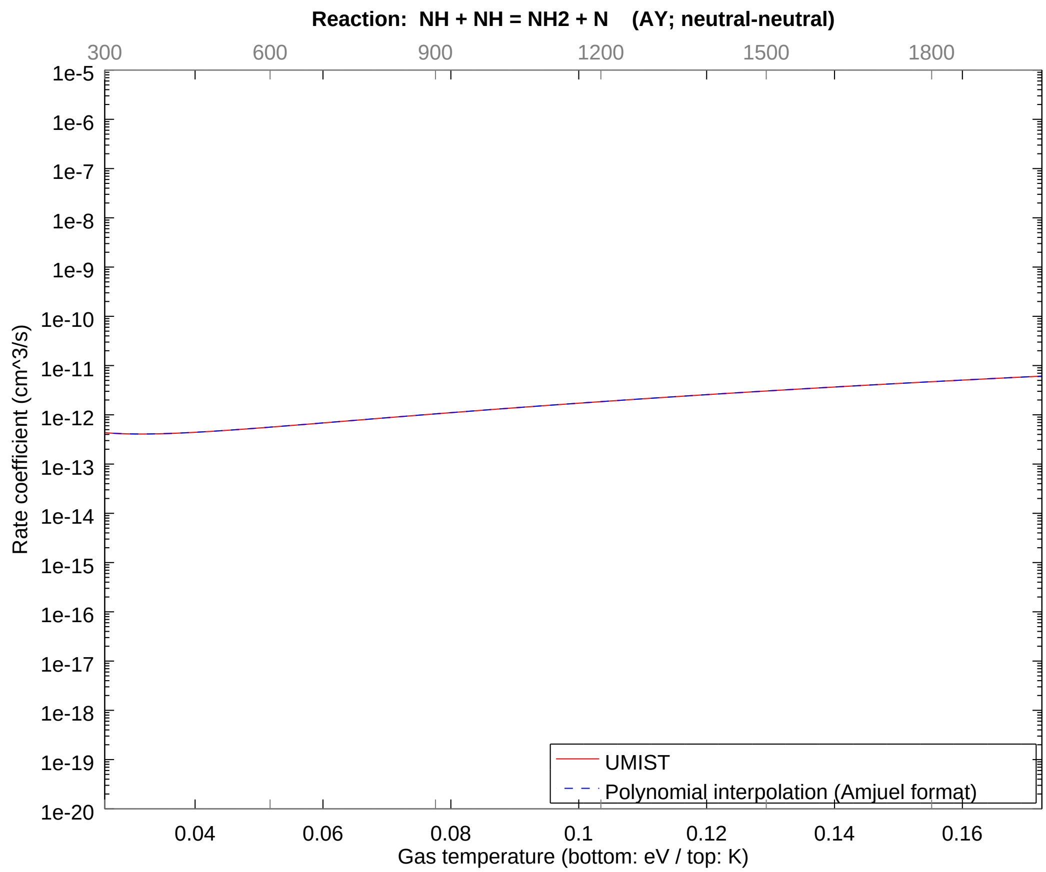
2.108 Reaction AX NH + NH = N2 + H + H

b0	-2.057484583183E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



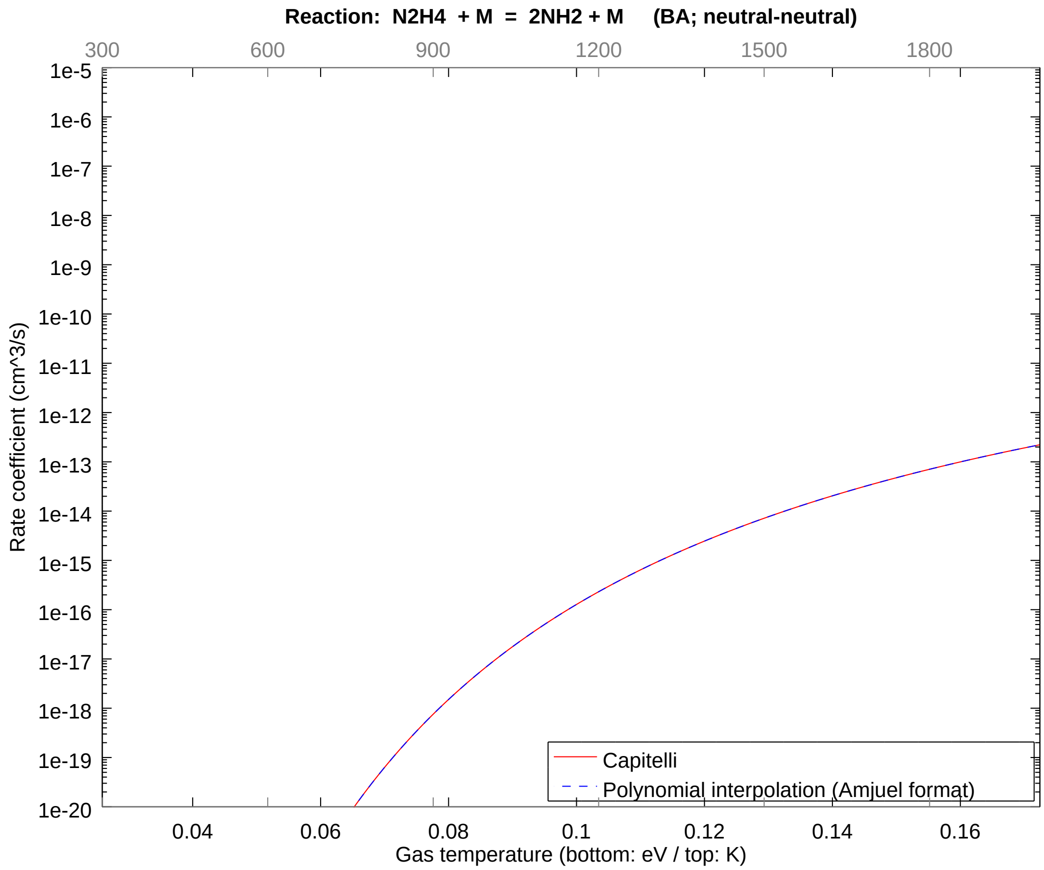
2.109 Reaction AY NH + NH = NH2 + N

b0	-2.087213986080E+01	b1	3.033324552294E+00	b2	1.402818964922E-01
b3	6.441276966251E-02	b4	4.853931554652E-02	b5	1.515385231802E-02
b6	3.799547234470E-03	b7	4.837915635689E-04	b8	3.598621695468E-05



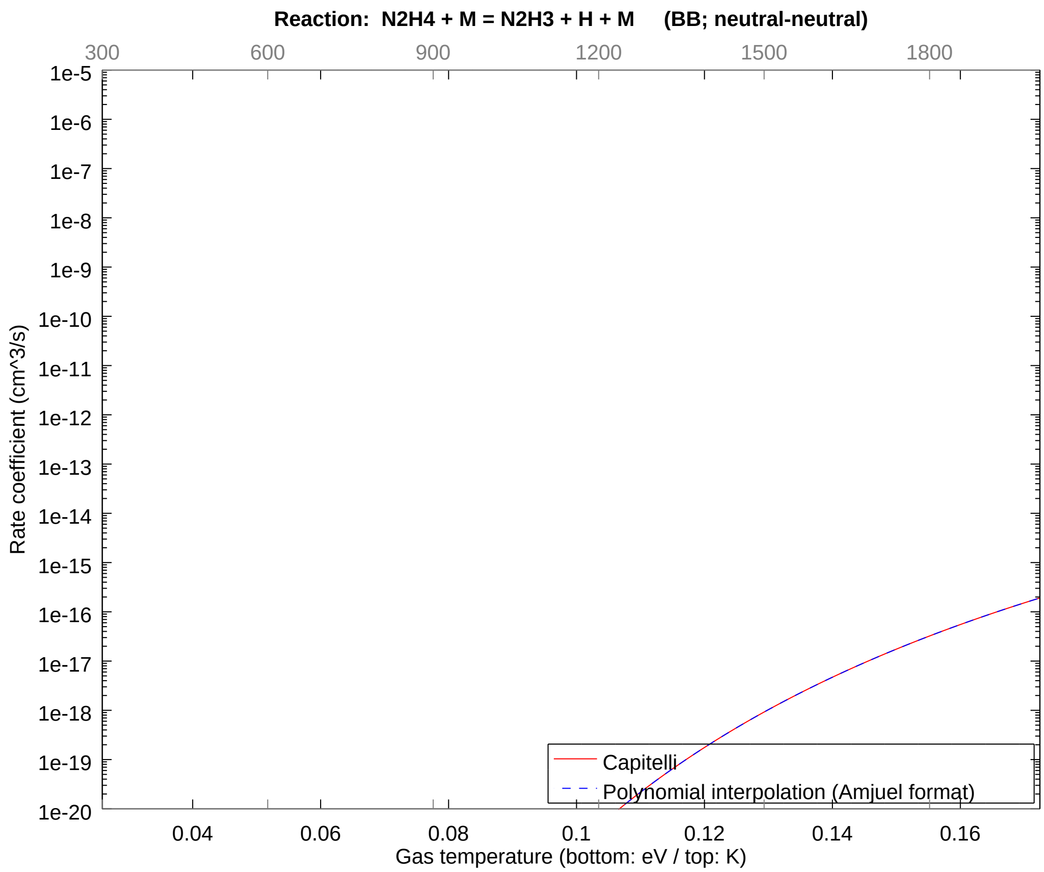
2.110 Reaction BA N2H4 + M = 2NH2 + M

b0	-2.092891758081E+01	b1	6.727642154328E-01	b2	-2.573292141534E+00
b3	-1.181569952382E+00	b4	-8.903917175497E-01	b5	-2.779780557347E-01
b6	-6.969783852452E-02	b7	-8.874537981438E-03	b8	-6.601211627768E-04



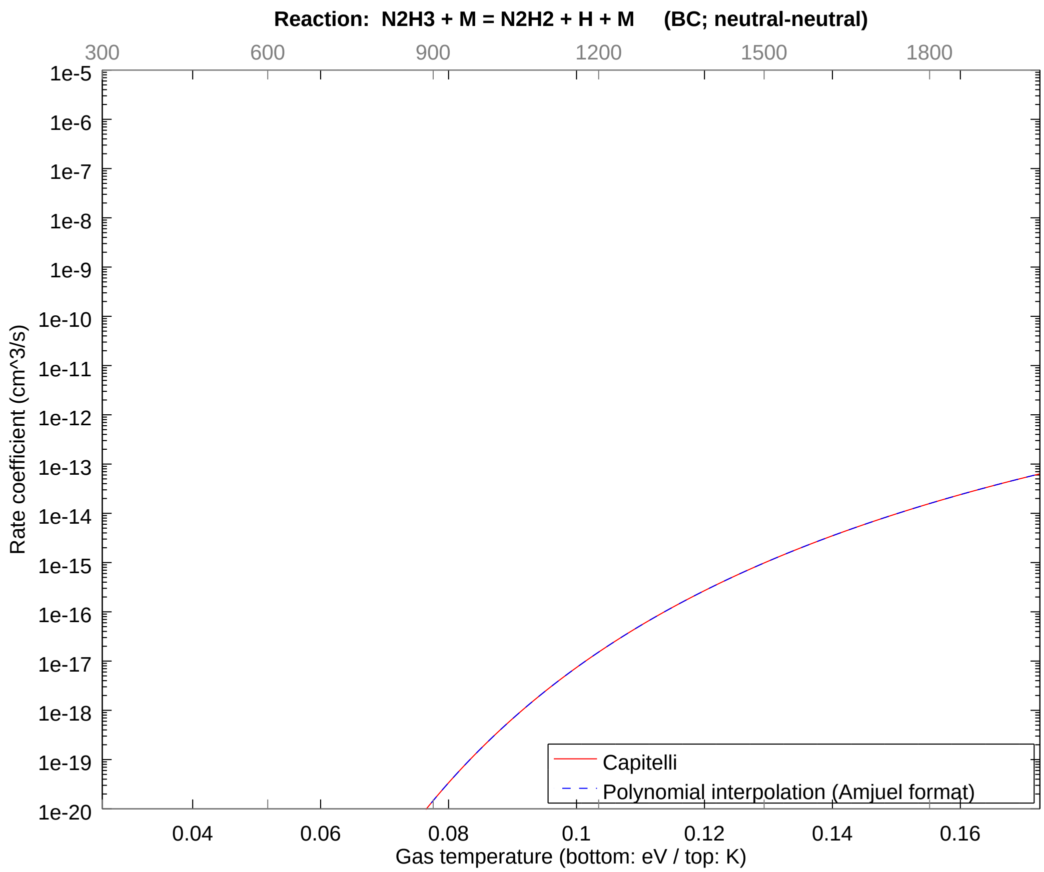
2.111 Reaction BB N2H4 + M = N2H3 + H + M

b0	-2.344346693000E+01	b1	1.045070638427E+00	b2	-3.997346999314E+00
b3	-1.835448457155E+00	b4	-1.383132759255E+00	b5	-4.318105702736E-01
b6	-1.082684867080E-01	b7	-1.378569003483E-02	b8	-1.025430930953E-03



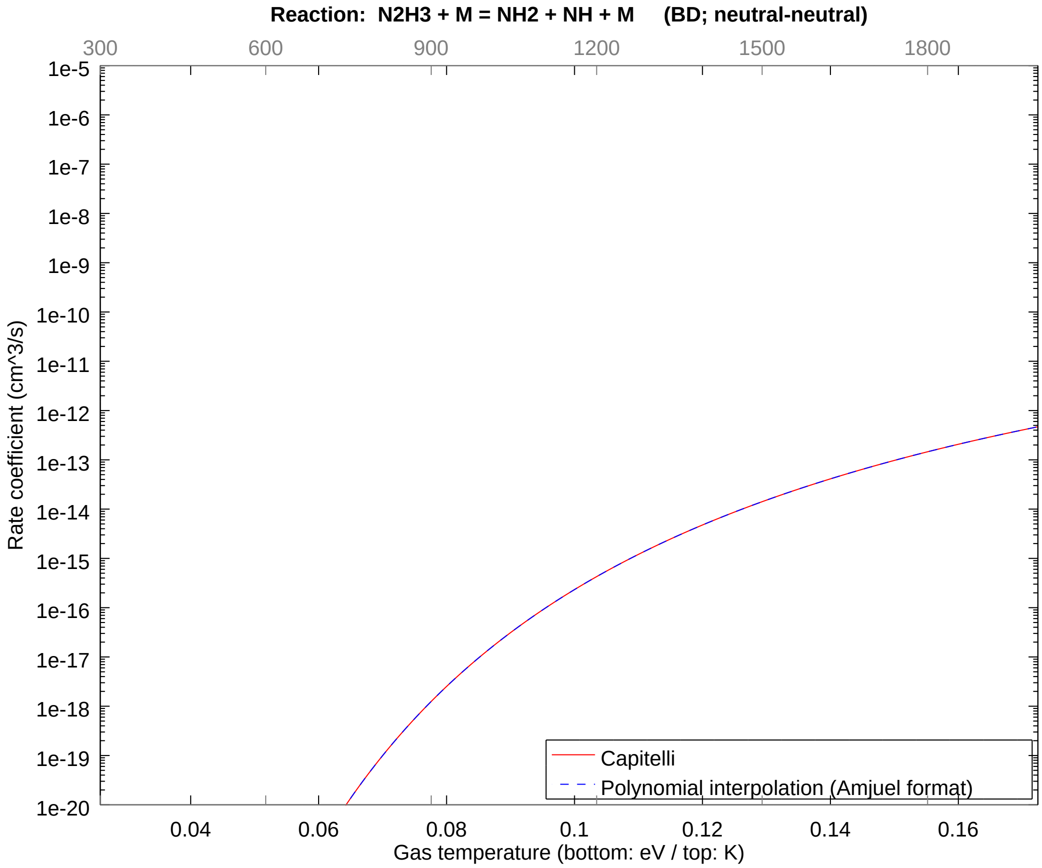
2.112 Reaction BC N2H3 + M = N2H2 + H + M

b0 -2.042976292036E+01	b1 8.164614287281E-01	b2 -3.122927353608E+00
b3 -1.433944115263E+00	b4 -1.080572472087E+00	b5 -3.373520092273E-01
b6 -8.458475546858E-02	b7 -1.077007036420E-02	b8 -8.011179159485E-04



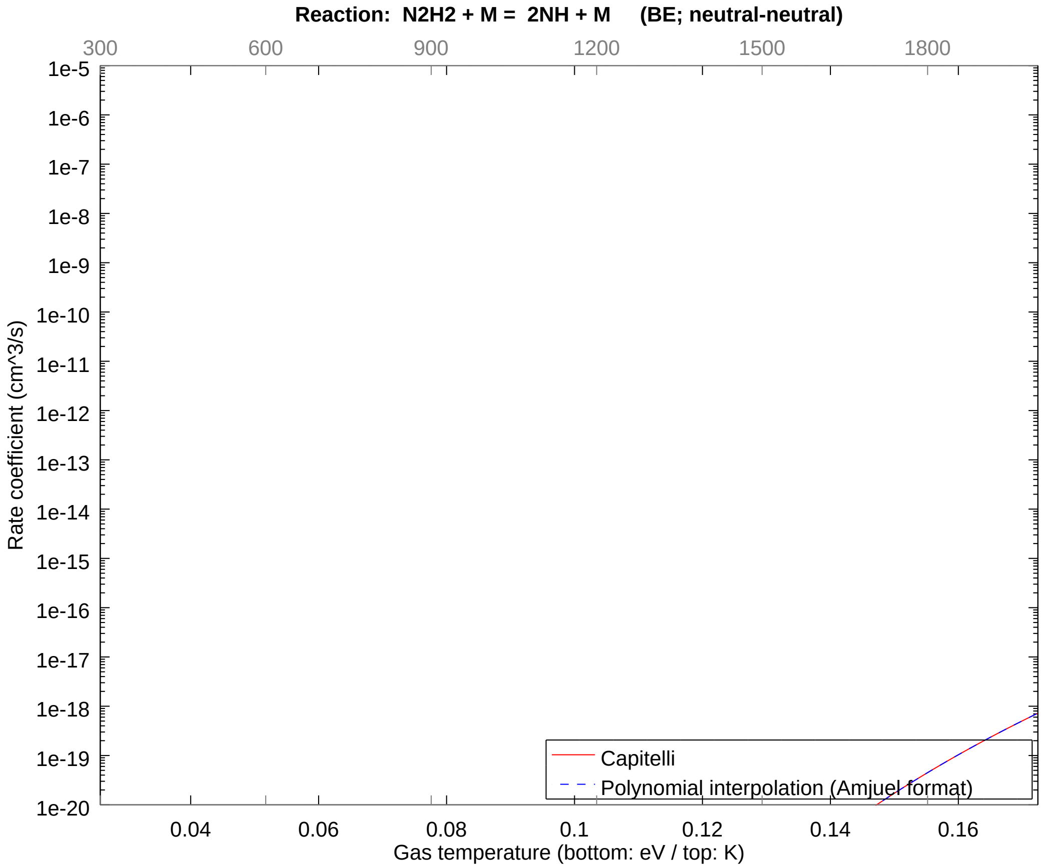
2.113 Reaction BD N2H3 + M = NH2 + NH + M

b0	-2.002340925171E+01	b1	6.858276010790E-01	b2	-2.623258975556E+00
b3	-1.204513055552E+00	b4	-9.076808758901E-01	b5	-2.833756875350E-01
b6	-7.105119455069E-02	b7	-9.046859101163E-03	b8	-6.729390491700E-04



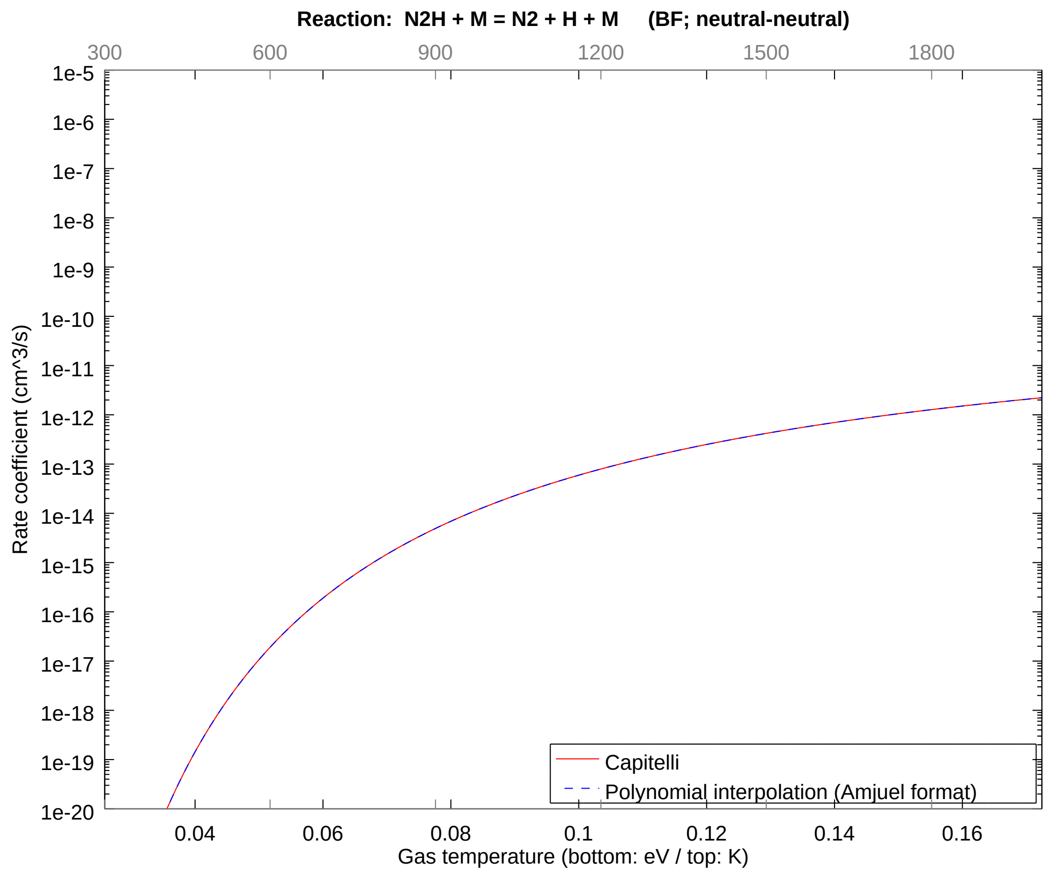
2.114 Reaction BE N2H2 + M = 2NH + M

b0 -2.185144297209E+01	b1 1.632922861522E+00	b2 -6.245854701319E+00
b3 -2.867888225723E+00	b4 -2.161144941771E+00	b5 -6.747040176978E-01
b6 -1.691695107906E-01	b7 -2.154014071243E-02	b8 -1.602235831147E-03



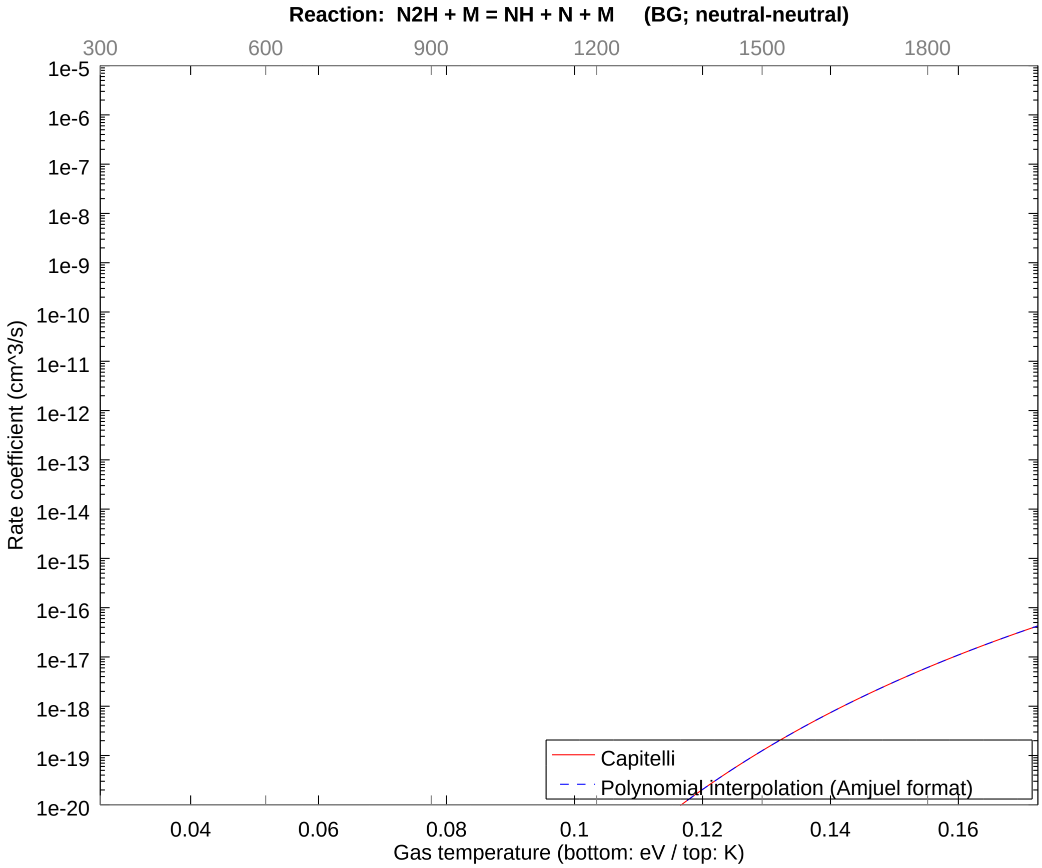
2.115 Reaction BF N2H + M = N2 + H + M

b0 -2.284781263330E+01	b1 3.265845688574E-01	b2 -1.249170944978E+00
b3 -5.735776487824E-01	b4 -4.322289900869E-01	b5 -1.349408040612E-01
b6 -3.383390225510E-02	b7 -4.308028152669E-03	b8 -3.204471666919E-04



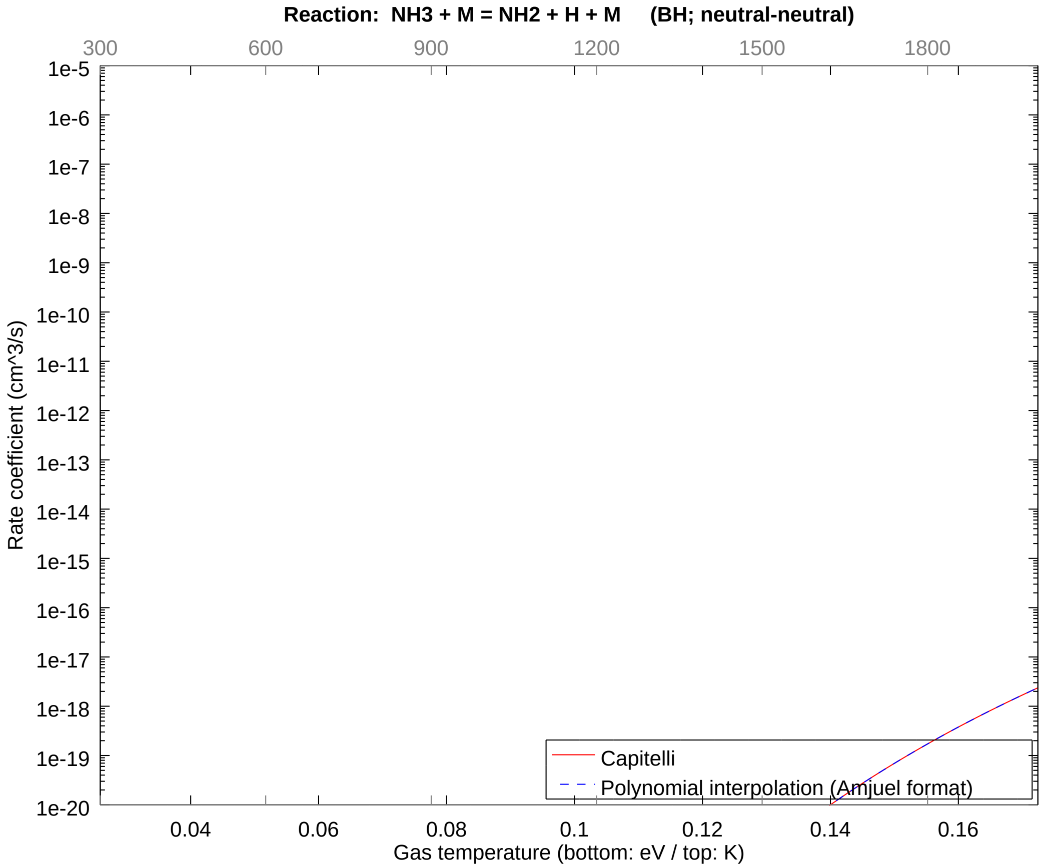
2.116 Reaction BG N2H + M = NH + N + M

b0	-2.374823218580E+01	b1	1.143045995553E+00	b2	-4.372098301507E+00
b3	-2.007521766427E+00	b4	-1.512801463379E+00	b5	-4.722928136757E-01
b6	-1.184186578008E-01	b7	-1.507809852559E-02	b8	-1.121565083068E-03



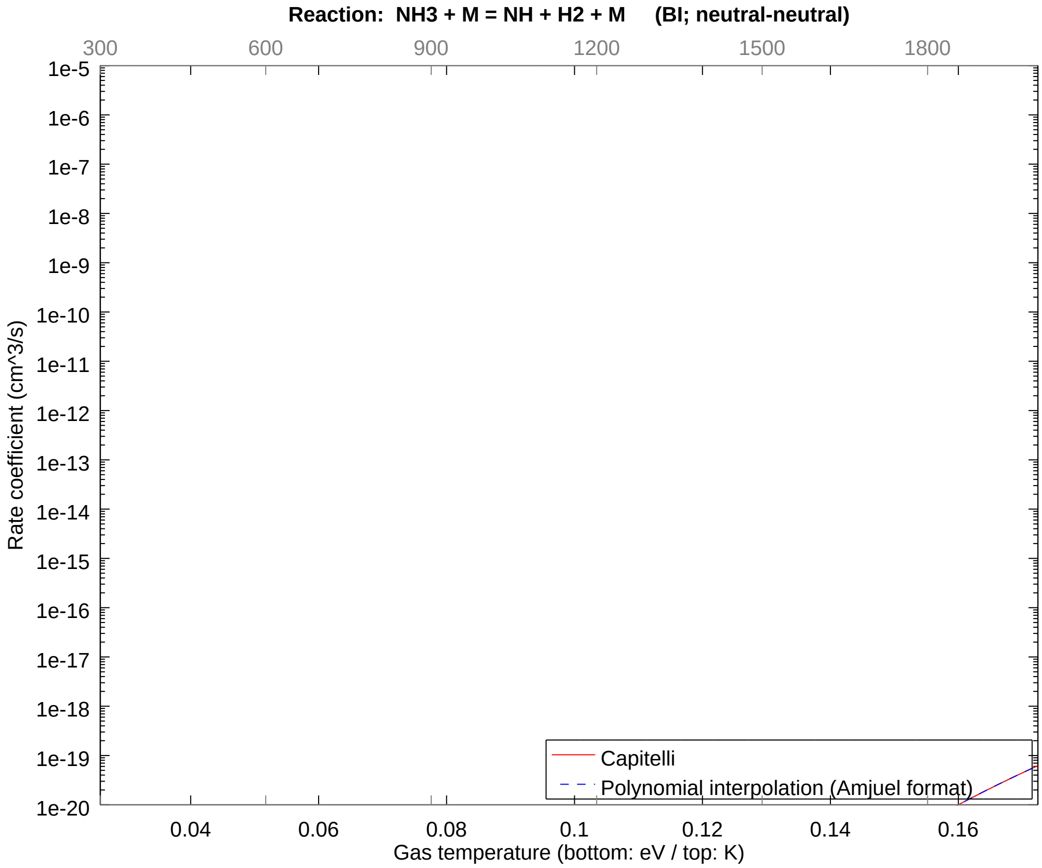
2.117 Reaction BH NH3 + M = NH2 + H + M

b0 -2.178056950826E+01	b1 1.541479169864E+00	b2 -5.896086853409E+00
b3 -2.707286496789E+00	b4 -2.040120830556E+00	b5 -6.369205943611E-01
b6 -1.596960184935E-01	b7 -2.033389286489E-02	b8 -1.512510626084E-03



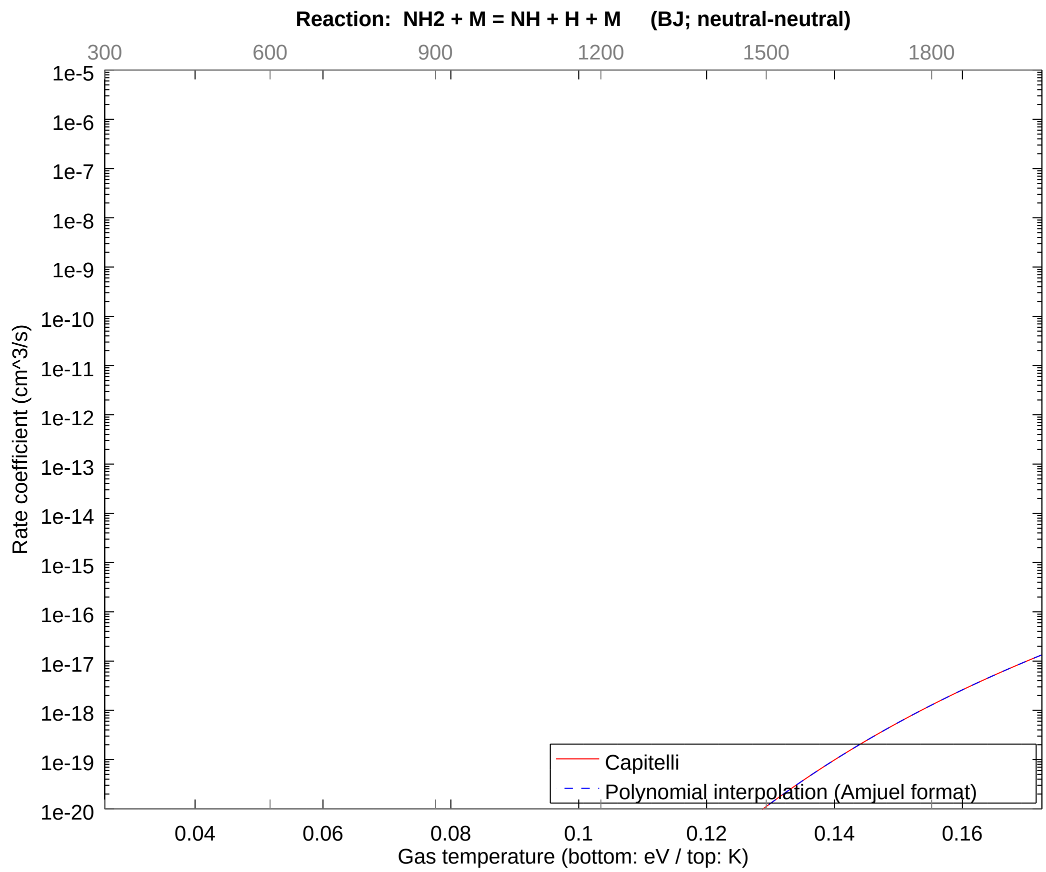
2.118 Reaction BI NH3 + M = NH + H2 + M

b0 -2.549792143959E+01	b1 1.534947489548E+00	b2 -5.871103419390E+00
b3 -2.695814932102E+00	b4 -2.031476245129E+00	b5 -6.342217765633E-01
b6 -1.590193401233E-01	b7 -2.024773226688E-02	b8 -1.506101681115E-03



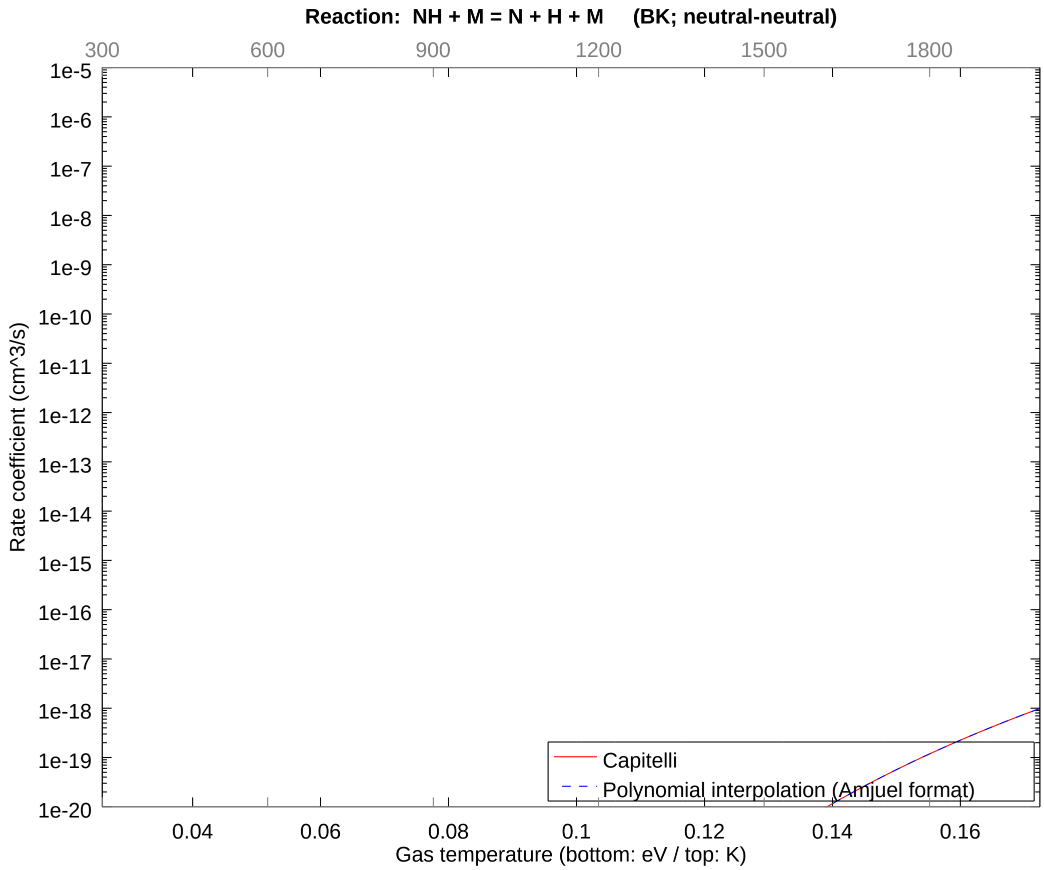
2.119 Reaction BJ NH2 + M = NH + H + M

b0 -2.404043242719E+01	b1 -4.977109602235E-01	b2 -5.746186315569E+00
b3 -2.638457160327E+00	b4 -1.988253342970E+00	b5 -6.207276952482E-01
b6 -1.556359497360E-01	b7 -1.981692943540E-02	b8 -1.474056963745E-03



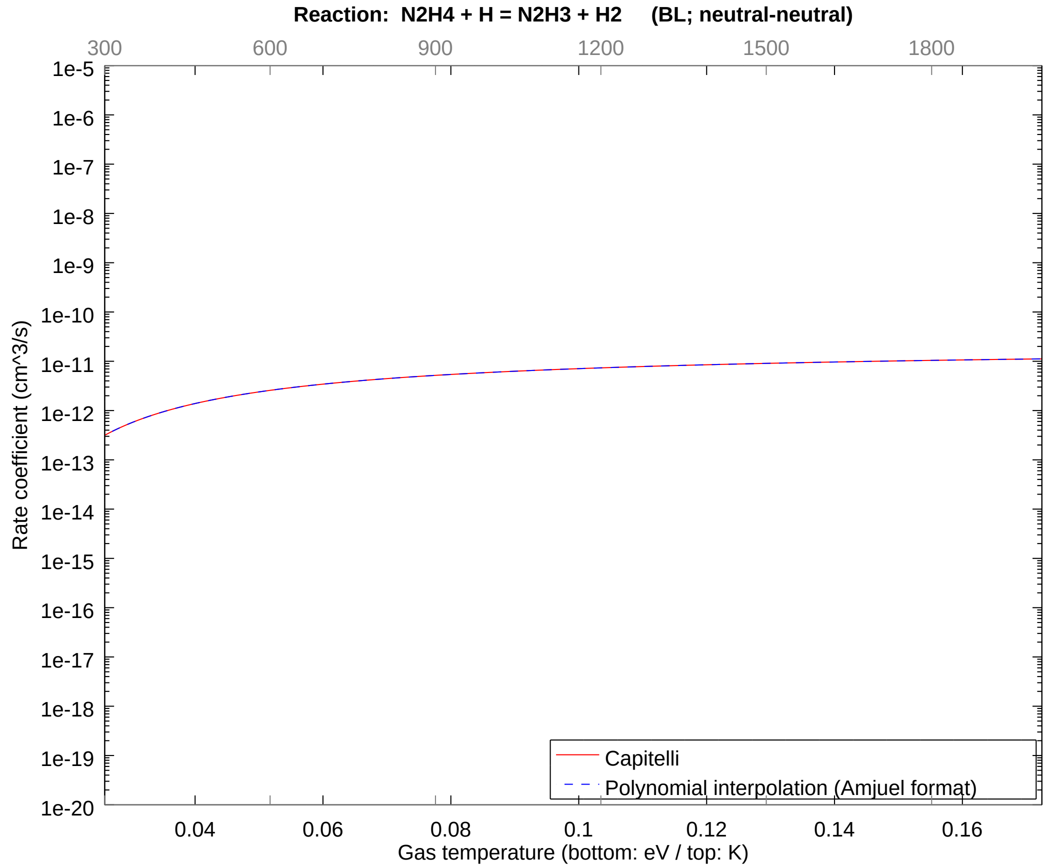
2.120 Reaction BK NH + M = N + H + M

b0 -2.823924894436E+01	b1 -6.283447873423E-01	b2 -5.246517936774E+00
b3 -2.409026100008E+00	b4 -1.815361746457E+00	b5 -5.667513734489E-01
b6 -1.421023887953E-01	b7 -1.809371816955E-02	b8 -1.345878096817E-03



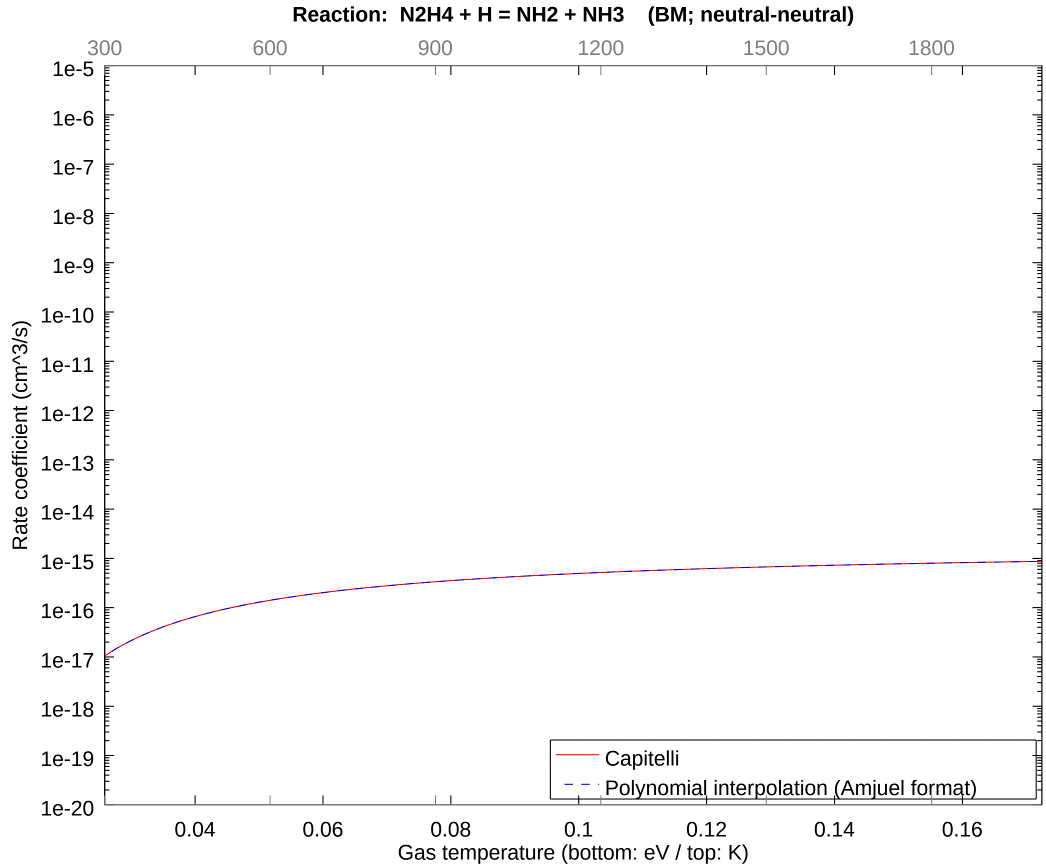
2.121 Reaction BL N2H4 + H = N2H3 + H2

```
b0 -2.471450008403E+01    b1  4.114965527105E-02    b2 -1.573955393912E-01
b3 -7.227078380576E-02    b4 -5.446085268013E-02    b5 -1.700254125772E-02
b6 -4.263071667420E-03    b7 -5.428115447081E-04    b8 -4.037634284977E-05
```



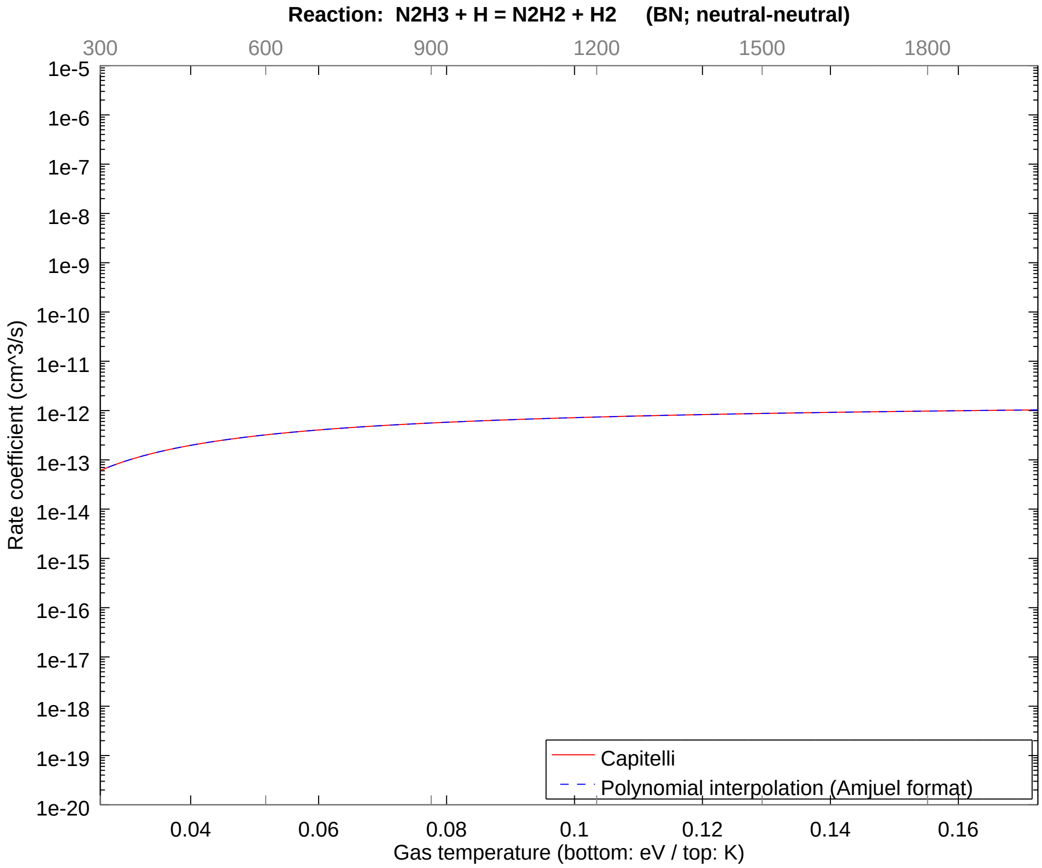
2.122 Reaction BM N2H4 + H = NH2 + NH3

b0	-3.405540044124E+01	b1	5.094718741288E-02	b2	-1.948706746692E-01
b3	-8.947811877476E-02	b4	-6.742772508512E-02	b5	-2.105076621879E-02
b6	-5.278088896147E-03	b7	-6.720524067615E-04	b8	-4.998975867108E-05



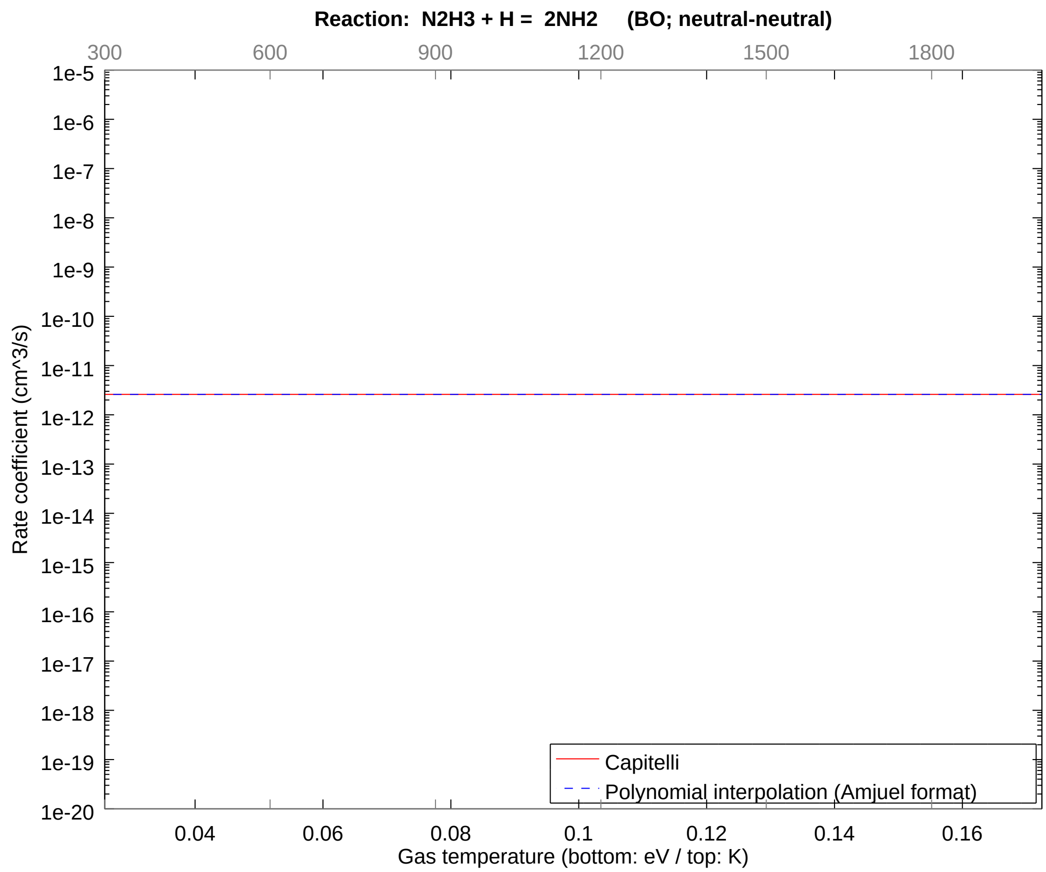
2.123 Reaction BN N2H3 + H = N2H2 + H2

b0	-2.720198128343E+01	b1	3.265845275819E-02	b2	-1.249170998387E-01
b3	-5.735776876858E-02	b4	-4.322290075117E-02	b5	-1.349408089682E-02
b6	-3.383390310207E-03	b7	-4.308028234413E-04	b8	-3.204471700596E-05



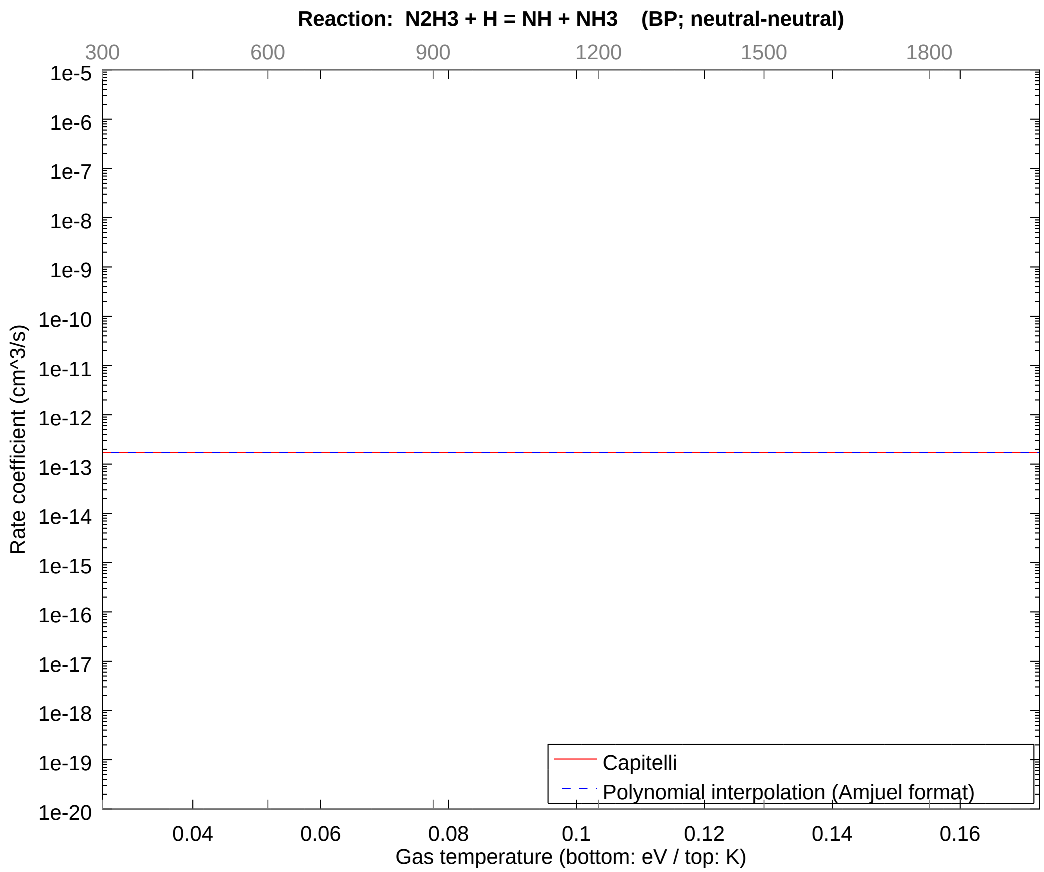
2.124 Reaction BO N2H3 + H = 2NH2

b0	-2.667550967090E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



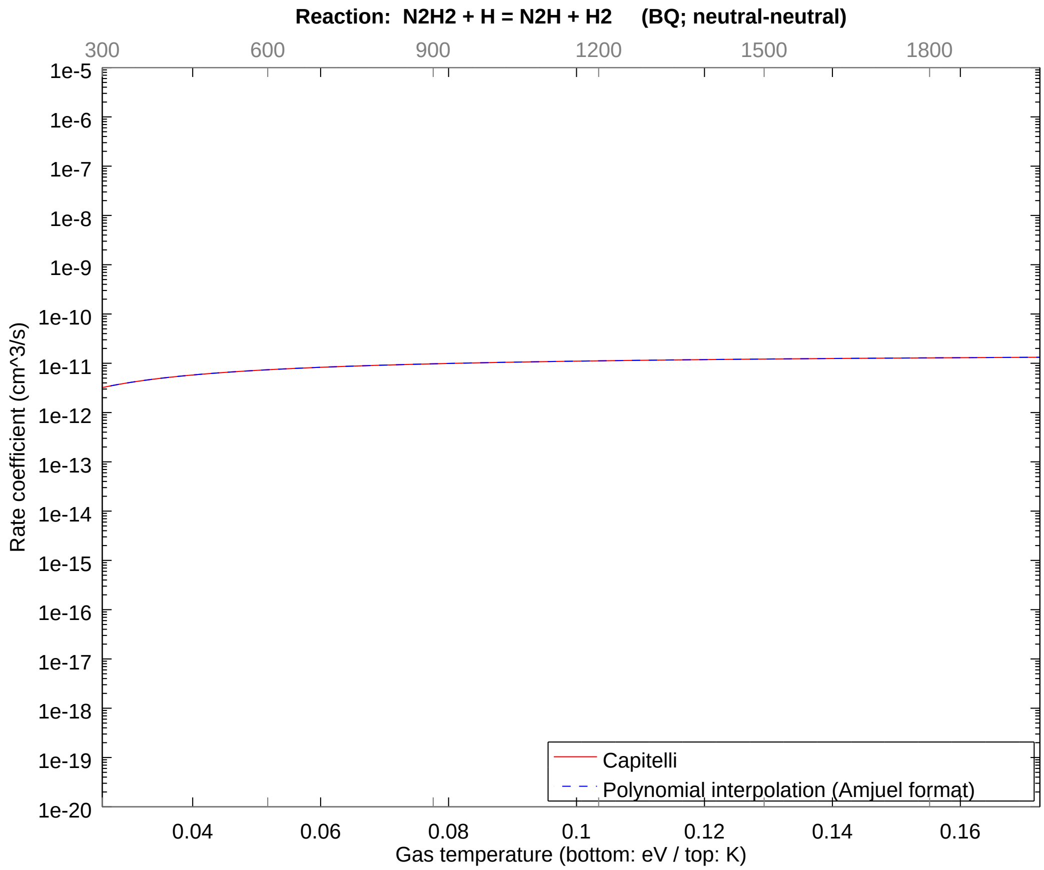
2.125 Reaction BP N2H3 + H = NH + NH3

b0	-2.940297795786E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



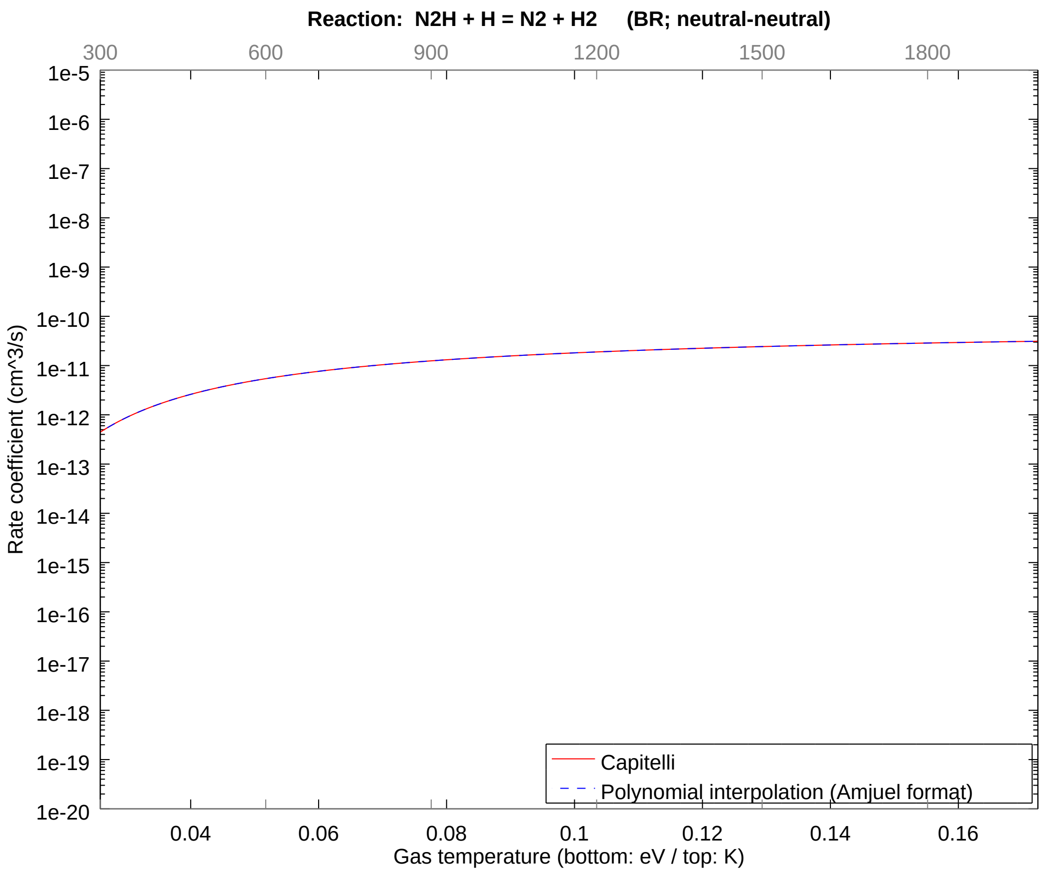
2.126 Reaction BQ N2H2 + H = N2H + H2

b0 -2.484860198207E+01	b1 1.632922359499E-02	b2 -6.245855355615E-02
b3 -2.867888706383E-02	b4 -2.161145159392E-02	b5 -6.747040798658E-03
b6 -1.691695217315E-03	b7 -2.154014179679E-04	b8 -1.602235877477E-05



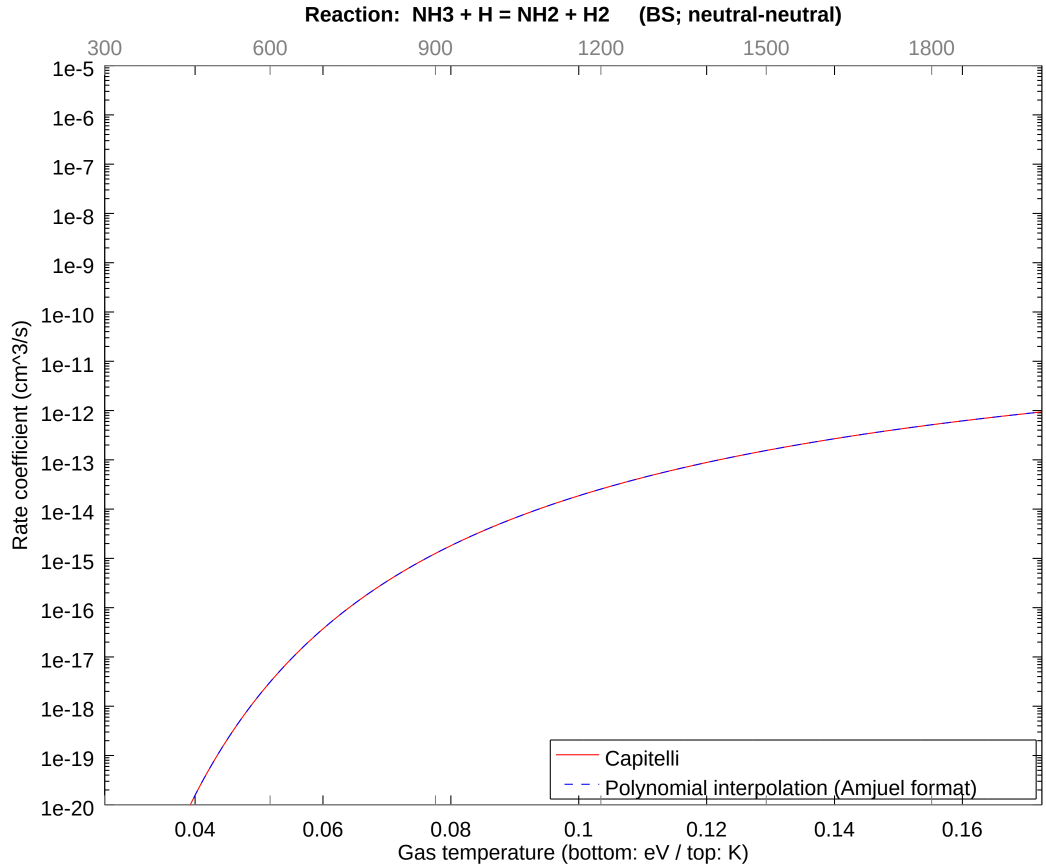
2.127 Reaction BR N2H + H = N2 + H2

b0 -2.359374900180E+01	b1 4.898767865117E-02	b2 -1.873756508272E-01
b3 -8.603665430488E-02	b4 -6.483435184066E-02	b5 -2.024112161349E-02
b6 -5.075085525879E-03	b7 -6.462042427391E-04	b8 -4.806707591320E-05



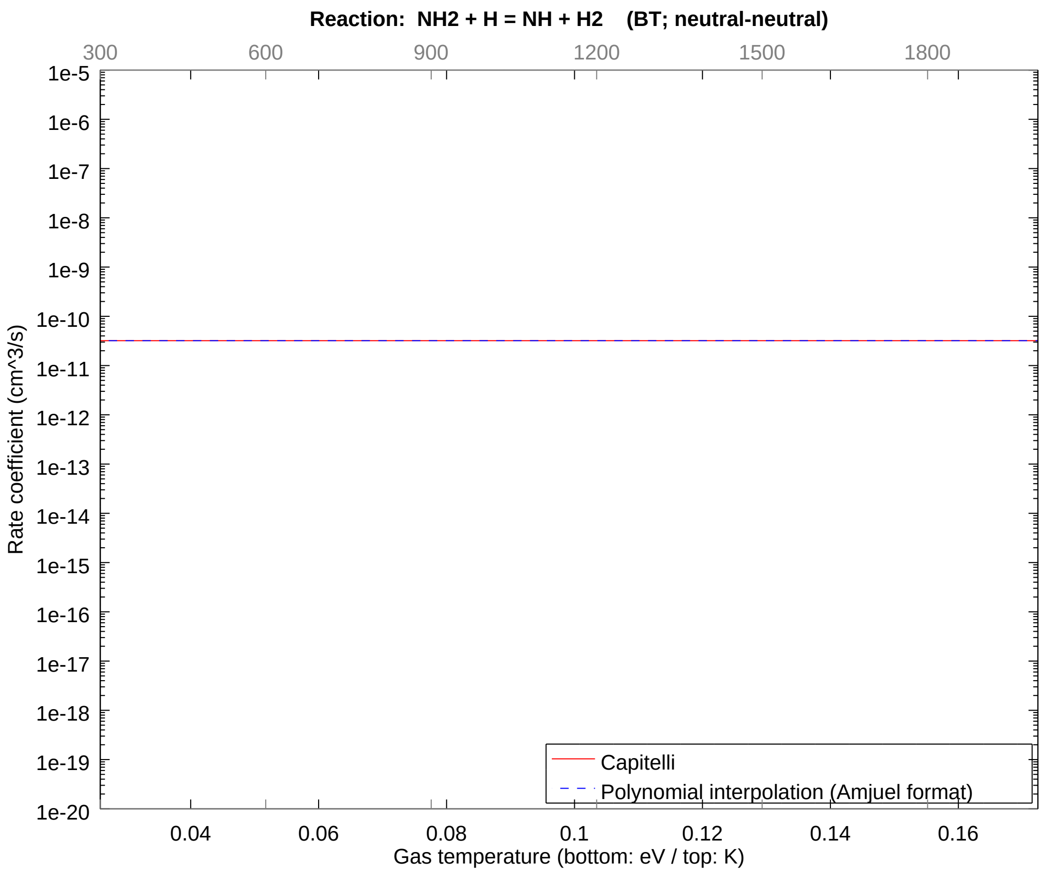
2.128 Reaction BS NH3 + H = NH2 + H2

```
b0 -2.338310025915E+01    b1  3.533645035060E-01    b2 -1.351602962390E+00
b3 -6.206110158630E-01    b4 -4.676717671848E-01    b5 -1.460059499565E-01
b6 -3.660828223069E-02    b7 -4.661286459931E-03    b8 -3.467238342890E-04
```



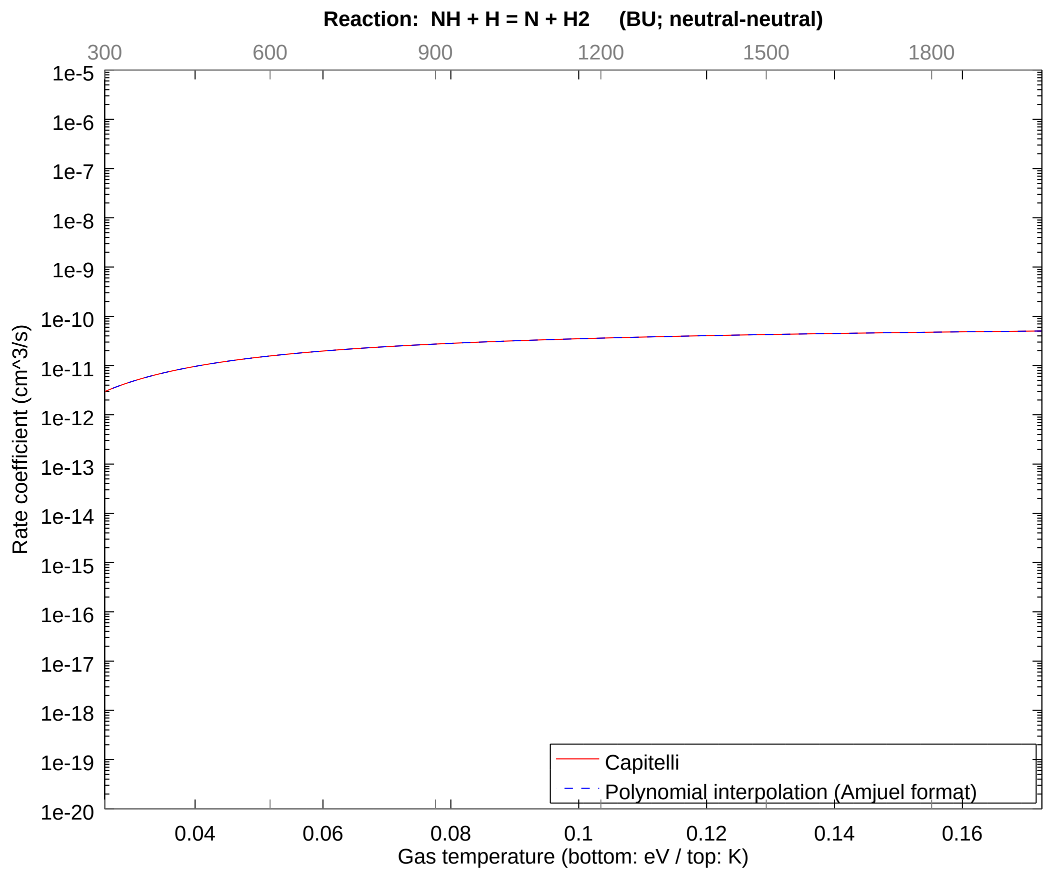
2.129 Reaction BT NH2 + H = NH + H2

b0	-2.416528521313E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



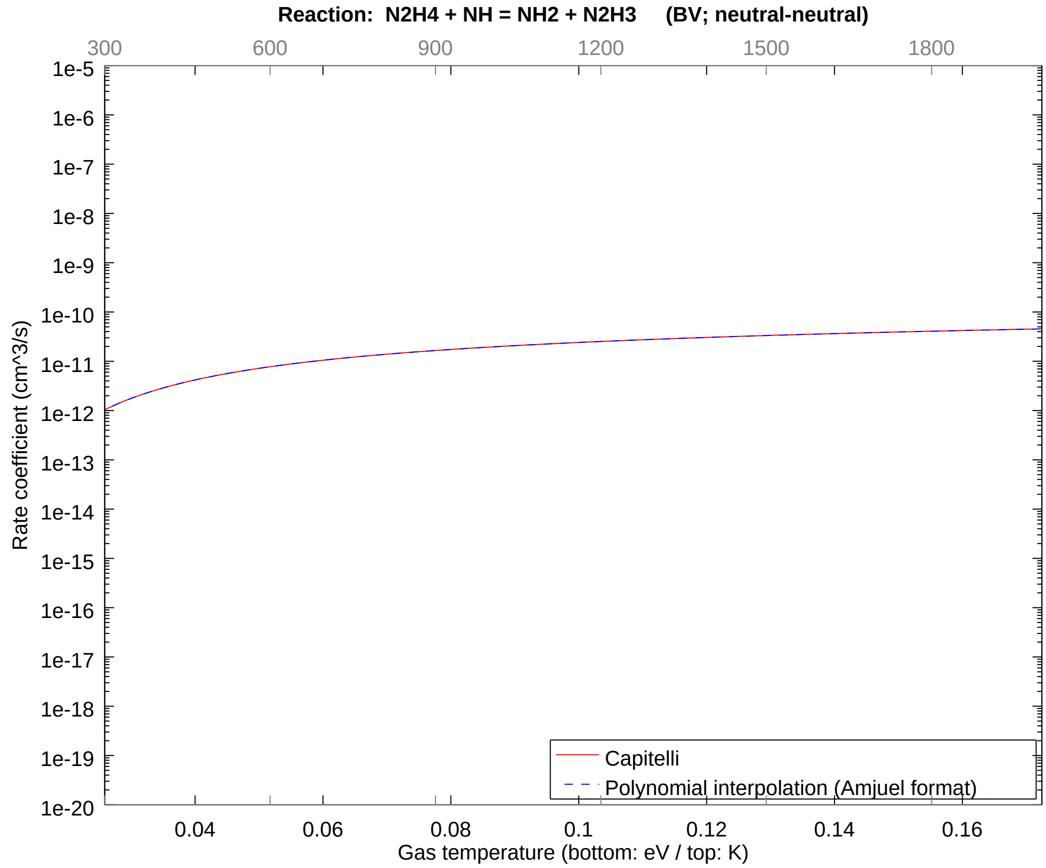
2.130 Reaction BU NH + H = N + H2

b0	-2.331376892600E+01	b1	3.265845470264E-02	b2	-1.249170975132E-01
b3	-5.735776723079E-02	b4	-4.322290014224E-02	b5	-1.349408075155E-02
b6	-3.383390290527E-03	b7	-4.308028221799E-04	b8	-3.204471698720E-05



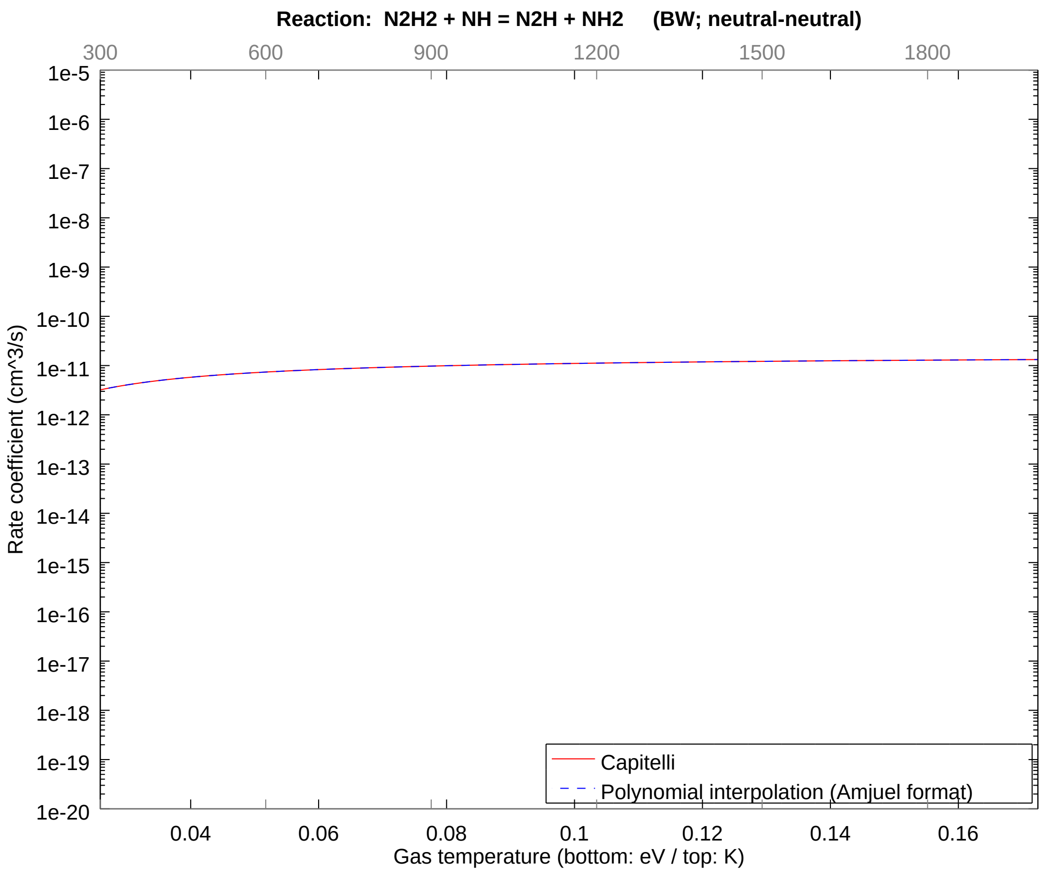
2.131 Reaction BV N2H4 + NH = NH2 + N2H3

b0	-2.253788555373E+01	b1	5.326584523650E-01	b2	-1.249171003025E-01
b3	-5.735776906945E-02	b4	-4.322290086803E-02	b5	-1.349408092450E-02
b6	-3.383390314085E-03	b7	-4.308028237328E-04	b8	-3.204471701487E-05



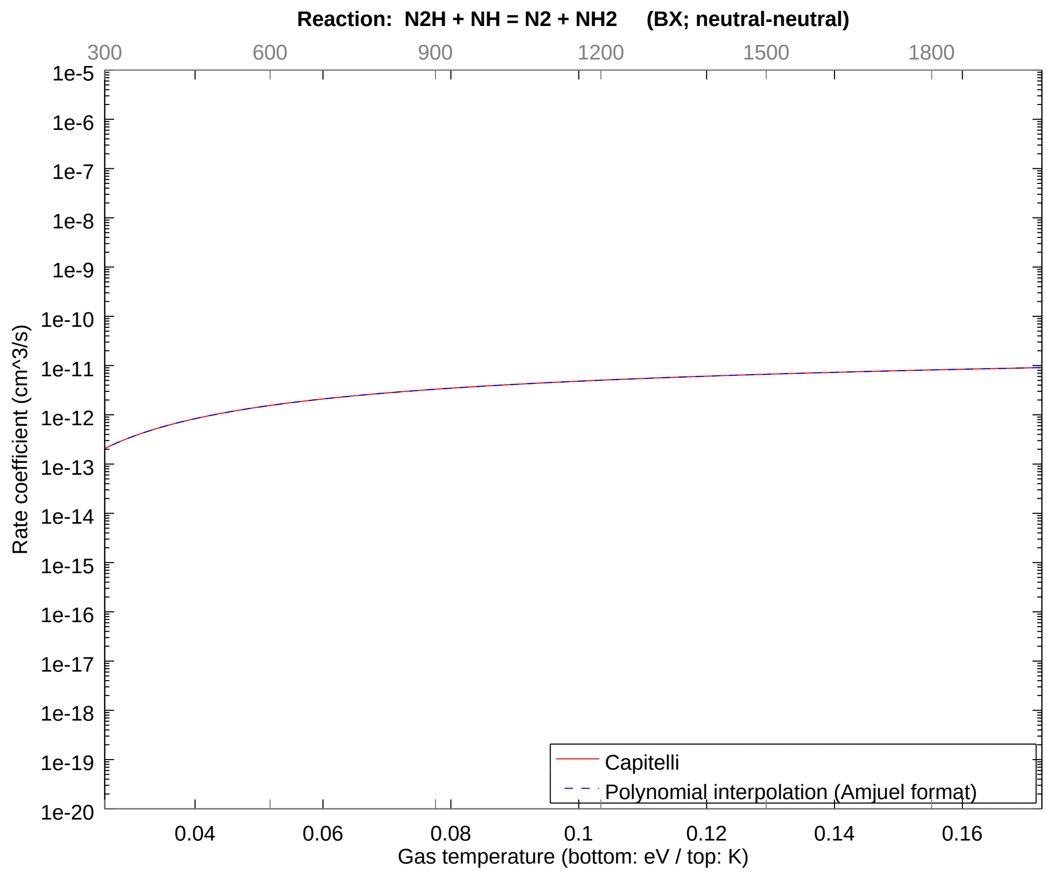
2.132 Reaction BW N2H2 + NH = N2H + NH2

b0 -2.484860198207E+01	b1 1.632922359499E-02	b2 -6.245855355615E-02
b3 -2.867888706383E-02	b4 -2.161145159392E-02	b5 -6.747040798658E-03
b6 -1.691695217315E-03	b7 -2.154014179679E-04	b8 -1.602235877477E-05



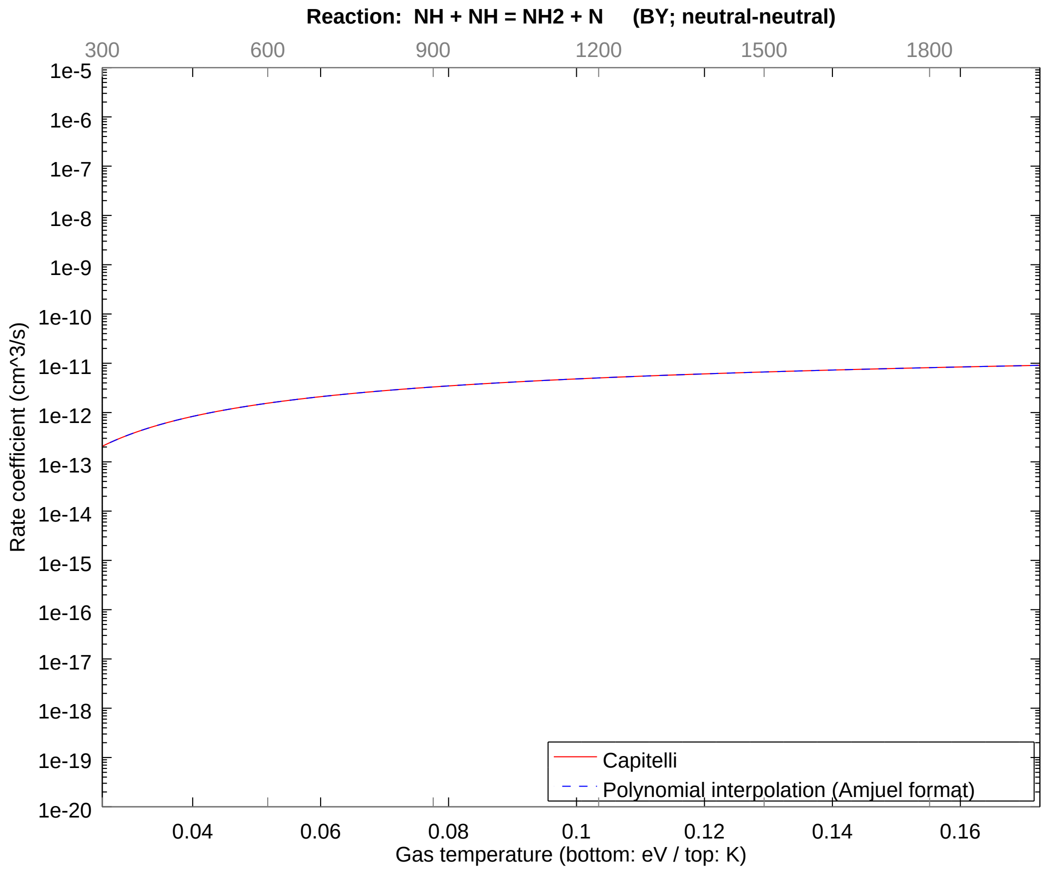
2.133 Reaction BX N2H + NH = N2 + NH2

b0 -2.414732346361E+01	b1 5.326584602754E-01	b2 -1.249170896738E-01
b3 -5.735776098318E-02	b4 -4.322289705700E-02	b5 -1.349407978481E-02
b6 -3.383390102820E-03	b7 -4.308028015319E-04	b8 -3.204471600200E-05



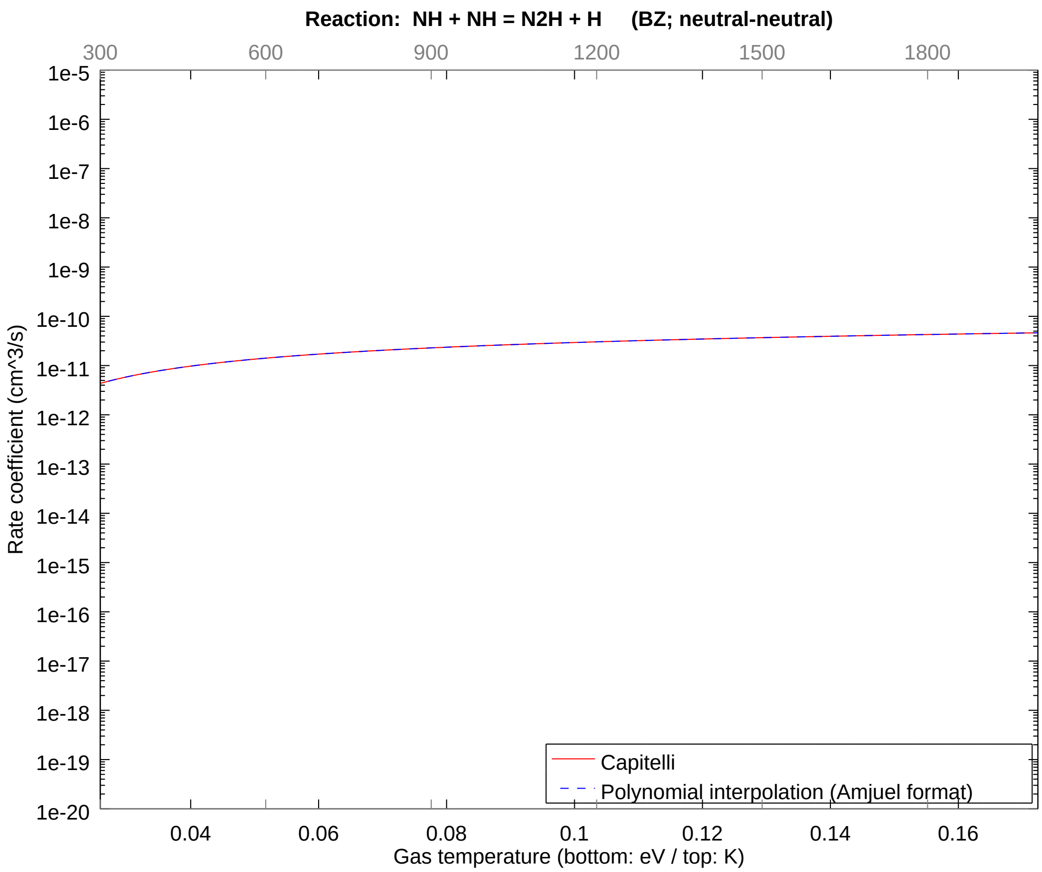
2.134 Reaction BY NH + NH = NH2 + N

b0 -2.414732346361E+01	b1 5.326584602754E-01	b2 -1.249170896738E-01
b3 -5.735776098318E-02	b4 -4.322289705700E-02	b5 -1.349407978481E-02
b6 -3.383390102820E-03	b7 -4.308028015319E-04	b8 -3.204471600200E-05



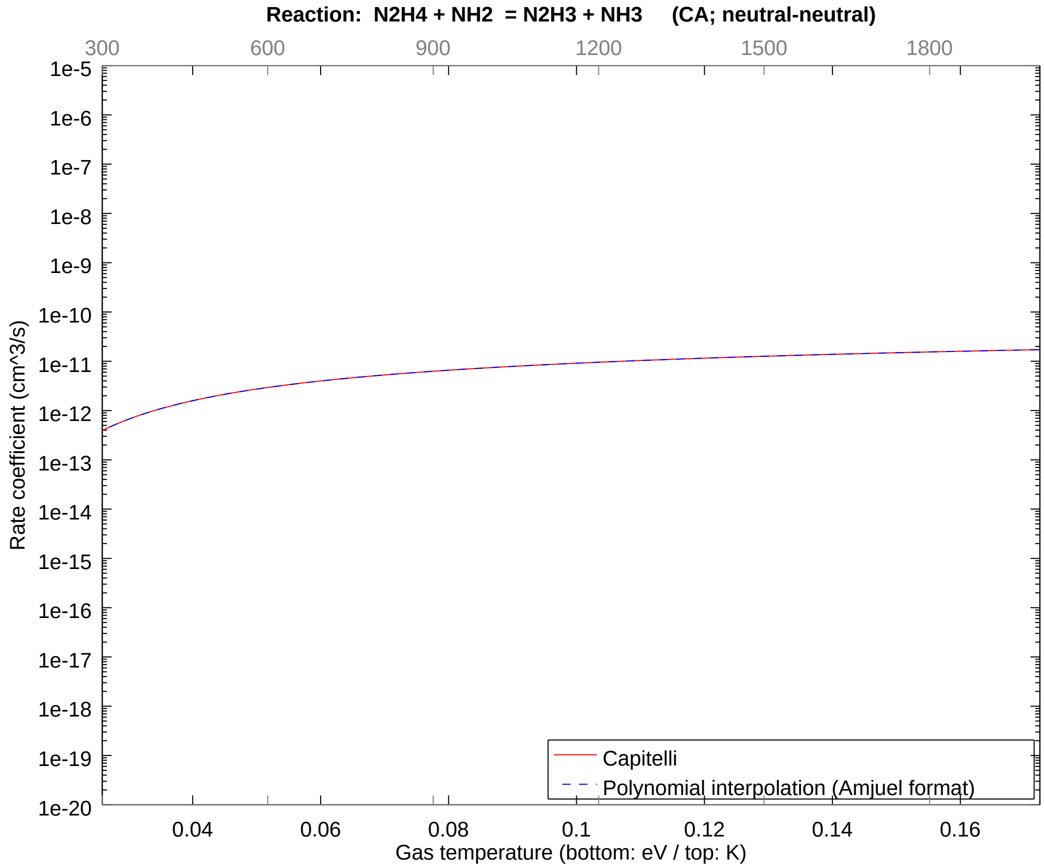
2.135 Reaction BZ NH + NH = N2H + H

b0	-2.271889295836E+01	b1	5.163292262952E-01	b2	-6.245855016915E-02
b3	-2.867888467247E-02	b4	-2.161145055453E-02	b5	-6.747040513874E-03
b6	-1.691695169278E-03	b7	-2.154014134061E-04	b8	-1.602235858798E-05



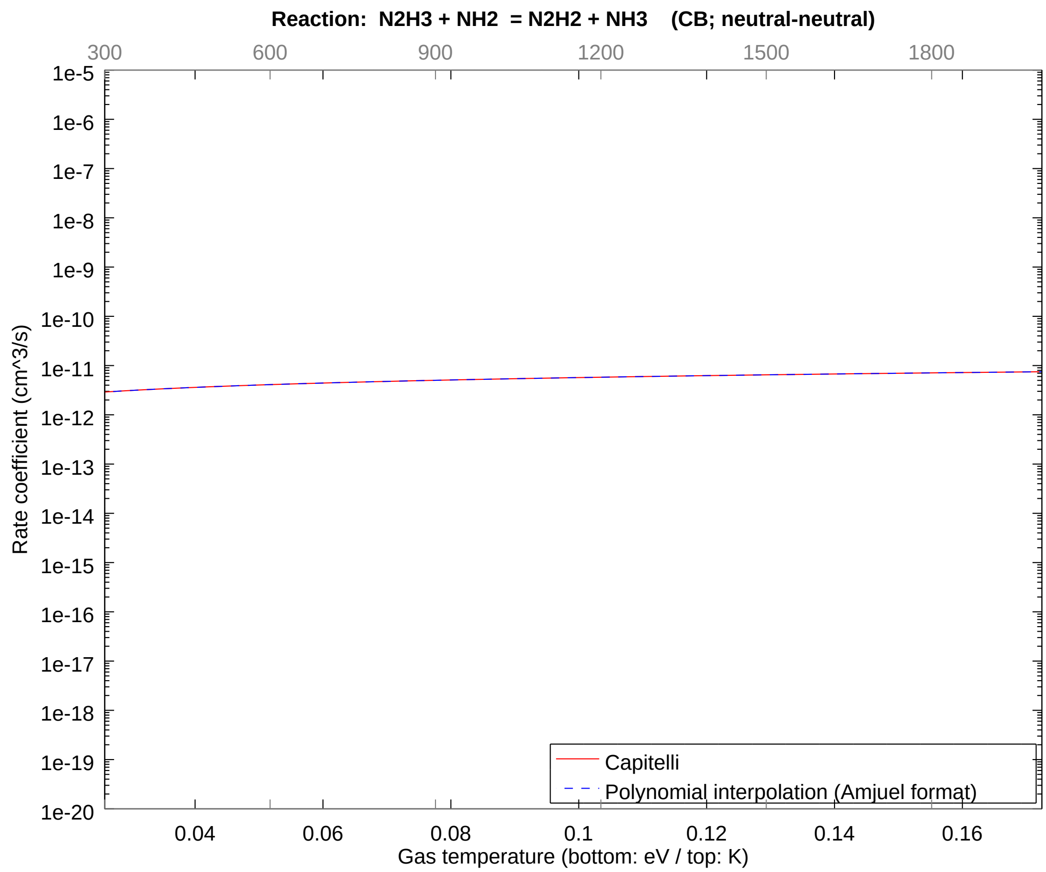
2.136 Reaction CA N2H4 + NH2 = N2H3 + NH3

b0 -2.350728610916E+01	b1 5.326584576680E-01	b2 -1.249170933627E-01
b3 -5.735776393498E-02	b4 -4.322289851799E-02	b5 -1.349408024277E-02
b6 -3.383390191598E-03	b7 -4.308028112622E-04	b8 -3.204471646367E-05



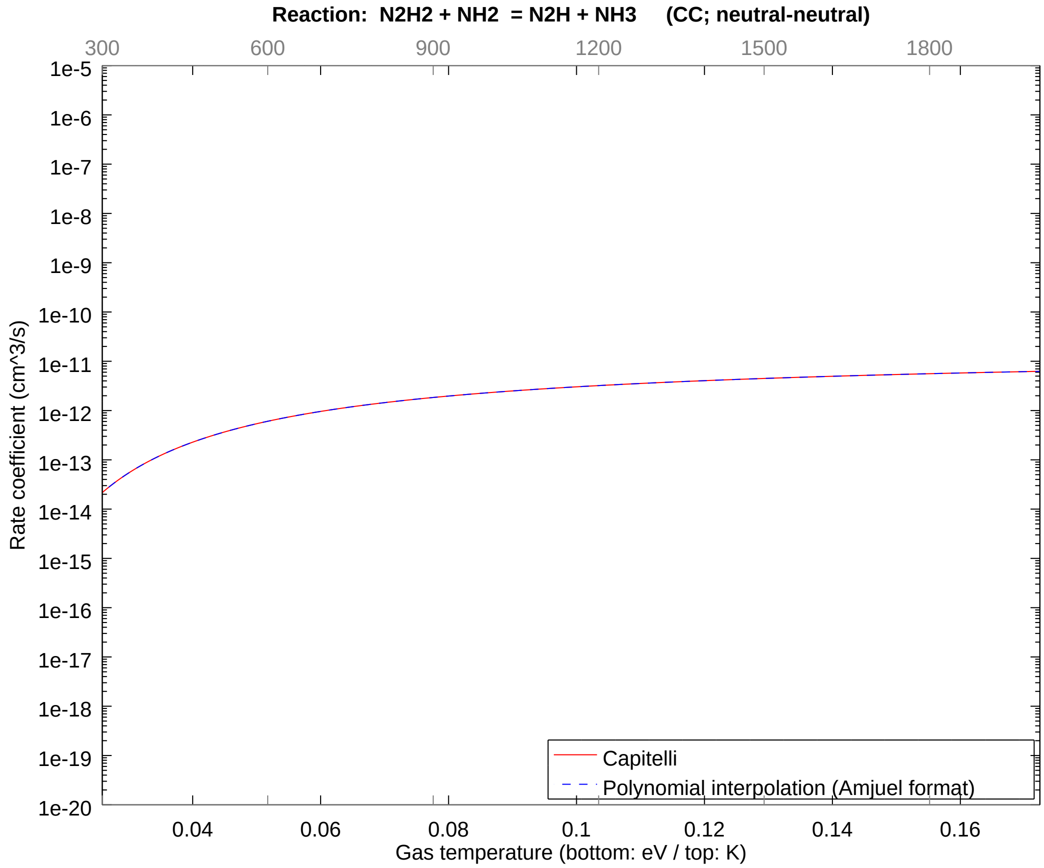
2.137 Reaction CB N2H3 + NH2 = N2H2 + NH3

b0 -2.473888223088E+01	b1 4.999999909009E-01	b2 -1.252288180630E-08
b3 -9.740196382601E-09	b4 -4.682862879700E-09	b5 -1.425228340047E-09
b6 -2.682081542161E-10	b7 -2.854101745411E-11	b8 -1.315306849704E-12



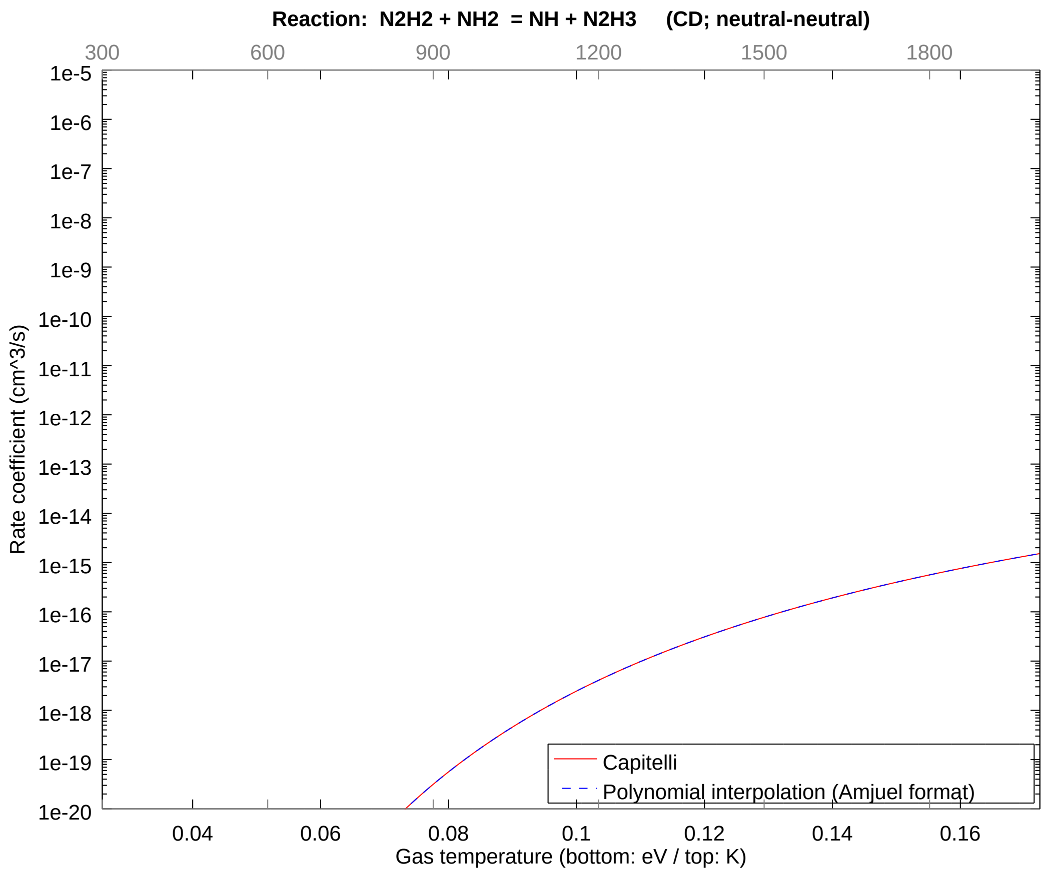
2.138 Reaction CC N2H2 + NH2 = N2H + NH3

```
b0 -2.500098460761E+01    b1  6.531690967044E-02    b2 -2.498341942865E-01
b3 -1.147155335862E-01    b4 -8.644579971549E-02    b5 -2.698816128360E-02
b6 -6.766780530776E-03    b7 -8.616056380250E-04    b8 -6.408943363558E-05
```



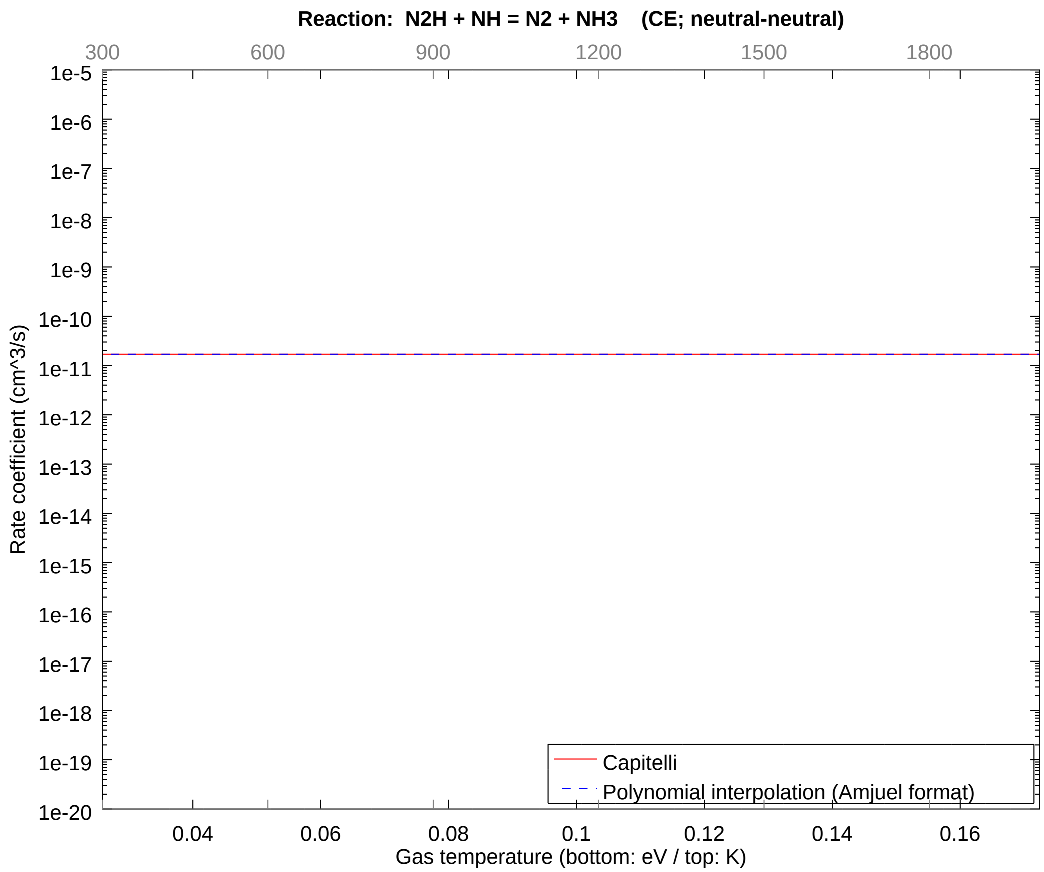
2.139 Reaction CD N2H2 + NH2 = NH + N2H3

b0 -2.646588531989E+01	b1 1.055193767911E+00	b2 -2.123590605188E+00
b3 -9.750820018544E-01	b4 -7.347892825853E-01	b5 -2.293993667175E-01
b6 -5.751763379524E-02	b7 -7.323647855047E-03	b8 -5.447601831483E-04



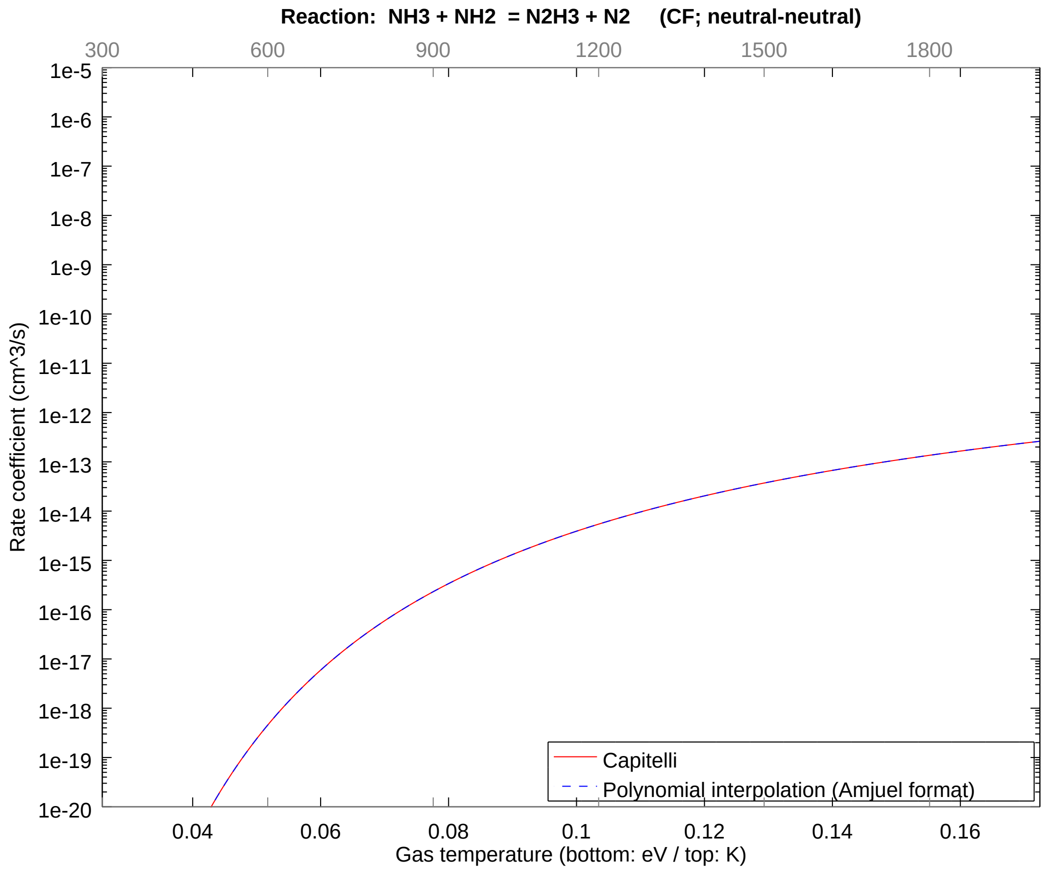
2.140 Reaction CE N2H + NH = N2 + NH3

b0	-2.479780777187E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



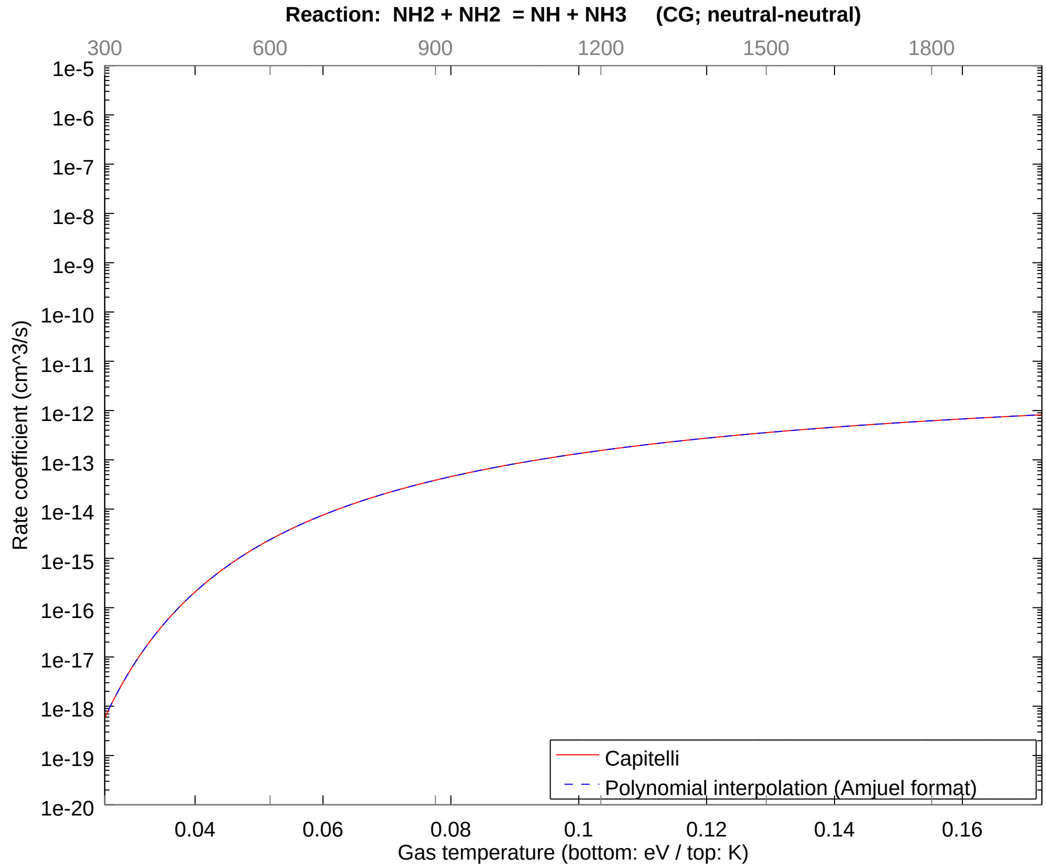
2.141 Reaction CF NH3 + NH2 = N2H3 + N2

b0	-2.377033307596E+01	b1	8.543442558806E-01	b2	-1.355350477056E+00
b3	-6.223317502405E-01	b4	-4.689684548518E-01	b5	-1.464107725851E-01
b6	-3.670978397938E-02	b7	-4.674210549023E-03	b8	-3.476851760129E-04



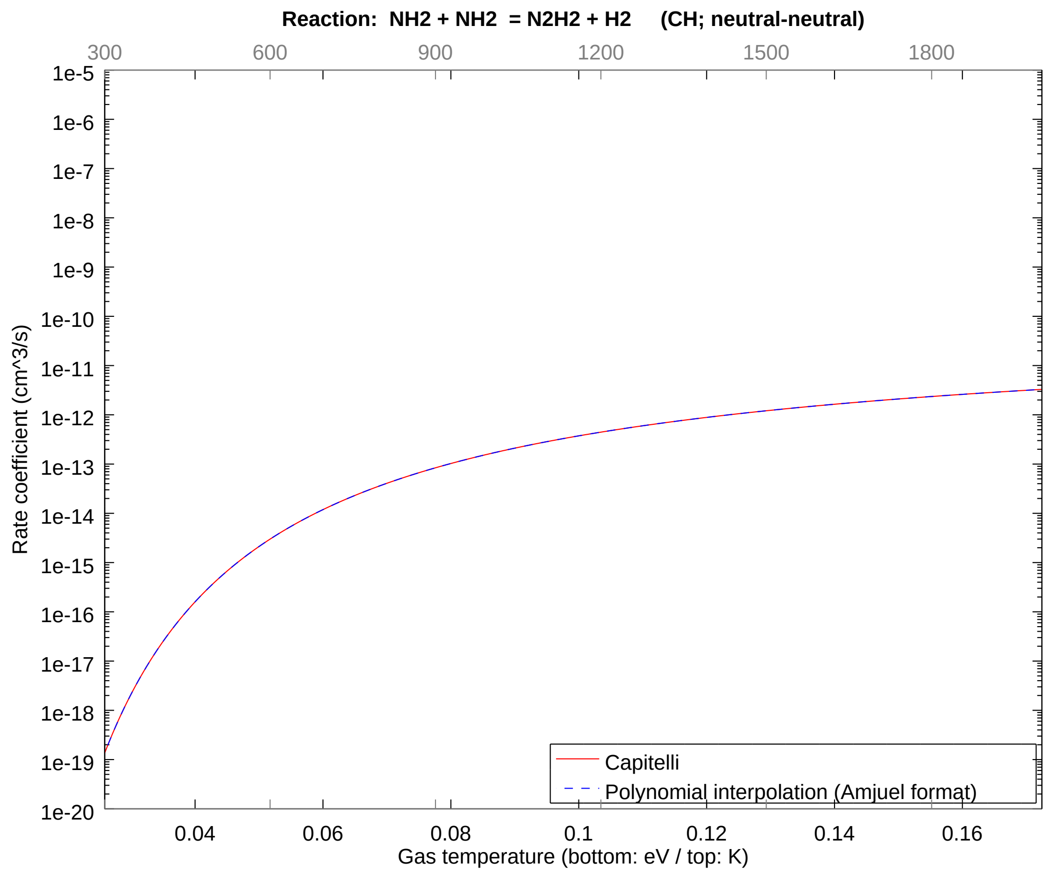
2.142 Reaction CG NH2 + NH2 = NH + NH3

b0 -2.583637811126E+01	b1 1.632922770044E-01	b2 -6.245854823952E-01
b3 -2.867888318610E-01	b4 -2.161144985246E-01	b5 -6.747040305731E-02
b6 -1.691695131471E-02	b7 -2.154014095617E-03	b8 -1.602235842058E-04



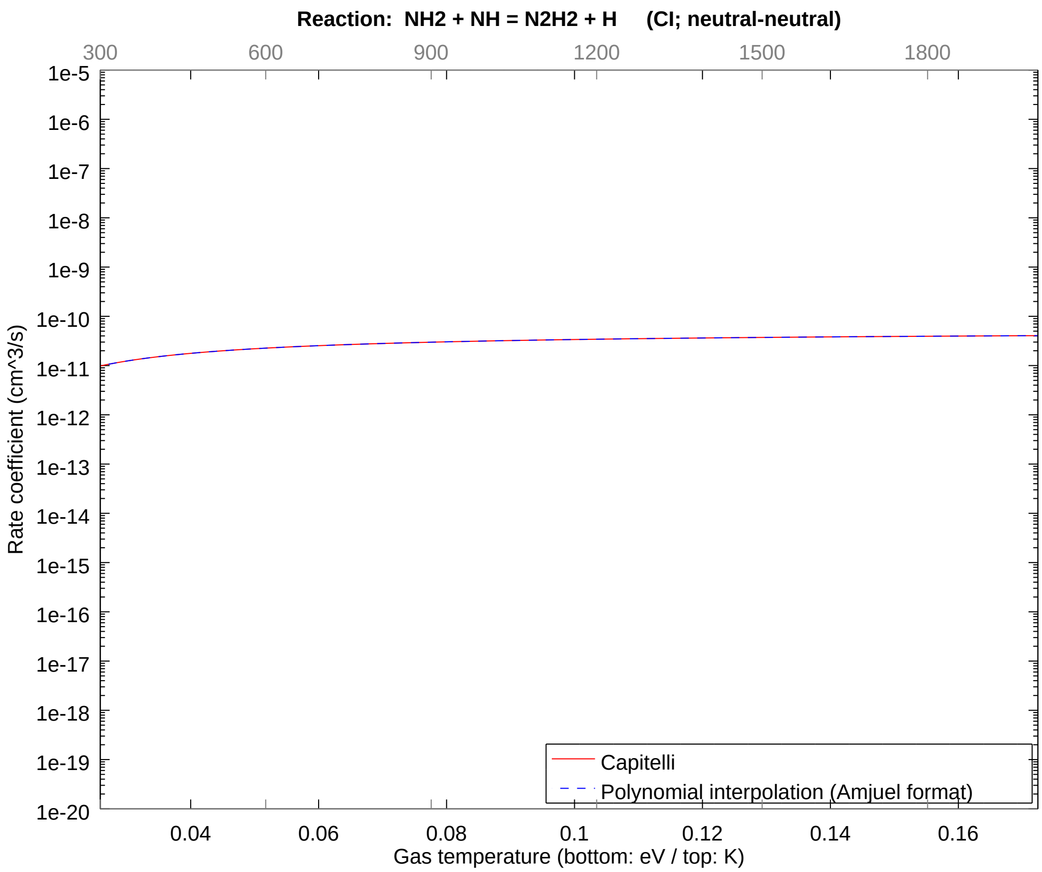
2.143 Reaction CH NH2 + NH2 = N2H2 + H2

b0 -2.405089687724E+01	b1 1.959507406339E-01	b2 -7.495025678272E-01
b3 -3.441465898399E-01	b4 -2.593373942811E-01	b5 -8.096448249087E-02
b6 -2.030034135994E-02	b7 -2.584816891937E-03	b8 -1.922683000102E-04



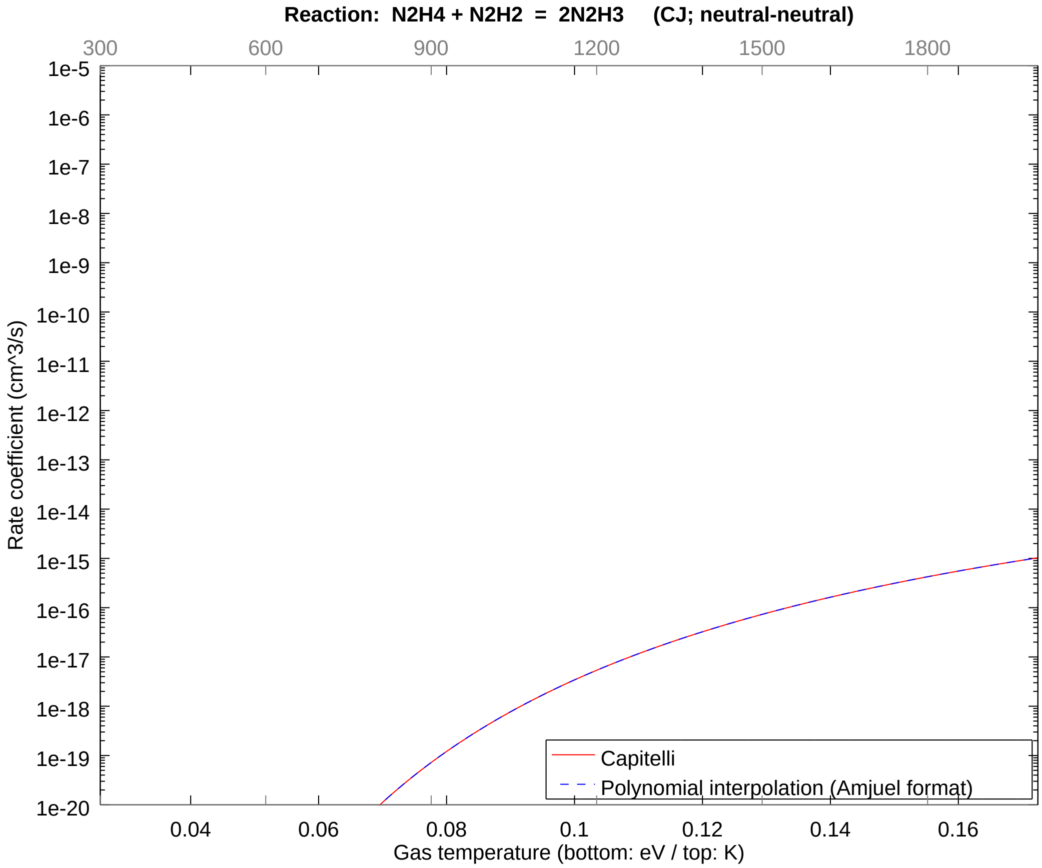
2.144 Reaction CI NH2 + NH = N2H2 + H

b0 -2.373057160757E+01	b1 1.632922319726E-02	b2 -6.245855454798E-02
b3 -2.867888818227E-02	b4 -2.161145229905E-02	b5 -6.747041064367E-03
b6 -1.691695276958E-03	b7 -2.154014253426E-04	b8 -1.602235916207E-05



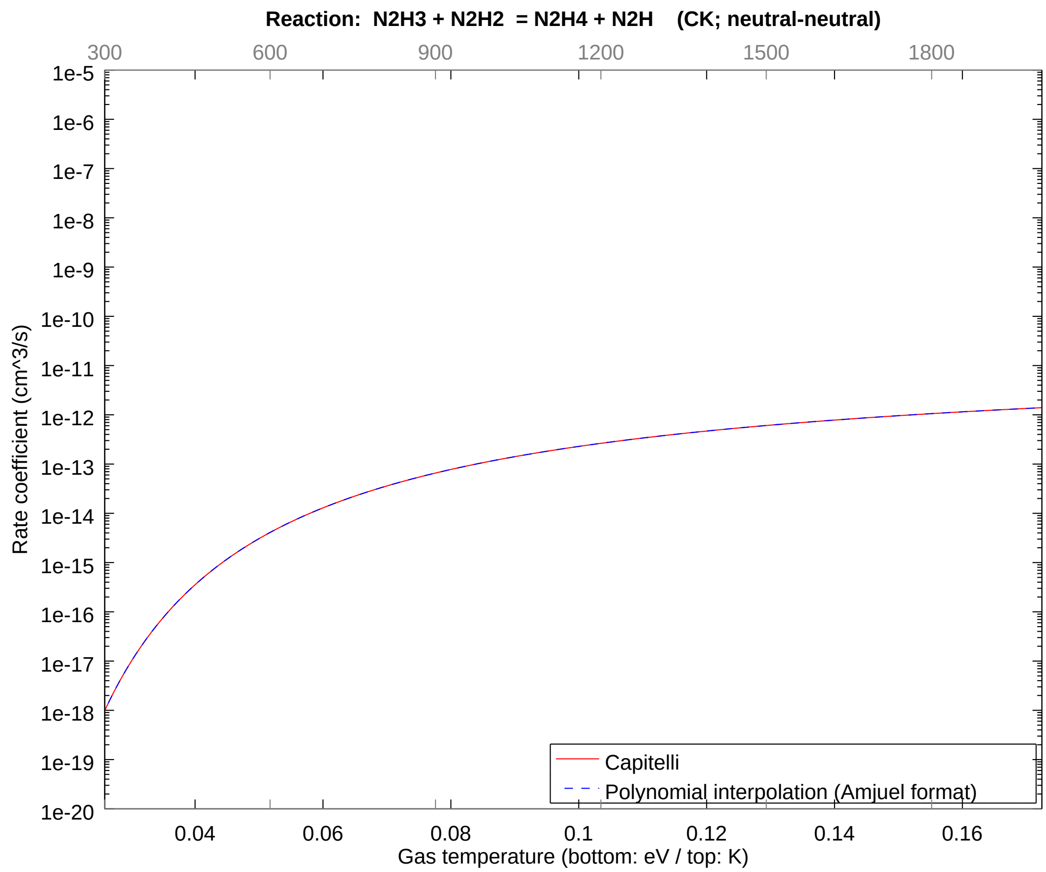
2.145 Reaction CJ N2H4 + N2H2 = 2N2H3

b0	-2.765592328912E+01	b1	9.898768553234E-01	b2	-1.873756414556E+00
b3	-8.603664708274E-01	b4	-6.483434839625E-01	b5	-2.024112057243E-01
b6	-5.075085331116E-02	b7	-6.462042221143E-03	b8	-4.806707496634E-04



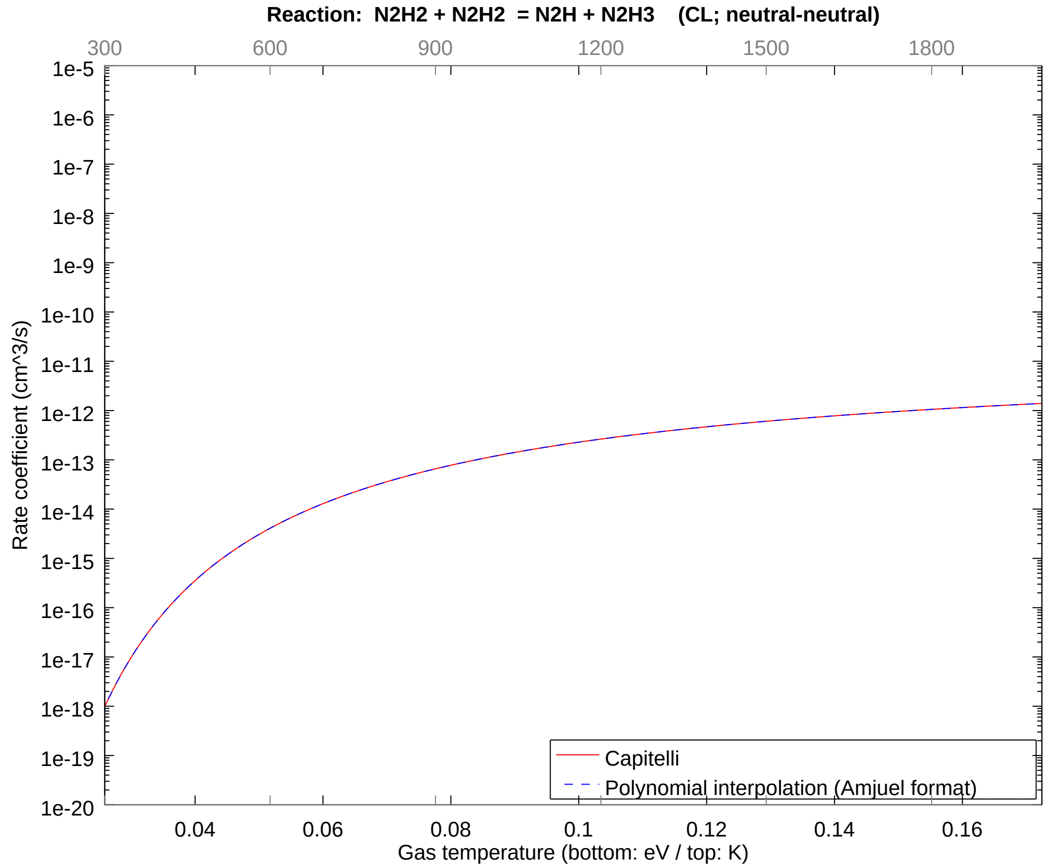
2.146 Reaction CK N2H3 + N2H2 = N2H4 + N2H

```
b0 -2.530574985861E+01    b1  1.632922819981E-01    b2 -6.245854756144E-01
b3 -2.867888266545E-01    b4 -2.161144960521E-01    b5 -6.747040231379E-02
b6 -1.691695117642E-02    b7 -2.154014081072E-03    b8 -1.602235835433E-04
```



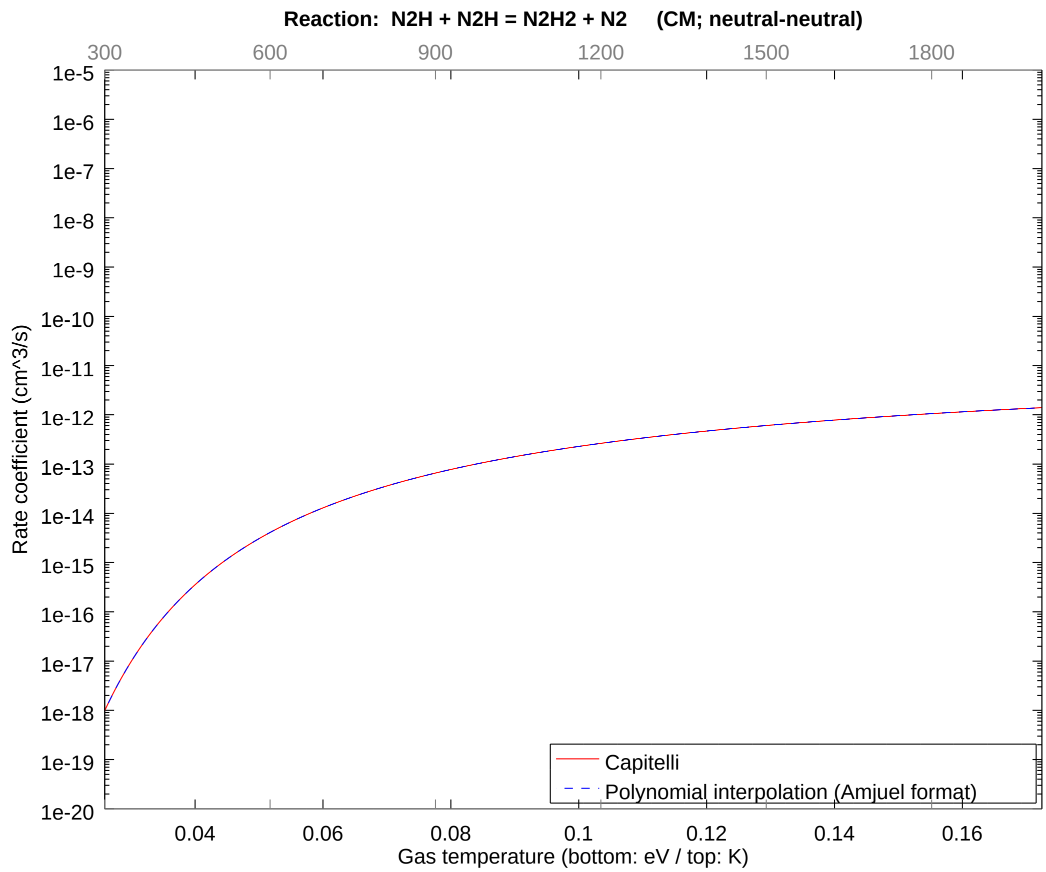
2.147 Reaction CL N2H2 + N2H2 = N2H + N2H3

b0 -2.530574985861E+01	b1 1.632922819981E-01	b2 -6.245854756144E-01
b3 -2.867888266545E-01	b4 -2.161144960521E-01	b5 -6.747040231379E-02
b6 -1.691695117642E-02	b7 -2.154014081072E-03	b8 -1.602235835433E-04



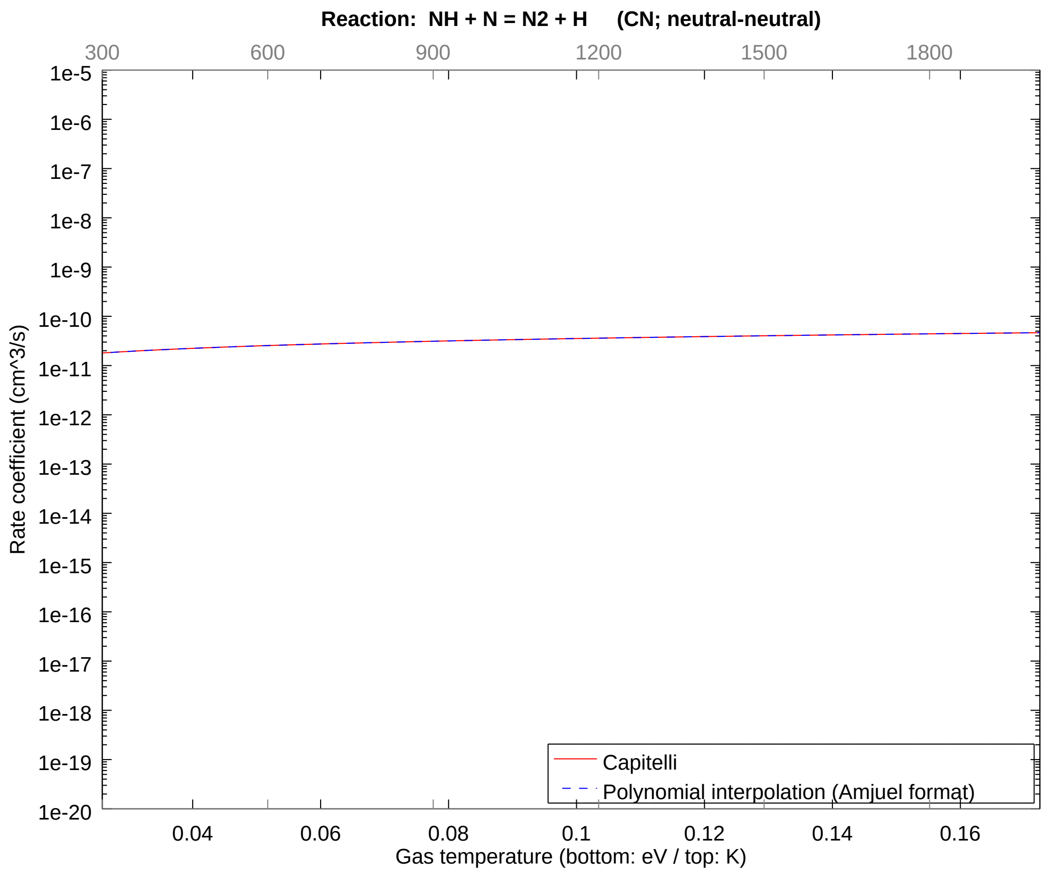
2.148 Reaction CM N2H + N2H = N2H2 + N2

b0 -2.530574985861E+01	b1 1.632922819981E-01	b2 -6.245854756144E-01
b3 -2.867888266545E-01	b4 -2.161144960521E-01	b5 -6.747040231379E-02
b6 -1.691695117642E-02	b7 -2.154014081072E-03	b8 -1.602235835433E-04



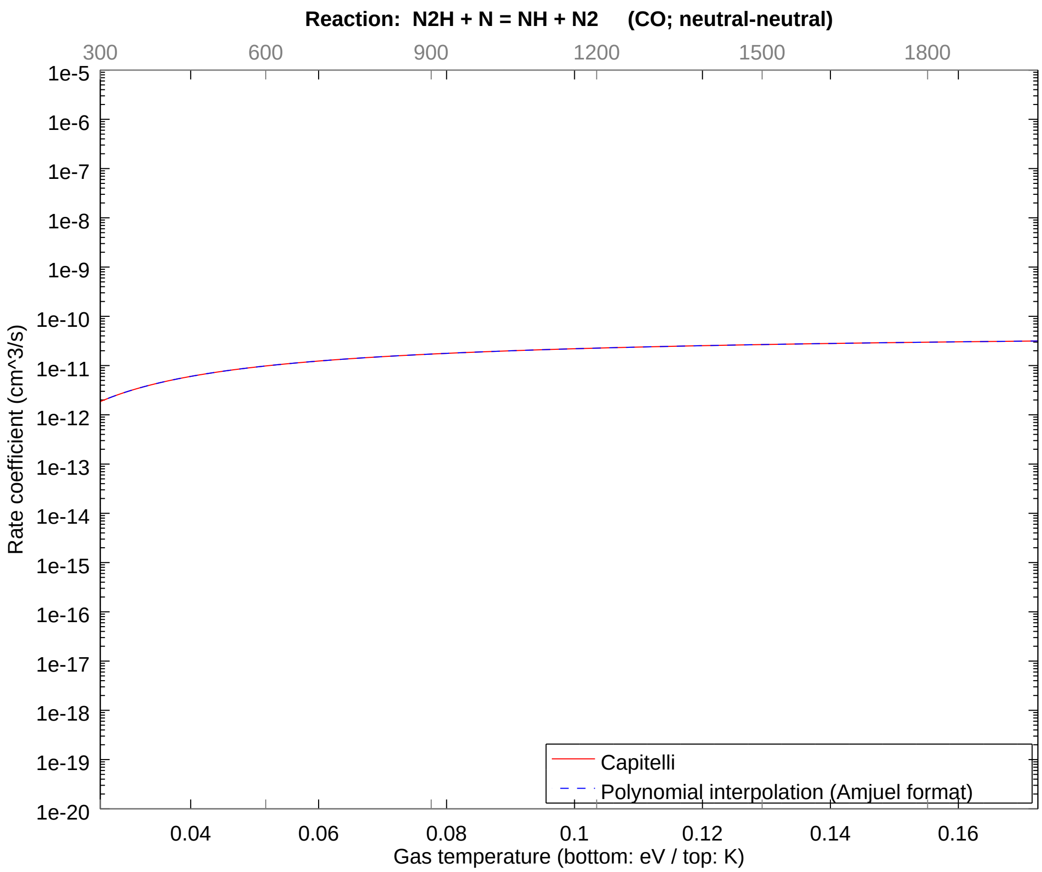
2.149 Reaction CN NH + N = N2 + H

b0 -2.291322120833E+01	b1 4.999999962924E-01	b2 -4.898825118178E-09
b3 -3.641896735617E-09	b4 -1.665571075976E-09	b5 -4.796655097632E-10
b6 -8.491375965166E-11	b7 -8.444185799009E-12	b8 -3.609433140001E-13



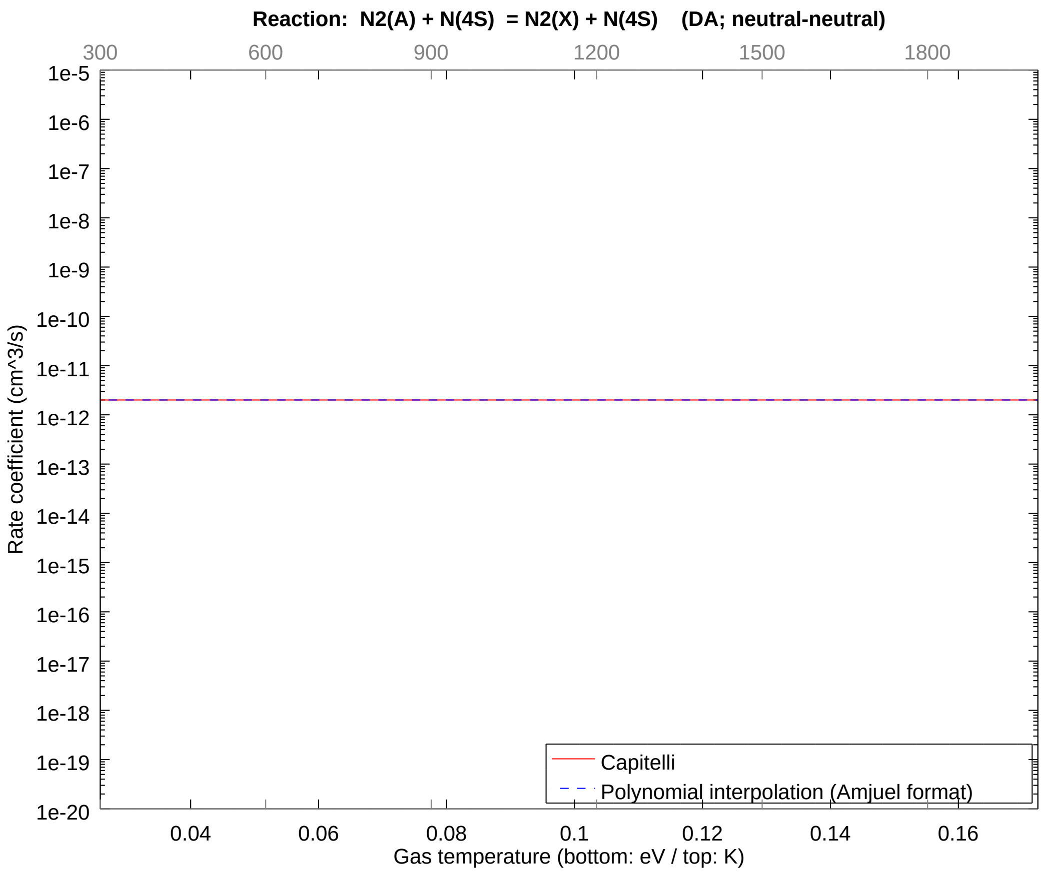
2.150 Reaction CO N2H + N = NH + N2

b0 -2.378136581554E+01	b1 3.265845386299E-02	b2 -1.249170984012E-01
b3 -5.735776771641E-02	b4 -4.322290027883E-02	b5 -1.349408076421E-02
b6 -3.383390287591E-03	b7 -4.308028213161E-04	b8 -3.204471692274E-05



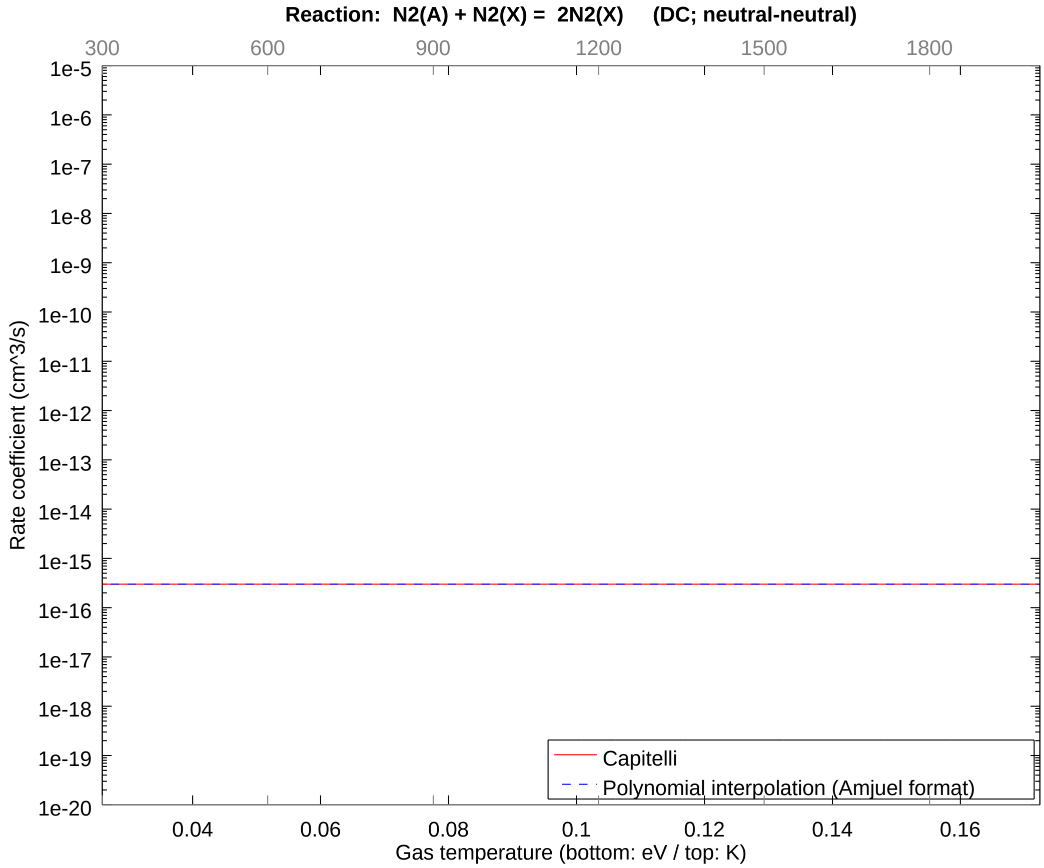
2.151 Reaction DA N2(A) + N(4S) = N2(X) + N(4S)

b0	-2.693787393537E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



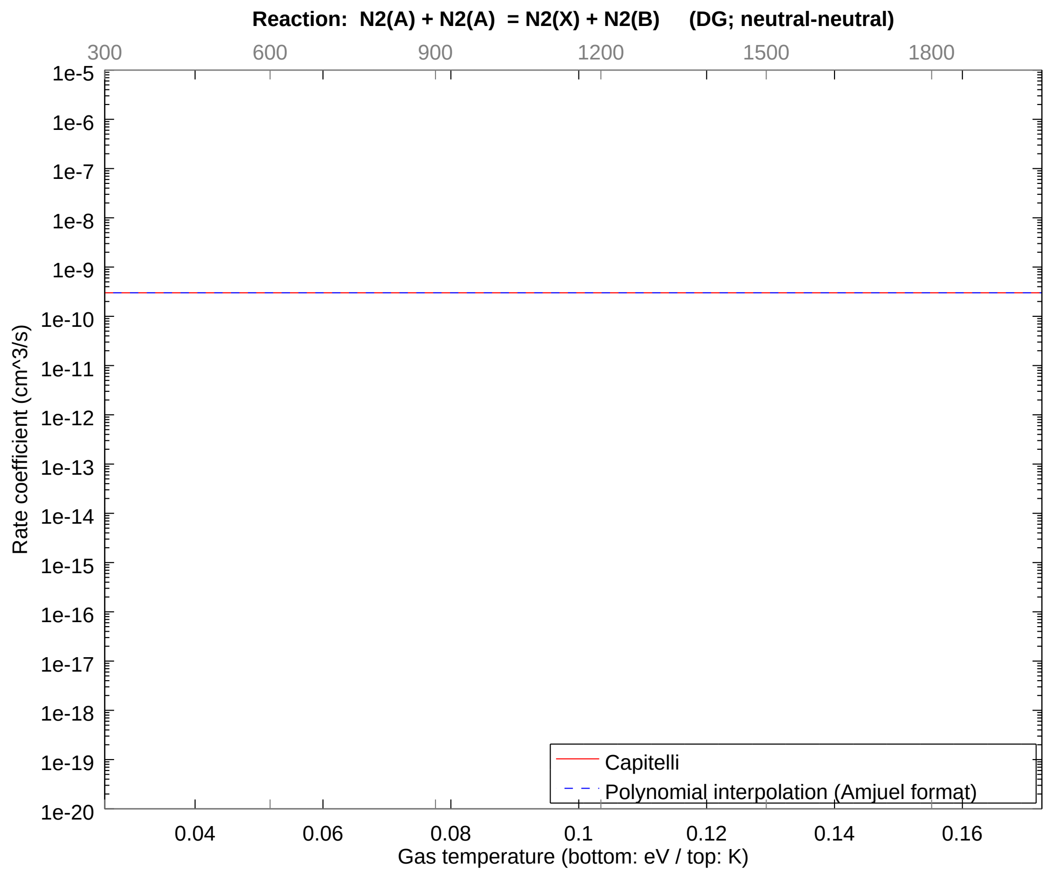
2.152 Reaction DC N2(A) + N2(X) = 2N2(X)

b0	-3.574274919924E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



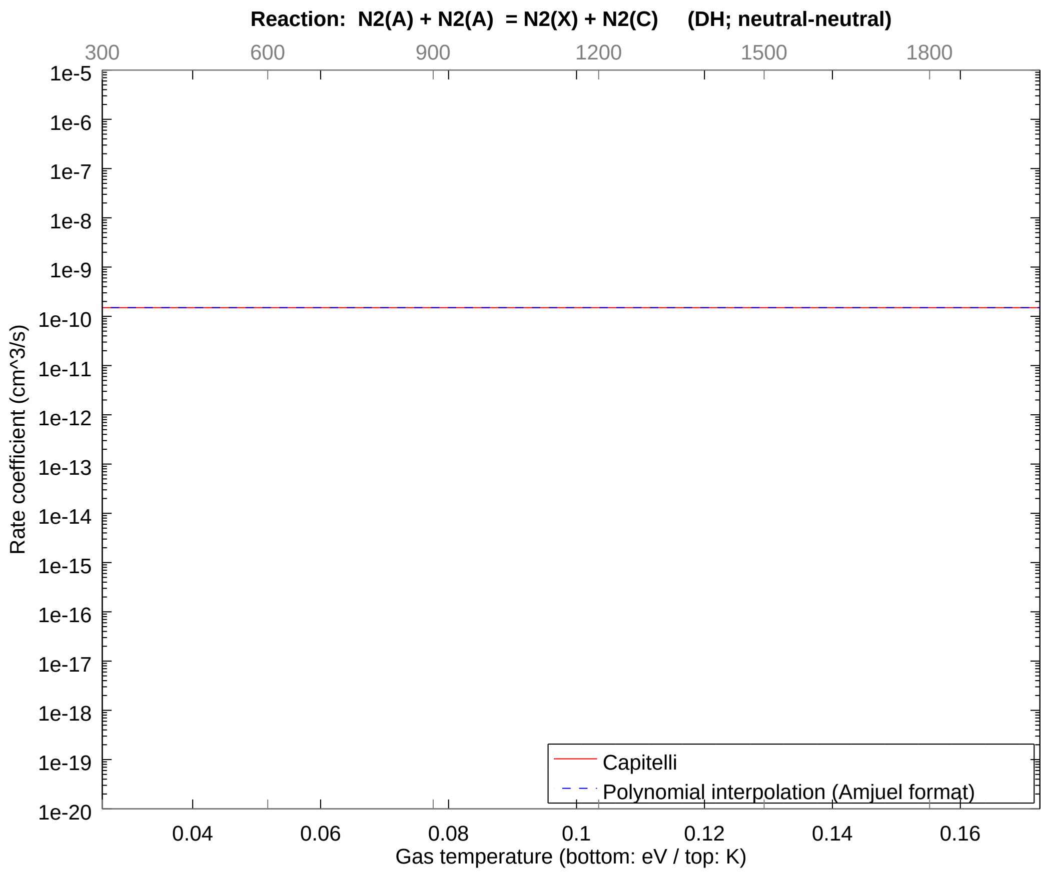
2.153 Reaction DG N2(A) + N2(A) = N2(X) + N2(B)

b0	-2.192723864127E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



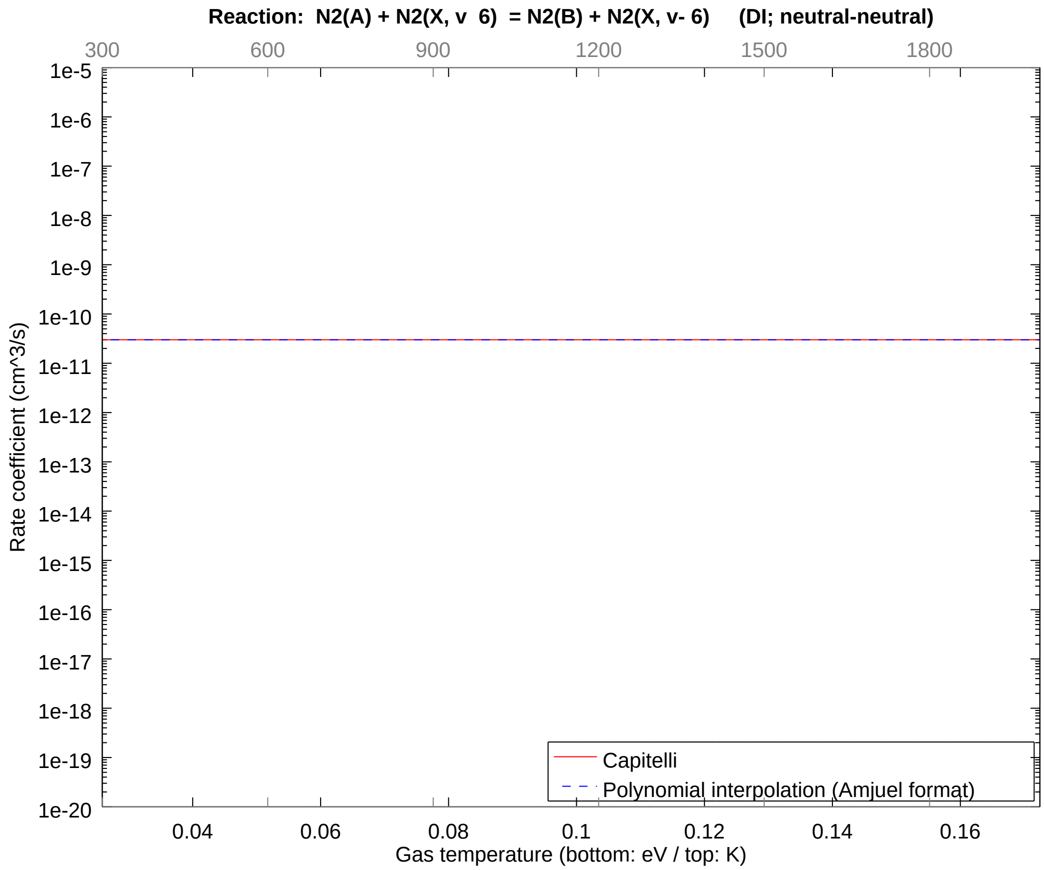
2.154 Reaction DH N2(A) + N2(A) = N2(X) + N2(C)

b0	-2.262038582183E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



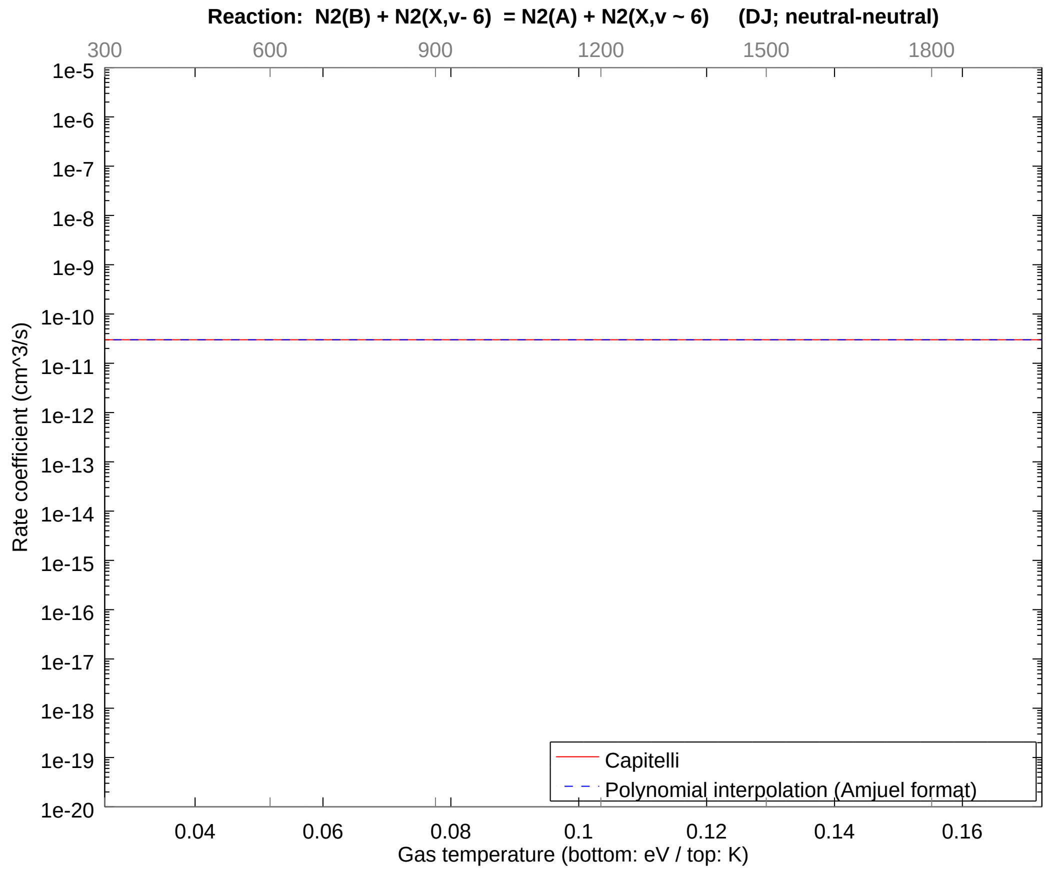
2.155 Reaction DI N2(A) + N2(X, v 6) = N2(B) + N2(X, v- 6)

b0	-2.422982373427E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



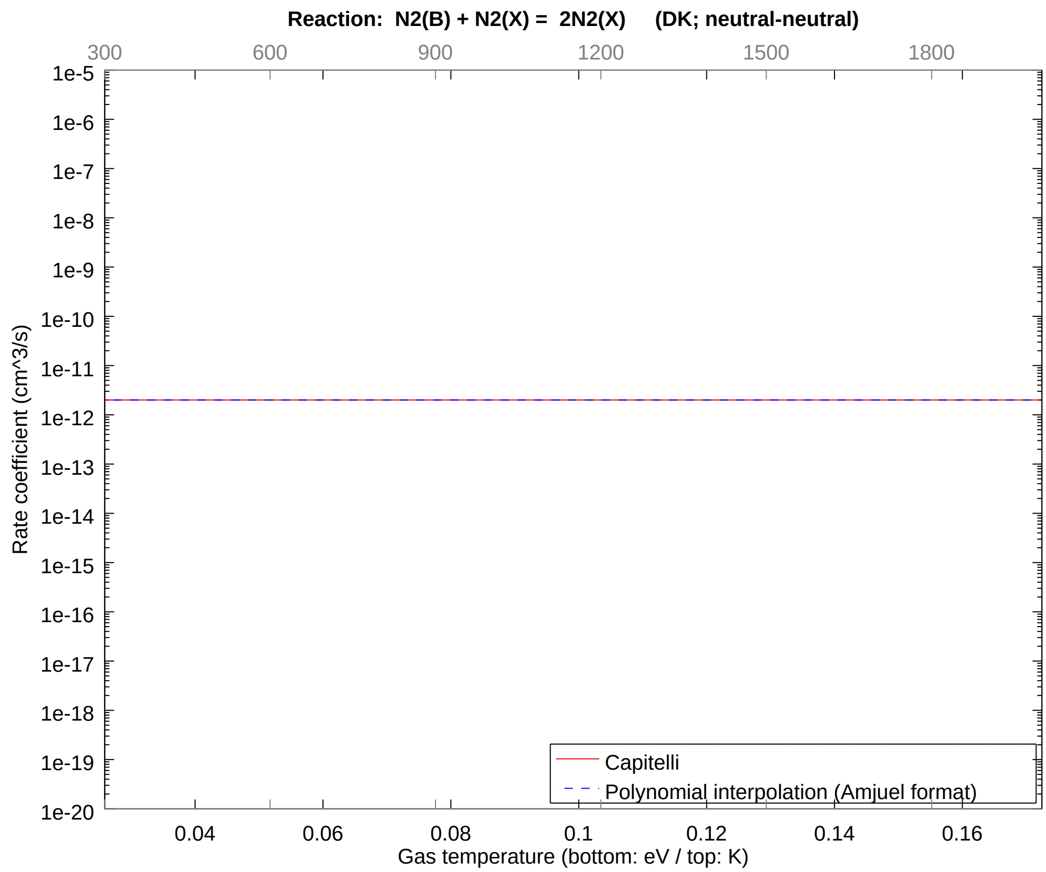
2.156 Reaction DJ N2(B) + N2(X,v- 6) = N2(A) + N2(X,v 6)

b0	-2.422982373427E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



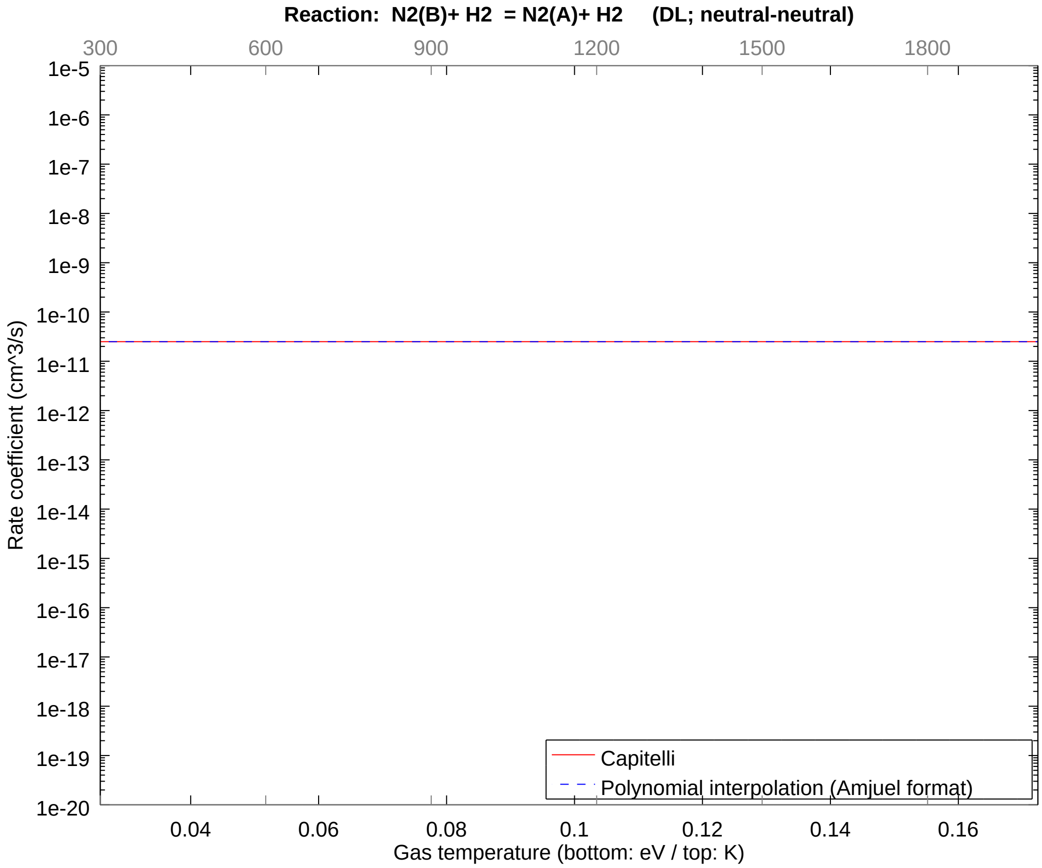
2.157 Reaction DK N2(B) + N2(X) = 2N2(X)

b0	-2.693787393537E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



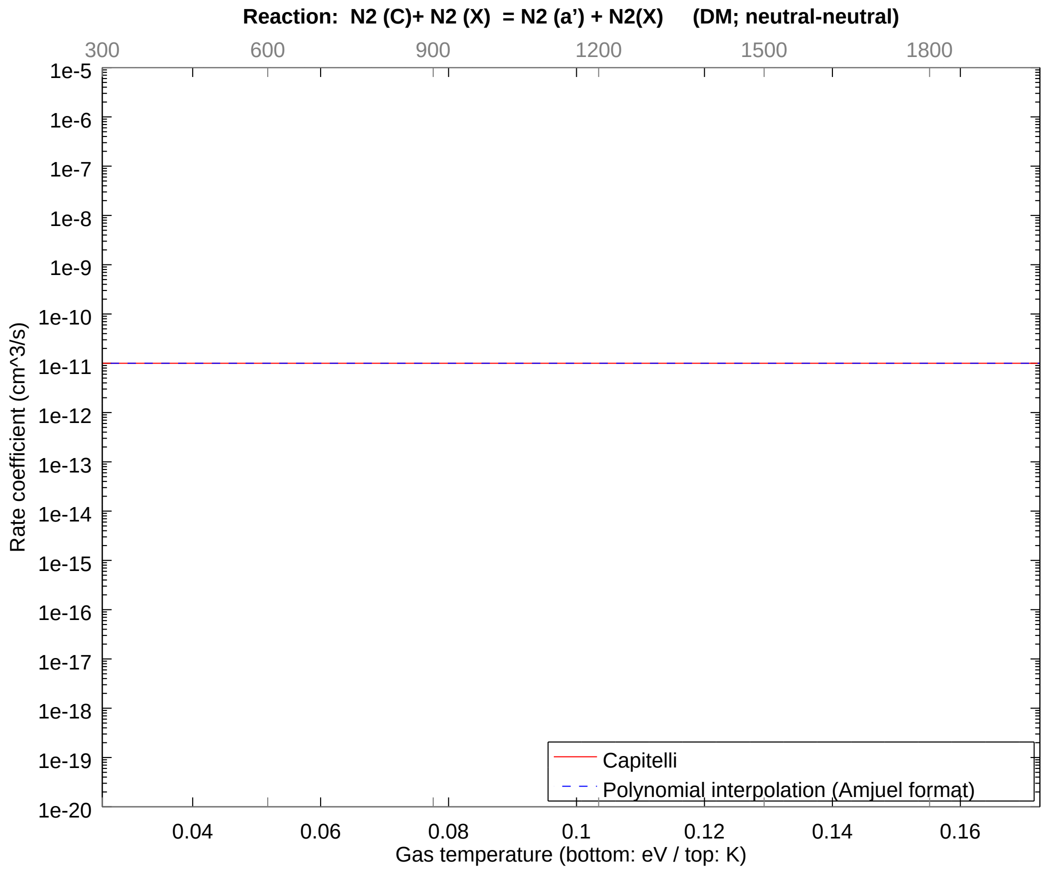
2.158 Reaction DL N2(B)+ H2 = N2(A)+ H2

b0	-2.441214529106E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



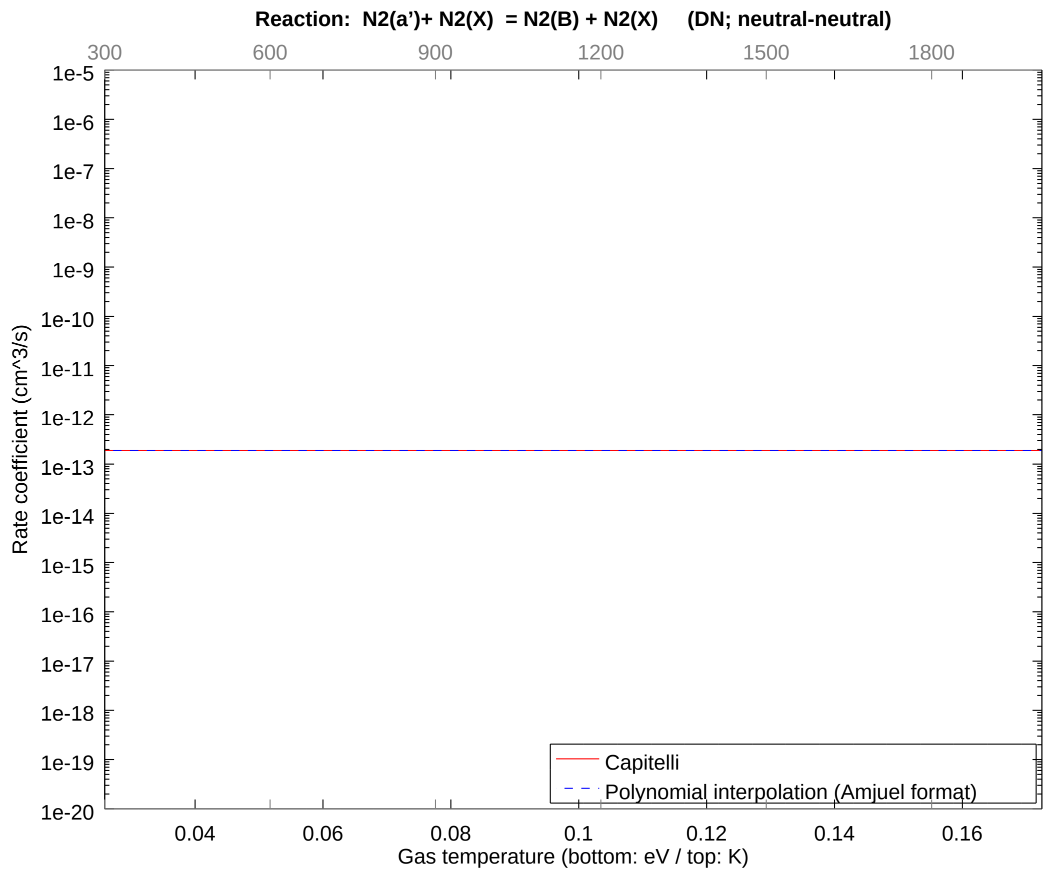
2.159 Reaction DM N2 (C)+ N2 (X) = N2 (a’) + N2(X)

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



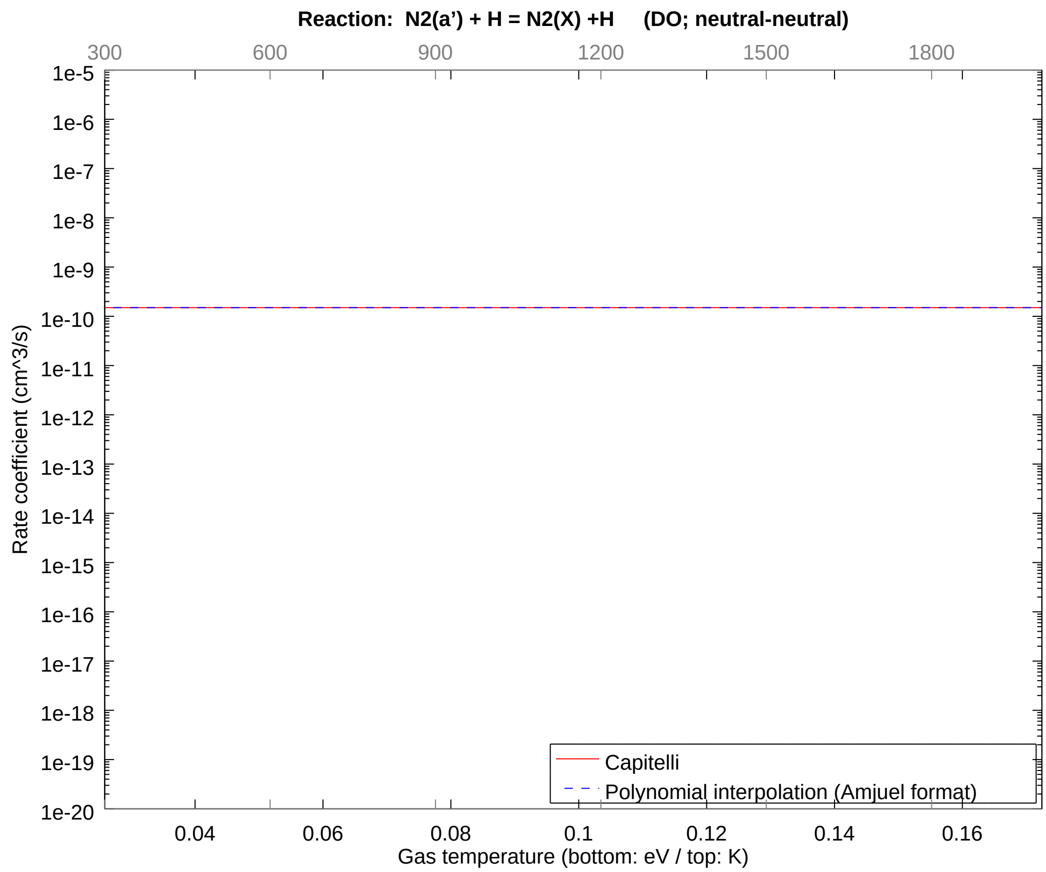
2.160 Reaction DN N2(a')+ N2(X) = N2(B) + N2(X)

b0	-2.929175232275E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



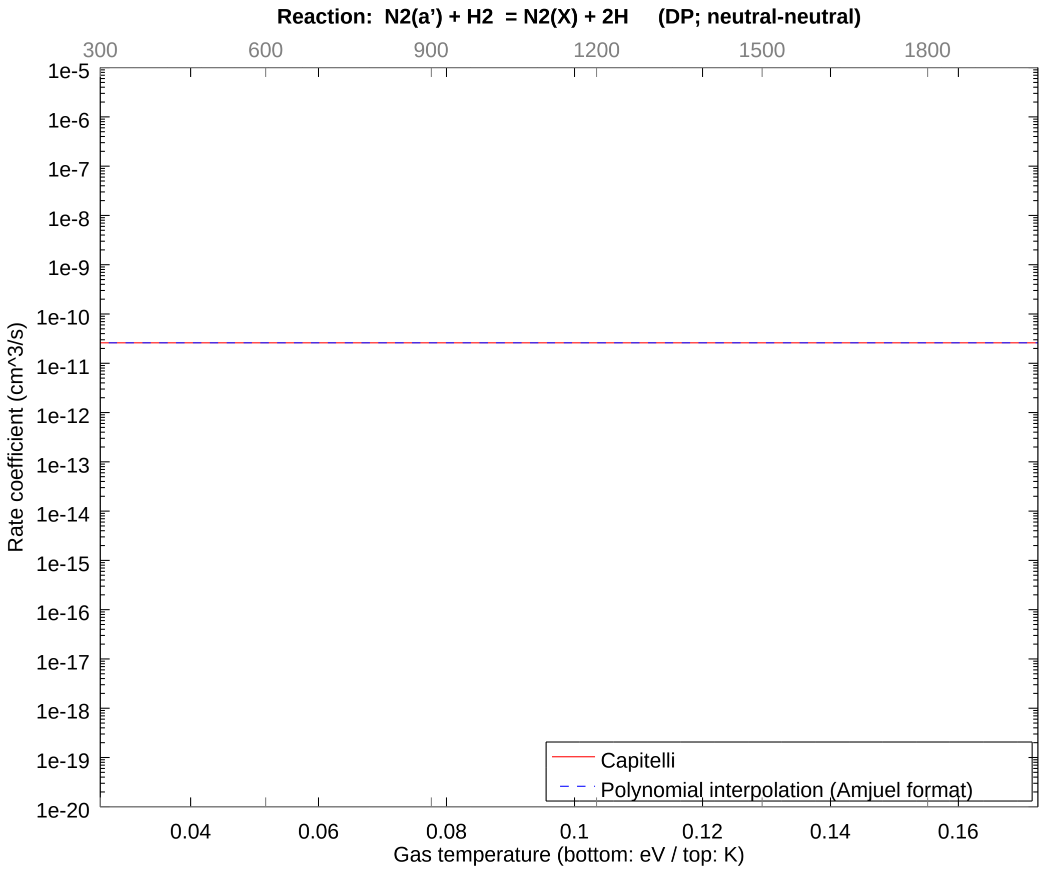
2.161 Reaction DO N2(a') + H = N2(X) +H

b0	-2.262038582183E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



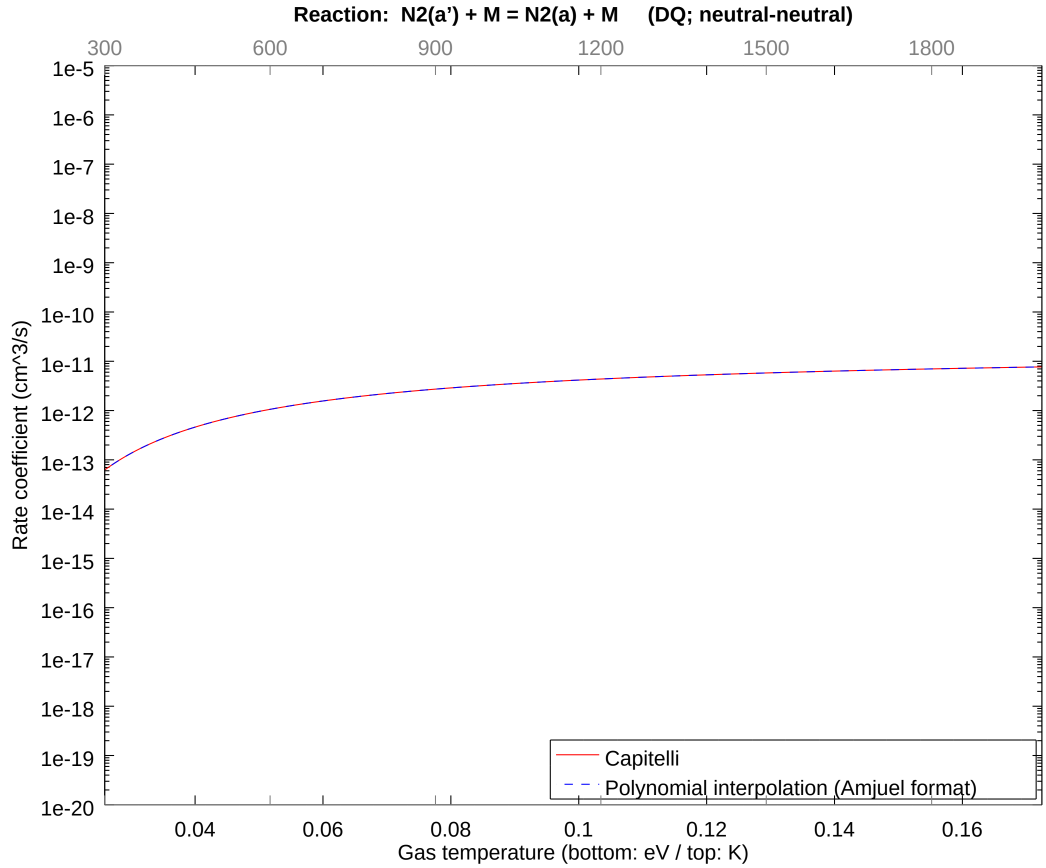
2.162 Reaction DP N2(a') + H2 = N2(X) + 2H

b0	-2.437292457791E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



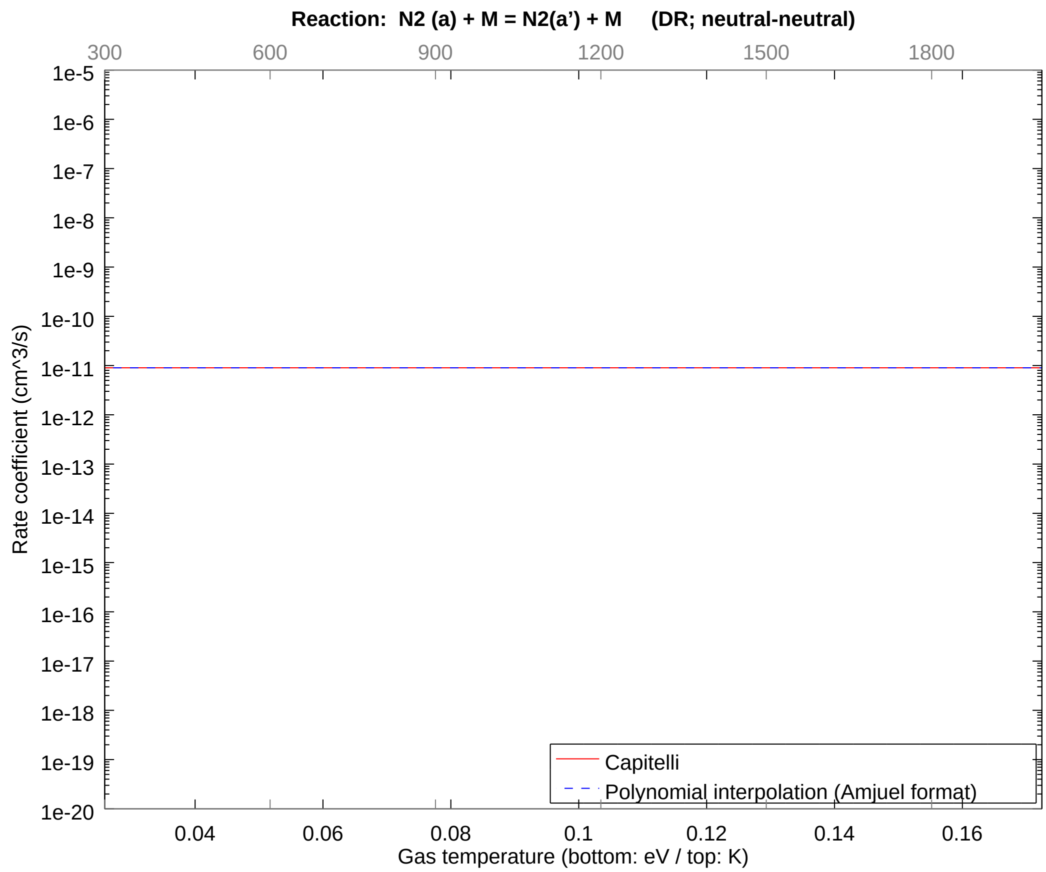
2.163 Reaction DQ N2(a') + M = N2(a) + M

```
b0 -2.491334966794E+01    b1  5.551937465612E-02    b2 -2.123590632336E-01
b3 -9.750820211534E-02    b4 -7.347892909455E-02    b5 -2.293993689667E-02
b6 -5.751763415969E-03    b7 -7.323647887183E-04    b8 -5.447601843037E-05
```



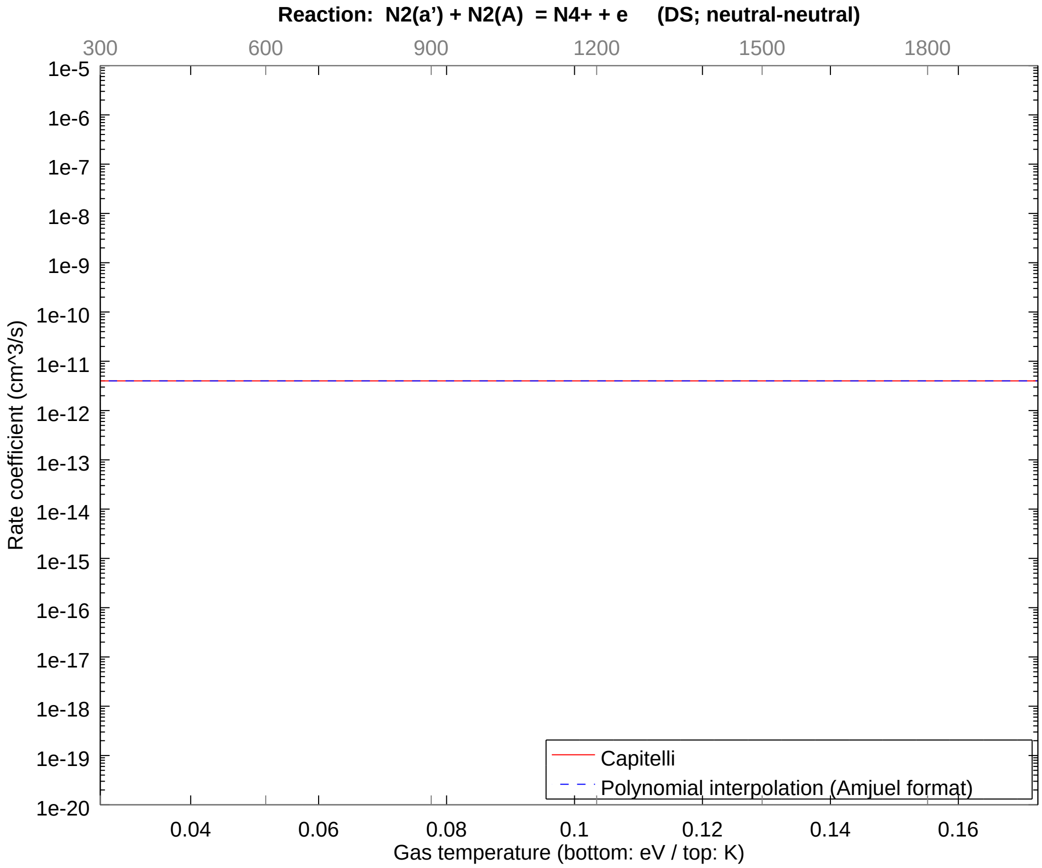
2.164 Reaction DR N2 (a) + M = N2(a') + M

b0	-2.543379653859E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



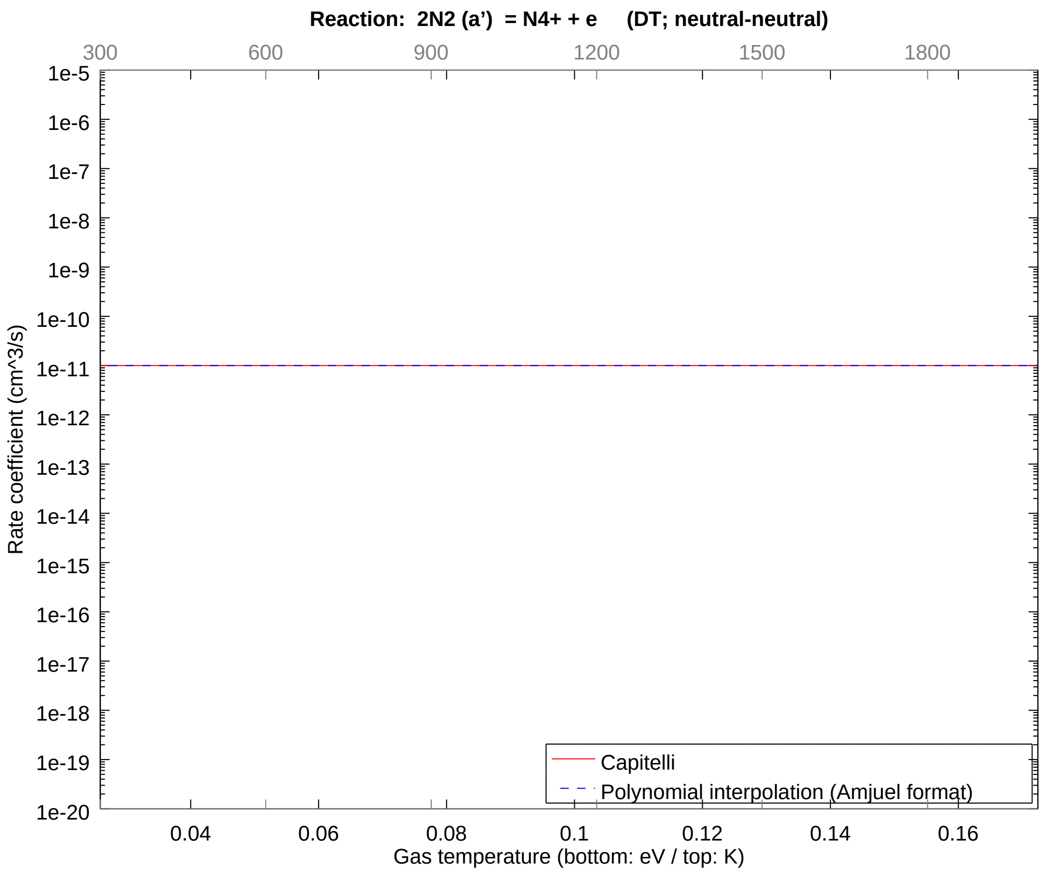
2.165 Reaction DS N2(a') + N2(A) = N4+ + e

b0	-2.624472675481E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



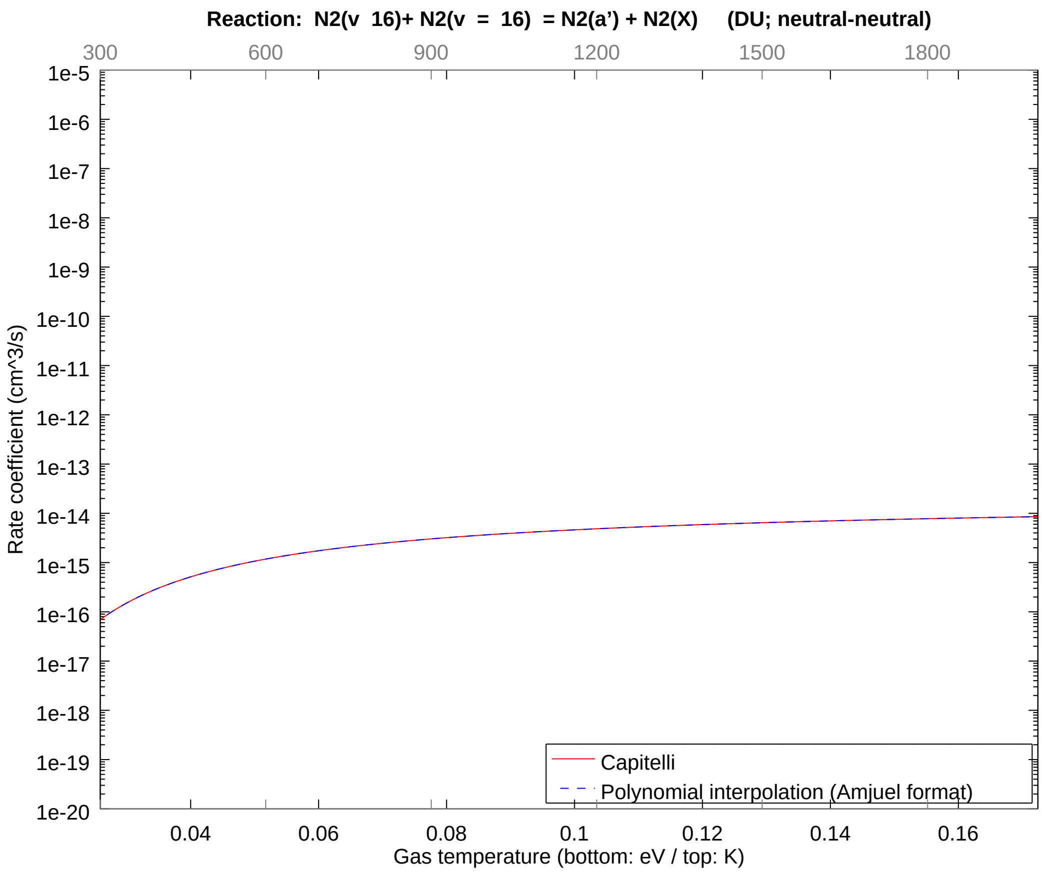
2.166 Reaction DT 2N2 (a') = N4+ + e

b0	-2.532843602293E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



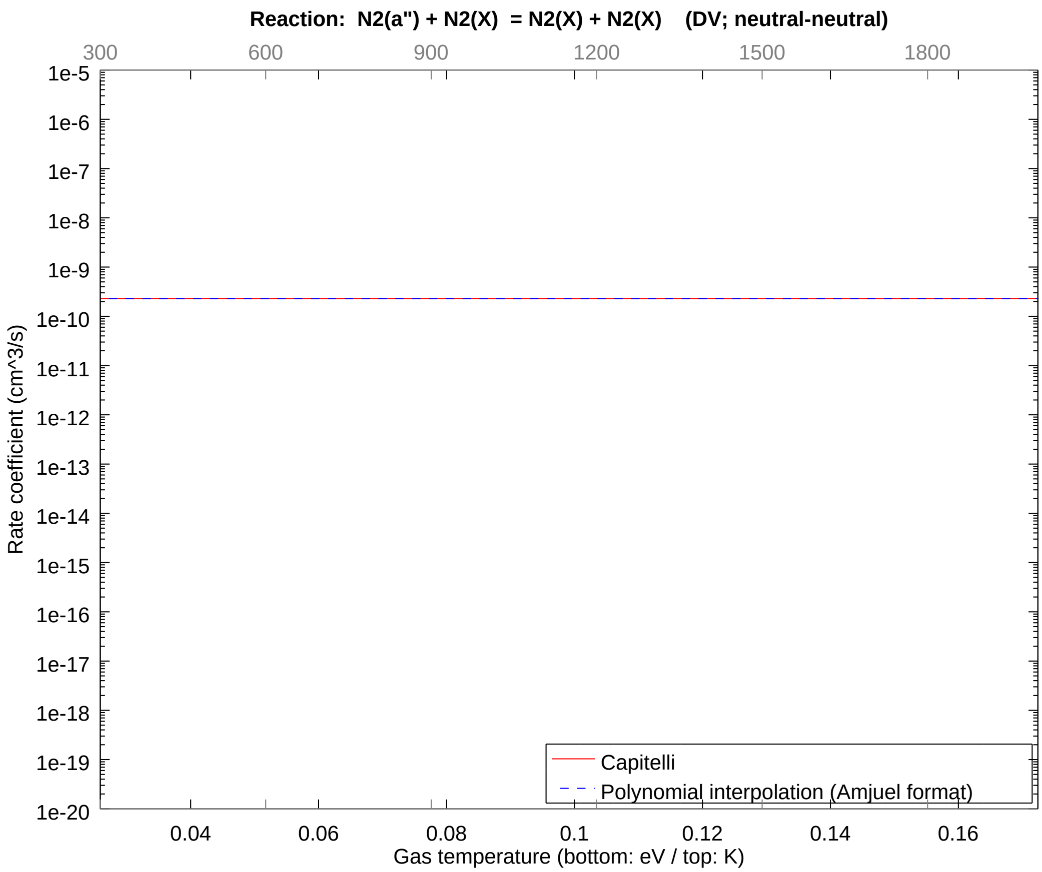
2.167 Reaction DU N2(v 16)+ N2(v = 16) = N2(a') + N2(X)

b0	-3.171574443248E+01	b1	5.551937076721E-02	b2	-2.123590686304E-01
b3	-9.750820635699E-02	b4	-7.347893115968E-02	b5	-2.293993753453E-02
b6	-5.751763538053E-03	b7	-7.323648019616E-04	b8	-5.447601905402E-05



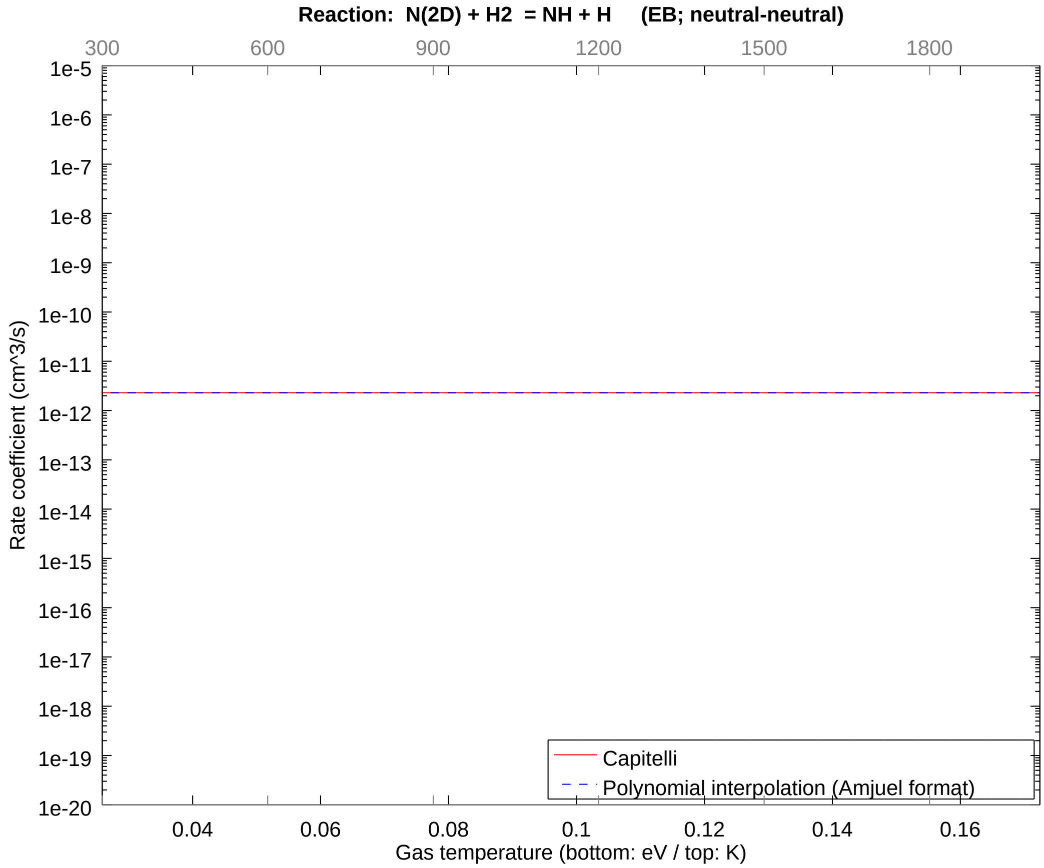
2.168 Reaction DV N2(a'') + N2(X) = N2(X) + N2(X)

b0	-2.219294180701E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



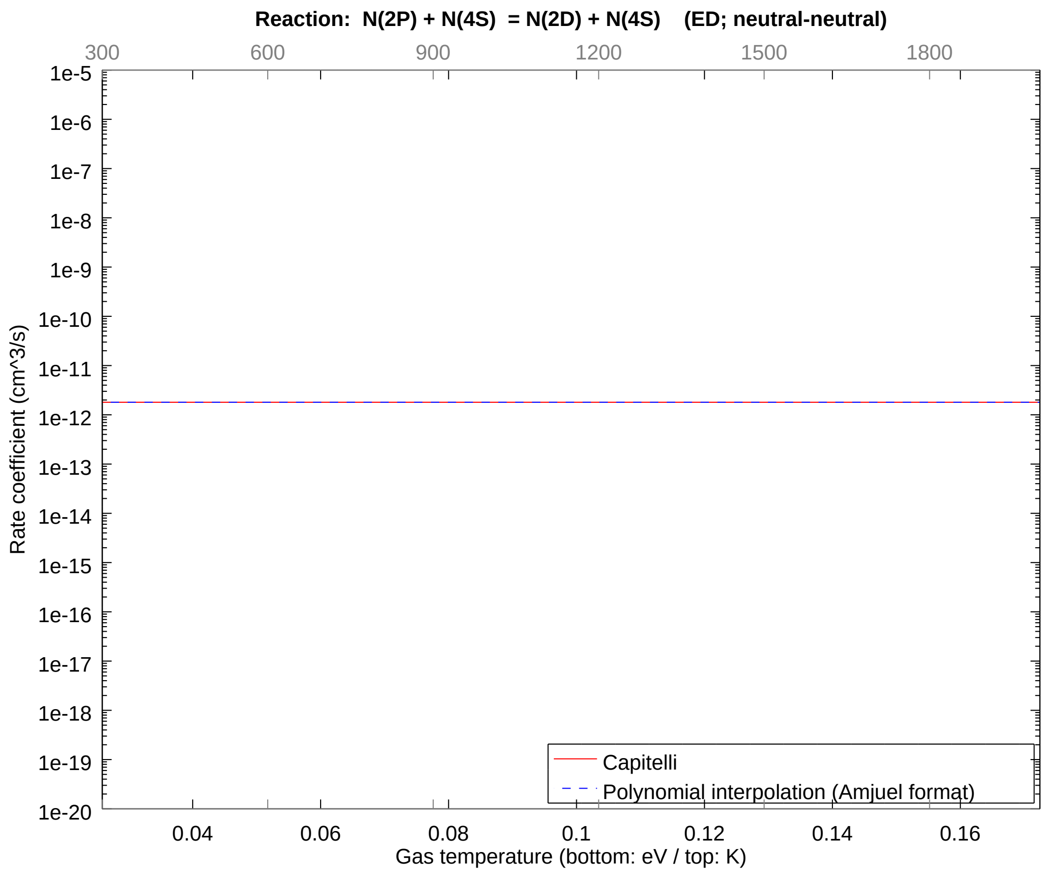
2.169 Reaction EB N(2D) + H2 = NH + H

b0	-2.679811199299E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



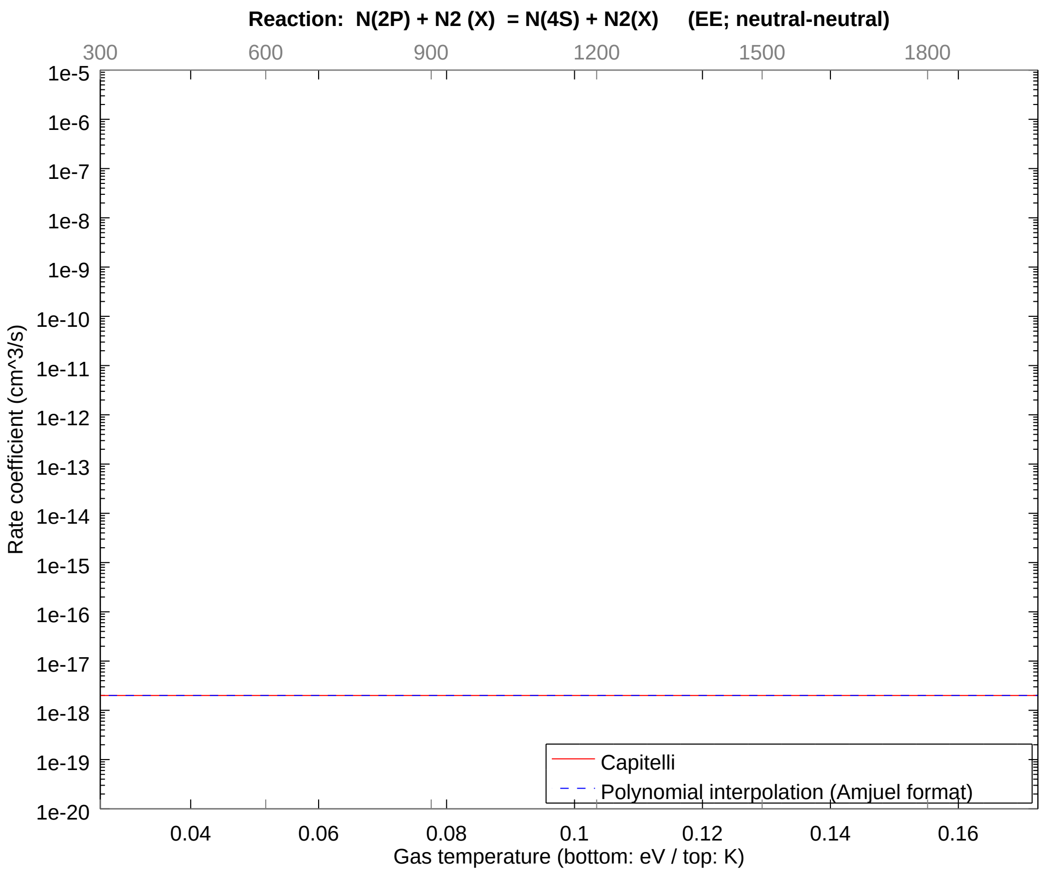
2.170 Reaction ED N(2P) + N(4S) = N(2D) + N(4S)

b0	-2.704323445103E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



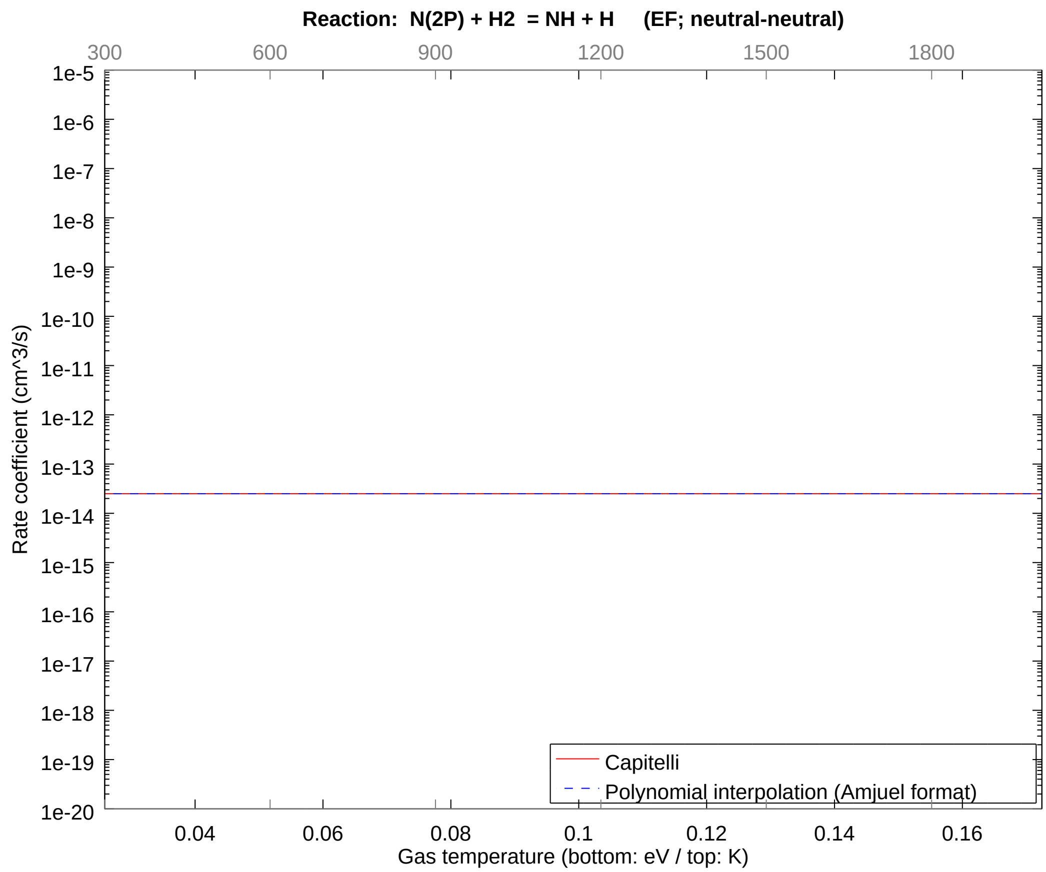
2.171 Reaction EE N(2P) + N2 (X) = N(4S) + N2(X)

b0	-4.075338449333E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



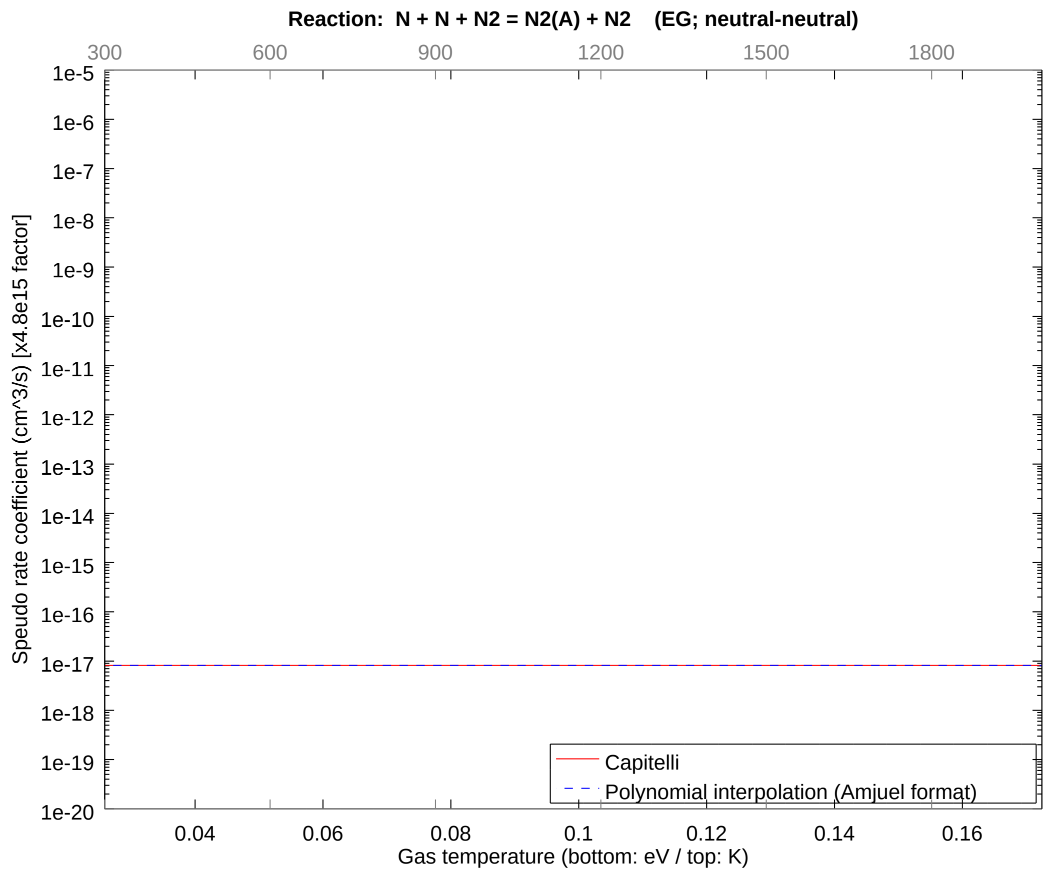
2.172 Reaction EF N(2P) + H2 = NH + H

b0	-3.131990057004E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



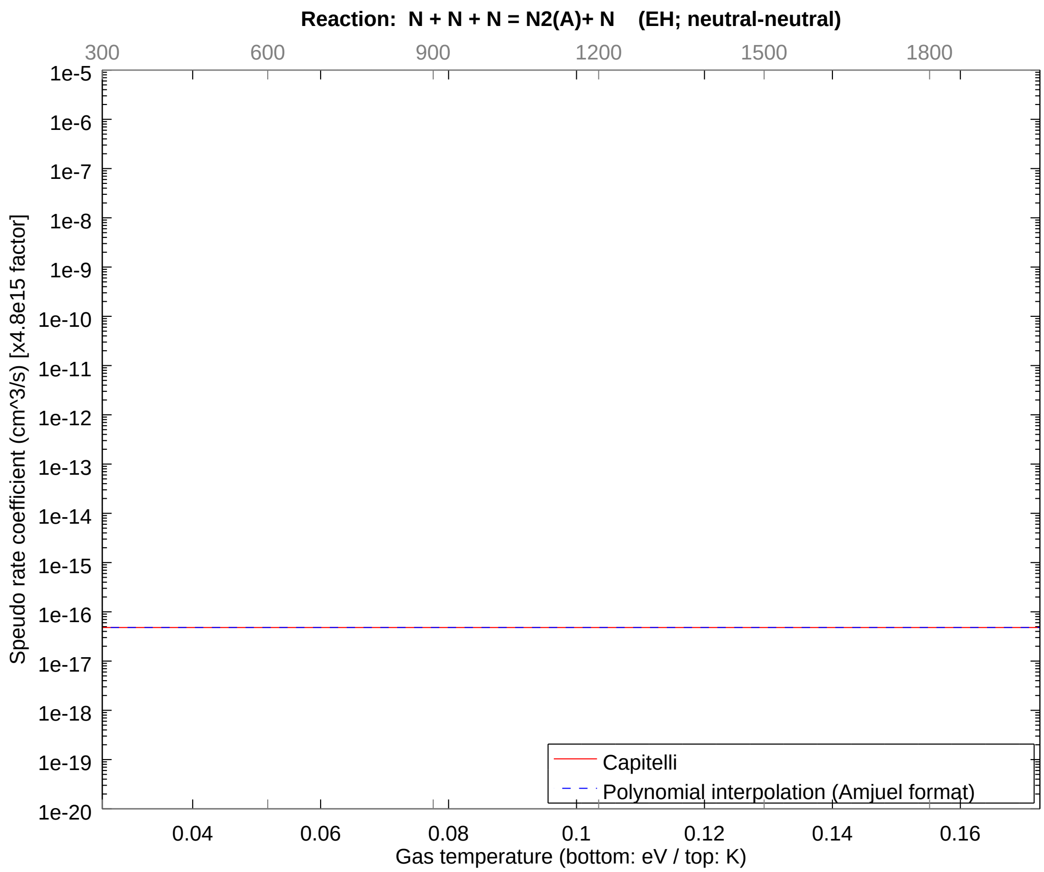
2.173 Reaction EG N + N + N2 = N2(A) + N2

b0	-3.934728750492E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



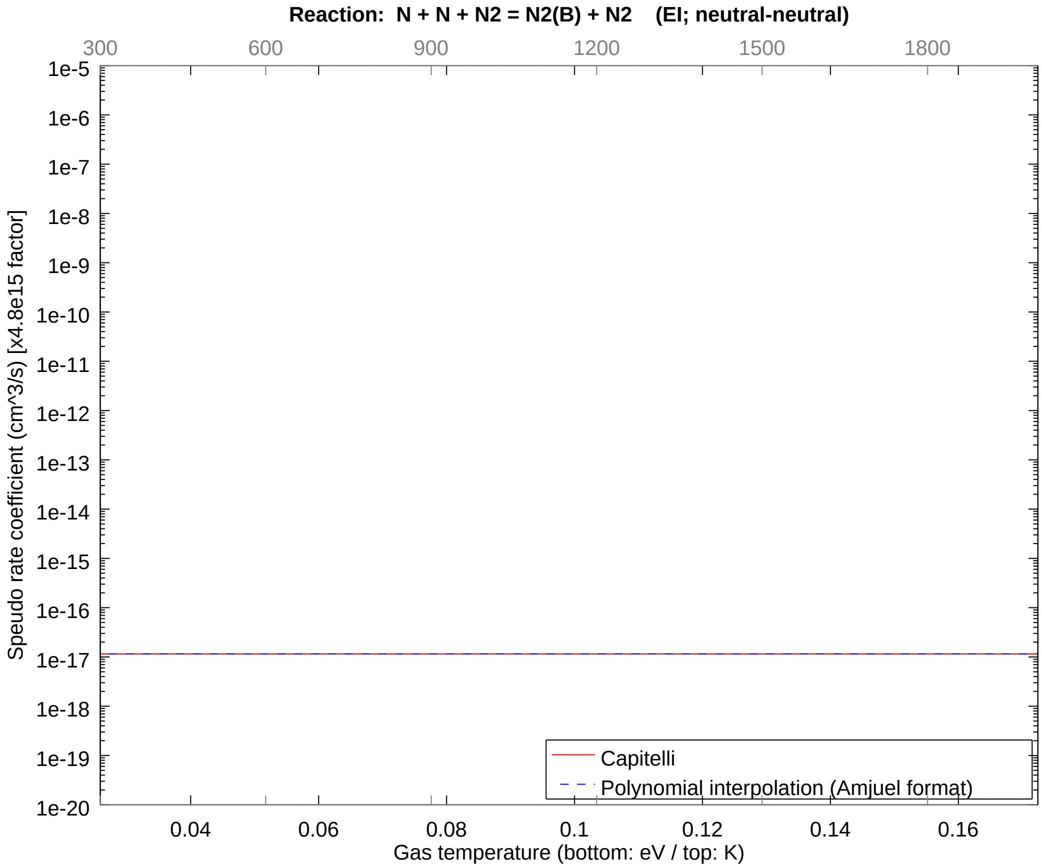
2.174 Reaction EH N + N + N = N2(A)+ N

b0	-3.757533066298E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



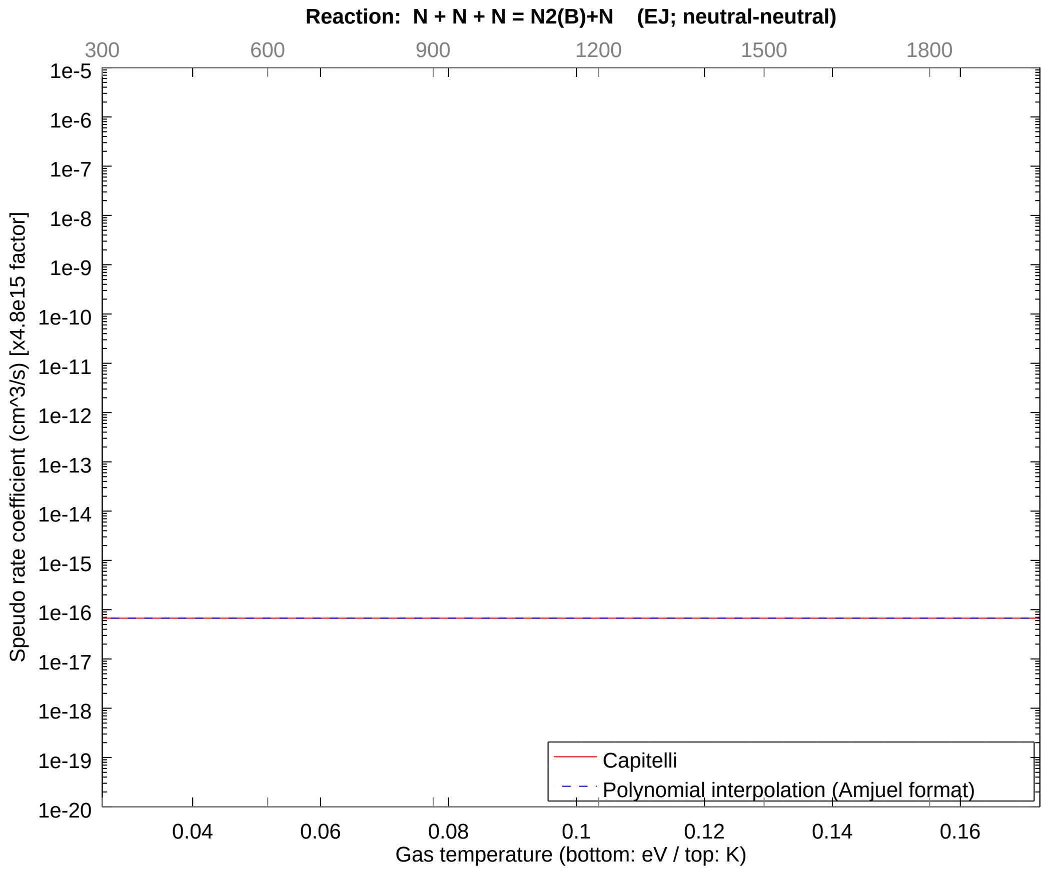
2.175 Reaction EI N + N + N2 = N2(B) + N2

b0	-3.900244701863E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



2.176 Reaction EJ N + N + N = N2(B)+N

b0	-3.723885842636E+01	b1	0.000000000000E+00	b2	0.000000000000E+00
b3	0.000000000000E+00	b4	0.000000000000E+00	b5	0.000000000000E+00
b6	0.000000000000E+00	b7	0.000000000000E+00	b8	0.000000000000E+00



3 Appendix

References

[1] A.B.Ehrhardt, W.D.Langer, PPPL 2477 (1987)

[2] R.K. Janev, B. Langer et al. Springer Series on Atoms+Plasmas, Vol 4, 1987